

Fenced Off: Export Processing Zones and the Local Economy in China

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Abstract

Place-based policies assume that activity concentrated in one place spreads to the economy around it. Export processing zones, the most common special economic zone, fence that activity off. We evaluate China's 60 export processing zones—enclaves inside existing development zones whose exports peaked at US\$130.2 billion—with the census of manufacturers, customs transactions, and business registrations, in an event-study design that tracks each location's exposure to overlapping zones, and with an estimated model of how development zones ration admission. Within 20 km of a zone, new domestic firms fell by 20 percent, concentrated among the smallest projects, even around fences that attracted almost no exports; incumbent employment, wages, and foreign entry did not change. The fence took about 19 percent of the host zone's admission slots and left little: domestic value added was at most 28 cents per exported dollar in the median zone, local suppliers did not expand, nearby exporters shifted into processing that builds no portable capability, and incumbents across the host prefecture's border became more likely to exit. The median zone needed about RMB 1,796 million of fenced exports a year for the wages it paid locally to cover the value added it displaced; outside five hubs, 84 percent of zones displaced more than they paid. Unlike development zones that bundle land, incentives, and linkages, a fenced zone's own activity says little about what it does to its host economy.

JEL: R11, R58, O14, O18, F14, F23. **Keywords:** place-based policy, special economic zones, export processing zones, firm entry, local linkages, spatial spillovers.

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1 Introduction

Place-based policies rest on a premise: activity concentrated in one place spreads to the economy around it, through agglomeration, local purchases, higher wages and land values, and the know-how that nearby firms pick up (Glaeser and Gottlieb 2008; Kline and Moretti 2014b; Neumark and Simpson 2015). Special economic zones are the most widespread instrument built on that premise—about 5,400 of them operated in more than 140 economies in 2019 (UNCTAD 2019)—and the most common type, the export processing zone (EPZ), is also the one that most directly contradicts it. An EPZ offers duty-free processing for export behind a customs fence, and the fence that makes it attractive to foreign processors separates their activity from local firms. At the same time, the land and approvals the zone occupies come out of an allocation that local firms would otherwise use. This paper asks what a fenced zone did to the economy around it, and why it did not work through the channels place-based policy usually relies on.

China’s EPZs make the question answerable at scale. Between 2000 and 2006 the State Council approved almost all of the 60 zones, each a fenced area carved out of an existing development zone, at a time when processing trade made up 53 percent of China’s exports. By 2006, 588 fenced firms—almost all foreign-invested, all processing imported materials—exported US\$53.7 billion, 5.6 percent of the country’s exports; fenced exports peaked at US\$130.2 billion in 2010. Because customs records identify the firms legally inside each fence, the zone’s own activity can be separated from the local firms around it, so that what the zone took and what it left can be measured separately.

We combine the census of above-scale manufacturing firms, geocoded to 0.1-degree cells, with the universe of customs transactions and business registrations. Two features of the setting shape the design. EPZs were built inside existing development zones and clustered in a few cities, so a location near one EPZ was usually near several special zones; we compare locations whose exposure to other zones evolves in the same way, stratifying on each location’s initial exposure and controlling for zones that open farther away, and show what a binary treatment would get wrong. And the zones differ from other development zones in specific rules; we write down a model of a zone that occupies admission capacity and restricts local transactions, derive its predictions against the usual explanations of crowding out, and estimate it.

Five findings. First, EPZs lowered local entry and left everything else in the local manufacturing economy unchanged. New domestic firms within 20 km fell by 20 percent (95 percent interval -36 to 1 with prefecture clustering, -34 to -3 with zone clustering), while

foreign entry, incumbents' employment, and total output did not move, and a modest decline in incumbents' output can be neither confirmed nor ruled out. The loss is not an artifact of the design. Entry near the zones had been rising before they opened, but the two comparison groups we use trend in opposite directions before opening and show the same decline after it, and extrapolating the main design's own pre-opening trend makes the decline larger rather than smaller. Treating each EPZ as a one-time binary event gives a similar loss of domestic entry but would misread it as a loss of all new firms, foreign ones included. And the loss is local: it is confined to roughly the first 20–40 km, while beyond 40 km the point estimates turn positive (Section 5).

Second, entry fell because the fence took admission capacity, not because it competed for workers or land. The loss fell on the smallest projects—entry in the smallest birth-size quintile fell by 44 percent while the largest quintile was unaffected—was as large around fences that attracted almost no exports as around busy ones, began when the fence opened rather than when it was approved, and came with no rise in wages. The lost entrants did not reappear within 40 km or inside the fence, and legal entities rarely relocate across prefecture borders (Section 6). Estimating the admission model on these patterns, the fence took 19 percent of the host zone's slots, and the queue followed the scale of a project's operation more closely than its recorded investment: a model in which it followed fixed investment fails a test it was not fitted to (Section 10.1).

Third, the fence left little, and none of the usual channels of a place-based policy opened. Domestic value added was at most 28 percent of fenced exports in the median zone, and local wages about 6 percent. Five independent designs find no productivity or agglomeration spillover to nearby firms; upstream industries did not expand, and no new local suppliers formed beyond logistics and trade agents; registrations of nontradable businesses rose by an imprecise amount; and allowing fenced firms to sell components locally in 2005 produced no response (Section 7).

Fourth, the zones reached nearby exporters, but in a form that builds no portable capability. Exporters across the host city—domestic ones at least as much as foreign ones—shifted into consigned processing, in which a foreign principal owns the goods and keeps the customer; for domestic firms, experience of that kind carries no predictive value for later entry into similar markets, unlike experience with imported materials, and the stock of portable export capability around the zones did not rise (Section 8).

Fifth, the zones' costs travelled across administrative borders but not along economic links. At 40–80 km, incumbents in a neighboring prefecture were 8.9 percentage points less likely to survive than incumbents at the same distance in the host prefecture, which controls land, credit, and admission; industry, input–output, political, and firm-network proximity to

the zone predict no spillover once placebo timing is imposed (Section 9).

What the zones traded. The estimated model values what the fence took. The projects kept out were small—worth about 43 percent of an average entrant—but each slot the fence occupied was worth about RMB 29.4 million of value added a year to the projects it would otherwise have admitted. For the median zone, what the fence left in local wages covered what it took only if its fenced exports exceeded about RMB 1,796 million a year. Five hubs produced 84 percent of fenced exports; outside them, 84 percent of zones displaced more value added than they paid to local workers. Siting the fences on land outside the development zones’ allocation would have raised local value added by about RMB 3,025 million a year across the 24 zones we can value, and approving only zones expected to reach break-even would have raised it by RMB 1,981 million (Section 10). City-level designs common in the literature find that zones raise foreign investment; we reproduce that result, show that it reflects reported investment, and find that manufacturing output in EPZ cities outside the hubs changed by -0.048 log points.

These results contribute to four literatures. Evaluations of place-based policies find that designation raises local activity when land, incentives, and investment arrive together (Busso et al. 2013; Kline and Moretti 2014a; Criscuolo et al. 2019), including in China (Wang 2013; Alder et al. 2016; Lu et al. 2019; Zheng et al. 2017); the EPZ separates what a zone takes from what it leaves, and shows that a zone can lower local entry through the admission capacity it occupies, a cost borne by small firms that never appear in a zone’s statistics. Work on special economic zones documents their spread and uneven performance (Farole 2011; Frick et al. 2019; Görg and Mulyukova 2024); we measure, zone by zone, the scale a fenced zone needs to add to its host economy. Work on foreign direct investment finds large gains for local suppliers where multinationals source locally (Javorcik 2004; Alfaro-Ureña et al. 2022); we measure the supplier and learning responses when rules restrict sourcing and control of transactions, and relate the portability of export experience (Morales et al. 2019; Bai et al. 2021) to who controls the transaction. Methodologically, when place-based treatments overlap, a binary event design estimates a weighted average that can have the wrong sign for some outcomes; building on work on spatial spillovers and overlapping exposure (Butts 2023; Borusyak and Hull 2023; Clarke 2017), we show how much a design that tracks exposure changes the answer and quantify how much of the comparison group the spillovers contaminate.

Section 2 describes the zones and how they differ from other development zones. Section 3 presents and estimates the model. Section 4 describes the data and design, Section 5 the effects on the local economy, and Sections 6–9 what the fence took, what it left, what nearby exporters learned, and where the effects travelled. Section 10 adds up, and Section 11

concludes.

2 What EPZs Are and How They Differ from Other Place-Based Policies

The place-based literature leads one to expect an export processing zone to work like the development zones studied by Wang (2013), Alder et al. (2016), and Lu et al. (2019): the government designates land, offers tax breaks and infrastructure, and attracts firms, and those firms hire, buy, sell, and are imitated nearby, so that the zone’s gains show up in the surrounding economy. If that is all an EPZ is, its local effects should resemble a development zone’s and the type of zone should not matter. This section shows that the EPZ differs from the development zones around it in three features, each with a prediction the paper tests. The fence took admission capacity from the development zone that hosted it (tested in Section 6); its customs rules cut the transactions through which zone activity usually reaches local firms (Sections 7 and 8); and zones were clustered, so that locations near an EPZ were rarely exposed to one zone only (Sections 4 and 5). We describe the policy in four steps, then compare it with the other zone types a prefecture could host and describe the firms inside the fence.

Why China built them. Export processing zones are the oldest and most common form of special economic zone: most of the roughly 5,400 zones operating in more than 140 economies in 2019 offer duty-free processing for export behind a customs fence (UNCTAD 2019; Farole 2011). By 2000 processing trade—imported inputs assembled for re-export—made up more than half of China’s exports but was scattered across thousands of bonded factories, each separately inspected and bonded, an arrangement that customs found hard to police (Figure 1). The State Council approved the first fifteen EPZs on April 27, 2000, to concentrate that trade inside fences under unified supervision; further approvals in 2001–2003 and 2005–2006, and one each in 2010 and 2011, brought the total to 60 (Figure 2). The fences held a slice of processing trade, not all of it: firms registered inside an EPZ shipped 5.8 percent of China’s exports over 2000–2011 (Appendix L.7).

From approval to operation. A prefecture applied through its province and the State Council approved. The host then built the fence, a customs office dedicated to the zone with round-the-clock supervision, and the infrastructure inside; customs and other central agencies inspected the zone jointly before it began operating, a median of 2 years after approval. The official date of that acceptance is recorded for 36 of the 59 zones; Section 4 explains how we

date the others. Each zone received its own customs district code, so a firm legally inside a fence is identified by its customs registration code rather than by its address (Section 4). The zone administration, an arm of the host government, then recruited firms and admitted them (Appendix L.2).

Where they were placed. The founding customs regulation allowed an EPZ only inside an existing development zone approved by the State Council, one EPZ per development zone, with a planned area of at most 3 km². Of the 60 zones in the directory, 44 sit inside a parent development zone, 13 adjoin one, and 2 stand alone; the sixtieth entry duplicates another zone under the name of an older development zone and is dropped (Appendix P). The median fence covers 2.8 km². The host prefecture supplied the fence’s land, infrastructure, and investment promotion out of the same development-zone allocation—plots, entry approvals, administrative attention—that it would otherwise have given to other firms. This is the first feature: *the fence draws on the host’s admission capacity*, the constraint at the center of the model in Section 3. If capacity is what the fence takes, entry near it should fall even where the fence stays nearly empty, and it should fall among the small projects that queue for the same allocation, without a rise in local wages or land prices (Appendix L.4).

What the fence restricted. Inside the fence, firms imported materials and equipment free of duty and exported without tax, but the founding regulation confined them to export processing: general trade, entrepôt trade, and retail were prohibited, and output could not be sold on the domestic market; the prohibition survived the 2011 revision. Selling components to processing firms outside the fence (“carry-over”) was piloted in two zones in 2002 and extended to all zones in May 2005; subcontracting a stage of production outside required a bond equal to the duties due; and domestic materials bought by zone firms qualified for export rebates only under a 2008 rule, and only for listed products. This is the second feature: *the rules closed or taxed the transactions through which a development zone’s firms usually reach the local economy*—selling to local customers, buying from local suppliers, and subcontracting to local firms. If these rules bind, the fence should leave little local value added, nearby suppliers should not expand, and neighbors should not learn to export; and the 2005 extension of carry-over, the only loosening inside the firm-survey window, is a direct test. The reforms that opened the fence came from 2015, when the EPZs were merged into comprehensive bonded zones, which numbered 168 in 2024 and handled about a fifth of China’s foreign trade (General Administration of Customs of China 2025). Appendix L documents each step (Table 63).

How the EPZ differs from other zones. Table 1 sets the EPZ beside the four other zone types a Chinese prefecture could host. National economic and technological development zones (from 1984) and high-technology zones (from 1988) are bundles of tax, land, and approval preferences whose content varies by place and year (Appendix Table 62). They are not customs special supervision zones: no customs line separates their firms from the rest of the economy, so those firms sell at home and buy from and subcontract to local firms without customs restriction. Bonded zones (from 1987) are fenced for bonded warehousing, processing, and trade. Among the types whose rules we can document, the EPZ is the one that both confined its firms to export processing and restricted their dealings with firms outside. Comprehensive bonded zones, which absorbed the EPZs, reopened those dealings step by step: processing orders from firms outside, creditable purchases from domestic suppliers, and domestic sales at selective tariffs. Neither the EPZ rules nor the later admission guidelines condition entry on local sourcing.

Who is inside the fence. If the fence merely relocated the processing firms a development zone would host anyway, the firms inside should look like processing exporters outside it. Table 2 compares fenced exporters with processing-trade and general-trade exporters in the same customs districts. By firm count, 75 percent of fenced exporters are foreign-invested, close to the 81 percent among processing exporters outside and far above the 28 percent among general-trade exporters; weighted by exports, the fence is almost entirely foreign (99 percent in 2000–2006). What sets the fence apart is organization, not ownership: imported-materials processing, in which the zone firm owns inputs and output, is 100 percent of fenced processing exports against 76 percent outside. Fenced firms are not selected on productivity—their mean within-industry TFP rank is 0.452, near the middle of the distribution—and they leave little locally: domestic value added is 0.291 of export value, an upper bound on local sourcing. The wage bill of the fenced firms the firm survey records is 1.1 percent of the median zone’s fenced exports, a lower bound because the survey records only some of the fenced exporters; relative to those firms’ own exports, wages are 6 percent in the median zone (Section 7). The industry mix depends on the weighting: electronics is 87 percent of fenced exports in 2000–2006 but only 21 percent in the median zone, because a few hub zones set the export-weighted share, so aggregate facts in this paper are reported zone by zone and without the hubs (Appendix L.8).

Overlap. The third feature is geographic. EPZs were placed where development zones and other special customs zones already were—17 of the 60, including all 5 export hubs, lie in the small Yangtze River Delta window of Figure 2—so that *a location near an EPZ was*

rarely exposed to one zone only. A design that treats each opening as a one-off event would attribute to the EPZ the effects of zones that arrived before or after it, and its controls would be exposed to zones of their own. Section 4 documents the overlap and builds the design around it; Section 5 shows how much it matters.

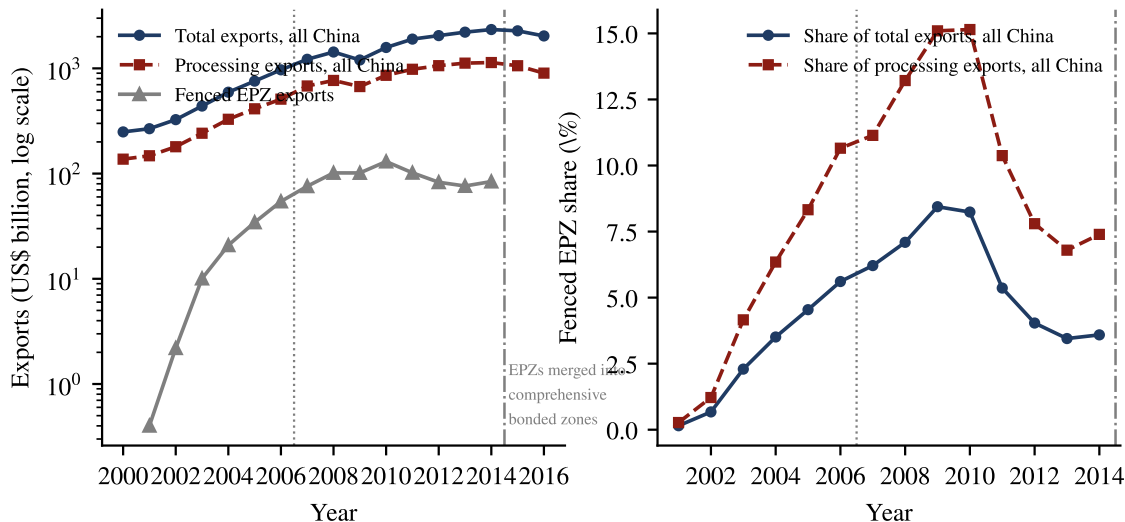


Figure 1: China's exports, processing exports, and exports from inside the EPZ fences

Notes: Customs data. Left: exports in US\$ billion, log scale. Right: fenced exports as a share of national total and processing exports. Fenced exports are exports of firms registered inside an EPZ in each year, read from the fifth digit of that year's customs registration code, through 2014; from 2015 the EPZs were merged into comprehensive bonded zones (dash-dotted line). After 2006 (dotted line) customs no longer separate the two processing regimes, and processing exports are exports other than general trade.

3 A Model of a Fenced Zone

The empirical sections ask what an EPZ took from the local economy and what it left there. This section writes down a model in which both are explicit. It builds on the enclave model of Rodríguez-Clare (1996), in which a foreign enclave helps its host through the local demand it creates and hurts it through the local activity it displaces. We keep the demand side of that model and change what is displaced. In Rodríguez-Clare's model the enclave competes with local firms for workers; in coastal China in the 2000s workers were supplied elastically by migrants, but admission to development zones—serviced plots and the approvals that come with them—was rationed. The model yields three propositions, each tied to a test in the sections that follow, and a set of predictions that separate rationed admission from the explanations usually offered for crowding out (Table 3).

Table 1: Five zone types a Chinese prefecture could host

Zone type (first year)	Customs line	Sales to the domestic market	Transactions with firms outside	Siting and admission
Economic and technological development zone (1984)	None: not a customs special supervision zone	Unrestricted (no customs line)	Unrestricted (no customs line)	—
High-technology development zone (1988)	None: not a customs special supervision zone	Unrestricted (no customs line)	Unrestricted (no customs line)	—
Bonded zone (1987)	Yes: bonded warehousing, processing, and trade	—	—	—
Export processing zone (2000)	Yes: dedicated customs office, round-the-clock supervision; export processing only	No: general trade and retail prohibited; selective-tariff pilots in a few zones from 2016	Restricted: carry-over sales to outside processors from 2005 (two pilots in 2002); outward processing only against a bond; rebates on listed domestic inputs from 2008	Inside a State-Council-approved development zone, one per zone, up to 3 km ² ; export processors and the warehousing and transport firms serving them
Comprehensive bonded zone (2006)	Yes: bonded processing, logistics, research, testing, and repair	Yes, paying duties on entry; selective tariffs in all zones from 2020	Processing orders from outside firms (2015, 2019); creditable purchases from domestic suppliers (2016 pilot)	The only type for new zones since 2012; EPZs converted from 2015; projects screened on input sources and output destinations

Notes: National-level zones, all approved by the State Council (development and high-technology zones also exist at the provincial level). First year: earliest approval year of a zone of that type in the official Catalogue of China’s Development Zones; for comprehensive bonded zones, the Suzhou Industrial Park pilot (December 2006). “Unrestricted (no customs line)”: development and high-technology zones are not customs special supervision zones, so no customs rule limits these transactions. “—”: no rule we could document from primary sources. No documented admission rule conditions entry on local sourcing. Approval and main preferences: Appendix Table 62; sources and detail: Appendix L and Table 63.

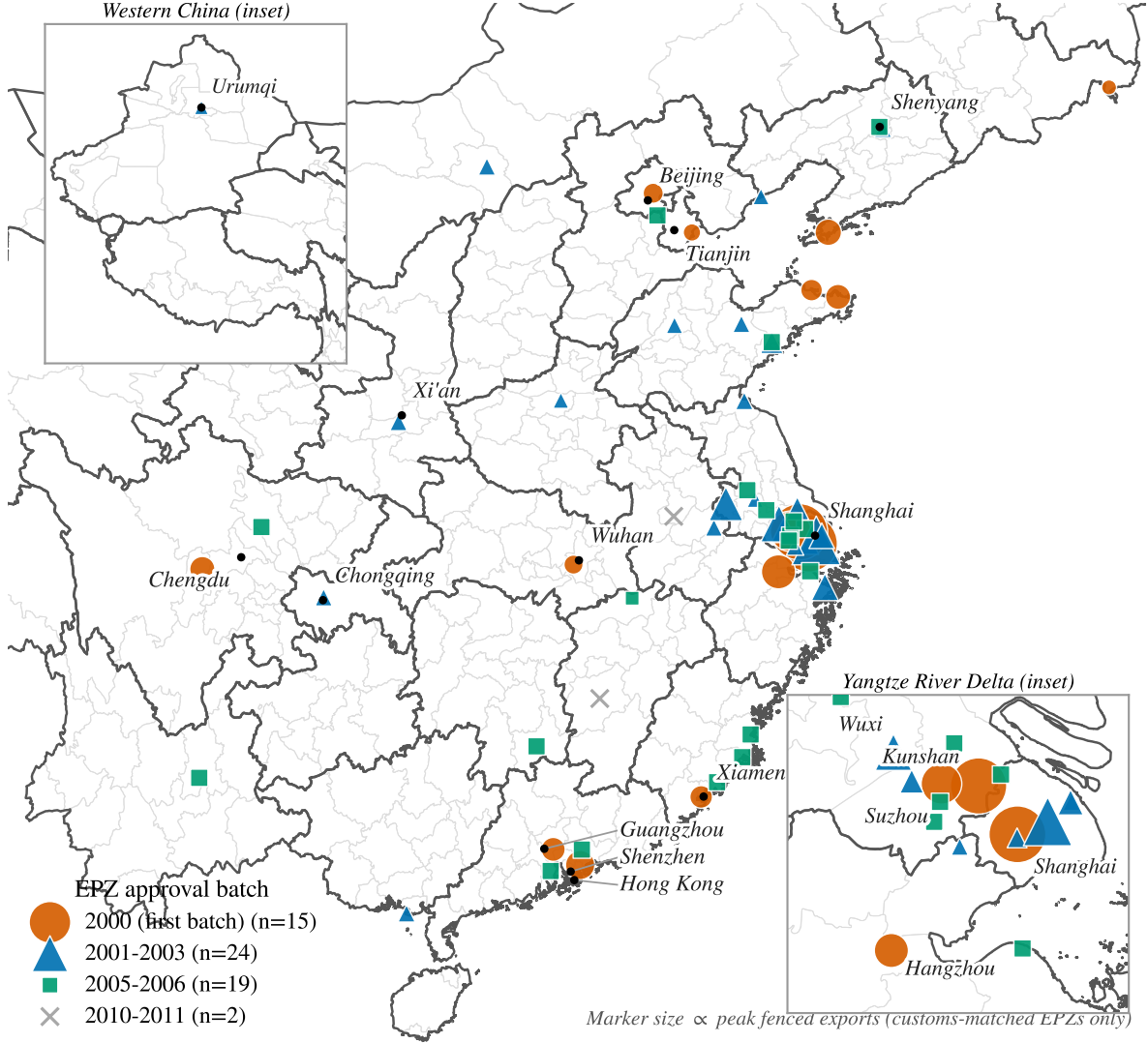


Figure 2: Export processing zones by approval batch

Notes: The 60 EPZs by approval batch (2000, 2001–2003, 2005–2006, and two approved in 2010–2011), with province and prefecture boundaries and major cities for reference. Marker size is proportional to a zone’s peak fenced exports for the 33 zones matched to customs records. Insets enlarge western China and the Yangtze River Delta, which holds 17 zones and all 5 export hubs.

3.1 Environment

Applicants. A host prefecture has a development zone to which N domestic manufacturing projects apply each year. The zone ranks applicants by a priority index x , drawn from a distribution F with density f . The index summarizes whatever moves a project up the queue; it rises with the project’s scale, so higher-ranked projects are larger on average. An admitted project with index x survives to age a with probability S_a and then produces value added $v_a(x)$, increasing in x . A refused project locates elsewhere in the prefecture with probability $\lambda \in [0, 1)$ and otherwise does not enter the prefecture. Whether it is then founded somewhere

Table 2: Who is inside the fence

	Fenced exporters	Outside-fence, processing trade	Outside-fence, general trade
Foreign-invested (% of firms)	75	81	28
Main export in HS 84, 85, or 90 (% of firms)	55	31	26
Imported-materials share of processing exports, 2000–2006 (%)	100	76	–
Median export value (US\$10,000)	41.6	38.6	14.9
Within-industry-year TFP percentile, own first four years (OP-ACF, mean)	0.481	0.503	0.533
DVAR, $1 - M/X$ (export-weighted)	0.291	–	–
Wage bill / exports, own first four years (%)	1.1	7.2	18.8

Notes: Customs data, 2000–2011. Fenced exporters: firms whose registration code places them inside an EPZ. The two comparison groups are exporters in the same customs districts that are in no special customs zone, split by whether processing is at least half of their exports. Each firm enters once, in its first year in the data. The first two rows count firms; the imported-materials row is export-weighted and uses firms first observed in 2000–2006, the years in which customs separate the two processing regimes. TFP rank: mean within-industry-year percentile of fenced firms matched to the firm survey, production function estimated by the control-function method. DVAR: one minus processing imports over processing exports of fenced firms, first four operating years, export-weighted across zones. Wage bill over exports: wage bill of the fenced firms matched to the firm survey over all of the zone’s fenced exports, median across zones; it is a lower bound on the fence’s wage share, because unmatched firms’ exports enter the denominator but not their wages. The wage share used in Sections 7 and 10 divides matched firms’ wage bill by their own exports. “–”: not available for the comparison groups.

else in China matters only for the national accounting of Section 10.7, where the share of blocked projects founded elsewhere is denoted ϕ .

Admission capacity. The zone can admit $\bar{K}$ projects a year. A slot is a serviced plot together with the approvals attached to it; the number of slots is fixed in the short run by the zone’s share of the annual construction-land quota and by its capacity to service land. Before the tender, auction, and listing rules of 2007, industrial land was granted mostly by negotiation at administered prices, so excess demand for slots was not cleared by the price of land; zones queued applicants instead. The criterion on paper was investment per unit of land, which the Ministry of Land and Resources codified in 2004 as minimum investment intensities by region and industry (Ministry of Land and Resources 2004). We take that rule as motivation only and do not tie x to any single measure of scale: estimated from the data, the index tracks the scale of a project’s output more closely than its fixed investment at birth (Section 10.1). When applicants outnumber slots, the zone admits every applicant above the threshold $\bar{x}$ defined by

$$N[1 - F(\bar{x})] = \bar{K}. \quad (1)$$

Factor prices. Labor is supplied elastically by migrants at the wage $\bar{w}$, and land is granted at the administered price $\bar{r}$. Neither responds to local demand in the short run.

The fence. An EPZ opens when customs accepts its fence. From then on the fence occupies m_z of the zone's slots, a share $m_z/\bar{K}$ of its capacity—the land for its plants and the approvals that go with it, committed by the zone's plan whether or not firms fill it. The fenced firms export Q_z . Per unit of exports they spend a share s_z on local inputs and pay a share σ_z to local factors; imported inputs enter in bond, and the remainder, π_z , is profit of the foreign owners. The domestic value-added ratio recorded by customs therefore has three parts,

$$\text{DVAR}_z = s_z + \sigma_z + \pi_z. \quad (2)$$

Local suppliers. As in [Rodríguez-Clare \(1996\)](#), local intermediate inputs are differentiated varieties combined with a constant elasticity of substitution, each produced at a fixed cost. With free entry, the number of varieties is proportional to local spending on intermediates, D_c , and each unit of local intermediate demand generates value added μ among local suppliers.

3.2 What the fence takes

Proposition 1 (Opportunity cost). *Suppose capacity binds. When the fence opens, the admission threshold rises from $\bar{x}$ to $\bar{x}'$, with $N[1 - F(\bar{x}')] = \bar{K} - m_z$, and*

- (a) *domestic entry into the prefecture falls by $(1 - \lambda)m_z$, a proportional loss of $\rho = (1 - \lambda)m_z/\bar{K}$, which equals the fence's share of the slots, $m_z/\bar{K}$, when refused projects do not relocate within the prefecture;*
- (b) *the projects lost are those with $x \in [\bar{x}, \bar{x}')$, the lowest-ranked of the projects that would have been admitted; because the index rises with scale, they are mostly small, and the average entrant becomes larger;*
- (c) *the loss depends on m_z and not on the fence's exports Q_z , and it begins when the fence opens;*
- (d) *wages, land prices, incumbent firms, and foreign entry outside the fence are unaffected.*

Panel (a) of Figure 3 illustrates. The fence does not compete with local firms in any market; it takes their place in the queue for plots. The zone keeps admitting its highest-ranked applicants, so the projects that no longer clear the bar are the marginal ones, and these are mostly small. Because the fence's slots are committed when customs accepts it, an empty fence costs the same entry as a busy one. And because neither wages nor land prices clear the queue, nothing in the prices that firms pay reveals the loss. The entry estimates measure ρ ; Section 6 finds that refused projects do not reappear elsewhere in the host prefecture, so λ is close to zero and ρ also measures the fence's slot share.

Two standard explanations of crowding out predict different patterns. If the fenced firms instead bid up local wages or land rents ([Moretti 2010](#)), entry falls in proportion to the fence's

activity, prices rise, incumbents' costs rise as much as entrants', and the loss falls hardest on labor- or land-intensive entrants of every size. If the fence raises the productivity of nearby firms (Greenstone et al. 2010), entry and incumbent output rise. Table 3 lists the predictions of the three explanations for each outcome we measure.

3.3 What the fence leaves

Proposition 2 (Local contribution). *The fence adds to local value added*

$$L_z = (\sigma_z + \mu s_z) Q_z \leq (1 + \mu) DVAR_z Q_z \quad (3)$$

per year, and

- (a) *local suppliers expand by $\mu s_z Q_z$, in proportion to what the fence buys locally, so they expand little when s_z is small, and not at all around an empty fence;*
- (b) *allowing the fence to sell to local firms raises L_z only if local firms can use its products as inputs.*

The bound in equation (3) is what makes the fence's contribution measurable: customs records give $DVAR_z$ for every fenced firm, and the wage bills of fenced firms give σ_z . What the bound cannot reveal is how much of the domestic value added is local purchases rather than foreign profit; Section 7 bounds s_z from the response of local suppliers.

3.4 The net effect

Proposition 3 (Break-even scale). *H years after opening, the zone changes local value added by*

$$\Delta VA_z = (\sigma_z + \mu s_z) Q_z - C(m_z), \quad C(m_z) = (1 - \lambda) \sum_{k=0}^H N \int_{\bar{x}}^{\bar{x}'} S_{H-k} v_{H-k}(x) dF(x), \quad (4)$$

where $C(m_z)$ is the value added the lost cohorts of entrants would have produced. Because the lost projects are the lowest-ranked and the index rises with scale, $C(m_z)$ is below m_z times the value added of an average entrant. The zone raises local value added only if its exports exceed the break-even scale $Q_z^* = C(m_z)/(\sigma_z + \mu s_z)$.

Panel (b) of Figure 3 shows the ledger. The cost is fixed once the fence is closed; the return rises with the fence's exports at the slope $\sigma_z + \mu s_z$. With activity concentrated in a few zones, most zones can fall short of Q_z^* even if the program as a whole exports a great deal. The proposition also says which rules would have changed the ledger: siting the fence

on land that does not come out of $\bar{K}$ sets $C(m_z)$ to zero, and letting the fence buy locally raises the slope.

The model leaves out two margins that the data can speak to: whether nearby firms acquired the fence’s export know-how (Section 8), and whether any of the cost crossed the prefecture’s border (Section 9). Section 10 estimates the admission queue from the effects on entry by size quintile reported in Sections 5–6, and uses it to value what the fence took; it also tests the estimated queue on moments it was not fitted to.

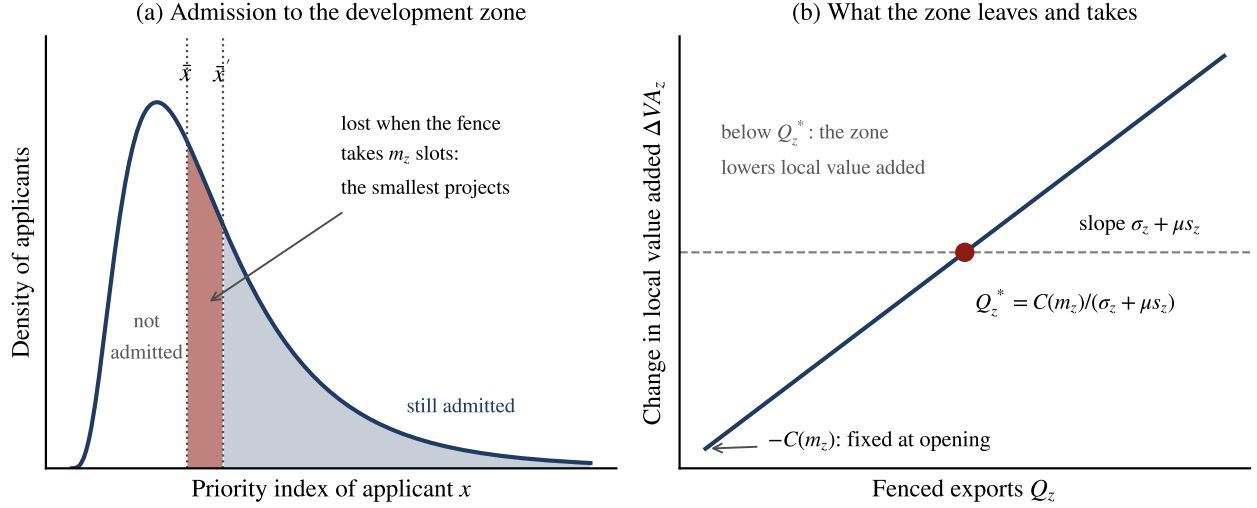


Figure 3: The model in two panels

Notes: Schematic. Panel (a): density of applicants’ priority index x , which rises with project scale. The zone admits applicants above $\bar{x}$; when the fence takes m_z slots the threshold rises to $\bar{x}'$, and the projects in between—the lowest-ranked of those that would have been admitted, mostly small—are lost (Proposition 1). Panel (b): the change in local value added against fenced exports. The cost of the lost entrants is fixed at opening; the fence’s local contribution rises with its exports (Propositions 2 and 3).

4 Data and Research Design

The standard evaluation of a place-based policy compares locations near a zone with locations farther away or not yet treated, before and after the zone opens, and treats the opening as a single event. That design presumes that a treated location is exposed to one zone and a control location to none. Around China’s EPZs neither presumption holds: zones sit inside older development zones, several special customs zones often open within commuting distance of one another, and a location that is a clean control one year may receive a zone of its own the next. A binary design would then attribute to the EPZ the effects of zones that arrived before or after it, and would compare treated locations with controls that were themselves being treated. This section describes the data, documents how common overlap is, and builds

Table 3: Predictions of rationed admission and of two alternative explanations

Outcome near the zone	Rationed admission (this model)	Factor-price competition	Agglomeration spillovers	Tested in
Domestic entry	Falls	Falls	Rises	Section 5
Which entrants are lost	The lowest-ranked, mostly small	Labor- or land-intensive, all sizes	None	Section 6
Scales with fence exports	No; empty fences cost as much	Yes	Yes	Section 6
Wages and land prices	Unchanged	Rise	Wages may rise	Section 6
Incumbent output	Unchanged	Falls	Rises	Section 5
Local suppliers	Expand only in proportion to local purchases: little, and not around empty fences	Fall with fence activity	Expand	Section 7.3
Net local value added	Positive only above Q_z^*	Ambiguous	Positive	Section 10

Notes: Rationed admission: Propositions 1–3. Factor-price competition: the fence raises local wages or land rents in proportion to its demand for them (Moretti 2010). Agglomeration spillovers: the fence raises the productivity of nearby firms (Greenstone et al. 2010). Appendix R derives each entry.

a design that tracks each location’s exposure; Section 5 compares its estimates with those of a binary design.

Data. Three microdata sets cover both sides of the fence. The annual survey of above-scale manufacturers (1998–2007; sales above RMB 5 million, Brandt et al. 2012), geocoded to 0.1-degree grid cells, records entry, exit, output, value added, employment, wages, fixed assets, profits, and subsidies of every firm around a zone. The universe of customs transactions (2000–2011, firm $\times$ product $\times$ destination $\times$ year) identifies the firms legally inside a fence from the fifth digit of their customs registration code and separates imported-materials from consigned processing. The 2002 input–output table links the fence’s industries to their suppliers and users. The zone directory gives the location and approval year of all 142 special customs zones, of which 60 are EPZs; it dates treatment and counts the zones each location is exposed to in each year. An EPZ opens when customs accepts its fence. The official acceptance date is recorded for 36 of the 59 EPZs with an approval date; for the others we set opening to the approval year plus two, the median lag among zones with an official date (Appendix M.11). For those zones the gap between approval and opening is two years

by construction, so only the zones with official dates carry information for comparing the approval and opening clocks. Industry codes are harmonized across the 2003 classification revision, and the 2004 census year is kept except for firms that appear only in that year (Appendix M); Appendix N lists these and six other data traps and how we handle each.

Why grid cells and Poisson. A median EPZ covers 2.8 km², against about 105 km² for a 0.1-degree cell and thousands for a prefecture. Geocoded firm addresses let us measure distance to the fence directly and compare locations within the same opening cohort; a median zone has 8.0 treated cells, enough to estimate the effect within 20 km and to test where it stops. Poisson keeps the many cell-years with no entrant and reports proportional effects that are comparable across outcomes of different scale. Log, inverse-hyperbolic-sine, and level specifications, 10 and 30 km radii, and coarser 0.2-degree cells or counties give the same conclusions (Appendix F).

Overlap is the rule. Three facts show that overlapping exposure is the common case, not an edge case. First, EPZs were placed inside or next to existing development zones (Section 2), so a fence arrives where a development zone is already at work. Second, special customs zones cluster: of the 82 prefectures that host one, 22 host two or more, and 6 of the 45 prefectures that host an EPZ host more than one EPZ. Third, of the 1360 grid cells within 20 km of a special customs zone, 77 percent have at least one more such zone 20 to 150 km away, 6.19 on average—the distances over which spillovers and controls are defined. As a result, 13 of the 60 EPZs opened within 40 km of a special customs zone that was already operating. Figure 4 shows what this looks like around Suzhou, where two national development zones, six EPZs, and the comprehensive bonded zones that later absorbed them lie within a few tens of kilometres of one another.

How the design handles overlap. For every 0.1-degree cell and year we count the special customs zones of any type operating within 40 km (the near ring) and 40 to 150 km away (the far ring). Four rules turn these counts into comparisons that isolate one EPZ’s arrival.

- (i) *First exposure.* A cell is treated in year g if its near-ring count rises in g , an EPZ is among the arrivals, and that EPZ lies within 20 km. In the main sample the EPZ is the first special customs zone within 40 km, so the estimate is the effect of a first fence, not of a second one layered on an earlier zone.
- (ii) *Initial-state stratification.* Each treated cell belongs to a sub-experiment defined by the opening year, the near-ring count and far-ring tier (0, 1–2, 3–5, 6–9, 10 or more zones) in the year before opening, the number of zones arriving in the opening year,

and the length of the post window. Controls come only from the same sub-experiment, so treated and control locations start from the same exposure state.

- (iii) *Control contamination.* A control cell stays in the sample only if its near-ring count does not change during the window, and in the main pool the next zone within 40 km of it opens more than three years after g : controls are places the state also chose for zones, not yet treated.
- (iv) *No dose accumulation.* A treated cell’s post window ends at the first later change in its own near-ring count, so a second zone arriving nearby does not enter the estimated effect of the first.

Far-ring exposure enters through the initial tier; its later year-to-year changes are not a regressor in the main specification, and adding them is one of the checks in Section 5. Controls are then reweighted by the odds of treatment estimated on pre-period firm density, output, foreign and electronics output shares, domestic entry rates, far-ring exposure, distances to the nearest national development zone and port, and an eastern-region indicator (Appendix O.2). The treated unit is thus a grid cell within 20 km of an EPZ once the zone operates, which yields 315 treated cells around 39 EPZs (Poisson estimation drops cells whose outcome is zero in every year, leaving, for example, 279 treated cells for new domestic firms). Table 4 shows why the reweighting matters and that it works: treated cells are denser, more foreign, and closer to national development zones (largest standardized difference 1.20), and after reweighting no difference exceeds 0.13.

Estimating equation. One equation delivers every location-level result in the paper. For outcome y in cell c of sub-experiment s in year t ,

$$E[y_{sct}] = \exp \left(\alpha_{sc} + \delta_{st} + \sum_{e \neq -1} \beta_e T_{sc} \mathbb{1}[t = g_s + e] \right), \quad (5)$$

estimated by Poisson on the stacked sub-experiments with the weights of Wing et al. (2024), so that coefficients are proportional effects, zeros and exits are kept (Wooldridge 2023; Chen and Roth 2024), and cell and sub-experiment-by-year effects absorb level differences and common shocks, including those shared by locations in the same exposure state. Pooling $e = 0, \dots, 3$ gives one coefficient per outcome; standard errors are clustered by prefecture.

Is exposure itself endogenous? Zones went where firms, ports, and older development zones already were, but predetermined geography sorts locations across the design’s exposure states, not within them: the R^2 of far-ring exposure on geography falls from 0.37–0.40 across

strata to 0.02–0.06 within them (Appendix Table 64).

Samples. Of the 60 EPZs, 39 anchor at least one treated cell in the main location sample: 13 drop out because every nearby cell was already exposed to an operating special customs zone, 7 opened after the firm survey ends, and 1 is a duplicate directory entry. The ledger of Section 10 also needs the fence’s customs records, which leaves 24 zones; the cross-border analysis of Section 9 uses 43. Of the 5 export hubs, 5 are in the location sample, 5 in the ledger sample, and 4 in the cross-border sample, so the entry decline and the fence’s exports are measured on overlapping sets of zones. Appendix P lists every zone and why it is in or out of each sample.

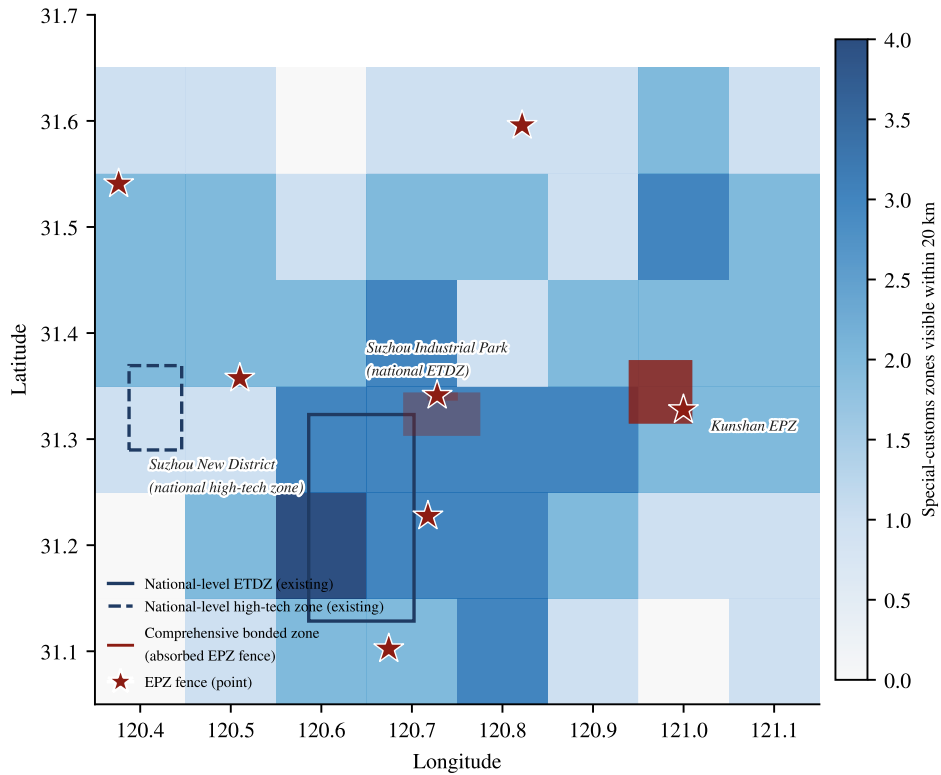


Figure 4: Overlapping exposure around Suzhou

Notes: Suzhou prefecture, including Kunshan. Shading: number of special customs zones of any type within 20 km of each 0.1-degree cell, counting every zone in the directory regardless of its opening year. Solid and dashed outlines: the national-level economic and technological development zone and high-technology zone that existed before the EPZs. Filled boxes: the comprehensive bonded zones that later absorbed EPZ fences. Stars: EPZ locations (no boundary exists for the original fences). The national facts in the text use the same point locations of all special customs zones.

Table 4: Treated and control locations before opening

Pre-period characteristic	Treated	Control	Std. diff. (raw)	Std. diff. (reweighted)
Number of firms (log)	3.134	1.724	0.846	0.133
Output (log)	14.153	12.270	0.803	0.104
Foreign-invested output share	0.290	0.156	0.470	0.063
Electronics output share	0.048	0.016	0.297	0.068
Domestic entry rate (per firm-year)	0.038	0.054	-0.297	-0.014
Distance to national development zone, km (log)	2.758	3.795	-1.201	-0.070
Distance to port, km (log)	5.015	5.336	-0.194	-0.037
Coastal province	0.670	0.563	0.181	0.064

Quantity	Count
Treated EPZs	39
Treated grid cells	315
Control cell $\times$ sub-experiment observations	8,565
Treated prefectures	42
Prefecture clusters	124
Sub-experiments	33

Notes: Top: means over pre-opening years of 0.1-degree cells within 20 km of an EPZ (treated) and of not-yet-treated control cells, main sample; standardized differences before and after odds reweighting. Bottom: sample sizes. Prefecture clusters counts every prefecture the design touches; the Poisson regressions drop cells whose outcome is zero in every year and cluster on the prefectures that remain.

5 What EPZs Did to the Local Economy

Place-based programs usually add activity where they land. China’s first economic zones raised entry, employment, and output in the villages they covered (Lu et al. 2019) and foreign investment in the cities that hosted them (Wang 2013); federal empowerment zones raised employment in their neighborhoods (Busso et al. 2013); a large new plant raises the productivity of the incumbents around it (Greenstone et al. 2010). An EPZ packed a large export operation into a median of 2.8 km², so the natural expectation is that it raised activity around it, if only through suppliers, imitators, and the foreign firms it drew. The model of Section 3 predicts instead that domestic entry falls while incumbents are untouched (Table 3). This section reports what happened and then takes up three questions that any such estimate must answer: whether a trend that began before opening produced it, whether it is an artefact of how overlapping zones are handled, and how local it is—including whether the comparison locations were themselves being treated.

5.1 One margin moves

Table 5 estimates equation (5) for the margins the model speaks to, one regression per row, and Figure 5 shows their dynamics. The outcomes come from the survey of above-scale manufacturers, which sees little of the fence itself: in the 24 zones with both records, firms that can be linked to the survey account for 7.6 percent of fenced exports in the first four operating years (Appendix Table 51). What follows therefore describes the manufacturing economy around the fence. One margin moves. New domestic firms within 20 km fall by 20 percent from the year the fence opens. New foreign-invested firms within 20 km do not rise (95 percent interval -22 to 32 percent): the zone did not draw foreign manufacturers to the area outside its fence. Domestic incumbents keep their employment and wage bill, and the pooled coefficient on their output is close to zero (interval -9 to 9 percent), although, as shown below, a modest decline in their output cannot be ruled out. Summed over all surveyed firms within 20 km, gross output and value added are unchanged, employment rises by 9 percent, and export delivery value rises by an imprecise 12 percent. The zones left the manufacturing economy around them no larger and reduced the flow of new local firms. That is the pattern Proposition 1 predicts, and not the one agglomeration predicts, under which entry and incumbent output would both rise. The same equation for exits, capital, profits, subsidies, productivity, and wages is in Appendix Table 18.

How firmly do the data reject an unchanged entry rate? With standard errors clustered by prefecture (117 clusters), the 95 percent interval for the pooled coefficient runs from -36 to 1 percent and just includes zero. A 20-km circle often straddles a prefecture line, so we also cluster by zone: 38 clusters hold the treated cells, grouped by the EPZ they surround, and 33 each hold the control cells of one sub-experiment, 71 in all; the interval becomes -34 to -3 percent. Because analytic intervals can be too narrow with this many clusters, we also invert score wild cluster bootstrap tests, which impose the null hypothesis and need no re-estimation of the Poisson model (Kline and Santos 2012), with Webb’s six-point weights and 9,999 draws. Clustered by zone, $p = 0.038$ and the 95 percent interval runs from -0.397 to -0.042 log points; clustered by prefecture, $p = 0.081$ and the interval runs from -0.426 to 0.017 (Appendix Table 20). The bootstrap thus confirms the analytic inference under both clusterings. The pooled coefficient is also the more conservative of the two summaries the design offers. It compares the post-opening years with the average pre-opening year; the event study compares them with the year before opening, and against that year the post-opening average is -0.427 log points (s.e. 0.123). The two differ because entry near future zones was rising before opening, and whether that rise, rather than the fence, produced the decline is the first question below.

Figure 5 shows the timing. Domestic entry turns down from the year the fence opens and

stays down; foreign entry moves around zero before and after. Gross output and incumbent employment drift upward over the pre-opening years, as entry does, and show no break at opening; the incumbent wage does not move. Incumbent output rises before opening and falls back after, so its pooled contrast is close to zero while its contrast with the year before opening is negative. We therefore put incumbents through the checks that Section 5.2 applies to entry (Appendix Table 21). Incumbent employment is flat under every reading. For incumbent output, extrapolating the pre-opening rise, which is not itself significant ($p = 0.31$), turns the post-opening average negative, -0.181 log points (0.075); but that reading does not survive any allowance for the slope of the trend to change: the smoothness interval of Rambachan and Roth (2023), which reproduces the detrended estimate when the trend is exactly linear, includes zero at the smallest positive bound we consider. We cannot rule out a modest decline in incumbent output. Its measured size also depends on the estimator: incumbent output falls by 8.4 percent with $\log(1 + y)$ and by 9.0 percent with the inverse hyperbolic sine, against 0.5 percent with Poisson, because the transformations put large weight on cells whose incumbent output drops to zero when their few incumbents exit, while Poisson measures the proportional change in total output (Chen and Roth 2024); none of the three is distinguishable from zero (Appendix F). No level outcome rejects the joint pre-opening test (Table 5, column 3). Value added comes close ($p = 0.06$) for a reason of sample composition rather than trend: the 2004 census did not collect value added, so in 24 of the 75 sub-experiments a pre-opening year drops out for value added but not for gross output, whose pre-opening test is far from rejecting ($p = 0.40$).

5.2 Did a pre-opening trend produce the decline?

The first panel of Figure 5 shows entry near future zones rising steadily over the four years before opening. The main design’s joint test that the three earliest coefficients are zero does not reject ($p = 0.23$), but with four noisy coefficients a test that fails to reject is weak reassurance (Roth 2022), and we do not rest on it. The threat is specific. A rising trend that simply continued after opening would make the decline larger, not smaller. The threat is a boom that would have ended at opening anyway—sites chosen for zones drawing entrants while the zone was being built—so that the fall after opening marks the end of the boom rather than a cost of the fence.

Four readings of the pre-opening rise bear on this threat (Appendix Figure 18 and Table 19). (i) *Timing*. The rise spans the construction period: a zone opened a median of 2 years after State Council approval, and approval started construction and investment promotion at the site (Section 2). Counted from approval, entry near the site jumps when

Table 5: Effects of an EPZ on the local manufacturing economy

Outcome	(1)	(2)	(3)	(4)
	Log points	Percent [95% CI]	Pre-opening p	Treated cells
<i>Entry</i>				
New domestic firms	−0.219 (0.118)	−20 [−36, 1]	0.23	279
New foreign-invested firms	0.012 (0.135)	1 [−22, 32]	0.39	219
<i>Incumbents (present before opening)</i>				
Output	−0.005 (0.046)	0 [−9, 9]	0.31	307
Employment	0.064 (0.035)	7 [0, 14]	0.46	307
Wage bill	0.034 (0.048)	3 [−6, 14]	0.24	307
<i>All firms within 20 km</i>				
Gross output	0.022 (0.035)	2 [−5, 9]	0.40	315
Value added	−0.022 (0.046)	−2 [−11, 7]	0.06	315
Employment	0.091 (0.040)	9 [1, 18]	0.25	315
Export delivery value	0.109 (0.060)	12 [−1, 26]	0.43	287

Notes: Pooled post-opening Poisson coefficients ($e = 0, \dots, 3$) from equation (5), one regression per row. Column (1): log points, standard errors clustered by prefecture in parentheses. Column (2): the implied percent effect and its 95 percent interval. Column (3): p -value of the joint test that the coefficients for $e = -4, -3, -2$ are zero, the test shown above each panel of Figure 5. Column (4): treated cells kept by Poisson, which drops cells whose outcome is zero in every year. Controls are not-yet-treated cells, reweighted. Counts and totals include every surveyed firm in the cell, among them the few fenced firms the survey records. Incumbents are domestic firms present in the cell in the year before opening; exits count as zero. Value added is missing in 2004. For new domestic firms, clustering by zone and wild cluster bootstrap inference are in Appendix Table 20.

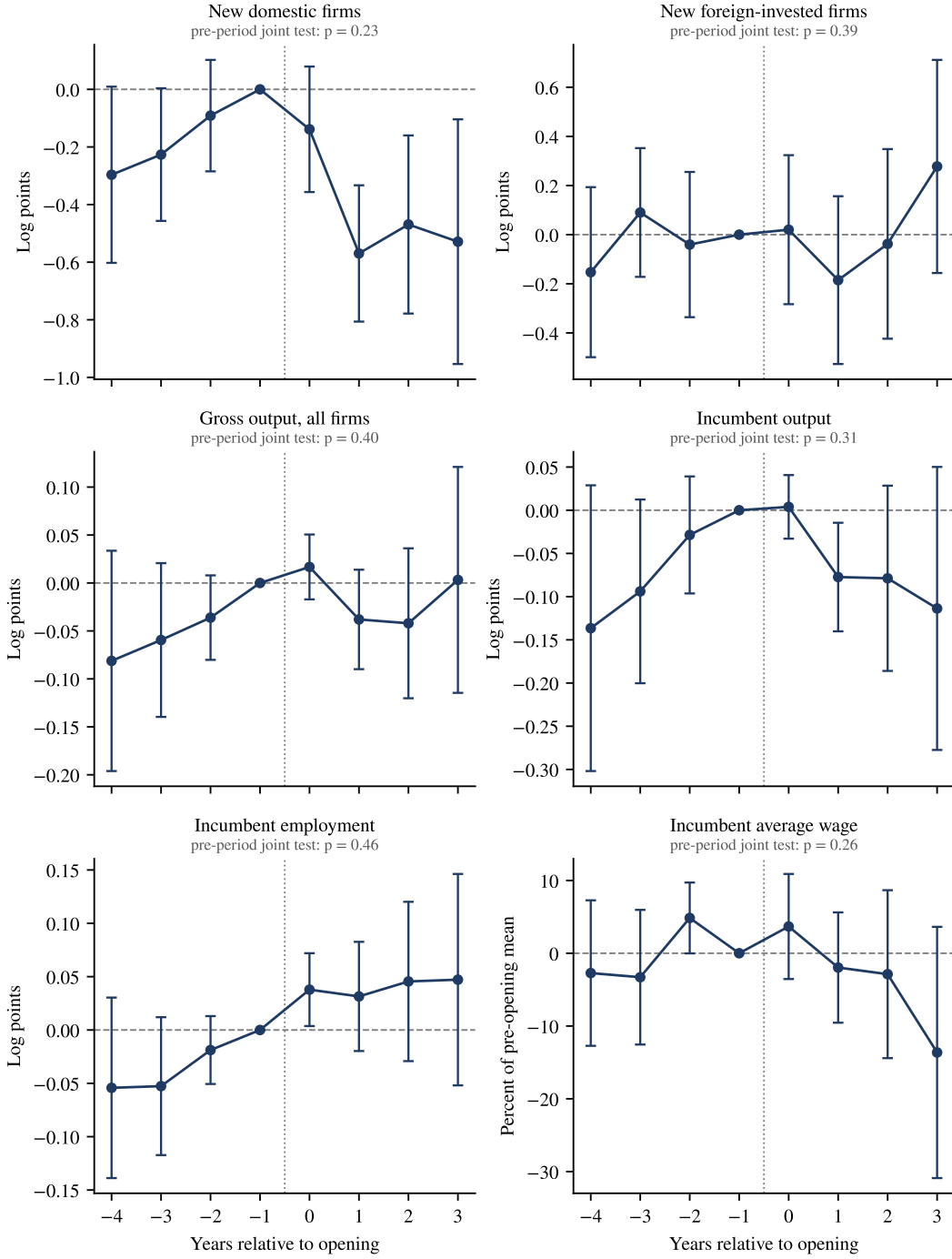


Figure 5: Event studies for the local economy around an EPZ

Notes: Coefficients β_e from equation (5) with the main comparison group (not-yet-treated cells, reweighted); $e = -1$ is the reference year. Counts and totals are Poisson coefficients in log points; the average wage is estimated in levels and shown as a percent of the treated cells' pre-opening mean. The p -value above each panel tests that the coefficients for $e = -4, \dots, -2$ are jointly zero. Bars are 95 percent confidence intervals clustered by prefecture. Vertical scales differ across panels. The same figure against future EPZ sites is in Appendix B.

the zone is approved and was already rising before, so the state chose sites where entry was growing (Section 6 reports these estimates). After opening, however, entry falls below its level four years before opening, before construction began, which a boom that simply ended would not produce. (ii) *A second comparison group*. Sites that later received an EPZ were chosen by the same state for the same kind of zone. Against them (Appendix Figure 17), entry near treated sites *falls* over the pre-opening years—the opposite slope—and then drops after opening by about as much as against the main controls (Appendix Table 19, first row). A pre-opening trend that changes sign with the comparison group while the post-opening decline does not is hard to reconcile with a decline produced by trends: the two groups bracket any counterfactual slope they share. (iii) *Detrending*. Extrapolating the main design’s pre-opening line makes the post-opening average more negative, not less: -0.683 log points (s.e. 0.179), a decline of 50 percent. In the main design the decline reverses the pre-opening trend rather than extending it. Against future sites, whose pre-opening slope has the other sign, the same extrapolation absorbs part of the decline and leaves -0.237 (s.e. 0.208), which cannot be distinguished from zero; the two detrended estimates bracket the unadjusted ones. (iv) *Bounds*. Under the bounds of Rambachan and Roth (2023), the 95 percent interval for the post-opening average excludes zero as long as the slope of a linear trend changes by no more than 0.054 log points a year, or as long as the year-to-year change in the departure from trend after opening is no more than 0.55 times the largest such change before opening; the second value comes from a plug-in approximation that overstates robustness (Appendix C.1).

The pre-opening rise is real, but it cannot produce the decline on its own: the fall takes entry below where it stood before construction began, appears against a comparison group whose pre-opening trend runs the other way, and grows when the main design’s own trend is extrapolated. What remains open is a boom-and-bust at treated sites that departs from trend after opening by more than the bounds allow, and Section 6 asks the two questions on which such a bust and the rationing of Section 3 part ways: which entrants were lost, and whether the loss scales with the fence’s activity.

5.3 Does the result depend on how overlapping zones are handled?

Most evaluations of economic zones treat each zone’s opening as a one-off event and compare locations near it with locations that are not. Section 4 explained why that design fits China’s EPZs poorly: zones overlap, so a binary design credits one zone with the effects of zones that came before or after it and compares treated places with controls that were themselves being treated. Whether the more elaborate design is what produces the answer can be checked directly. Table 6 estimates the same equation three ways, in the spirit of the side-by-side

comparisons of [Donaldson and Hornbeck \(2016\)](#) and [Borusyak and Hull \(2023\)](#). Column (1) is a pure binary design: the EPZ’s opening is a one-off event, cells within 20 km are treated, and every other special customs zone is ignored. Column (2) adds the year-to-year change in far-ring exposure as a regressor. Column (3) is the main design.

For domestic entry the binary design finds a somewhat larger decline, 23 percent against 20, and the far-ring control changes nothing; the gap is well within sampling error, and the intervals of the two designs’ event-study paths overlap in every year ([Appendix Figure 21](#)). The difference that matters is foreign entry. The binary design puts a decline on new foreign-invested firms nearly as large as the one on domestic firms, though less precisely estimated (column 1); the main design finds none (column 3). Incumbent output is unchanged under both. A binary design would conclude that the zone depressed all new manufacturing around it; the main design says it cut domestic entry only. The two conclusions point to different explanations, and [Section 6](#) starts from the second.

Where does the gap come from? The binary design treats 426 cells where the main design treats 279: it also treats cells that already had a special zone within 40 km, follows every treated cell through later arrivals, and pools cells that started from different exposure states. Admitting the already-exposed cells into the main design by itself changes little (a decline of 19 percent against 20; [Appendix Table 39](#)). A Shapley decomposition over the rules of [Section 4](#) attributes most of the gap in domestic entry to stratifying by initial exposure (0.087 log points) and to ending the post window when another zone arrives nearby (0.043); excluding contaminated controls moves the estimate the other way (−0.041), and the far-ring control does nothing (−0.001; [Appendix C.4](#)). The binary design makes the decline look larger and broader because it credits the first zone with later zones that arrive nearby and compares cells already surrounded by zones with cells that were not. [Section 4](#) showed that geography predicts exposure across the design’s strata but not within them, so stratification removes the non-random part of exposure that the binary design leaves in. Entering both treatment definitions in one regression cannot adjudicate between them, because in the main sample they are nearly collinear ([Appendix C.4](#)). The case for the main design is the one [Section 4](#) made: it answers the question this paper asks—what one fence does to a location that had none—whereas the binary design answers a mixture of that question and the effect of the zones that followed.

5.4 How local is the loss, and were the controls treated?

If the fence took admission capacity from the development zone it sits in, the loss should be confined to the zone’s own catchment; a shock common to the region—a provincial policy,

Table 6: Binary and exposure designs

	(1)	(2)	(3)
	Pure binary	+ far-ring control	Main design
New domestic entrants	−0.267 (0.117)	−0.270 (0.120)	−0.219 (0.118)
New foreign entrants	−0.220 (0.155)	−0.217 (0.147)	0.012 (0.135)
Incumbent output	−0.017 (0.022)	−0.017 (0.022)	−0.005 (0.046)
Treated grid cells	426	426	279

Notes: Pooled post-opening Poisson coefficients ($e = 0, \dots, 3$) from equation (5), log points; standard errors clustered by prefecture in parentheses. Column (1): a cell is treated if it lies within 20 km of an EPZ in the year the EPZ opens; other special customs zones are ignored (no stratification by initial exposure, no exclusion of controls exposed during the window, no truncation of the post window when a later zone arrives); controls are not-yet-treated cells reweighted on firm density and initial far-ring exposure. Column (2): column (1) plus the change since the year before opening in the number of special customs zones 40–150 km away. Column (3): the main design (Section 4; Table 5). Treated grid cells: cells kept by Poisson for new domestic firms. Decomposition of the gap: Appendix C.4.

an export boom—would show up at every distance. And if control cells were themselves near zones that opened during the window, the design would compare treated locations with partly treated ones, and the estimate would shrink toward zero in proportion to the contaminated share (Butts 2023). Figure 6 re-estimates the design ring by ring, each ring with its own controls and pre-trend test, and overlays a continuous distance profile, following the pairing of discrete rings and a continuous kernel in Diamond and McQuade (2019) and the cross-validated bandwidth of Clarke (2017).

The decline in domestic entry is confined to roughly the first 20–40 km; beyond 40 km the point estimates turn positive. Entry falls by 19 percent in the innermost ring (0–5 km) and by 27 percent at 10–20 km, while the 5–10 km ring is close to zero (−0.032); ring by ring the estimates are noisy, and the 10–20 km ring fails its own pre-trend test ($p < 0.01$). Between 20 and 40 km the coefficient is zero (0.011, s.e. 0.115) and the pre-trend test passes. Between 40 and 80 km it is positive, 17 percent (s.e. 0.097), but imprecise and with a marginal pre-trend ($p = 0.08$); whether it reflects entrants displaced outward is one of the displacement tests of Section 6. The continuous profile crosses zero between 20 and 40 km. Cross-validation picks the widest bandwidth on its grid (40 km), so the fitted profile is close to a straight line: the 5-km bins are too noisy to resolve more than a slope. A search for a single break in the same profile puts it at 45 km rather than at 20 km (Appendix Figure 22); with one break allowed, the flat 20–40 km ring is pooled with the decline inside it, and the largest shift in the profile comes where the positive outer coefficients begin, not where the decline ends. Incumbent

output is close to zero at every distance. Incumbent employment rises modestly at every distance, 40–80 km included, so the small gain in incumbent employment in Table 5 does not trace a local effect of the fence.

The control pool is clean within 40 km: because the design drops any control whose next zone within 40 km opens within three years, 0 percent of retained controls see a special customs zone open within 20 km during the window and 0 percent within 20–40 km. At 40–80 km, beyond the rule’s reach, 16 percent do, which would scale the estimate toward zero by a factor of 0.84 even in the worst case, in which those controls received the full treatment effect (Appendix C.6).

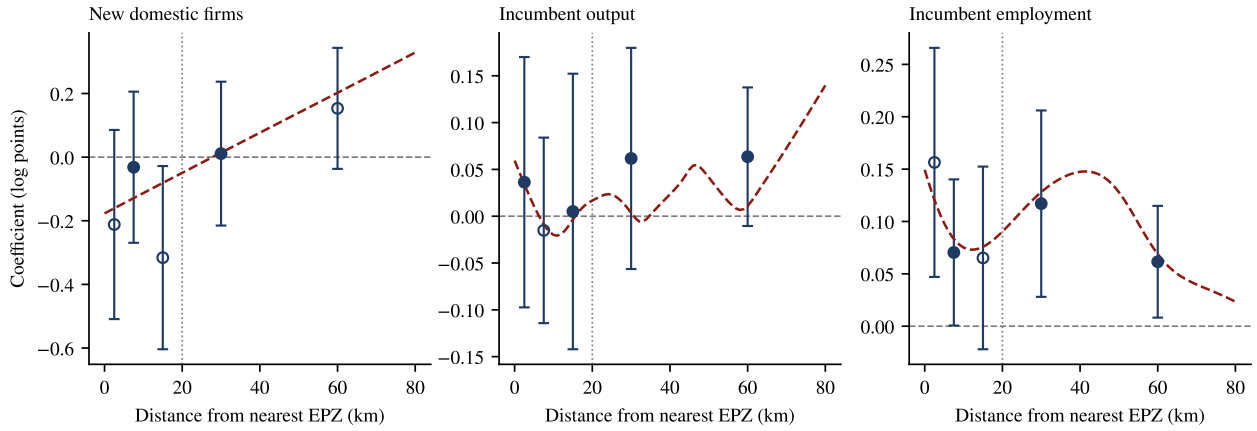


Figure 6: How local is the effect? Distance rings and a continuous profile

Notes: Each point is the pooled post-opening coefficient from equation (5) estimated separately for cells in one distance ring (0–5, 5–10, 10–20, 20–40, 40–80 km from the nearest EPZ), each ring with its own not-yet-treated, reweighted controls; bars are 95 percent intervals clustered by prefecture. Filled markers: the ring’s joint pre-trend test has $p \geq 0.10$; hollow markers: $p < 0.10$. Dashed curve: local-linear fit through sixteen 5-km bins estimated with the same design, Gaussian kernel, bandwidth chosen by leave-one-out cross-validation. The dotted line marks the main design’s 20-km radius. Incumbents are domestic firms present in the year before opening. Full estimates and the break search: Appendix C.5.

Summary. The EPZs cut domestic entry within 20 km by 20 percent and left incumbent employment and local output unchanged; a modest decline in incumbent output cannot be ruled out. The decline is not the continuation of a pre-opening trend: it reverses the main design’s own trend, appears against a comparison group whose trend runs the other way, and survives moderate departures from parallel trends. It is not an artefact of how overlapping zones are handled: a binary design finds a similar decline in domestic entry but would extend it, with a similar point estimate, to foreign entry. It is confined to roughly the first 20–40 km, beyond which the point estimates turn positive, and the control pool is clean within 40 km. With wild bootstrap inference the pooled coefficient has $p = 0.038$ clustered by zone and $p = 0.081$ clustered by prefecture, and the argument that follows does not rest on it alone.

Section 6 asks why entry fell.

6 Why Entry Fell: Admission Rationed by Project Size

Among local firms, the response to an EPZ is on the entry margin, the margin through which China’s economic zones generated their gains (Lu et al. 2019). This section asks where the loss comes from. Three families of explanations compete. The fence may compete with local firms in markets, bidding up wages or land rents until marginal entrants stay away, as in the enclave model of Rodríguez-Clare (1996) and the standard account of local crowding out (Moretti 2010). It may take part of the development zone’s admission capacity, as in the model of Section 3. Or nothing may have been lost: the missing entrants may have located elsewhere, or local firms may have moved into the fence and been counted there—the displacement that turns many place-based gains into transfers between places (Neumark and Simpson 2015). The explanations differ in which entrants are lost, whether the loss grows with the fence’s activity, when it starts, what happens to prices, and where the missing firms turn up. We take these in the order of Proposition 1, then check displacement, and eliminate the remaining candidates one at a time (Table 7).

Who is kept out. Part (b) of Proposition 1 says that a zone that keeps admitting the applicants highest in its priority index loses the lowest-ranked projects it would otherwise have admitted. Because the index rises with scale, the loss should sit at the bottom of the size distribution of entrants and spare the top. We do not observe the plot or the investment each entrant brought, so we rank entrants by output at birth, the measure of scale that the estimated priority index tracks most closely (Section 10.1), dividing domestic entrants nationwide into quintiles within two-digit industry and entry year. The left panel of Figure 7 shows the gradient. Entry in the smallest quintile falls by 44 percent (95 percent interval -61 to -19) and in the second by 33 percent, while entry in the largest quintile is unchanged (1 percent; -17 to 23); the gap between the top and bottom quintiles is 0.583 log points (s.e. 0.184). The entrants that do appear are correspondingly larger: survivors are 19 percent larger at birth. Ranked by employment or by assets, the largest quintile is again unaffected, but the loss spreads across the lower four quintiles instead of concentrating in the smallest (Appendix Table 25, Panel A). A size gradient alone does not settle the mechanism, because higher local costs would also push out the least productive entrants first (Hopenhayn 1992). What separates the explanations is whether the loss follows the fence’s activity and whether prices move.

The loss does not need fenced activity. Part (c) says that the loss depends on the slots the fence occupies, not on what it attracts. Every explanation that runs through the fenced firms themselves—their demand for workers, land, or inputs, or their competition for buyers—predicts the opposite: where the fence is idle, nothing is lost. Fences differ enormously in what they attracted, and some exported almost nothing in their first years of operation, which gives a direct test. We group treated cells by fenced exports in operating years 0–3 relative to the base-year output of incumbents within 20 km: empty (below 0.1 percent), small (0.1 to 2 percent), and large (above 2 percent). Around empty fences, entry in the smallest quintile falls by 46 percent (–68 to –9); around large ones, by 65 percent (–84 to –22; right panel of Figure 7). All domestic entry shows the same pattern less precisely, falling by 24, 17, and 33 percent around empty, small, and large fences (Appendix Table 26). The point estimate is largest around the busiest fences, so we cannot exclude that activity adds to the loss. But no channel that runs through the fenced firms can produce a loss around fences that exported almost nothing, and that loss is large and its interval excludes zero. The fence costs entry by being there.

The loss starts at opening, not at approval. A zone’s life has two dates that separate admission from anticipation. State Council approval announces the zone and starts construction and investment promotion at the site; customs acceptance, a median of 2 years later, opens the fence and, in the model, commits its slots. If would-be entrants stayed away because they expected the zone to raise their costs or crowd them out, or if the zone set its plots aside as soon as it was approved, entry would fall from approval. Rationing by an operating fence predicts a fall only from opening. Re-estimating equation (5) with event time counted from approval, entry near the site is 42 percent higher in the approval year and 25 percent higher in the year after, relative to the year before approval (Appendix Table 25, Panel C): construction and promotion drew entrants in rather than keeping them out. On the operating clock, the decline appears once the fence is open (Figure 5). For EPZs without a recorded acceptance date, opening is set two years after approval, so for them the two clocks differ by construction. Restricting the treated cells to those around EPZs with an official acceptance date (36 EPZs have one; 23 of them anchor treated cells in the estimation sample) leaves both findings in place: on the operating clock the pooled coefficient is –0.238 (0.122), a change of –21 percent against –20 percent in the full sample, and on the approval clock entry still rises in the approval year (Appendix Table 27). The approval clock also shows entry near future sites rising in the years before approval—the proportional difference four years before approval is –41 percent—the same upward drift that Section 5.2 examines; the decline after opening reverses that drift rather than extending it.

Prices did not clear the queue. The standard account of crowding out runs through prices: a large new employer or land user bids up local wages or rents, and the entrants that can least afford them stay away (Moretti 2010). It predicts that wages or land prices rise, that the loss grows with the fence’s demand, and that incumbents, who pay the same wages, lose too. Part (d) of Proposition 1 predicts none of this: migrant inflows made labor supply close to perfectly elastic at the scale of a zone (Appendix L) and industrial land was granted largely at administered prices (Section 3), so excess demand for plots was met by a queue rather than a price. City wages do not rise: the estimate is -0.037 log points (s.e. 0.022), and increases above 0.6 percent are ruled out. Incumbents near the fence keep their employment, and the pooled coefficient on their output is close to zero; a modest decline in their output cannot be ruled out (Section 5.1), but nothing like the fall in entry. We have no land-price data covering the years in which the EPZs studied here opened: comprehensive parcel records begin in 2007, and the only constant-quality land-price estimate we have comes from later zones (Appendix D). The testable implications of a land-price channel are therefore the empty fences and the incumbents. A price driven by the fence’s demand for land would grow with the fence’s activity, and the loss is already large around empty fences. A price rise caused by the plots the fence withdrew from supply would bear hardest on land-hungry entrants of every size, and the loss is not larger in industries with high fixed assets per worker, our closest proxy for land intensity (Table 7). Even such a rise would run through the plots the fence occupies, the premise of the model. What remains open is whether the queue for plots was cleared entirely by rule or partly by price, not whether the fence’s claim on the zone’s land caused the loss.

Not a reshuffle. The missing entrants may still have located elsewhere in the city or in another city, or local firms may have moved into the fence and reappeared there as fenced exports. Evaluations of China’s zones meet the mirror-image concern—that a zone’s gains are taken from other places—by checking whether the gains come at the expense of untreated places nearby (Lu et al. 2019) or elsewhere in the province (Wang 2013). The Reallocation group of Table 7 applies the same checks to a loss, one row for each place the missing entrants could have gone. Just beyond the fence’s reach, in the 20–40 km ring, entry does not rise (0.011, s.e. 0.115). Farther out, at 40–80 km, the point estimate is positive (0.153, s.e. 0.097), but it is imprecise and entry there was already moving before opening (pre-opening $p = 0.08$), so some would-be entrants may have been founded farther away; the ledger in Section 10 therefore reports the local cost alongside its counterpart under different shares of relocated projects. Output of all manufacturers within 20 km does not fall (0.022, s.e. 0.035), as expected when the lost projects are small. The fence did not absorb the host city’s firms:

tracing every fenced exporter by name and by survey identifier to firms active in the host prefecture in the two years before opening, only 5.6 percent of fenced exports in the first four operating years comes from firms that were already there (24 zones with customs records of fenced firms). And legal entities rarely cross prefecture borders: in a conditional-logit model of the destinations chosen by relocating manufacturers, a destination in another prefecture of the same province is chosen with a relative probability of 0.024 compared with an otherwise identical destination in the firm’s own prefecture. We have no estimate of entry in other prefectures of the same province, the diversion check of Wang (2013); Section 9 finds that incumbents across the border lose rather than gain. Within 40 km the missing entrants do not reappear, inside the fence or outside it; in the terms of Proposition 1, the share λ of refused projects that locate elsewhere in the prefecture is small, and the only room for relocation is farther away. Appendix D describes how each row is built.

The remaining explanations. Each of the remaining rows of Table 7 predicts a trace that the data do not show. Fiscal favors shifted toward the fence would cut entrants’ subsidies or raise incumbents’ taxes, and neither moves; had domestic firms merely registered inside the fence, dropping fence-registered entrants would close the gap, and the estimate is unchanged (-0.220 against -0.219); foreign entrants did not take the place of the missing domestic ones (0.012 log points, s.e. 0.135). Competition for workers through non-wage means would concentrate the loss in labor-intensive industries and grow with the fence’s activity, yet the high-minus-low labor-intensity difference is -0.098 (s.e. 0.105) and labor-intensive entry falls about as much around empty fences as around busy ones, while competition in product markets is excluded by the fence’s own rules, which allowed its output to be sold only abroad.

What the evidence adds up to. Each test alone leaves room for another story; together they leave room for one. The loss falls on the smallest projects and spares the largest. It appears around fences that attracted almost nothing, which rules out every channel that runs through the fenced firms. It starts when the fence opens, not when the zone is announced. Wages do not move and incumbents keep their employment, so prices did not clear the queue. And the missing entrants do not reappear inside the fence or elsewhere nearby. Of the explanations in Table 7, only one predicts all five patterns: the fence took admission capacity that the development zone would otherwise have allocated to domestic projects, and the zone, still admitting the applicants highest in its priority index, turned away the lowest-ranked, most of them small. On paper, development zones screened projects by investment per unit of land, a criterion the Ministry of Land and Resources codified nationally in 2004 as minimum investment intensities by region and industry (Ministry of Land and Resources 2004). That

rule arrives in the middle of our sample, and for earlier years we have no documentary evidence of how individual zones screened; the size pattern is evidence of rationing that rises with scale, not of any particular text. The explanation also says what the fence cost: the value added the lost projects would have produced, $C(m_z)$ in Proposition 3. Whether a zone repaid that cost depends on what the fence gave back—the local income, supplier demand, and spillovers through which place-based policies are expected to pay for themselves. Section 7 turns to what the fence left.

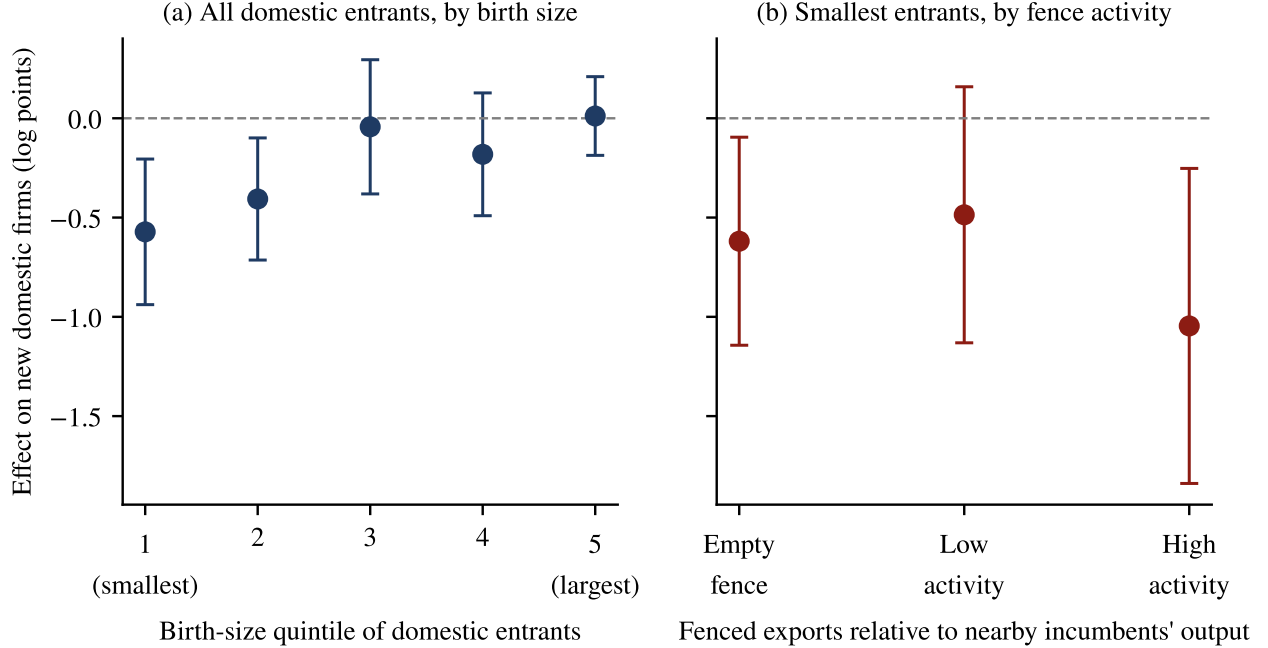


Figure 7: Which entrants are kept out

Notes: Equation (5), main design. Left: new domestic firms by quintile of birth output, quintiles defined within two-digit industry and entry year over all domestic entrants nationwide. Right: new domestic firms in the smallest quintile, by fence activity, measured as fenced exports in operating years 0–3 relative to the base-year output of incumbents within 20 km: empty fence (below 0.1 percent), low activity (0.1 to 2 percent), high activity (above 2 percent). Bars are 95 percent confidence intervals clustered by prefecture.

7 What the Fence Left: The Usual Channels of Place-Based Policy

Section 6 put a price on the fence: the value added of the small projects it kept out. A place-based policy is expected to repay such a cost through a short list of channels (Neumark and Simpson 2015; Kline and Moretti 2014a): (i) income paid to local workers; (ii) agglomeration economies, the productivity that nearby firms gain from density (Greenstone et al. 2010; Lu et al. 2019); (iii) backward linkages, demand for local suppliers; (iv) forward linkages,

Table 7: Why did local entry fall? Explanations, predictions, and evidence

Explanation	Prediction	Estimate	Verdict
<i>Who is kept out, and does it take fenced activity?</i>			
Admission rationed by size	Smallest projects are turned away first	Top–bottom birth-size quintile: 0.583 (0.184); survivors 19% larger (0.073)	Supported
Any channel through fenced activity	Loss grows with fenced exports	Smallest quintile, empty fence: −46% (95% CI −68 to −9); high activity: −65% (−84 to −22)	Rejected
Labor competition for workers	Loss largest where fenced firms hire most	High–low labor intensity −0.098 (0.105); labor-intensive entry, empty fence −0.387 (0.261), high activity −0.453 (0.273)	Not detected
Product-market competition	Fenced output takes local sales	Output must be exported; incumbent output −0.005 (0.046), employment 0.064 (0.035)	Rejected
<i>Prices</i>			
Wage competition	Fenced employers bid up wages	City wage −0.037 (0.022) [$\leq 0.6\%$]	Rejected
Land/capital intensity	Loss largest among land-intensive entrants	High–low fixed assets per worker 0.176 (0.105)	Not detected
<i>Reallocation: where did the missing entrants go?</i>			
Just beyond the fence	Entry rises at 20–40 km	New domestic firms, 20–40 km: 0.011 (0.115); pre-opening p 0.11	Not detected
Farther away	Entry rises at 40–80 km	New domestic firms, 40–80 km: 0.153 (0.097); pre-opening p 0.08	Inconclusive
Activity left the area	Output near the fence falls	Output of all manufacturers within 20 km: 0.022 (0.035)	Not detected
Local firms moved into the fence	Fenced exports come from firms already in the prefecture	Share of fenced exports ($e = 0-3$) from firms in the host prefecture before opening: 5.6% (24 zones)	Rejected
Firms moved to another prefecture	Relocations cross prefecture borders freely	Relative probability of a destination in another prefecture of the same province: 0.024	Rejected
<i>Other explanations</i>			
Fiscal favors shifted to the fence	Fewer subsidies for entrants, more tax for incumbents	Entrants' subsidy rate 1.655 pp (1.474); incumbents' effective tax rate −0.0020 pp (0.0075)	Not detected
Relabeling into the fence	Entrants register inside the fence instead	Excluding fence-registered entrants −0.220 (0.118), against −0.219 (0.118)	Rejected
Anticipation of the zone	Entry falls from approval, before opening	Entry against the year before approval: +42% at approval, +25% a year later (s.e. 0.111, 0.102); four years before approval −41%	Rejected
Foreign firms crowding in	Foreign entrants replace domestic ones	New foreign-invested firms 0.012 (0.135)	Not detected

Notes: Equation (5), main design, unless noted; log points unless marked pp or percent; standard errors in parentheses. Brackets hold only upper bounds ($\leq x\%$, the upper end of the 95 percent interval); full 95 percent intervals are written out in the cell. Regressions: Appendix Table 25; Reallocation rows: Appendix D. *Supported:* predicted sign, and the 95 percent interval excludes zero. *Rejected:* an estimate, placebo, or institutional rule contradicts the prediction. *Not detected:* the interval includes zero and is narrow enough to be informative. *Inconclusive:* too imprecise, or too weak a pre-opening test, to tell a small effect from a large one.

inputs for local users (Javorcik 2004); and (v) the local multiplier, the non-tradable activity that tradable jobs create through local spending (Moretti 2010). Each channel meets one of the fence’s rules, and Proposition 2 turns the rules into predictions. Inputs entered in bond and profits went to foreign owners, so local income lies between the wage bill and domestic value added (equation (3)). Walls and trading restrictions cut the contact through which density raises productivity (Table 3). Duty-free imports, with export rebates on domestic inputs only from 2008, kept local purchases s_z small, so suppliers should expand little, and not at all around an empty fence (part (a)). Output had to be exported, with carry-over sales to processors outside the fence allowed nationally only from May 2005, so forward linkages were closed until then and should open afterwards only if local firms could use the fence’s components (part (b)). Only the multiplier is left open—fenced workers live and spend outside the fence—but it scales with the wage bill $\sigma_z Q_z$, which the rules keep small.

The subsections test these predictions in turn: local income (Section 7.1), agglomeration and productivity (Section 7.2), linkages and new suppliers (Section 7.3), the local multiplier (Section 7.4), and the one loosening of the rules inside our window, the 2005 extension of carry-over (Section 7.5). Table 9 at the end of the section collects every channel, its prediction, the estimate that tests it, and the verdict.

7.1 Local income

Jobs and wages are the first benefit that place-based policies are credited with (Busso et al. 2013; Kline and Moretti 2014a). For a fence the question is how much of the export value stays in the host economy. Customs records answer it from above. The domestic value-added ratio of fenced exports, one minus processing imports over processing exports (Kee and Tang 2016), contains local inputs s_z , local factor payments σ_z , and the profits of foreign owners π_z (equation (2)), so it bounds the local share from above; the wage bill of fenced firms matched to the firm survey measures σ_z directly.

Nationally, the domestic value-added ratio of fenced exports rose from 0.27 in 2003 to 0.48 in 2006 as operations matured, in line with the national rise documented by Kee and Tang (2016); fenced exports in these years came entirely from processing with imported materials (Appendix Table 40 follows the fences year by year). The aggregate is dominated by the large hubs. In the median zone, domestic value added over the first four operating years was at most 28 percent of exports, and the wage bill of fenced firms was 6 percent (interquartile range 3.5–9.2 percent across the 17 zones whose fenced firms can be matched to the survey). Of every 100 yuan the typical fence exported, about 6 were paid to local workers, and at most 28 could have stayed in the host economy in any form. This is the slope of the ledger in

Proposition 3; Section 10 sets it against the cost of the lost entrants.

7.2 Agglomeration and productivity

The central promise of a place-based policy is that concentrated activity makes the firms around it more productive. Large plant openings raise the productivity of incumbents in the same county (Greenstone et al. 2010), and China’s economic zones have been found to raise activity in and around the zones (Lu et al. 2019). If a fence worked this way, nearby incumbents’ productivity and output would rise, exporting would spread, and the gains would show up in the city’s output and wages (Table 3). If the fence’s walls cut the contact through which density pays, none of this would move. A single null would not separate the two: it could reflect low power, a short horizon, the wrong spatial unit, or the wrong outcome. We therefore bring four designs that differ in data, unit, and estimator, and add the capitalization of wages; a fifth, which follows nearby exporters for eight years, belongs with the learning evidence of Section 8.3.

(1) Incumbent firms near the zone, linked across the customs and firm records, show a change in industry-adjusted total factor productivity of 0.127 log points (s.e. 0.102). (2) The design of Lu et al. (2019), rerun on our zip-code panel, gives a distance-free spillover of 0.067 (s.e. 0.0518) for employment and 0.023 (s.e. 0.0622) for output. (3) A larger local stock of exporters in a product–destination market does not raise a firm’s own entry into that market once city $\times$ product $\times$ destination effects absorb comparative advantage (elasticity -0.0019 , s.e. 0.0019). (4) Cities that won a comprehensive bonded zone, the successor to the EPZ, grow no faster than the runners-up of the same application round, whether measured by city GDP (0.022, s.e. 0.029) or by night lights near the site (-0.047 , s.e. 0.060). Wages, finally, do not capitalize the zone (-0.037 , s.e. 0.022). No estimate is significant at the 5 percent level, and the largest in absolute value, for wages, is negative (Appendix Table 41 reports each design in its own units). The main design agrees: incumbents’ output (-0.005 , s.e. 0.046), wages, and labor productivity do not change (Appendix Table 18). The fence did not work as an agglomeration device, and the entry loss of Section 5 was not offset by productivity gains among the firms that stayed—what Proposition 1(d) predicts and the agglomeration account denies.

7.3 Linkages and new suppliers

Foreign plants help local firms mainly through the supply chain. Javorcik (2004) separates three channels: the same industry (horizontal), which foreign firms have little reason to help; local suppliers (backward), which they have every reason to help; and local users of their

output (forward). In the enclave model of Rodríguez-Clare (1996), the backward channel is as strong as the enclave’s local purchases. The fence’s rules predict a specific pattern: little backward response, because inputs came in duty-free; no forward response before 2005, because output could not be sold locally; and in the fence’s own industry, nothing beyond the local growth that drew the zone there in the first place.

To separate the fence’s pull on an industry from everything else that happens in a location, we compare two-digit industries j within the same cell and year,

$$E[y_{scjt}] = \exp \left(\alpha_{scg(j)} + \delta_{sct} + \mu_{sg(j)t} + \beta T_{sc} \text{Post}_{st} X_j \right), \quad (6)$$

where Post_{st} marks the years from opening, $g(j)$ groups industries into the fence’s own industry (electronics), its top-quartile suppliers, and the rest, and the cell-by-year effects δ_{sct} absorb anything common to the location, including the entry loss. X_j is the industry’s exposure to the fence, fixed before opening from the 2002 input–output table and the industry mix of fenced output: its share of fenced output (horizontal, H_j), its weight in the fence’s input basket (backward, B_j), or the weight of the fence’s industries in its own inputs (forward, F_j), each standardized across industries (Appendix H.3). Cells, reweighting, and clustering are those of the main design. Two checks target the concern that an industry’s exposure picks up where it was already heading: an event-study version tests the pre-opening coefficients jointly, and a placebo moves each opening three years earlier and uses only years before the true opening, so that no year it treats is treated.

Table 8 reports the three channels one at a time. *Backward*: industries that supply the fence do not expand. Their employment changes by 0.016 (s.e. 0.020) and their output by -0.069 (s.e. 0.045) per standard deviation of exposure; with a binary indicator for the top quartile of suppliers, output changes by 0.022 (s.e. 0.041; upper bound 11 percent) and employment by -0.081 (s.e. 0.043; Appendix Table 42). *Horizontal*: the fence’s own industry grows faster near the zone, by 0.043 (s.e. 0.016) in employment and 0.049 (s.e. 0.023) in output, but the placebo three years early is larger still (0.130 and 0.074), and the pre-opening employment coefficients are jointly different from zero ($p = 0.018$). Electronics was already expanding where zones were placed; the fence did not start it. *Forward*: industries that use the fence’s products gain 0.076 (s.e. 0.030) in employment and 0.071 (s.e. 0.035) in output, and their pre-opening coefficients pass the joint test ($p = 0.292$ and 0.661). The placebo, however, is no smaller: 0.126 (s.e. 0.086) and 0.077 (s.e. 0.065). It is imprecise, but a forward effect that is as large three years before opening cannot be attributed to the fence. The rules point the same way: before May 2005, fenced firms outside the two pilot zones could not sell components to local processors at all, and when the rule changed, the users did not respond

Table 8: Horizontal, backward, and forward exposure to the fence: estimates and placebo checks

	Estimate	Opening moved three years earlier	Pre-opening joint p
	(1)	(2)	(3)
<i>Panel A. Employment</i>			
Horizontal H_j (own industry)	0.043 (0.016)	0.130 (0.056)	0.018
Backward B_j (fence buys from j)	0.016 (0.020)	0.021 (0.106)	0.913
Forward F_j (j buys from the fence)	0.076 (0.030)	0.126 (0.086)	0.292
Observations	458,623	220,912	458,623
<i>Panel B. Output</i>			
Horizontal H_j (own industry)	0.049 (0.023)	0.074 (0.051)	0.502
Backward B_j (fence buys from j)	−0.069 (0.045)	−0.180 (0.131)	0.193
Forward F_j (j buys from the fence)	0.071 (0.035)	0.077 (0.065)	0.661
Observations	458,623	220,912	458,623

Notes: Poisson estimates of equation (6) with one exposure measure at a time; cell $\times$ industry $\times$ year panel of firms in the firm survey, main design (not-yet-treated controls, reweighted). Exposures are standardized across two-digit industries: horizontal H_j is the industry’s share of fenced output, backward B_j its weight in the fence’s input basket, forward F_j the weight of the fence’s industries in its own inputs (2002 input–output table). Column (1): pooled post-opening coefficient, $e = 0, \dots, 3$. Column (2): the same regression with each opening moved three years earlier, estimated on years before the true opening only. Column (3): p -value of the joint test that the event-time coefficients $e = -4, \dots, -2$ (relative to $e = -1$) are zero. New domestic entrants are in Appendix Table 44. Standard errors clustered by prefecture in parentheses.

(Section 7.5). We read the forward estimate as a sign that electronics-using industries were growing near the zones, not as evidence of a linkage. Entering the three exposures together cannot separate them, because electronics ranks first on all three and the horizontal measure is correlated 0.94 with the backward and 0.95 with the forward measure (Appendix Table 43); new entrants, whose estimator does not converge in the placebo sample, are in Appendix Table 44. A zip-code version of the race, on a different panel, loads the loss onto the backward interaction (Appendix K.2).

Linkages can also show up as new firms rather than larger ones. We count births near the zones in three sources that share no firm identifier—the firm census, first-time exporters in the customs records, and business registrations—relative to reweighted comparison cells before and after opening (Appendix Table 46). Births of electronic-component firms were

already 50 percent above comparison cells before opening and did not rise further (change after opening, in units of the comparison mean, -0.419 , s.e. 0.330); new exporters of the products that fenced firms import (-0.025 , s.e. 0.183) and registrations of electronics suppliers (-0.002 , s.e. 0.305) do not change. Only logistics and customs brokerage (0.252 , s.e. 0.150) and import–export trading (0.291 , s.e. 0.184) rise, both imprecisely: what formed around the zones was the service layer of a customs platform, not a supply base, the pattern part (a) of Proposition 2 predicts when local purchases are small.

7.4 The local multiplier

Tradable jobs create non-tradable ones: workers and firms spend locally, and restaurants, shops, and services follow (Moretti 2010); around China’s industrial parks, new consumption amenities appear (Zheng et al. 2017). This is the one channel the fence’s rules leave open, but its size should track the wage bill, about 6 percent of exports (Section 7.1). The prediction is a positive response of modest size.

We classify business registrations by their stated line of business into non-tradable services (retail and wholesale, catering and accommodation, resident services, real estate, and entertainment), producer services, and manufacturing, and count new registrations per above-scale incumbent firm in the base year. The comparison uses a broader version of the main design that also counts EPZs arriving near a zone already in operation (Appendix H.4). Figure 8 shows the differences year by year; none is normalized away. Non-tradable registrations are level with comparison cells four to two years before opening, begin to rise in the year before opening, while the zone is being built (Section 5.2), and stay higher afterwards. In the four years after opening they are 44 percent above comparison cells (relative effect 0.436 , s.e. 0.316), and 0.326 (s.e. 0.269) in years 4–6; both intervals include zero. Averaged over the four years before opening, including the construction year, the difference is 22 percent (relative effect 0.219 , s.e. 0.326 ; joint pre-opening $p = 0.081$). Producer services (0.260 , s.e. 0.443) and manufacturing (0.141 , s.e. 0.213) show no clear change.

The point estimate is positive but imprecise, and part of the rise precedes the fenced jobs that would drive a multiplier: fenced firms begin to export only at opening, so registrations in the construction year respond to building the zone or to its announcement rather than to fenced wages. The size is modest in any case: the post-opening estimate implies about one additional non-tradable registration for every 5.8 fenced jobs, an order of magnitude that counts surviving firms rather than jobs or sales and so cannot be compared with the multipliers of Moretti (2010). The one channel the rules left open works, if at all, at the scale the small wage bill allows.

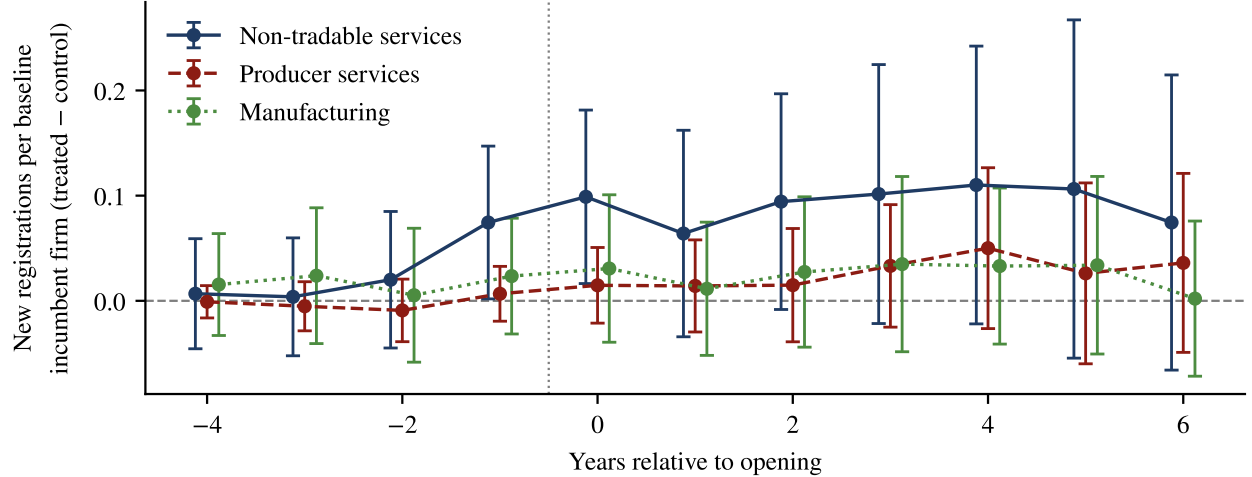


Figure 8: New business registrations near an EPZ, by sector

Notes: Difference between treated and reweighted comparison cells in new business registrations per above-scale incumbent manufacturing firm in the base year, by year relative to opening; every year, including $e = -1$, is estimated, and none is normalized to zero. Treated cells lie within 20 km of an arriving EPZ, including arrivals near a zone already in operation; comparison cells are not yet treated (Appendix H.4). Sectors are assigned from the registered line of business. Bars are 95 percent confidence intervals from a bootstrap over prefectures (199 draws). Relative effects and pre-opening tests are in Appendix Table 47.

7.5 Opening the fence

If the rules were what kept the fence apart, lifting them should create linkages; if local firms had little use for the fence’s products, lifting them should change nothing (part (b) of Proposition 2). The rules changed once inside our window. In May 2005 the carry-over rule, which let fenced firms sell components to processing firms outside the fence for further processing and re-export, was extended from two pilot zones to all zones—the one change in our window that raised the fence’s potential local sales. It opened the forward linkage only; it did not change the terms on which local suppliers sold into the fence. We compare downstream user industries with other industries near zones already operating, before and after 2005, and use cells near the older bonded zones, to which the rule did not apply, as a placebo (Appendix H.5).

Eligible downstream industries did not respond in 2005–2007. Their output changes by -0.067 (s.e. 0.140) and their employment by -0.026 (s.e. 0.061) relative to other industries; the estimates rule out employment gains above 10 percent and output gains above 23 percent, and the event study shows no break in 2005 (Appendix Figure 24). Permission to buy from the fence did not create local buyers. Together with the forward estimate of Section 7.3, which was as large before opening as after, this places the constraint on the local side: local processors could not use what the fence made, at least not in the years 2005–2007 that the data cover.

What the fence left. Of the five channels through which place-based policies are expected to pay for themselves, the fence’s rules closed or narrowed four, and none shows a gain (Table 9): local income was bounded by a small wage bill, productivity did not spill over, suppliers did not expand or form, and users grew before the zones and did not respond when allowed to buy. The one channel left open, the local multiplier, has a positive but imprecise point estimate of the size the wage bill allows. One channel remains, the one by which EPZs are most often defended—whether nearby firms learned to export from the zones—and Section 8 takes it up.

8 Learning: Export Experience near the Zones

Place-based policies are often defended by what concentrated activity teaches the firms around it. For an export processing zone the natural lesson is exporting: local firms that subcontract for, supply, or simply watch foreign processors learn what sells where and how to reach buyers, and later export on their own (Aitken et al. 1997; Alfaro-Ureña et al. 2022). On this view a fence that gathers foreign processing in one place works as a school for the surrounding economy, and the export know-how it spreads is a gain that the ledger of Section 10 does not count. Section 7 found no gains through suppliers or through sales across the fence; this section asks whether the zones left export capability behind. The answer is that the zones did reach nearby exporters, but by drawing them into consigned processing, in which the foreign principal keeps the customer, so the portable export capability around the zones did not rise. The customs records make the question answerable because they record who controlled each export transaction.

Chinese processing trade took two legal forms (Feenstra and Hanson 2005). Under processing with imported materials, the processor buys the imported inputs, owns and finances the goods, and sells to buyers it finds itself. Under consigned processing (processing with supplied materials), a foreign principal owns the inputs and the output throughout, chooses the market, and pays the processor for the processing step (Appendix M.7). The difference is who holds the customer. An imported-materials processor is like a tailor who buys cloth and sells suits: every sale teaches it something about who buys suits and where. A consigned processor is like a tailor who sews cloth a client brings in and hands the suit back: it learns to sew but never meets the people who wear the suits. Fenced exports themselves came almost entirely from foreign-invested firms processing imported materials (Appendix L.3). What matters for the local economy is which kind of experience the zones added outside the fence.

The argument has four steps. We first establish, across all Chinese exporters, which kind

Table 9: What the fence left: channels, predictions, and evidence

Channel	Prediction	Estimate	Verdict
<i>Local income (Section 7.1)</i>			
Wages and domestic value added	Wages paid inside the fence raise local income	Wage bill of fenced firms, median zone: 6% of exports (IQR 3.5–9.2%); domestic value added, median zone: at most 28%	Bounded
<i>Agglomeration and productivity (Section 7.2)</i>			
Productivity and agglomeration spillovers	Density near the fence raises nearby firms' productivity, city output, and wages	Incumbent TFP 0.127 (0.102); Lu–Wang–Zhu replication, employment 0.067 (0.0518); peer-network entry -0.0019 (0.0019); city GDP of contest winners 0.022 (0.029); city wage -0.037 (0.022)	Not detected
<i>Input–output linkages (Section 7.3)</i>			
Backward (local suppliers)	Industries that supply the fence expand, and new suppliers form	Top-quartile suppliers: output 0.022 (0.041) [$\leq 11\%$], employment -0.081 (0.043); backward exposure, employment 0.016 (0.020); births of electronic-component firms -0.419 (0.330)	No expansion
Horizontal (own industry)	The fence's own industry expands near the zone because of the fence	Employment 0.043 (0.016); opening moved three years earlier 0.130 (0.056); pre-opening p 0.018	Predates the zone
Forward (local users)	Industries that use the fence's products expand	Employment 0.076 (0.030), output 0.071 (0.035); opening moved three years earlier 0.126 (0.086) and 0.077 (0.065); pre-opening p 0.292	Suggestive only
<i>Local multiplier (Section 7.4)</i>			
Non-tradable services	Fenced workers' spending raises non-tradable activity nearby	New non-tradable registrations relative to comparison cells: after opening 0.436 (0.316), before 0.219 (0.326); one registration per 5.8 fenced jobs	Positive, imprecise
<i>Opening the fence (Section 7.5)</i>			
Carry-over extended (2005)	Downstream users expand once they may buy from the fence	Output -0.067 (0.140) [$-29, 23\%$]; employment -0.026 (0.061) [$-14, 10\%$]	Bounded

Notes: Each row is a channel through which fenced activity could benefit the host economy, its prediction, the estimate that tests it, and the verdict; panel headings give the section with the full evidence. Coefficients are in log points, except the linkage exposures (per standard deviation of exposure) and the multiplier row (relative to the comparison mean); standard errors in parentheses. Brackets give 95 percent bounds in percent: $[\leq x]$ is the upper bound, $[a, b]$ the confidence interval. Verdicts: *not detected*, the interval includes zero and excludes large effects; *bounded*, only a bound is informative; *no expansion*, a tight null on a predicted gain; *predates the zone*, the placebo with openings moved three years earlier is as large and the pre-opening coefficients are jointly different from zero; *suggestive only*, passes the pre-opening test but the placebo is as large; *positive*, *imprecise*, positive point estimate whose interval includes zero.

of experience carries over to new markets (Section 8.1). We then ask which kind the zones added around them (Section 8.2), whether the stock of portable experience near the zones rose (Section 8.3), and whether nearby firms learned through direct contact with fenced firms (Section 8.4).

8.1 Which export experience carries over to new markets

Figure 9 gives the basic fact before any regression, in the spirit of the transition matrices of Morales et al. (2019). For each exporter, product, and year, it shows how often the firm enters, under ordinary trade, a large destination it did not serve with that product last year, depending on whether the destination resembles one the firm served last year under a given form. Pooling all firms, ordinary-trade experience separates similar from dissimilar destinations sharply: firms enter 6.2 percent of similar candidates and 3.1 percent of dissimilar ones. Imported-materials experience also separates them (6.2 against 4.1 percent). Consigned experience barely does (4.9 against 4.5 percent). The figure shows domestic and foreign-invested firms separately rather than these pooled rates: the gap shrinks in the same order in both groups, and in both the consigned bars are the only pair whose confidence intervals overlap.

The raw comparison mixes three things. Firms doing consigned work may be smaller or less able; similar destinations may simply be the markets that were growing; and a firm’s products determine which markets are open to it. The regression removes all three. For firm f , HS6 product p , and year t , the risk set pairs the firm with each candidate destination d defined as in Figure 9, and entry depends on whether the firm served d itself under form r in $t - 1$ (D^r) and whether it served a destination similar to d (S^r):

$$\text{entry}_{fpt} = \sum_r a_r D_{fd,t-1}^r + \sum_r b_r S_{fd,t-1}^r + \alpha_{fpt} + \delta_{pdt} + \varepsilon_{fpt}. \quad (7)$$

Firm $\times$ product $\times$ year effects hold fixed how much the firm expands, so the coefficients describe only where the expansion goes; product $\times$ destination $\times$ year effects absorb demand. The destination-specific weight of form r is $\kappa_r = a_r/a_O$ and its portability weight is $\omega_r = b_r/b_O$, both relative to ordinary trade. The model’s extension in Appendix R.5 predicts $\omega_A \approx 0$ for firms that do not control the transaction, because their experience in a destination is information about their principal’s buyers rather than about demand in similar markets, and $\omega_I > 0$ for firms that do.

Both forms of processing build a relationship with the destination itself: having served a destination under either form predicts serving it later under ordinary trade about as strongly as ordinary-trade experience does ($\kappa_I = 1.19$, $\kappa_A = 0.80$; Table 10, Panel A). Only one form

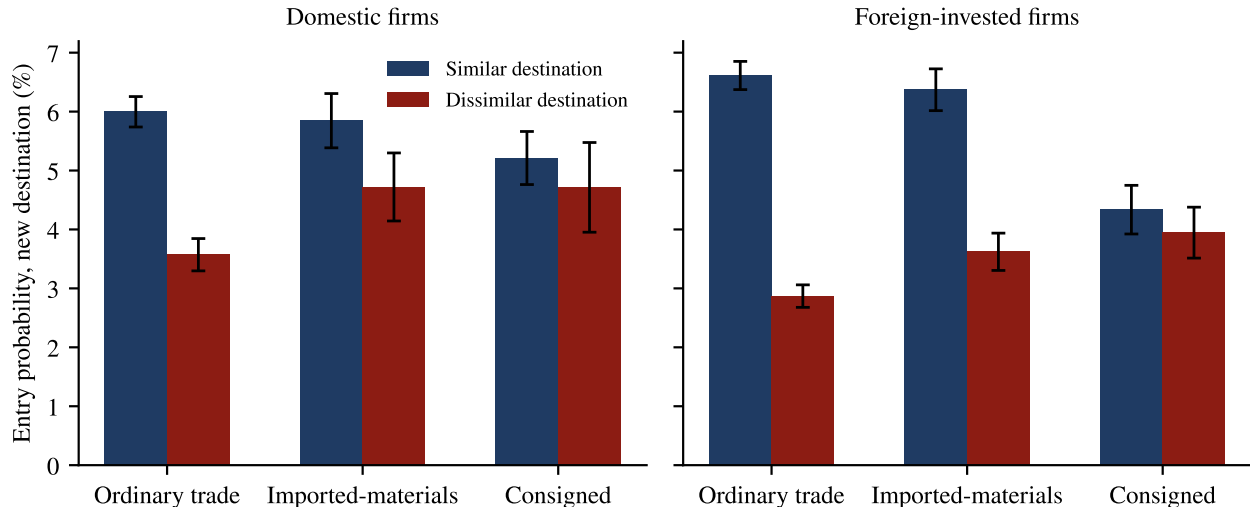


Figure 9: Entry into new destinations by kind of export experience

Notes: Share of candidate destinations that a firm enters with a product under ordinary trade in year t , 2002–2006, among firm $\times$ product $\times$ year observations of firms that exported under the form on the horizontal axis in $t - 1$. Candidates are the 40 largest destinations of Chinese exports in $t - 1$ that the firm did not serve with the product under ordinary trade in $t - 1$. A candidate is similar if it shares a continent or an official language with, or lies within 2,000 km of, another destination the firm served under that form in $t - 1$. The sample is the 5 percent random sample of non-intermediary exporters of Table 10, restricted as there to firm $\times$ product $\times$ year groups with at least one entry; firms with experience under several forms appear in several groups. Raw means without fixed effects; bars are 95 percent confidence intervals clustered by firm. Ownership from the sixth digit of the customs code (Appendix E.1).

carries over to new markets. Imported-materials experience has a portability weight of 0.36 (s.e. 0.15), well below ordinary trade ($p < 0.001$ for $\omega_I = 1$); consigned experience has none ($\omega_A = -0.44$, s.e. 0.18), and the two weights differ with $p < 0.001$. Among foreign-owned firms the consigned weight is negative (-0.78), as expected when a principal assigns markets to its processors. Among domestic firms it is 0.03 (s.e. 0.19) against 0.38 (s.e. 0.18) for imported materials, a difference that is not statistically significant ($p = 0.230$), so the conclusion for local firms below rests on the bound on the capability stock in Section 8.3, not on this difference alone.

Three further results support the reading and are detailed in Appendix E. Consigned experience records a current contract: it stops predicting entry after one year, whereas imported-materials experience keeps a share of 0.38 of its effect (Appendix Table 31). The weights also account for the finding of Bai et al. (2021) that processing experience predicts later ordinary exporting: in their design, the effect in our data comes from imported-materials processing (0.043, s.e. 0.008) rather than consigned processing (0.008, s.e. 0.009; Appendix Table 32). And the ranking $\omega_I > \omega_A$ holds across candidate sets, similarity definitions, and a structural conditional logit, with similarity by official language alone the one exception (Appendix E.2). Organizational form is chosen by the firm and its partner, so we read the

Table 10: Relative predictive weights of export experience (ordinary trade = 1)

	All firms (1)	Domestic (2)	Foreign-owned (3)
<i>Panel A. Same destination, $\kappa_r = a_r/a_O$</i>			
Imported-materials processing, κ_I	1.189 (0.109)	1.017 (0.167)	0.826 (0.067)
Consigned processing, κ_A	0.802 (0.221)	0.605 (0.138)	1.307 (0.175)
<i>Panel B. Similar new destinations (portability), $\omega_r = b_r/b_O$</i>			
Imported-materials processing, ω_I	0.362 (0.154)	0.377 (0.179)	0.130 (0.155)
Consigned processing, ω_A	-0.442 (0.181)	0.032 (0.191)	-0.776 (0.203)
Candidate tuples	1,161,607	632,524	423,498
Firms	4,231	1,910	2,321
Entry rate	0.058	0.058	0.057

Notes: Linear probability model of equation (7), 2002–2006, on a 5 percent random sample of non-intermediary exporters; candidates are the 40 largest destinations of Chinese exports in $t - 1$. Similar destinations are those on the same continent, with the same official language, or within 2,000 km. Relative weights with delta-method standard errors clustered by firm in parentheses. Tests of equal weights and the ordinary-trade coefficients used as the normalization are in Appendix Table 28. Ownership from the sixth digit of the customs code.

weights as the predictive content of each kind of experience, not as the causal effect of control. Export experience, in short, predicts entry into new markets only where the firm controls the transaction.

8.2 What the zones added: consigned processing

If the zones spread export know-how, the experience they added around them should be of the kind that travels. Two readings predict otherwise, and they differ in who switched. Under the first, the zones drew local firms into processing for foreign principals, the form that leaves nothing portable. Under the second, which Table 10 invites, any shift toward consigned work happened among foreign-invested firms, whose consigned experience already carries a negative weight, and local firms were untouched.

We follow every firm that exported in the year before a zone opened, in areas within 40 km of the arriving zone and in not-yet-treated comparison areas of the same sub-experiment, reweighted by the odds of treatment on pre-opening firm characteristics (Appendix E.4). Fenced firms and trade intermediaries are excluded, so the outcome describes firms outside

Table 11: Starting consigned processing after a zone opens, by distance and ownership

Distance to the zone	All firms		By ownership in the year before opening		
	Pooled	Within city	Domestic	Foreign	Foreign – domestic
	(1)	(2)	(3)	(4)	(5)
0–5 km	0.022 (0.008)	–0.005 (0.010)	0.044 (0.012)	0.018 (0.010)	–0.019 (0.020)
5–10 km	0.009 (0.006)	–0.009 (0.008)	0.017 (0.011)	0.008 (0.006)	–0.003 (0.011)
10–20 km	0.011 (0.005)	–0.007 (0.009)	0.012 (0.006)	0.012 (0.006)	0.004 (0.012)
20–40 km	0.011 (0.003)	–0.006 (0.006)	0.013 (0.005)	0.011 (0.004)	–0.006 (0.007)
Firms	236,266	235,963	44,297	177,984	222,281
City $\times$ sub-experiment FE	No	Yes	No	No	No

Notes: Linear probability model on firms that exported in the year before a zone opened (openings 2001–2006), excluding fenced firms and trade intermediaries, in treated areas and in not-yet-treated comparison areas of the same sub-experiment, weighted by the odds of treatment within sub-experiments. The outcome is one if the firm did no consigned processing in the year before opening and exports under consigned processing in the post-opening window; its comparison mean is 3.4 percent. Coefficients on indicators for the distance from the firm to the arriving zone; all regressions control for the change in exposure to zones 40–150 km away and include sub-experiment fixed effects. Column (2) adds city $\times$ sub-experiment fixed effects. Columns (3)–(4): ownership from the sixth digit of the customs code; domestic are state-owned and private firms, foreign are joint ventures and wholly foreign-owned firms, including those invested from Hong Kong, Macao, and Taiwan; other ownership types enter only columns (1)–(2). Column (5) is the coefficient on the interaction of each distance bin with a foreign indicator in one regression on domestic and foreign firms, so it need not equal column (4) minus column (3). Standard errors clustered by prefecture in parentheses.

the fence; because treated and comparison firms face the same years, a nationwide drift between trade forms is differenced out. The outcome is starting consigned processing: none in the year before opening, some in the post-opening window.

Nearby exporters became more likely to start consigned processing: within 20 km of the zone by 1.2 percentage points (s.e. 0.5) against a comparison mean of 3.4 percent, while the probability of starting imported-materials processing did not change (coefficient -0.001 , s.e. 0.007). By distance, the estimates are in column (1) of Table 11. The zones added the kind of experience that does not travel.

Columns (3)–(5) show who switched. Within 5 km of the zone, domestic firms became 4.4 percentage points (s.e. 1.2) more likely to start consigned processing and foreign-invested firms 1.8 points (s.e. 1.0). The foreign-minus-domestic difference, estimated in one regression that interacts each distance bin with a foreign indicator rather than by subtracting the two columns, is -1.9 points (s.e. 2.0, $p = 0.340$). At 20–40 km the two estimates are 1.3 and 1.1

points. Within 10 km the point estimates for domestic firms exceed those for foreign-invested firms, beyond 10 km the two are similar, and no difference is statistically significant. The shift is therefore not a foreign-firm phenomenon: domestic exporters were drawn into consigned work at least as strongly, and for domestic firms consigned experience has no predictive content for new markets ($\omega_A = 0.03$). The zones reached local firms, in the form that leaves them no portable experience. Domestic firms that switched entered more new markets afterwards than those that did not, but that comparison reflects which firms take on new business, not what consigned work does for them (Appendix E.6).

Where the shift happened tells whether it is learning from neighbours. Learning from nearby firms should fade with distance and should survive a comparison within the same city. Neither holds. For all firms, the effect at 20–40 km (1.1 points) is at least as large as at 5–10 km (0.9) and 10–20 km (1.1); only the 0–5 km bin is larger (2.2). Compared within the same city, every estimate becomes small and insignificant (between -0.9 and -0.5 points; column (2)). Among domestic firms the estimate is also largest within 5 km, but it remains positive and precisely estimated at 20–40 km. The zones reorganized processing across the host city rather than seeding it next door. The reorganization is also small in aggregate: even if all of it were attributed to the zones, the weights of Table 10 imply a change in nearby ordinary-trade entry of about -0.1 percent.

Why would firms take on work that teaches them little about markets? Because it asks little of them. A consigned processor needs neither working capital for inputs nor buyers of its own, and credit-constrained Chinese firms sort into exactly this form (Manova and Yu 2016). The city-wide pattern fits a zone that brings foreign principals to a city and, with them, a supply of such contracts to the firms around it. The contracts pay for processing; they do not teach a firm to export on its own.

8.3 The stock of portable experience did not rise

A shift on one margin does not settle the net effect: nearby firms could have added ordinary-trade destinations, imported-materials destinations, or new exporters while some of them took on consigned work. The portable-experience stock adds up all margins for the non-fenced, non-intermediary exporters of a 0.1-degree cell c ,

$$K_{ct} = \sum_{f \in c} \left(N_{ft}^O + \hat{\omega}_I N_{ft}^I + \max\{\hat{\omega}_A, 0\} N_{ft}^A \right), \quad (8)$$

where N_{ft}^r is the number of destinations firm f serves under form r and the weights come from equation (7) estimated on a 5 percent sample of firms that is excluded from this analysis.

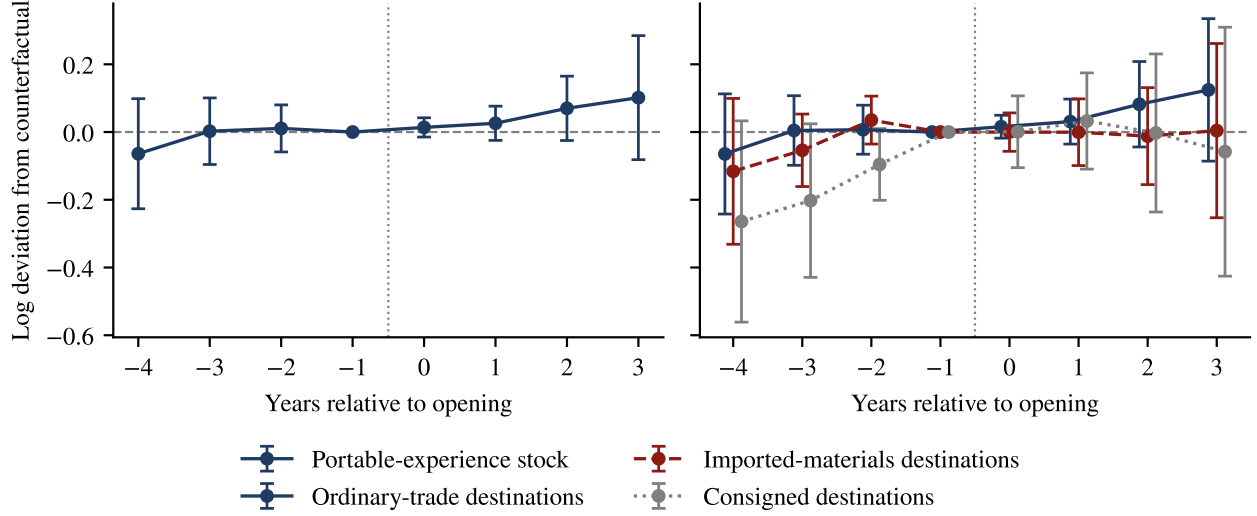


Figure 10: Portable export experience of nearby firms around the opening of an EPZ

Notes: Event-time estimates of the log gap between treated cells and the counterfactual implied by not-yet-treated comparison cells, relative to $e = -1$, for cells within 40 km whose count of special customs zones rises with the arrival of an EPZ, 2001–2006. Left: the portable-experience stock of equation (8). Right: destinations served under each trade form. Consigned destinations, whose estimates rise toward zero before opening, enter the stock with weight zero, so their path does not move the left panel. Outcomes are cell totals over non-fenced, non-intermediary exporters. Bars are 95 percent confidence intervals from a bootstrap over prefectures.

Setting the negative consigned weight to zero keeps the stock from falling mechanically when firms switch to consigned work. If proximity to a fence taught local firms to export, K should rise after opening. We estimate its path with the customs counterpart of the main design: cells whose count of special customs zones within 40 km rises with the arrival of an EPZ, compared with not-yet-treated cells that start in the same exposure state and receive a zone later, reweighted by the odds of treatment, with a proportional estimator that keeps cells without exporters and inference by bootstrap over prefectures (Appendix E.4).

The stock did not measurably rise (Figure 10). Before opening the estimate is -0.017 (s.e. 0.045). Over the first four years after opening the stock rises by 0.053 log points (s.e. 0.041), and the 95 percent confidence interval excludes increases above 13 percent. Ordinary-trade destinations move by 0.063, and imported-materials and consigned destinations by -0.002 and -0.007 . Restricting treatment to cells within 20 km, as in the main design, gives 0.034 (s.e. 0.040). Set against this, fenced exports in the fifth year after opening equal 39 percent of the pre-opening exports of nearby firms, summed over the 37 EPZs that opened in 2001–2006 and so are observed five years later: a large new body of exporting next door left no measurable trace on the portable experience of the firms around it.

Learning could be slow. Trade forms are recorded only through 2006, so the stock can be followed for four years. Extending the window to 2011 with ordinary-trade outcomes,

which are recorded throughout, the number of nearby ordinary-trade exporters changes by 2 percent (s.e. 13) four to eight years after opening, and the number of new product–destination markets that survive into the following year by 8 percent (s.e. 13). The window ends in 2011 because customs records from 2012 no longer carry the postcode that locates new firms. These long-run intervals are wide: they show no gain but cannot exclude a moderate one. We therefore state the result as no measurable increase: effects above 13 percent are ruled out over the first four years, and the long-run intervals are wider. Other comparison pools, the timing of arrival, and splits by distance and electronics intensity are in Appendix E.4.

8.4 Direct contact with fenced firms

The remaining route is direct contact. A local firm that sees a fenced firm sell its product in a market learns that the market is viable, even without portable experience of its own (Aitken et al. 1997). This route predicts that a fenced firm’s sales of a product to a destination raise nearby firms’ entry into the same product–destination market, and that it matters in proportion to how often fenced and local firms sell the same things. We add three signals from year $t - 1$ to a conditional-logit version of the entry model: fenced firms of an operating zone within 20 km sold the same four-digit product to d ; as a placebo, fenced firms sold only other products to d ; and other exporters in the same city sold the product to d (Appendix E.5).

The fence signal is product-specific but rare. Its coefficient is 0.14 (s.e. 0.07), about 0.13 of a firm’s own experience with the destination, while fenced sales of other products carry -0.04 (s.e. 0.04). Markets that fenced firms enter only in the following year carry a coefficient of 0.07 (s.e. 0.07), so we treat 0.14 as an upper bound on the information in the signal. Only 0.46 percent of local entry decisions face such a signal, against 43 percent that face one from other local exporters: fenced and local firms rarely sold the same products. At the upper-bound coefficient, switching the fence signal off lowers nearby firms’ ordinary-trade entries by about 0.5 percent; even if fenced firms overlapped with local firms’ markets as often as local exporters overlap with each other, entries would be only about 6.2 percent higher. Signals among local exporters are larger (0.37), but a placebo built from local firms that first sell the product to d only in the following year has the same coefficient (0.37). A signal that the future predicts as well as the past measures shocks common to a city’s exporters, not information passed between them, so the peer route cannot be credited either.

Together these results are a negative test of localized peer learning on the two signatures such learning should leave. Learning from neighbours should be local: the consigned shift shows no gradient beyond the first 5 km and vanishes when firms are compared within the same city. It should run through shared markets: fenced firms supply a same-product signal

Table 12: Export know-how: channels, predictions, and evidence

Channel	Prediction	Estimate	Verdict
<i>Which experience carries over (Section 8.1)</i>			
Portability	Experience in either processing form predicts a firm’s entry into similar new markets	Portability weight, imported materials ω_I : 0.36 (0.15); consigned ω_A : -0.44 (0.18); $p(\omega_I = \omega_A) < 0.001$	Imported materials only
<i>What the zones added (Sections 8.2–8.3)</i>			
Shift in trade form	Nearby exporters take up a form that builds portable experience	Start consigned processing, within 20 km: 1.2 pp (0.5; comparison mean 3.4%), 0–5 km: 2.2 pp (0.8), domestic firms 0–5 km: 4.4 pp (1.2); start imported materials, within 20 km: -0.001 (0.007); within city: -0.9 to -0.5 pp	Consigned, city-wide
Capability stock	The portable-experience stock of nearby exporters rises	Within 40 km, years 0–3: 0.053 (0.041) [$\leq 13\%$]	Bounded
<i>Direct contact (Section 8.4)</i>			
Signals from fenced firms and peers	Nearby firms follow fenced firms and local peers into their markets	Fenced-firm signal present in 0.46% of entry decisions, city-peer signal in 43%; peer signal 0.37, lead placebo 0.37	Rare; peer signal fails placebo

Notes: Each row is a way the zones could have spread export know-how, its prediction, the estimate that tests it, and the verdict; panel headings give the section with the full evidence. Portability weights ω_r are relative to ordinary trade. The shift in trade form is in percentage points (standard errors in parentheses) except the within-20 km imported-materials coefficient, which is a probability; the capability stock is in log points, with its 95 percent upper bound in percent in brackets. Verdicts: *imported materials only*, only that form’s experience predicts entry into similar new markets; *consigned, city-wide*, the zones added consigned rather than portable experience, and the shift does not fade with distance or survive a within-city comparison; *bounded*, only an upper bound is informative; *rare; peer signal fails placebo*, the fence signal reaches few entry decisions and the peer signal is as large when measured from the following year.

to only 0.46 percent of local entry decisions, and the signal among local firms fails its time placebo. Each piece alone could be explained away; jointly they reject localized peer learning as an account of what changed around the zones.

Summary. The zones did reach local exporters. After a zone opened, more of them—domestic firms at least as much as foreign ones—took on work across the host city, but they took it on as consigned processing, in which the foreign principal keeps the customer. That form records a current contract, not a market the firm can find again or build on, and the stock of portable export experience around the zones did not measurably rise; the direct route, selling the same products to the same markets as fenced firms, was rarely open. Table 12 collects the evidence. For place-based policy the lesson is that the export activity a zone brings teaches the local economy only if local firms carry it out in a form they control.

9 Where the Effects Travel: Borders and Channels beyond Geography

Sections 5–8 looked within the host prefecture; the zone’s costs need not stop there. They could travel along two maps. On the administrative map, the host prefecture approves, finances, and runs the zone and controls the resources around it, so what happens to a firm depends on which side of the host’s border it sits. On the economic map, the zone reaches whoever is linked to it—by industry, by input–output flows, by political rivalry between officials, by public finance, or by firm networks—wherever they are. Designs that compare the two sides of an administrative border assume that the far side is untouched; when economic links cross the border, that side is where part of the effect lands (Jardim et al. 2022). Section 9.1 follows incumbent manufacturers on the two sides of the host’s border; Section 9.2 asks why the host’s own incumbents are shielded; Section 9.3 asks whether anything travels along the economic links instead.

9.1 Across the prefecture border

The zones kept out firms that would have entered (Section 6); firms that were already in place near the zone were not hurt: within 20 km, incumbent output changes by -0.005 (s.e. 0.046) and incumbent employment by 0.064 (s.e. 0.035) in the grid-level design (Table 5). Rationing of development access predicts exactly this, because incumbents already hold their land.

The grid-level design does not distinguish incumbents by jurisdiction, and the labor market it draws on does not stop at the border: a worker can commute from, and a firm can buy from or sell to, a neighboring prefecture without any change in registration (Appendix L.11). If the zone’s costs travel through such markets, incumbents across the border should be hurt at least as much as host incumbents at the same distance. If instead the host shields its own firms, the neighbor bears more. If the costs travel by firms moving into the host, legal entities should cross the border. We follow every above-scale manufacturer in operation within 80 km of a zone in the year before it opened and compare incumbents at the same distance on the two sides of the host prefecture’s border. Controls are incumbents around sites not yet *approved* for a special zone, reweighted on site and firm characteristics (Appendix O). This design dates the control pool by approval rather than by opening, as the main design does, because an approved site is under construction: a site approved but not yet open may already be changing the incumbents around it, and would contaminate a not-yet-open control (Appendix J.2). A second sample restricts the comparison to incumbents within 25 km of the

border, on either side, so that the two sides are compared where they meet rather than at equal distance from the zone.

Table 13: Incumbent output and survival by side of the border and distance to the zone

	(1)	(2)	(3)
	Aggregate output	Survival	Log output survival
Control group	Not yet approved, reweighted		
<i>Panel A. Host prefecture</i>			
0–20 km	−0.012 (0.062)	−0.005 (0.017)	−0.031 (0.048)
20–40 km	−0.091 (0.079)	0.024 (0.014)	−0.077 (0.071)
40–80 km	−0.022 (0.063)	0.054 (0.014)	−0.083 (0.039)
80–150 km	0.181 (0.104)	0.047 (0.024)	0.082 (0.045)
<i>Panel B. Neighboring prefecture, same province</i>			
20–40 km	−0.038 (0.098)	0.017 (0.032)	−0.036 (0.052)
40–80 km	−0.081 (0.051)	−0.035 (0.017)	−0.015 (0.033)
80–150 km	0.097 (0.039)	0.022 (0.011)	0.044 (0.031)
<i>Panel C. Neighboring minus host prefecture, same distance</i>			
20–40 km	0.053 (0.127)	−0.007 (0.033)	0.041 (0.087)
40–80 km	−0.058 (0.085)	−0.089 (0.020)	0.068 (0.047)

Notes: Post-period averages ($e = 0, \dots, 3$) from a stacked event study of incumbents present in the year before opening, with comparison cells defined by side of the border $\times$ distance bin $\times$ two-digit industry $\times$ prior exposure and province $\times$ year effects. Aggregate output is $Y_{it}/Y_{i,g-1}$ with exits set to zero; survival indicates that the firm remains above scale; both are weighted by base-year output. Log output is conditional on survival. Controls are odds-reweighted incumbents around sites not yet approved for a special zone; other control pools are in Appendix J. Panel C reports the difference between neighboring and host incumbents at the same distance, from the same regression. The neighboring side has too few incumbents within 20 km to report. Standard errors clustered by prefecture in parentheses.

Table 13 reports the result. Inside the host prefecture, incumbents are shielded: aggregate output within 20 km changes by -0.012 (s.e. 0.062), and 40–80 km from the zone the probability of remaining above scale rises by 5.4 percentage points (s.e. 1.4). Across the border the picture depends on distance. At 40–80 km, where most neighboring incumbents

are located, the neighbors' probability of remaining above scale falls by 3.5 percentage points (s.e. 1.7), and they are 8.9 percentage points (s.e. 2.0) less likely than host incumbents at the same distance to remain above scale. At 20–40 km the two sides do not differ (-0.007 , s.e. 0.033). The gap is a 40–80 km result, not a statement about every distance, and the corresponding gap in aggregate output (-0.058 , s.e. 0.085) has the same sign but is imprecise.

Survivors' output (column 3) points the same way. Neighboring survivors' output does not change at 40–80 km (-0.015 , s.e. 0.033). Host survivors' output at 40–80 km falls by 0.083 log points (s.e. 0.039), and within the border band it changes by -0.099 (s.e. 0.046). A host that keeps marginal firms above the reporting threshold adds small firms to its pool of survivors, so the average survivor produces less even if no firm shrinks: the fall in survivors' output is the counterpart of the rise in survival, not a separate loss. This coefficient is the least robust of the host's estimates—its pre-opening average is 0.070 (s.e. 0.032), and it survives only a small allowance for pre-trends ($\bar{M}^* = 0.05$)—so we use it only as a consistency check.

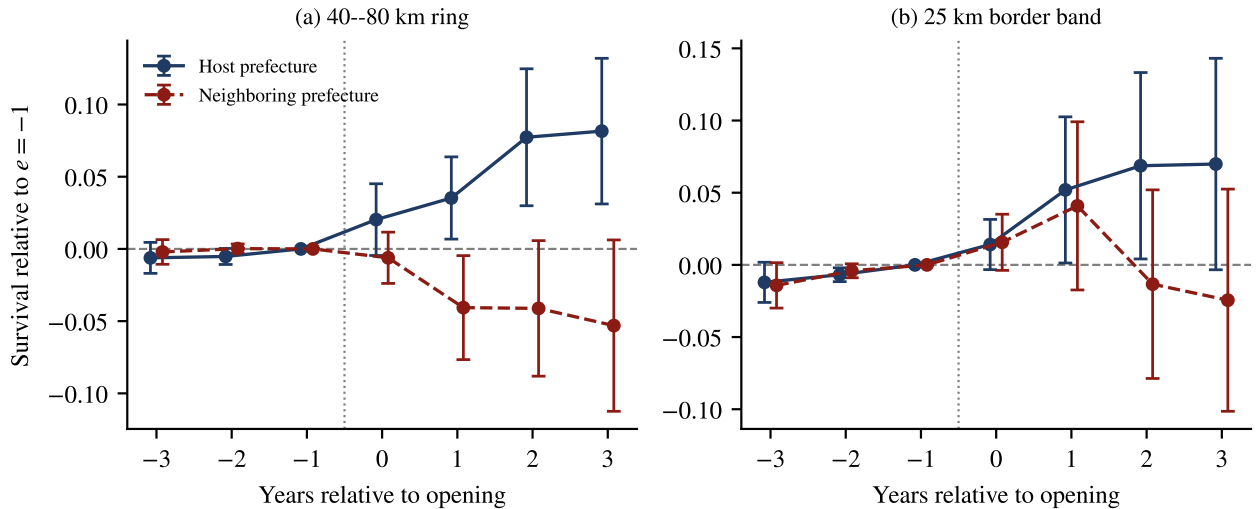


Figure 11: Survival of incumbents on the two sides of the host prefecture's border

Notes: Event-time coefficients on the probability that an incumbent remains above scale, relative to the year before opening, with 95% confidence intervals (prefecture clusters); host and neighboring incumbents from the same regression. Panel (a): incumbents 40–80 km from the zone. Panel (b): incumbents within 25 km of the land border between the host prefecture and an adjacent prefecture of the same province, on either side. Because incumbents are defined by operating in the year before opening, pre-period coefficients compare earlier presence in the above-scale universe rather than differential exit. Controls: incumbents around sites not yet approved for a special zone, odds-reweighted.

Figure 11 shows how the gap opens. At 40–80 km, host and neighboring incumbents move together before opening and part within a year of it: host survival climbs and neighboring survival falls, and neither path reverts within the window. Along the border itself the gap is 4.6 percentage points (s.e. 1.5), and it comes from the host side: host incumbents within

25 km of the border are 5.1 percentage points (s.e. 2.1) more likely to remain above scale, while neighboring incumbents' survival does not change (0.005, s.e. 0.023). The two samples define comparable firms by different geometry—distance to the zone, and distance to the border—and both place the asymmetry on the extensive margin with the host side ahead.

In money, the base-year output of incumbents retained above scale changes by RMB 42 billion (s.e. 13) on the host side and by −55 billion (s.e. 29) on the neighboring side when both sides are restricted to 20–80 km from the zone, a difference of −97 billion (s.e. 27); totals of aggregate output are imprecise on both sides (Appendix Table 56). What the host government can see—its own incumbents—shows no loss. The one precisely estimated money total is the difference between the two sides, and its losing side lies outside the host's jurisdiction.

The output coefficients cannot bear much weight: host incumbents were declining relative to controls before opening (pre-period averages 0.043, 0.063, and 0.091 at 0–20, 20–40, and 40–80 km, relative to the year before opening), and none of the output contrasts is significant even under exact parallel trends, so no breakdown value is defined for them (Appendix Table 55). The claims of this section therefore rest on survival, whose pre-period coefficients only record entry into the above-scale universe; the relevant check, the output of the same neighboring firms, is flat before opening (−0.002, s.e. 0.028). Relative-magnitude robust sets (Rambachan and Roth 2023), computed with a plug-in implementation that overstates robustness, keep the host's retention at 40–80 km up to $\bar{M} = 1.98$ but the neighbor's loss, taken alone, only up to $\bar{M} = 0.42$, so the robust statement is the one both samples share: host incumbents are more likely than their neighbors to stay in business.

The asymmetry is not produced by firms moving from the neighbor into the host. Conditional on relocating, a firm chooses a destination in another prefecture of the same province with a relative probability of 0.024 compared with a destination at the same distance in its own prefecture (Appendix Table 59): legal entities, and the tax base registered with them, stay where they are. The cost crosses the border through markets, not through firms. Among the alternative control pools, the two broad pools reproduce the result and the two narrow pools do not; the narrow pools remain unbalanced on pre-opening growth after reweighting or contain sites already under construction (Appendix J).

9.2 Why the host's incumbents are shielded

A shock that works through a labor market spanning the border should hit incumbents on both sides alike at the same distance. An asymmetry that tracks the border itself needs something that stops at the border. The obvious candidate is the host government. The

prefecture is the level that allocates land and development-zone admission, runs investment promotion, sets most industrial policy, and receives the zone’s tax revenue; it is also the level at which officials are evaluated (Section 2; Appendix L.10). Firms pay tax where they are registered, and moving a registration across a prefecture border requires re-filing it (Appendix L.11), so a host that keeps its incumbents in business keeps their tax base, while a neighbor’s firms cannot reach the host’s instruments without leaving home. Chinese local governments are also documented to extend credit and tax favors to firms they want to keep, favors that end at the jurisdiction’s edge (Bai et al. 2020). The same government that hands the fence its admission capacity (Section 3) thus controls the land and support of its own incumbents, and could steer toward them the demand for services that the zone generates.

Together with a labor market that spans the border, these instruments make three predictions. The asymmetry should sit on the extensive margin, since land and credit decide whether a marginal firm stays in business rather than how much a survivor produces: the gap is in survival, and host survivors’ average output, if anything, falls (Table 13). The cost should land where the zone draws its workers, which is down the wage gradient: in the 20–80 km sample the survival gap is concentrated where the neighbor pays lower wages than the host (-0.075 , s.e. 0.021) and reverses where it pays more (0.062 , s.e. 0.033). And firms should not escape by moving, which they do not. The first and third predictions hold in both samples; the wage gradient holds in the distance-bin sample but not within the border band (Appendix J).

If the shielding were a strategic choice, it should be stronger where the host has more reason to compete or the neighbor less capacity to respond. We test the standard measures of jurisdictional competition, with the decision rule written down before estimation: the neighbor’s fiscal capacity, and three measures of rivalry between the two prefectures’ leaders—adjacency in the province’s per-capita GDP ranking, overlap in promotion opportunities (Yu et al. 2016), and appointment within a year of each other (Li and Zhou 2005). None predicts the gap robustly. The difference between fiscally weak and strong neighbors is -0.028 (s.e. 0.036) in the 20–80 km sample and 0.006 (s.e. 0.038) in the border band. The gap is smaller between rank-adjacent rivals (0.075 , s.e. 0.041), but the sign reverses in the band, and no rivalry measure has the same significant sign in both samples. Rivals expand their own industrial land more slowly after the host’s zone opens, not faster (-0.792 , s.e. 0.333). Of the 12 group differences in survival, none of the fiscal or rivalry contrasts clears a Bonferroni threshold ($|z| > 2.87$); details are in Appendix J and Appendix K.

What the evidence supports is therefore a jurisdictional asymmetry, not a strategy. The host’s own incumbents are shielded, the cost travels outward through the labor market to the neighbor, and nothing in fiscal need or political rivalry predicts how much. We do not observe

the shielding actions themselves—which firm received land, credit, or an introduction—and the account rests on the combination of an institutional fact (the instruments are the host’s) with three implications that the data confirm. Two tests that would bear on the shielding more directly are not done here: an indirect test in the spirit of [Bai et al. \(2020\)](#), asking whether shielded incumbents sell more at home while exposed ones turn to exports, and a reverse test of whether a zone’s size predicts the promotion of the host’s officials.

9.3 Channels beyond geography

The border result says that the administrative map matters. It does not rule out that the zone’s effects also travel along economic links that ignore both distance and borders. Following the practice of measuring channels explicitly and racing them on common ground ([Greenstone et al. 2010](#); [Javorcik 2004](#)), Table 14 lists the candidates—geography, industry, input–output, political rivalry, public finance, and firm networks—with one measure and one estimate for each; the designs, the validity checks, and a figure that places every estimate on a common scale are in Appendix K.

No non-geographic channel survives its own placebo; prefecture-level adjacency effects are not robust to the estimator. Same-sector exposure does not shape the loss near the zone, and the one industry coefficient that does—a larger loss where local industries would have sold to the zone’s sector—stays within 20 km. Upstream industries become more likely to be present near a zone but do not grow, downstream users do not respond, rival border pairs look like other pairs, financing-vehicle issuance already differed before opening, and local peers’ export experience does not predict entry once the market in each city is held fixed. The one estimate that passes its placebo, employment in adjacent prefectures of the same province (0.066, s.e. 0.025), falls to 0.000 (s.e. 0.045) in a long-difference two-way fixed-effects version on the same panel. It also measures a different object from the border design: total employment in a whole prefecture, entrants and non-manufacturing included, rather than the survival of incumbent manufacturers near the zone. The prefecture panel was built on a broader treatment set as well: the 174 national-level zones in the development-zone catalogue whose names identify them as bonded or export processing zones, dated by approval, rather than the 142 special customs zones of Section 4, of which 60 are EPZs, dated by customs acceptance.

The nulls have institutional reasons. The fenced firms import their inputs duty-free and can sell only abroad, and no admission guideline conditions entry on local sourcing (Appendix L.3), so industry and input–output links have little to carry. Rivalry cannot move land or credit across a border, because those instruments belong to one jurisdiction. And the

Table 14: Channels beyond geography

Channel	Measure	Estimate (s.e.)	Conclusion
Geography: adjacent prefecture, same province	Employment of prefectures adjacent to a host prefecture (prefecture panel)	0.066 (0.025); long-difference two-way fixed effects 0.000 (0.045)	Passes its placebo; not robust to the estimator
Industry	Same-sector and backward input–output exposure to the zone’s sector, within 20 km (zip-code panel)	Same sector 0.057 (0.038); backward -0.120 (0.039)	No competition; the backward loss stays within 20 km
Upstream suppliers	Employment and presence of supplier industries (zip-code cells)	Employment 0.032 (0.024); presence +1.8 pp (0.42)	Presence only; no growth
Downstream users	Employment of user industries (zip-code cells)	0.005 (0.040)	Not detected
Political rivalry	Neighbor-minus-host survival gap, rival versus other border pairs	Rank adjacency 0.075 (0.041); promotion overlap 0.034 (0.039); same cohort -0.024 (0.040)	Not detected
Public finance	Host’s financing-vehicle bond issuance; incumbents’ effective tax rate	Issuance 0.943 (0.223), pre-trend $p = 0.0002$; tax rate -0.0020 (0.0075)	Issuance fails its pre-trend; tax rate unchanged
Firm networks	Entry into a product–destination market on local peers’ experience in it	-0.0019 (0.0019)	Not detected

Notes: Standard errors in parentheses; pp: percentage points. Geography: prefecture panel of the 174 national-level bonded areas and export processing zones, effect on employment of adjacent prefectures in the same province, with its long-difference two-way fixed-effects version on the same panel. Industry and input–output: zip-code panels, effects within 20 km of a zone. Political rivalry: the border design of Table 13, difference in the neighbor-minus-host survival gap between rival and other border pairs. Public finance: prefecture panel; pre-trend p is the joint test of no difference before opening. Firm networks: firm $\times$ product $\times$ destination panel with city $\times$ product $\times$ destination effects. Definitions, samples, and placebo checks: Appendix K.

fenced firms’ export know-how stays within their own processing networks (Section 8). What travels beyond the ring is therefore not supplier demand or knowledge but the incumbents’ survival, and it travels across the administrative border rather than along economic links.

9.4 Summary

The zones’ effects follow the administrative map rather than the economic one. Within the host prefecture, incumbents are shielded and entrants bear the cost; across the border at 40–80 km, incumbents exit, and the neighbor-minus-host difference in the base-year output of

incumbents retained in business, 20–80 km from the zone on both sides, is RMB –97 billion (s.e. 27). This is what a fence that cuts local links, placed under a host government that controls admission and the resources of its own incumbents, would produce. It also locates a cost that the host’s decision does not count, the standard argument for approving such zones above the level of the jurisdiction that hosts them (Oates 1972; Besley and Coate 2003).

10 The Net Effect: What the Zones Traded

Sections 5–9 measured the two sides of Proposition 3 one at a time: what the fence took from the local economy and what it left there. This section puts them on one scale with the admission model of Section 3. We first estimate the model from the entry effects by project size (Section 10.1), then say what the ledger counts (Section 10.2), lay it out zone by zone (Section 10.3), value an admission slot and the rules that would have changed the ledger (Section 10.4), ask who gained and who paid (Section 10.5), set the ledger against city totals and the literature (Section 10.6), and ask what it implies for the country (Section 10.7).

Every flow in this section is a *slot-year value*: the value added in the fourth operating year generated by one admission a year, which equals one project’s cumulative value added over its first four years, ages zero to three. It is the unit of Proposition 3 with $H = 3$. For comparison with annual figures, a domestic entrant near the zones produces RMB 18.94 million of value added a year on average over ages zero to three; a slot-year value adds up those four years weighted by the probability that the project survives, so it is less than four times the annual figure.

10.1 Estimating the admission model

Sections 5 and 6 established that the loss fell on small projects: entry near the fence fell most in the smallest national size quintiles and not in the largest. What the lost projects were worth, what a slot in the queue was worth, and what another allocation rule would have produced require the admission model of Section 3. We estimate it from reduced-form estimates already reported, changing none of them, and judge it on moments it was not fitted to (Kline and Moretti 2014a; Busso et al. 2013).

Estimable form and identification. A project’s size a is its birth size, a normal score within its two-digit industry and birth year nationally, so the national size quintiles are fixed intervals; applicants’ sizes are $a \sim \mathcal{N}(m, s^2)$. The priority index of Section 3 is $x = a + u$, where $u \sim \mathcal{N}(0, \sigma_u^2)$ captures everything other than size that moves a project in the queue.

The fence raises the admission threshold on x from k_0 to $k_0 + \Delta$, so entry in size quintile q changes by

$$\beta_q = \log \frac{\Pr(a \in B_q, a + u \geq k_0 + \Delta)}{\Pr(a \in B_q, a + u \geq k_0)}, \quad (9)$$

where B_q is the q -th national quintile. We estimate $(s, \sigma_u, k_0, \Delta)$ by minimum distance from nine moments: the five quintile effects of Section 6 and the shares of pre-opening entrants in four quintiles. The level of the β_q pins down Δ and, with the shares, the entry loss ρ ; their slope in size pins down σ_u relative to s ; the shares locate the threshold. The share of applicants admitted before opening is not identified, because only admitted projects are observed; we set it to one half, and ρ moves little over the range 0.05 to 0.95 (Appendix R.4).

The queue follows output, not fixed investment. The 2004 intensity standards suggest that zones ranked projects by fixed investment, so our first specification measures size by birth fixed assets. It fails a test it was not fitted to: it predicts that the loss falls only slightly with output size (top minus bottom 0.15), whereas the data show a steep gradient (0.58, $p = 0.024$). With size measured by birth output, the model predicts the fixed-asset gradient it was not fitted to (predicted 0.22, data 0.34, $p = 0.511$) and the size of surviving entrants. This is the main specification: whatever zones wrote down as investment intensity, the index that governed admission followed the scale of a project’s operation, its output, not the fixed investment recorded at birth (Figure 12).

Estimates and further tests. Entry near the fence falls by 19 percent (95 percent interval 8–32), against 20 percent implied by the reduced-form entry coefficient β as $1 - e^\beta$ (Appendix R.1); the agreement is by construction, not a test. Because refused projects do not reappear elsewhere in the host prefecture (Section 6), λ is close to zero and this entry loss is also the fence’s slot share, $m_z/\bar{K}$. The priority index correlates 0.81 with project size (95 percent interval 0.50–0.95, and 0.62–0.92 over the admission rates the data cannot distinguish): size governs the queue, but not alone. Two predictions are tested. The threshold shift depends on the slots taken, not on the fence’s exports: predicted and observed losses agree around empty and busy fences (joint $p = 0.84$). The model does not generate the imprecise larger loss in less capital-intensive industries (predicted 0.005, $p = 0.11$), which more likely reflects screening standards set by industry. We now use the estimated model to value the projects the fence blocked.

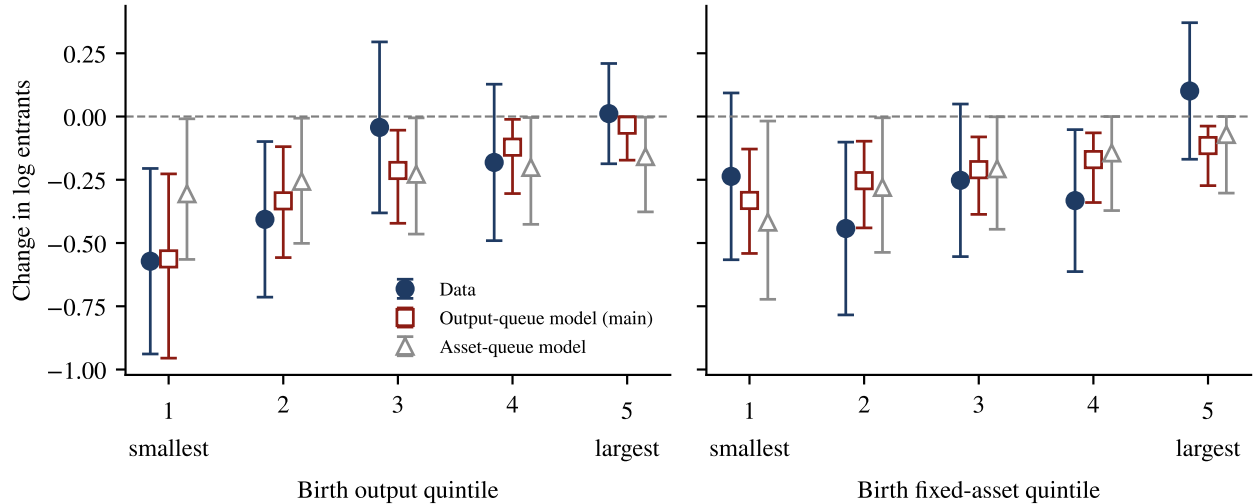


Figure 12: The admission model against the data

Notes: Change in log domestic entrants near an EPZ by birth-size quintile. Circles: reduced-form estimates (Section 6) with 95 percent intervals. Squares: the main specification, in which the priority index follows birth output; it is fitted to the output quintiles (left) and predicts the fixed-asset quintiles (right). Triangles: the alternative in which the index follows birth fixed assets; it is fitted to the fixed-asset quintiles and predicts the output quintiles. Model intervals from 499 bootstrap replications over zones. Estimates and tests in Appendix Table 68.

10.2 What the ledger counts

The ledger is a local value-added footprint, not a welfare calculation. For each zone it counts two flows within the host prefecture: the income the fence pays to local workers, and the value added of the domestic projects the fence kept out of the development zone. It leaves out the fiscal cost of the zones—duties and import-stage taxes not collected on bonded imports, and the cost of servicing the land—the profits of the fenced firms’ owners, gains to consumers and to the rest of the country, what the crowded-out projects produced if they were founded elsewhere, and their growth after their fourth year. Section 10.5 says what can be priced of the omitted items; Section 10.7 discusses the rest.

We keep the local scope for two reasons. Local value added is what the zones were approved to raise and what the host government, which allocated the land, can observe. And it is what the evidence identifies: every estimate in this paper compares places near a zone with places that do not yet have one, and such comparisons identify a zone’s effect on its surroundings relative to the controls, not its effect on the country (Kline and Moretti 2014b).

The fence’s side is $\sigma_z Q_z$, where Q_z is the zone’s mean annual fenced exports over its first operating years (up to four) and σ_z is the wage bill of fenced firms per unit of exports: 6 percent in the median of the 17 zones whose fenced firms can be matched to the firm survey (Section 7.1). Local suppliers do not expand with the fence (Section 7.3), so we set $s_z = 0$; the customs domestic value-added ratio, at most 28 percent of exports in the median zone,

gives the bound most favorable to the zone. The cost side, $C(m_z)$, is the slot-year value of the projects kept out by the fence's slots: the slots it took, m_z , times the value of the projects the estimated queue places below the raised threshold. Because the lost projects are the smallest, $C(m_z)$ is well below the value of as many average entrants. Appendix I.1 values the lost entrants instead with the reduced-form entry effect in each size quintile; that version is a robustness check.

10.3 The ledger zone by zone

The fences' exports (Figure 1) suggest that the zones paid their way. The alternative is that the program's exports were produced in a few zones while its cost, which does not depend on activity (Section 6), was paid in every zone. The two readings differ in the distribution across zones: under the first the typical zone clears its break-even scale; under the second the program total can be positive while most zones fall short.

Figure 13 plots, for each of the 24 zones with both entry and customs data, the fence's local labor income minus the slot-year value of the projects it kept out, against its fenced exports. Activity was concentrated. We call hubs the five ledger zones with the largest cumulative fenced exports over 2001–2006: Songjiang, Kunshan, Suzhou Industrial Park, Hangzhou, and Wuxi. Nanjing and Wuxi are nearly tied for fifth (US\$2.46 and 2.39 billion); we keep Wuxi, which the ledger observes for 4 operating years against Nanjing's 2. Both clear their break-even scale, so which of the two counts as a hub does not change the share of non-hub zones that fall short. The hubs produced 84 percent of the ledger zones' annual fenced exports, and the median hub exported RMB 4,761 million a year.

The break-even scale of the median zone is RMB 1,796 million of fenced exports a year (US\$219 million), and RMB 1,183–3,072 million across the interquartile range of fenced firms' wage shares. The median of the other 19 zones exported RMB 114 million a year, paid about RMB 7 million to local workers, and kept out projects worth RMB 88 million. Outside the hubs, 84 percent of zones displaced more value added than they paid to local workers, and 71 percent of all zones did. The non-hub share moves little across the interquartile range of wage shares (89 percent at the lower quartile, 84 at the upper), and 84 percent of non-hub zones (67 percent of all zones) fall short even if all of the fence's domestic value added had stayed local.

Two features of the figure limit what it can bear. One income share is applied to every zone, so the figure shows how far the typical zone falls from break-even, not which individual zones are above it: at a fixed share, larger fences clear it by construction. And each zone's break-even scale depends on how many projects its own fence kept out, so the dotted line,

drawn at the median, is not a zone-by-zone boundary: a zone to its left comes out ahead if its fence displaced few projects, and a zone to its right, one of the hubs among them, falls short if its fence displaced many. What the figure does establish is that the program's exports are not a sufficient statistic for its local effect. The few zones that reached scale carried the total; the typical zone did not repay what it took.

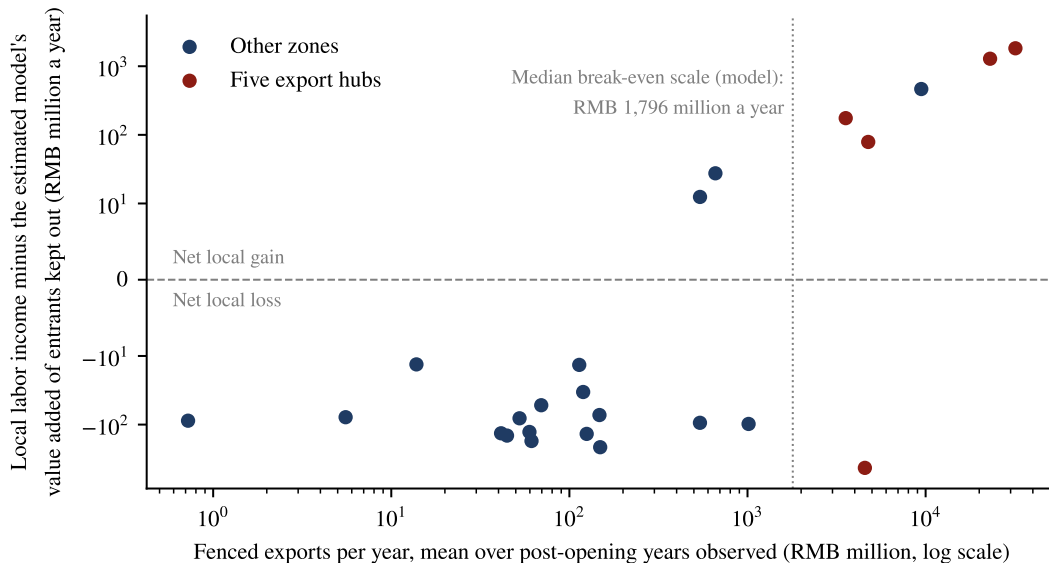


Figure 13: The local ledger of each export processing zone

Notes: Each point is an EPZ with both entry and customs data (24 zones). Horizontal axis: mean annual fenced exports over the zone's first operating years (up to four), log scale. Vertical axis, slot-year values in RMB million: local labor income of the fence (6 percent of fenced exports, the median wage share of fenced firms matched to the firm survey) minus the value added of the projects its slots kept out, valued with the estimated admission model (Section 10.1). The vertical axis is linear between -10 and 10 and logarithmic beyond, so that zones with small and large balances can be read on one scale. Dashed line: zero net effect. Dotted line: the median of the zones' break-even scales, RMB 1,796 million a year. The median break-even line is not a zone-by-zone boundary: each zone's break-even scale depends on how many projects its own fence kept out, so individual zones need not change sign at this line. Hubs: the five ledger zones with the largest cumulative fenced exports in 2001–2006 (Section 10.3). The reduced-form version is Appendix Figure 25.

10.4 What a slot was worth, and what other rules would have yielded

The model values the blocked projects from the whole distribution that the estimated queue implies, not from the quintile point estimates, of which the largest is both the least precise and slightly positive. A higher threshold can only lower entry in every size class, so the model does not let the fence admit more of the largest projects. Table 15 reports the valuation.

The projects the fence blocked were small. The average blocked project has a slot-year value of RMB 26.2 million, 43 percent of the RMB 61.5 million of an average entrant before

opening; 39 percent of blocked projects come from the smallest national output quintile and 5 percent from the largest. The marginal blocked project—the shadow value of a slot—is worth RMB 29.4 million (95 percent interval 17.6–47.4). Against this value, a slot held by the fence returned little: in the median zone the fence’s local labor income per slot it occupies is 0.14 times the value of a blocked project, and 0.06 times outside the hubs.

Summed over the 24 zones, blocked projects would have produced RMB 3,025 million of value added a year (820–8,657), against RMB 4,916 million of local labor income paid by the fences. The local efficiency loss—the value of blocked projects in excess of the fence’s local income, summed over the zones where it is positive—is RMB 1,981 million a year (350–6,996). The program total is positive, RMB 1,891 million a year, but only because of the hubs. The five hubs earn 84 percent of the fences’ local labor income, while the other zones account for 61 percent of the value of blocked projects; without the hubs the net change is RMB –1,048 million a year (–5,022 to 765). The totals are imprecise, mostly because the bootstrap carries the sampling error of the reduced-form quintile effects into them; the share of zones that fall short is estimated more tightly.

Two assumptions move the totals; neither moves the share of zones that fall short much. The first is how a zone’s capacity is measured. The main specification measures it in the years the fence occupies it, from post-opening entry and the entry estimate. Measuring it by mean annual entry before opening lowers the value of blocked projects from RMB 3,025 million to 1,208 million, the efficiency loss from 1,981 million to 545 million, and the net change outside the hubs to RMB –44 million; the share of zones that fall short is 58 percent, and 74 percent outside the hubs, against 71 and 84 percent in the main specification. Entry into the treated locations grew over the period, so pre-opening entry understates the capacity the fence took. The second is how many large projects are among those blocked: the model places 5 percent of them in the largest quintile, whose reduced-form effect is the least precise, and the specification in which admission follows fixed assets, which places more there, nearly doubles the value of blocked projects (Appendix Table 52). The unidentified admission rate before opening matters less: over its range from 0.05 to 0.95 the efficiency loss lies between RMB 1,749 and 2,240 million and the net change outside the hubs between RMB –1,247 and –870 million.

Proposition 3 says which rules would have changed the ledger, each by changing one term of equation (4); Panel C of Table 15 prices them. Siting the fence on land outside the development zone’s allocation, rule (a), sets $C(m_z)$ to zero and would have raised local value added by RMB 3,025 million a year (820–8,657); raising the zone’s capacity by the fence’s slots at approval, rule (c), has the same effect and would have required about 115 additional admissions a year across the zones. These rules matter most for the many small

Table 15: The value of an admission slot, the ledger, and other allocation rules

	Estimate	95 percent interval
<i>Panel A. Blocked projects and the slot-year value of a project (RMB million of value added)</i>		
Entry loss near the fence, $\rho \approx m_z/\bar{K}$ (percent)	19	[8, 32]
Average entrant before opening	61.5	[51.8, 69.1]
Average blocked project	26.2	[17.3, 43.1]
Marginal blocked project (shadow value of a slot)	29.4	[17.6, 47.4]
Blocked projects in the smallest output quintile (percent)	39	[23, 62]
Blocked projects in the largest output quintile (percent)	5	[0, 15]
<i>Panel B. The ledger of the 24 zones (RMB million a year)</i>		
Value added of blocked projects, total	3,025	[820, 8,657]
Local labor income of the fences, total	4,916	[1,132, 9,311]
Net change in local value added	1,891	[−4,798, 6,745]
excluding the five hubs	−1,048	[−5,022, 765]
Local efficiency loss, $\sum_z \max\{0, C_z - L_z\}$	1,981	[350, 6,996]
Zones where blocked value exceeds local income (percent)	71	[54, 88]
excluding the five hubs (percent)	84	[68, 95]
Local return per slot / value of a blocked project, median zone	0.14	[0.03, 0.74]
Break-even fenced exports, median zone	1,796	[474, 5,174]
<i>Panel C. Other rules: change in local value added (RMB million a year)</i>		
(a) Fence on land outside the admission capacity	3,025	[820, 8,657]
(b) Slots taken from the smallest admitted projects	1,015	[−33, 5,572]
(c) Capacity raised with the approval	3,025	[820, 8,657]
admission slots added a year	115	[38, 245]
(d) Fence sources locally like processing firms outside it	1,720 to 7,308	
(e) Only zones above their own break-even scale	1,981	[350, 6,996]
one break-even threshold for all zones	1,514	[327, 6,216]

Notes: Estimated admission model, main specification (admission follows birth output), for the 24 zones with entry and customs data. Values are slot-year values: the value added in the fourth operating year generated by one admission a year, equal to one project's value added over ages zero to three, in RMB million. ρ is the proportional loss of domestic entry near the fence; it equals the fence's share of admission slots, $m_z/\bar{K}$, because refused projects do not reappear elsewhere in the host prefecture. Admission rate before opening normalized to one half; 95 percent intervals from 499 bootstrap replications over zones, each redrawing the reduced-form quintile effects. Local labor income is 6 percent of fenced exports; the efficiency loss sums the value of blocked projects in excess of local income over zones. Rule (b) takes the fence's slots from the smallest projects admitted before opening. Rule (d) does not use the model and is a range because the fence's purchases are bounded only from above. Rule (e) assumes that exports are known at approval. Sensitivity, the reduced-form ledger, and national accounting: Appendix Tables 52, 51, and 49.

zones, whose ledger is dominated by the fixed cost. Taking the fence’s slots from the smallest admitted projects, rule (b)—the aim of the auction rules introduced for industrial land in 2007—would have added only RMB 1,015 million (−33 to 5,572), 34 percent of rule (a), because the queue already ranked projects largely by scale (Section 10.1): most of the loss comes from how many slots the fence took, not from which projects lost them. Letting the fence buy locally as much as processing exporters outside it, rule (d), would have added RMB 1,720 to 7,308 million, a range because the fence’s current purchases are bounded only from above; this rule does not use the model and matters most for the hubs, where most of the value leaks out. Approving only zones whose exports would reach their own break-even scale, rule (e), would have recovered the whole efficiency loss, RMB 1,981 million, with 7 of the 24 zones approved, and a single threshold at the median break-even scale, which does not require knowing each zone’s cost, RMB 1,514 million; both assume that exports were known at approval and are upper bounds. The rules differ in what they ask of the host: (a) and (c) need land or construction quota the host did not have, (e) needs foresight, and only (d) changes the fence’s rules rather than its site or its number.

10.5 Who gained and who paid

Following Busso et al. (2013), Table 16 lists each party the zones affected that this paper’s estimates can price, the effect on it in money, and its sensitivity. The rows are not additive: they differ in unit (the median non-hub zone, a pooled total) and we do not sum them.

In the median non-hub zone, local workers received about RMB 7 million a year in wages from the fence (4.0 to 10.5 million across the interquartile range of wage shares), while the projects kept out would have produced RMB 88 million of value added a year, valued with the estimated model; the table also reports the reduced-form and average-entrant valuations. Local suppliers gained nothing measurable (Section 7.3). The fenced firms, mostly foreign-invested (Table 2), retained at most RMB 25.0 million a year of domestic value added beyond the wage bill, an upper bound that contains their profits. Across the prefecture border, the base-year output of incumbents 20–80 km from the zone that stay above scale changed by RMB −54.6 billion (s.e. 29.4), pooled over all zones and post-opening years; the difference from host incumbents on the same distance range, RMB −97 billion (s.e. 27), is the one money total of Section 9 that is precisely estimated.

We do not estimate fiscal costs. The central government’s forgone tariffs and import-stage value-added tax on bonded imports, and the host’s revenue from industrial land leases, have been priced only with a calibrated model on other samples; Appendix Table 50 reports both and explains why neither enters the ledger.

Read as a whole, the table places the zones’ local costs on parties that do not appear in a zone’s own statistics. The item that zone statistics record, the fence’s wage bill, is a gain; the items they do not record—projects that were never admitted and incumbents across the border—are losses. An evaluation that counts activity inside the zone sees only the first.

Table 16: Who gained and who paid

Party	Item (RMB million a year per zone, unless noted)	Sensitivity	Source
Local workers	Wage bill of the fence: +7.0 (median non-hub zone)	4.0 to 10.5 across the interquartile range of fenced firms’ wage shares	This paper
Domestic entrants kept out	Value added of the entrants kept out, in the fourth operating year: −88 (median non-hub zone, estimated admission model)	Valued with the reduced-form size quintiles: −59; valued as average entrants: −223	This paper
Local suppliers	No money estimate: supplier entry, employment, and output do not rise with the fence	Point estimates and placebo checks in the linkage analysis	This paper
Foreign owners	Domestic value added beyond the wage bill, an upper bound that contains profits: at most 25.0 (median non-hub zone)	The fenced firms also receive the tax expenditure counted in appendix Table 50	This paper
Incumbents in neighboring prefectures	Change in base-year output of incumbents 20–80 km away that stay above scale: −54.6 billion (s.e. 29.4), all zones and post-opening years pooled	Neighbor minus host, both sides 20–80 km: −97 billion (27); aggregate output instead: −141.9 billion (92.1)	This paper

Notes: Signs from the point of view of each party; all rows are estimated in this paper. Median non-hub zone: median over the 19 zones outside the five hubs, slot-year values in RMB million (Figure 13); the entrant row values the lost entrants with the reduced-form quintile effects, with the estimated model (Table 15), and as average entrants. Foreign owners: customs domestic value-added ratio minus the wage share, times fenced exports. Neighboring prefectures: base-year output times the post-opening effect, summed over treated incumbents 20–80 km from the zone (Appendix Table 56); the neighbor-minus-host difference in the sensitivity column compares with host incumbents at 0–80 km, and with host incumbents at 20–80 km it is RMB −97 billion (s.e. 27). Rows differ in unit and are not added. Fiscal items: Appendix Table 50.

10.6 City totals and the literature

City-level evaluations of China’s zones find that designation raises foreign investment and output (Wang 2013; Alder et al. 2016). Our null could come from our choices—a narrow zone definition, or an estimator that compares treated cities only with cities not yet treated—rather than from the zones. If so, the literature’s broad zone category and two-way fixed effects should show gains on every outcome, and moving to EPZs or to our estimator should remove

them. If instead the gain is confined to what cities report as foreign investment, it should appear for reported investment and not for the output of foreign-invested firms or of industry as a whole, whatever the zone definition or estimator.

Table 17 crosses the two zone definitions with the two estimators on our city panel. Reading across columns (1) and (2), or (3) and (4), changes only the estimator; reading (1) against (3), or (2) against (4), changes only the zone definition. Column (1) reproduces the literature: on national-level development zones, reported utilized foreign investment rises by 0.208 log points (s.e. 0.089). Our estimator on the same zones gives 0.140 (s.e. 0.081), smaller and less precise but of the same sign, so the estimator is not what separates us from the literature; the yearbook sample of EPZ cities is too small to estimate this outcome. The difference is the outcome. The output of foreign-invested firms, measured in the firm survey rather than reported by the city, does not rise in any column (0.101, s.e. 0.125, in column 1; -0.038 , s.e. 0.055, for EPZs with our estimator). Neither does industrial output: the one coefficient distinguishable from zero is negative, for EPZ cities under two-way fixed effects (-0.090 , s.e. 0.043). Customs exports do not rise either; for this outcome the broad category is the special customs zones as a whole, on which two-way fixed effects give 0.053 (s.e. 0.089). The positive result in the literature is a result about reported investment, which the activity measured in firm and customs records does not match.

Manufacturing output of EPZ cities changes by -0.036 log points relative to cities that had not yet received a special zone (s.e. 0.026; 41 cities), and by -0.048 outside the five hubs (95 percent interval -10.0 to 0.3 percent). The ledger was not fitted to city totals, and adding up the fence's output, the output of blocked entrants, and the upstream response predicts a change in host-city output of -0.20 percent of pre-opening output, the same sign as the city estimates and inside their intervals (Appendix I). City totals are too noisy to test the ledger sharply, but they rule out gains in manufacturing output larger than 1.6 percent and, outside the hubs, point the same way as the ledger.¹

10.7 Local and national

Place comparisons cannot say whether the country lost anything. They difference out whatever is common to treated and control places, including any change in the national level of activity—the missing intercept of spatial designs (Kline and Moretti 2014b). Two national margins are invisible to them. On the cost side, a project kept out of the host's

¹These comparisons concern proportional changes relative to cities not yet treated. Counterfactuals that extrapolate each treated city's own pre-opening trend in levels answer a different question and, in fast-growing coastal cities, can attribute growth in foreign-invested output to a zone. On our city panel, designs that adjust for each city's own pre-opening path produce significant effects at a placebo approval date two years early, so we do not use them.

Table 17: The literature’s city-level setting and ours

Outcome	(1) Broad zones, city TWFE	(2) Broad zones, this paper’s estimator	(3) EPZ, city TWFE	(4) EPZ, this paper’s estimator
Reported utilized FDI (city yearbook)	0.208 (0.089)	0.140 (0.081)	n.a.	n.a.
Foreign-invested firms’ output (firm survey)	0.101 (0.125)	−0.105 (0.065)	−0.068 (0.084)	−0.038 (0.055)
Industrial output, all sectors (firm survey)	0.014 (0.066)	0.068 (0.074)	−0.090 (0.043)	−0.020 (0.032)
Customs exports	0.053 (0.089)	−0.016 (0.068)	−0.056 (0.091)	−0.051 (0.065)

Notes: Effects of a city’s first zone of the given category, dated by approval year, on log city outcomes; standard errors in parentheses. Broad zones, by data source: for foreign investment reported in city yearbooks, national-level economic-and-technological and high-technology development zones; for firm-survey outcomes, a city’s first national-level zone of any type; for customs exports, all special customs zones (EPZs and the bonded zones). City TWFE: two-way fixed effects (city and year), all other cities as controls, standard errors clustered by city. This paper’s estimator: [Callaway and Sant’Anna \(2021\)](#) with cities not yet treated as controls, averaging the first three years after approval for firm-survey outcomes and the first six for the others; city cluster bootstrap. n.a.: the yearbook sample of EPZ cities is too small to estimate. Firm-survey outcomes aggregate above-scale firms by prefecture, 1998–2007; exports come from customs records.

development zone may have been founded elsewhere in China; if a share ϕ was, the national cost is $1 - \phi$ times the local one. On the fence’s side, fenced exports may have displaced processing exports elsewhere or may have come to China only because of the fence, and the central tax expenditure is a national cost that the local ledger does not see.

The evidence speaks only to the host prefecture. The missing entrants do not reappear in untreated locations of the host city, and relocating firms rarely cross a prefecture border (Section 6), so the within-prefecture relocation share λ is close to zero; a project founded in another prefecture from the start leaves no trace in our data, so ϕ is not identified. Scaling the value of blocked projects by $1 - \phi$, the efficiency loss falls from RMB 1,981 million a year at $\phi = 0$ to 775 million at $\phi = 0.5$ but stays positive, because in the zones furthest below break-even the fence displaced more than it left even at half the cost (Appendix Table 49).

We do not attach a sign to the national effect. It would require the intercept on both sides of the ledger—where blocked projects went, and what the fences’ exports displaced or created elsewhere in the country—which place comparisons cannot deliver; [Busso et al. \(2013\)](#) draw the same line between local incidence and national efficiency. What the evidence establishes is the statement a host government needs: an EPZ added to its host’s value added only if it reached a scale that most zones never reached.

11 Conclusion

This paper measures what China’s export processing zones did to the local economies that hosted them. What it measures is a local footprint—the value added the zones created and displaced within their host prefectures—not welfare: it omits the value accruing to foreign owners and to the national economy, the fiscal cost of the zones, where crowded-out projects went if they went anywhere, and the growth those projects would have achieved after their first years.

Within that scope the answer is clear. A zone that is fenced off from the local economy contributes to it only through what it leaves behind, and it takes something away regardless of what it attracts. The fences left little: a small share of each exported dollar as local wages, no expansion of local suppliers, no productivity spillover to nearby firms, and a kind of export experience that does not carry to new markets. They took admission to the development zones that hosted them, and the projects they kept out were small local ones that never appear in a zone’s own statistics. The cost did not stay inside the host’s boundary: incumbents across the prefecture border became more likely to exit, while those inside it, whose land and credit the host controls, were insulated.

The lesson reaches beyond export processing zones. Place-based policies are usually evaluated by the activity they attract. When a policy draws on a locally rationed resource—land, approvals, the attention of a zone administration—that activity is not a sufficient statistic for local development, because the resource’s alternative use does not show up in the zone’s statistics, and because the rules that make the zone attractive may be the same rules that keep its activity from spreading.

Two design rules follow. First, a zone that cannot expect to reach its break-even scale—about RMB 1,796 million of exports a year for the median zone we can value—should not occupy capacity that local firms would otherwise use; siting fences on land outside the development zones’ allocation, or approving only zones expected to reach that scale, would have raised local value added by RMB 3,025 million and RMB 1,981 million a year across the 24 zones we can value (Section 10). Second, where zones do reach scale, the rules that fence them off determine how much of it stays local. Permitting sales to local processors was not enough in 2005; the reforms after 2015 that merged the zones into comprehensive bonded zones and opened them to domestic sales are the margin to study next.

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A Background Exhibits

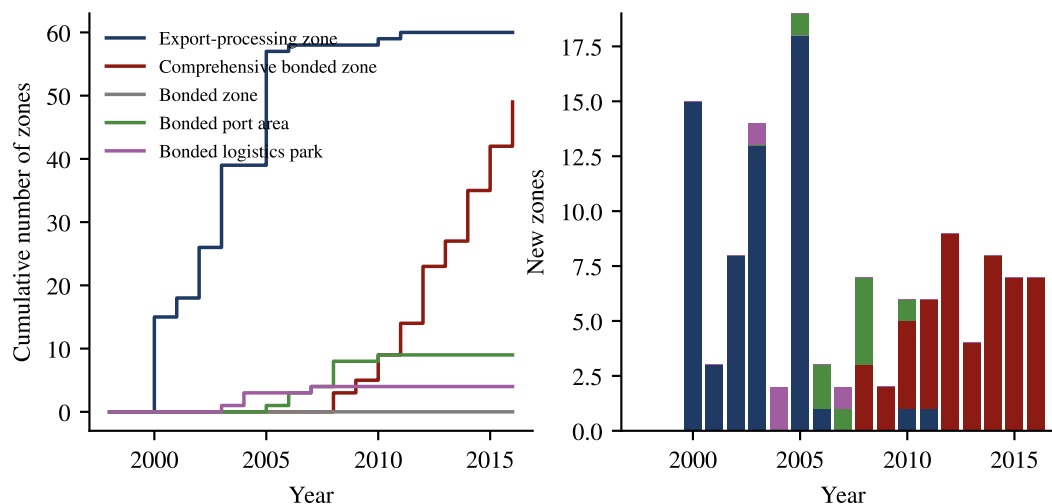


Figure 14: Special customs supervision zones by type and year

Notes: Zone directory. EPZs are identified by name. Comprehensive bonded zones absorbed most EPZs after 2015.

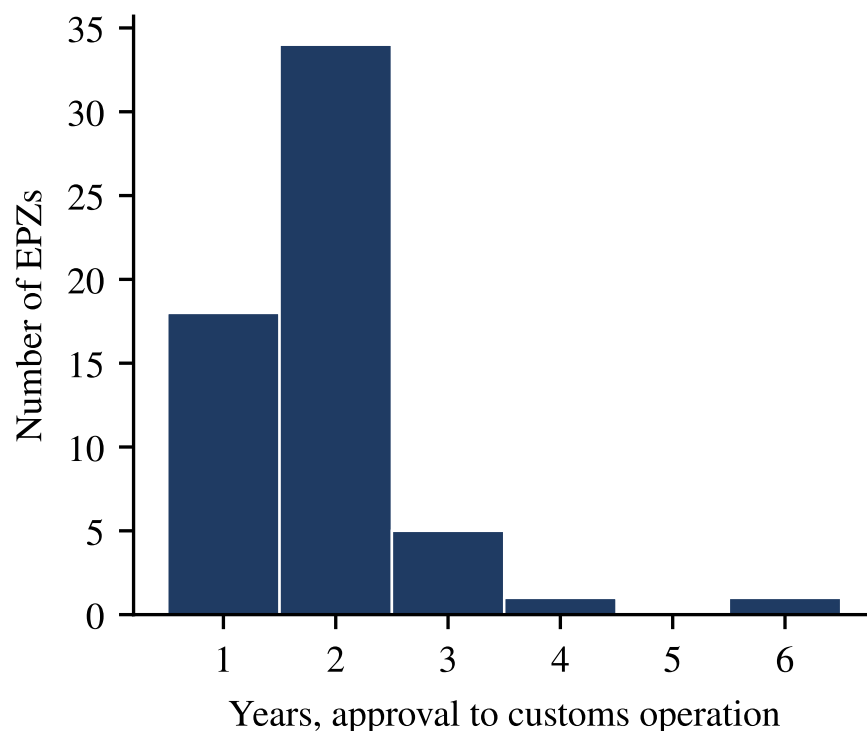


Figure 15: Years from State Council approval to customs operation

Notes: 59 EPZs; operation is the official customs acceptance year where available (36 zones), otherwise approval plus two years. Median 2 years.

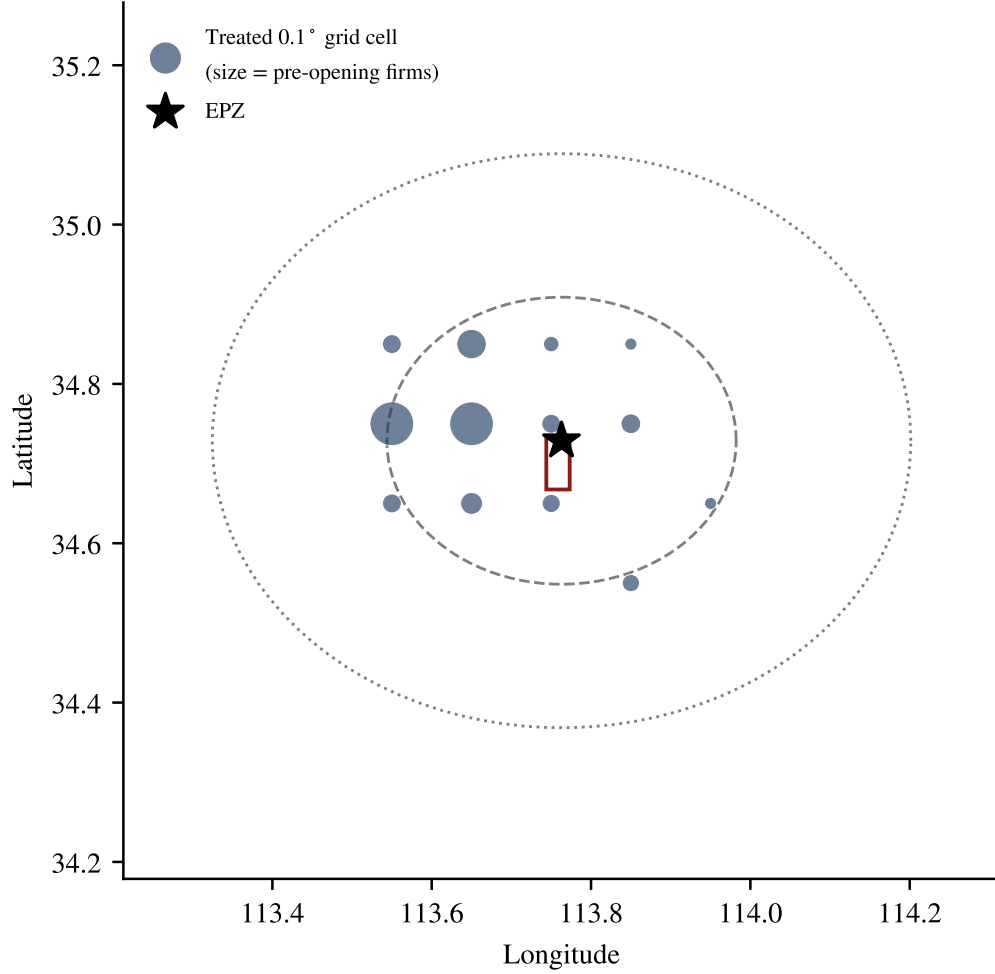


Figure 16: An example: the Zhengzhou EPZ and its surroundings

Notes: EPZ location, its parent development zone, 20 km and 40 km rings, and the 13 treated 0.1-degree cells shaded by pre-opening firm density.

B All Location-Level Outcomes

Table 18 estimates equation (5) for every location-level outcome the survey supports, under the main comparison group (column 1) and against cells around sites that later host an EPZ (column 2). The future-site group was chosen by the same state for the same kind of zone, but it is smaller and less balanced on pre-opening characteristics, so it serves as a check rather than as a second main estimate. Against it, domestic entry falls by more (39 percent, interval -53 to -20) and incumbent exits rise. Of the joint pre-opening tests in column (1), those for active firms ($p = 0.02$) and incumbents' subsidy income ($p = 0.01$) reject, and value added comes close ($p = 0.06$), because the 2004 census did not collect value added and the pre-opening sample for that outcome therefore differs from the one for gross output (Appendix C.3); none of these outcomes carries the argument of the paper. Figure 17

overlays the two groups' event studies. For new domestic firms their pre-opening paths slope in opposite directions and their post-opening paths nearly coincide, the second reading of the pre-opening rise in Section 5.2.

C Design Checks for the Local Effects

This appendix documents the checks of Section 5: bounds on the pre-opening trend in domestic entry (§C.1), inference for the entry estimate (§C.2), pre-opening trends in incumbent outcomes (§C.3), the comparison of binary and exposure designs (§C.4), the distance rings, continuous profile, and break search (§C.5), and the contamination of the control pool (§C.6).

C.1 The pre-opening trend in domestic entry

Target. Every object in Table 19 concerns the average post-opening event-study coefficient for new domestic firms, $\theta = \frac{1}{4} \sum_{e=0}^3 \beta_e$, measured against the year before opening. Unadjusted, it is -0.427 (s.e. 0.123) in the main design. It differs from the pooled coefficient of Table 5 (-0.219) because the pooled regression measures the post-opening years against all pre-opening years rather than against $e = -1$ alone. The joint pre-opening test does not reject in either design ($p = 0.23$ in the main design, $p = 0.37$ against future sites). Figure 18 plots the four readings of Section 5.2.

Linear extrapolation. A line through the coefficients for $e = -4, -3, -2$, pinned at zero in the reference year, is fitted by generalized least squares using their joint covariance matrix from the event-study regression, extrapolated to $e = 0, \dots, 3$, and subtracted before averaging; the standard error is by the delta method, holding the weights fixed. The detrended average is -0.683 (s.e. 0.179) in the main design and -0.237 (s.e. 0.208) against future sites.

Bounds. Write each event-study coefficient as $\beta_e = \tau_e + \delta_e$, with $\tau_e = 0$ before opening and $\delta_{-1} = 0$. We use the two restrictions on the trend δ proposed by Rambachan and Roth (2023) and implemented both constructions ourselves, following their paper, because the authors' software could not be run in our computing environment.

- *Smoothness*, $|\delta_{e+1} - 2\delta_e + \delta_{e-1}| \leq M$ for every e from -4 to 3 . At $M = 0$ any linear trend is allowed; a larger M lets the slope change by up to M a year. For this class Rambachan and Roth recommend a fixed-length confidence interval, which we compute directly: an affine estimator of θ whose pre-opening weights make it invariant to linear trends, with weights chosen to minimize the worst-case bias plus 1.96 standard errors, the worst-case

Table 18: Effects of an EPZ on 21 local outcomes

	(1)	(2)
	Main design	Future-site pool
<i>Panel A. Firm dynamics</i>		
New domestic firms	−0.219 (0.118)	−0.492 (0.135)
New foreign-invested (incl. HK/Macau/Taiwan) firms	0.012 (0.135)	0.046 (0.201)
Exits of pre-existing domestic firms	0.124 (0.145)	0.395 (0.155)
Active firms	0.027 (0.040)	−0.134 (0.061)
<i>Panel B. Local activity (all firms in the cell)</i>		
Gross output	0.022 (0.035)	−0.057 (0.046)
Value added	−0.022 (0.046)	−0.098 (0.057)
Employment	0.091 (0.040)	−0.033 (0.050)
Wage bill	0.057 (0.050)	−0.054 (0.046)
Fixed assets	0.039 (0.049)	−0.009 (0.048)
Export delivery value	0.109 (0.060)	0.114 (0.067)
<i>Panel C. Incumbents (present in year $g - 1$)</i>		
Output	−0.005 (0.046)	−0.067 (0.039)
Employment	0.064 (0.035)	0.013 (0.043)
Wage bill	0.034 (0.048)	−0.036 (0.038)
Fixed assets	0.068 (0.046)	0.084 (0.045)
Profits (Poisson on the positive part, $\max(\text{profit}, 0)$)	−0.102 (0.084)	−0.137 (0.090)
Subsidy income	−0.017 (0.150)	−0.052 (0.184)
<i>Panel D. Incumbent ratios (weighted by base-year employment)</i>		
Average wage (wage bill / employment)	−0.000 (0.001)	−0.001 (0.001)
Labor productivity (value added / employment)	−0.012 (0.009)	−0.020 (0.011)
Capital intensity (fixed assets / employment)	−0.003 (0.008)	−0.000 (0.005)
Profit rate (profit / sales)	−0.000 (0.005)	0.005 (0.005)
Subsidy rate (subsidy / sales)	−0.001 (0.001)	−0.001 (0.001)
Treated grid cells	279	279
Controls	Not yet treated, reweighted Future EPZ sites	

Notes: Pooled post-opening coefficients ($e = 0, \dots, 3$) from equation (5), one regression per cell. Panels A–C: Poisson, coefficients are log points. Panel D: least squares on cell-level ratios of sums, weighted by base-year employment. Column (1): not-yet-treated controls, reweighted; column (2): controls restricted to sites that later host an EPZ. Counts and totals in Panels A and B include every surveyed firm in the cell, among them the few fenced firms the survey records (Section 5.1). Incumbents are domestic firms present in the year before opening; exits count as zero. Value added is missing in 2004. The joint pre-trend test rejects for active firms and subsidies in column (1). Standard errors clustered by prefecture in parentheses.

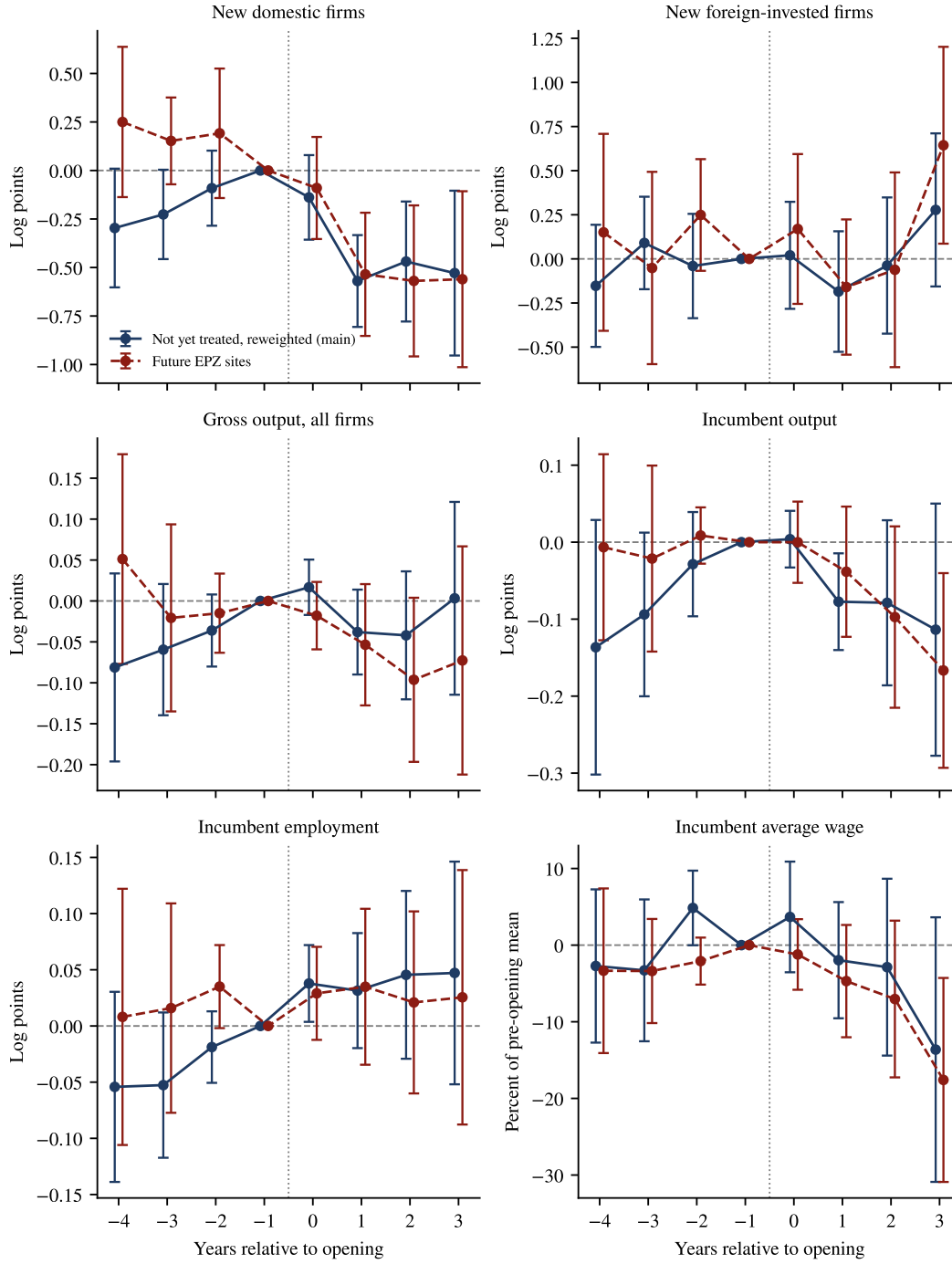


Figure 17: Event studies against the two comparison groups

Notes: Coefficients β_e from equation (5); $e = -1$ is the reference year. Solid: not-yet-treated cells, reweighted (the main text). Dashed: cells around sites that later host an EPZ, unweighted. The two groups differ in their pre-opening paths—for new domestic firms the slopes have opposite signs—but give similar post-opening paths. Bars are 95 percent confidence intervals clustered by prefecture.

bias over the class being a linear program. Our half-length adds 1.96 standard errors to the worst-case bias instead of using their folded-normal critical value, so our intervals are slightly wider than theirs.

- *Relative magnitudes*: after opening, the year-to-year change in δ is at most $\bar{M}$ times the largest year-to-year change before opening. Rambachan and Roth invert a moment-inequality test for this class. We fix the largest pre-opening change at its point estimate instead, which turns the restriction into a bound on each post-opening change and gives a closed-form worst-case bias for θ ; the interval is $\hat{\theta}$ plus or minus that bias and 1.96 standard errors. Because it ignores the sampling error in the largest pre-opening change, this plug-in interval is narrower than theirs, and the breakdown value it implies overstates robustness.

The breakdown values—the largest M or $\bar{M}$ at which the interval still excludes zero—are 0.054 and 0.55 in the main design, and 0.000 and 0.33 against future sites. Against future sites the smoothness interval includes zero even for an exactly linear trend, because the detrended estimate is itself imprecise.

Approval clock. The squares in Figure 18 re-estimate the main design with event time counted from State Council approval rather than opening. The coefficients are in Appendix Table 25, Panel C, and are discussed in Section 6; the same estimates for EPZs with an official acceptance date are in Table 27.

C.2 Inference for the entry estimate

Clusters. Prefectures are the default cluster (117 in the main sample). The zone clustering groups the treated cells of each sub-experiment by the EPZ they surround and gives the control cells of each sub-experiment their own cluster: 38 treated clusters and 33 control clusters, 71 in all. Splitting the treated clusters further by sub-experiment gives 85.

Score wild bootstrap. For a hypothesized coefficient β_0 on $T_{sc}Post_{st}$ we fit the Poisson model with β_0 imposed as an offset, keeping the fixed effects and weights of equation (5). With the restricted fitted values $\hat{\mu}_{0i}$, the score contribution of observation i is $\psi_i = w_i \tilde{T}_i (y_i - \hat{\mu}_{0i})$, where $\tilde{T}_i$ is the treatment indicator residualized on the fixed effects by least squares weighted by $w_i \hat{\mu}_{0i}$. The cluster sums $S_g = \sum_{i \in g} \psi_i$ are multiplied by independent draws v_g from the six-point Webb distribution, 9,999 times, and β_0 is rejected at 5 percent when $|\sum_g S_g|$ exceeds the 95th percentile of $|\sum_g v_g S_g|$ (Kline and Santos 2012). Because the null is imposed and only the scores are reweighted, the Poisson model is re-estimated once per hypothesized

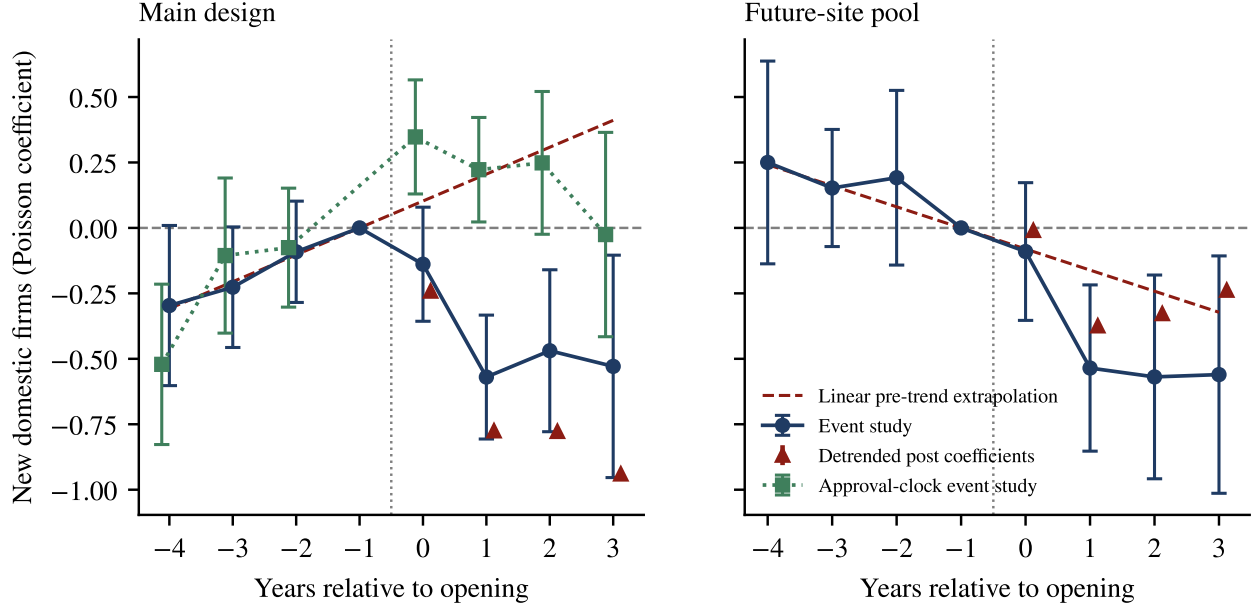


Figure 18: Four readings of the pre-opening rise in domestic entry

Notes: New domestic firms within 20 km. Left: main design (not-yet-treated cells, reweighted). Right: controls restricted to cells around sites that later host an EPZ, shown only as a diagnostic. Circles: event-study coefficients from equation (5), $e = -1$ the reference year, with 95 percent intervals clustered by prefecture. Dashed line: a line fitted to the pre-opening coefficients by generalized least squares, pinned at zero in the reference year, and extrapolated. Triangles: post-opening coefficients minus the extrapolated line. Squares (left panel): the main-design event study with event time counted from State Council approval instead of opening; for this series the horizontal axis reads years relative to approval. Bounds of [Rambachan and Roth \(2023\)](#) for both panels: Table 19.

value, not once per draw. The 95 percent interval collects the values of β_0 that are not rejected on a grid with a step of one quarter of the analytic standard error, and the p -value for $\beta_0 = 0$ is interpolated along the grid. Clustered by zone, $p = 0.038$ with interval -0.397 to -0.042 ; clustered by prefecture, $p = 0.081$ with interval -0.426 to 0.017 . Both are close to the analytic inference with the same clusters (Table 20).

The pairs bootstrap. The last row of Table 20 resamples whole zone clusters with replacement and re-estimates the model on each of 99 samples. It neither imposes the null nor reweights fixed scores, and with 99 draws its quantiles are imprecise; its standard error (0.067) is smaller than the analytic one (0.098) while its p -value (0.061) is larger. It is shown only for comparison, and the text relies on the wild bootstrap.

C.3 Pre-opening trends in incumbent outcomes

Table 21 and Figure 19 apply to incumbent output and employment the extrapolation and bounds that §C.1 applies to entry. The unadjusted post-opening average is the mean of the

Table 19: Pre-opening trends in domestic entry: extrapolation and bounds

	(1) Main design	(2) Future-site pool
<i>Unadjusted (no pre-trend adjustment)</i>		
Post average, $e = 0..3$	-0.427 [-0.667, -0.186]	-0.439 [-0.722, -0.156]
<i>Rambachan–Roth (2023): relative magnitudes $\bar{M}$</i>		
$\bar{M} = 0$	-0.427 [-0.667, -0.186]	-0.439 [-0.722, -0.156]
$\bar{M} = 0.5$	-0.427 [-0.836, -0.017]	-0.439 [-0.961, 0.084]
$\bar{M} = 1$	-0.427 [-1.006, 0.152]	-0.439 [-1.201, 0.323]
$\bar{M} = 1.5$	-0.427 [-1.175, 0.321]	-0.439 [-1.440, 0.563]
$\bar{M} = 2$	-0.427 [-1.344, 0.490]	-0.439 [-1.680, 0.803]
Breakdown $\bar{M}$ (plug-in bound)	0.55	0.33
<i>Rambachan–Roth (2023): smoothness M (FLCI)</i>		
$M = 0$	-0.681 [-1.032, -0.331]	-0.251 [-0.656, 0.154]
$M = 0.02$	-0.687 [-1.173, -0.201]	-0.246 [-0.788, 0.295]
$M = 0.05$	-0.693 [-1.364, -0.022]	-0.239 [-0.980, 0.502]
$M = 0.1$	-0.687 [-1.648, 0.274]	-0.226 [-1.284, 0.832]
Breakdown M	0.054	0.000
<i>Linear pre-trend extrapolation</i>		
Detrended post average	-0.683 [-1.034, -0.333]	-0.237 [-0.645, 0.171]
Joint pre-trend p ($e \in \{-4, -3, -2\} = 0$)	0.23	0.37

Notes: New domestic firms within 20 km. Each entry is the average of the event-study coefficients for $e = 0, \dots, 3$, measured against $e = -1$, with its 95 percent interval in brackets. Column (1): main design (not-yet-treated cells, reweighted). Column (2): controls restricted to cells around sites that later host an EPZ. Relative magnitudes: the post-opening year-to-year change in the trend is at most $\bar{M}$ times the largest pre-opening change, the latter fixed at its estimate (a plug-in approximation that overstates robustness). Smoothness: the slope of the trend changes by at most M a year; fixed-length intervals. Breakdown: the largest $\bar{M}$ or M at which the interval excludes zero. Linear extrapolation: generalized-least-squares line through the pre-opening coefficients, pinned at zero in $e = -1$. Standard errors clustered by prefecture.

event-study coefficients for $e = 0, \dots, 3$ against $e = -1$; it differs from the pooled coefficient of Table 5, which measures the post-opening years against all pre-opening years. Incumbent employment is indistinguishable from zero under every reading and against both comparison groups. Incumbent output rises, insignificantly, before opening in the main design ($p = 0.31$), so extrapolating that rise turns the post-opening average negative (-0.181 (0.075)); against future sites the pre-opening path is flat ($p = 0.87$) and the detrended average, -0.060 (0.065), is not significant. The smoothness interval, which equals the detrended estimate when the trend is exactly linear, includes zero once the slope may change at all, and the relative-magnitude interval includes zero already under parallel trends (breakdown value

Table 20: Inference for the entry estimate

Method	Estimate (SE)	p	95% CI	Clusters
<i>Panel A. By prefecture</i>				
Analytic CRV1	−0.219 (0.118)	–	[−0.451, 0.012]	117
Score wild bootstrap (Webb, 9999 reps)	−0.219	0.081	[−0.426, 0.017]	117
<i>Panel B. By EPZ zone (71 clusters – see text)</i>				
Analytic CRV1	−0.219 (0.098)	–	[−0.412, −0.027]	71
Score wild bootstrap (Webb, 9999 reps)	−0.219	0.038	[−0.397, −0.042]	71
Naive whole-cluster bootstrap (B=99, superseded)	−0.219 (0.067)	0.061	[−0.272, −0.002]	71

Notes: Pooled post-opening Poisson coefficient for new domestic firms within 20 km, main design (Table 5), log points. Analytic CRV1: cluster-robust standard error and normal-approximation interval. Score wild bootstrap: restricted score bootstrap of Kline and Santos (2012) with Webb weights, 9,999 draws; the interval inverts the test on a grid of hypothesized values. Zone clusters: treated cells grouped by the EPZ they surround, plus one cluster for the control cells of each sub-experiment. Last row: whole-cluster pairs bootstrap with re-estimation, 99 draws, shown for comparison only.

0.00). We therefore read incumbent output as possibly declining modestly, not as clearly unchanged.

Value added in 2004. The 2004 economic census did not collect value added, so value added is missing for every cell in 2004 while gross output is not. For sub-experiments that open in 2006–2008, 2004 is one of the pre-opening years $e = -4, \dots, -2$; this is the case for 24 of the 75 sub-experiments. The pre-opening coefficients for value added are therefore estimated on a different set of sub-experiments from those for gross output, and the joint pre-opening test for value added ($p = 0.06$) reflects that change in composition; for gross output, whose sample does not change, $p = 0.40$.

C.4 Binary and exposure designs

Four switches. The pure binary design of Table 6, column (1), differs from the main design in four rules, each of which can be switched on separately: (a) *contamination exclusion*, which keeps a control cell only if its near-ring count does not change during the window; (b) a *far-ring control*, the change since the year before opening in the number of special zones 40–150 km away, entered as a regressor; (c) *initial-state stratification*, which defines sub-experiments by opening year, near-ring count, and far-ring tier in the year before opening rather than by opening year alone; and (d) *truncation of dose accumulation*, which ends a treated cell’s post window at the next change in its own near-ring count. The main design has (a), (c), and (d) on and (b) off (§O.3). We estimate all sixteen combinations and attribute the difference between all four off and all four on to each switch by its Shapley value, the

Table 21: Pre-opening trends in incumbent output and employment: extrapolation and bounds

	(1) Main design	(2) Future-site pool
<i>Panel. Incumbent output (present in year $g - 1$)</i>		
<i>Unadjusted (no pre-trend adjustment)</i>		
Post average, $e = 0..3$	-0.066 [-0.147, 0.015]	-0.076 [-0.156, 0.005]
<i>Rambachan–Roth (2023): relative magnitudes $\bar{M}$</i>		
$\bar{M} = 0$	-0.066 [-0.147, 0.015]	-0.076 [-0.156, 0.005]
$\bar{M} = 0.5$	-0.066 [-0.229, 0.096]	-0.076 [-0.193, 0.042]
$\bar{M} = 1$	-0.066 [-0.311, 0.178]	-0.076 [-0.231, 0.080]
$\bar{M} = 1.5$	-0.066 [-0.392, 0.260]	-0.076 [-0.268, 0.117]
$\bar{M} = 2$	-0.066 [-0.474, 0.341]	-0.076 [-0.306, 0.155]
Breakdown $\bar{M}$ (plug-in bound)	0.00	0.00
<i>Rambachan–Roth (2023): smoothness M (FLCI)</i>		
$M = 0$	-0.182 [-0.328, -0.036]	-0.058 [-0.184, 0.068]
$M = 0.02$	-0.167 [-0.440, 0.106]	-0.052 [-0.288, 0.185]
$M = 0.05$	-0.139 [-0.575, 0.297]	-0.054 [-0.441, 0.333]
$M = 0.1$	-0.138 [-0.824, 0.548]	-0.054 [-0.691, 0.583]
Breakdown M	0.005	0.000
<i>Linear pre-trend extrapolation</i>		
Detrended post average	-0.181 [-0.329, -0.034]	-0.060 [-0.188, 0.068]
Joint pre-trend p ($e \in \{-4, -3, -2\} = 0$)	0.31	0.87
<i>Panel. Incumbent employment (present in year $g - 1$)</i>		
<i>Unadjusted (no pre-trend adjustment)</i>		
Post average, $e = 0..3$	0.040 [-0.018, 0.099]	0.028 [-0.041, 0.096]
<i>Rambachan–Roth (2023): relative magnitudes $\bar{M}$</i>		
$\bar{M} = 0$	0.040 [-0.018, 0.099]	0.028 [-0.041, 0.096]
$\bar{M} = 0.5$	0.040 [-0.060, 0.141]	0.028 [-0.085, 0.140]
$\bar{M} = 1$	0.040 [-0.103, 0.184]	0.028 [-0.129, 0.184]
$\bar{M} = 1.5$	0.040 [-0.145, 0.226]	0.028 [-0.173, 0.228]
$\bar{M} = 2$	0.040 [-0.187, 0.268]	0.028 [-0.216, 0.272]
Breakdown $\bar{M}$ (plug-in bound)	0.00	0.00
<i>Rambachan–Roth (2023): smoothness M (FLCI)</i>		
$M = 0$	0.001 [-0.067, 0.070]	0.071 [-0.018, 0.159]
$M = 0.02$	-0.007 [-0.187, 0.174]	0.110 [-0.091, 0.312]
$M = 0.05$	-0.007 [-0.337, 0.324]	0.115 [-0.236, 0.467]
$M = 0.1$	-0.007 [-0.587, 0.574]	0.115 [-0.486, 0.717]
Breakdown M	0.000	0.000
<i>Linear pre-trend extrapolation</i>		
Detrended post average	-0.005 [-0.075, 0.065]	0.079 [-0.010, 0.169]
Joint pre-trend p ($e \in \{-4, -3, -2\} = 0$)	0.46	0.19

Notes: Incumbents: domestic firms present in the cell in the year before opening; exits count as zero. Each entry is the average of the event-study coefficients for $e = 0, \dots, 3$ against $e = -1$, log points, with its 95 percent interval in brackets. Column (1): main design; column (2): controls restricted to cells around sites that later host an EPZ. Bounds and extrapolation as in Table 19; the relative-magnitude breakdown uses the plug-in approximation. Standard errors clustered by prefecture.

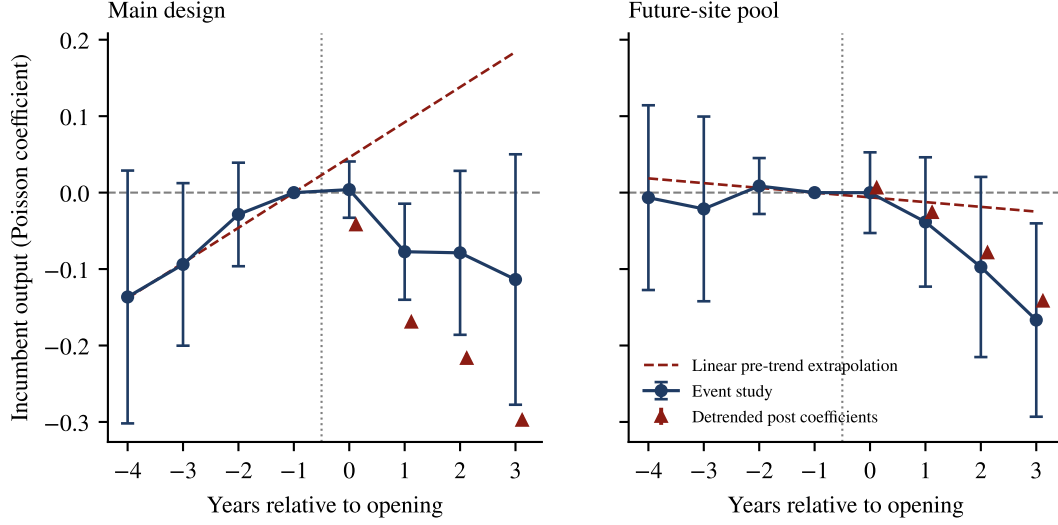


Figure 19: Incumbent output around opening: event study and linear extrapolation

Notes: Output of domestic incumbents within 20 km. Circles: event-study coefficients from equation (5), $e = -1$ the reference year, with 95 percent intervals clustered by prefecture. Dashed line: generalized-least-squares line through the pre-opening coefficients, pinned at zero in $e = -1$, and extrapolated. Triangles: post-opening coefficients minus the extrapolated line. Left: main design. Right: controls restricted to cells around sites that later host an EPZ.

average of its marginal contribution over all orderings of the other three.

Reconstruction. Switching (a) off requires a broader candidate pool than the main sample contains, so the decomposition rebuilds the design from the raw exposure counts. It holds the treated set fixed at every cell within 20 km of an opening EPZ, whatever its initial exposure, and reweights controls on two characteristics (firm density and initial far-ring exposure) rather than the main design’s nine. Its all-on estimate for domestic entry is therefore -0.180 , not the main estimate -0.219 ; Table 22 and Table 6 report the main design’s own estimate for comparison.

Results. For domestic entry (Table 22, Figure 20), initial-state stratification (0.087) and truncation (0.043) move the estimate toward zero, contamination exclusion moves it away (-0.041), and the far-ring control does nothing (-0.001). For foreign entry the same two switches move a decline of about the size of the domestic one most of the way to zero; the rest of the distance to the main design’s estimate reflects the reconstruction’s broader treated set and narrower reweighting. Figure 21 overlays the event-study paths of the pure binary and main designs for domestic entry. Their intervals overlap in every year. The binary design’s pre-opening path is flatter and its post-opening coefficients are, if anything, less negative; its larger pooled coefficient comes from the flatter pre-opening path against which the pooled contrast is measured.

Both treatments in one regression. We add to the main design an indicator for any special customs zone operating within 20 km of the cell in the current year, the naive treatment of a binary design. Alone, the main treatment’s coefficient is -0.219 (s.e. 0.118). With the indicator it becomes -0.850 (s.e. 0.223), and the indicator’s coefficient is 0.639 (s.e. 0.239). In the main sample a treated cell has no zone within 40 km before opening and its EPZ within 20 km after, so the two regressors are highly correlated; the standard error on the main treatment roughly doubles, and the two coefficients take large offsetting values. The regression recovers their sum, which stays close to the single-treatment coefficient, but not how to divide it. We do not read the split as evidence for either definition. Whether predetermined geography explains exposure is examined in Table 64.

Table 22: From a binary to the exposure design: Shapley decomposition

	New domestic entrants (1)	New foreign entrants (2)
<i>Shapley contribution of each design choice (log points)</i>		
Contamination exclusion	-0.041	0.015
Far-ring time-varying control	-0.001	-0.002
Initial-state stratification	0.087	0.089
Dose-accumulation truncation	0.043	0.047
Pure binary (all four off)	-0.267	-0.220
All four on (reconstructed)	-0.180	-0.070
Main design (for comparison)	-0.219	0.012

Notes: Pooled post-opening Poisson coefficients for new domestic and new foreign-invested firms within 20 km. Top panel: the average change in the coefficient when each rule is switched on, over all orderings of the other three rules (Shapley value); the four contributions sum to the difference between the all-off and all-on rows. “All four on (reconstructed)” differs from the main design because the decomposition holds the treated set fixed across initial exposure states and reweights on two characteristics. Standard errors clustered by prefecture in every regression.

C.5 Distance rings, continuous profile, and break search

Rings. Each ring (0–5, 5–10, 10–20, 20–40, and 40–80 km) is a separate stacked design: a cell is treated when an EPZ opens at a distance within the ring, and treated and control cells, weights, fixed effects, and clustering are built as in the main design. So that an arrival up to 80 km away registers as an arrival, the near ring that defines arrivals and the candidate control universe is widened from 40 to 80 km. Pooling the three inner rings reproduces the main estimate (-0.217 , s.e. 0.120), but under the wider candidate universe its pre-trend test is less favorable (Table 23, last row). The rings therefore locate the effect; they do not replace the main estimate.

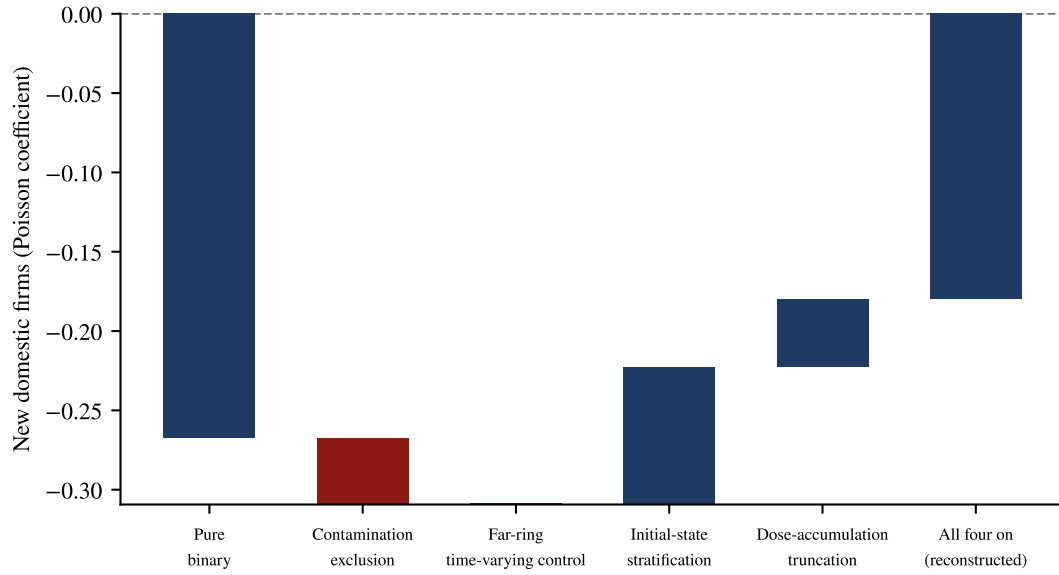


Figure 20: Contribution of each design rule to the domestic-entry estimate

Notes: Waterfall from the pure binary design (first bar) to the design with all four rules on (last bar); intermediate bars are the Shapley contributions of Table 22, column (1). A contribution that makes the estimate more negative is shown in red. Poisson coefficients, log points.

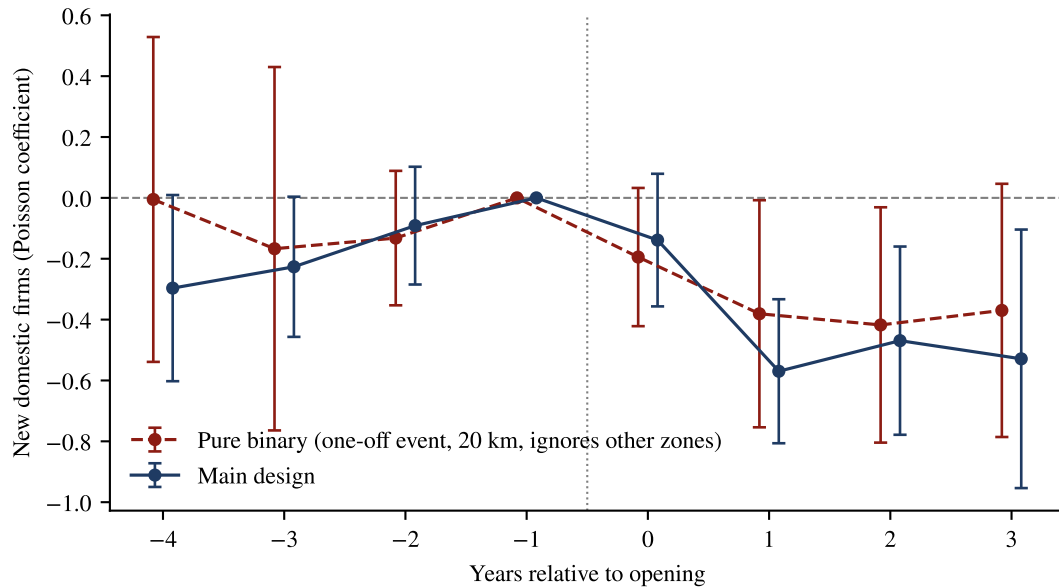


Figure 21: Event-study paths of the binary and exposure designs

Notes: New domestic firms within 20 km. Dashed: pure binary design (Table 6, column 1). Solid: main design (column 3). Coefficients β_e from equation (5), $e = -1$ the reference year, with 95 percent intervals clustered by prefecture; the two series are offset horizontally for legibility.

Continuous profile. Re-weighting cells continuously by distance inside one regression would mean rebuilding the stacked design at every evaluation point. We instead estimate the same design on sixteen 5-km bins from 0 to 80 km (pooled coefficients only) and fit a local-linear regression through the bin estimates with a Gaussian kernel and inverse-variance weights, the kernel-and-ring pairing of [Diamond and McQuade \(2019\)](#). The bandwidth is chosen by leave-one-out cross-validation over a grid from 5 to 40 km, as in [Clarke \(2017\)](#): 40 km for new domestic firms, the upper end of the grid, 5 km for incumbent output, and 10 km for incumbent employment.

Break search. For each candidate boundary between bins we fit a separate constant on each side of the new-entry profile by inverse-variance weighted least squares and record the weighted sum of squared residuals (Figure 22). The minimum is at 45 km. A boundary at 20 km fits better than every candidate except the two between it and 45 km, where the split separates the negative and zero bins from the positive outer ones.

Table 23: Effects by distance ring

Ring	Coefficient (SE)			Pre-trend p		
	Entry	Inc. output	Inc. employment	Entry	Inc. output	Inc. employment
0–5 km	−0.212 (0.152)	0.036 (0.068)	0.156 (0.056)	0.07	0.18	0.00
5–10 km	−0.032 (0.121)	−0.015 (0.051)	0.070 (0.036)	0.29	0.00	0.73
10–20 km	−0.316 (0.147)	0.005 (0.075)	0.065 (0.045)	0.00	0.18	0.01
20–40 km	0.011 (0.115)	0.062 (0.060)	0.117 (0.045)	0.11	0.34	0.13
40–80 km	0.153 (0.097)	0.064 (0.038)	0.062 (0.027)	0.08	0.48	0.36
0–20 km (pooled)	−0.217 (0.120)	0.004 (0.053)	0.073 (0.035)	0.03	0.00	0.01

Notes: Pooled post-opening Poisson coefficients ($e = 0, \dots, 3$), one regression per ring and outcome, each ring with its own not-yet-treated, reweighted controls; standard errors clustered by prefecture in parentheses. Entry: new domestic firms. Incumbents: domestic firms present in the year before opening. Pre-trend p : joint test that the coefficients for $e = -4, -3, -2$ are zero. Last row: the three inner rings pooled, estimated with the ring design’s wider candidate universe.

C.6 Contamination of the control pool

For each opening cohort g and each candidate control cell, we find the nearest special customs zone of any type that becomes operational between g and $g + 3$ and record whether it lies

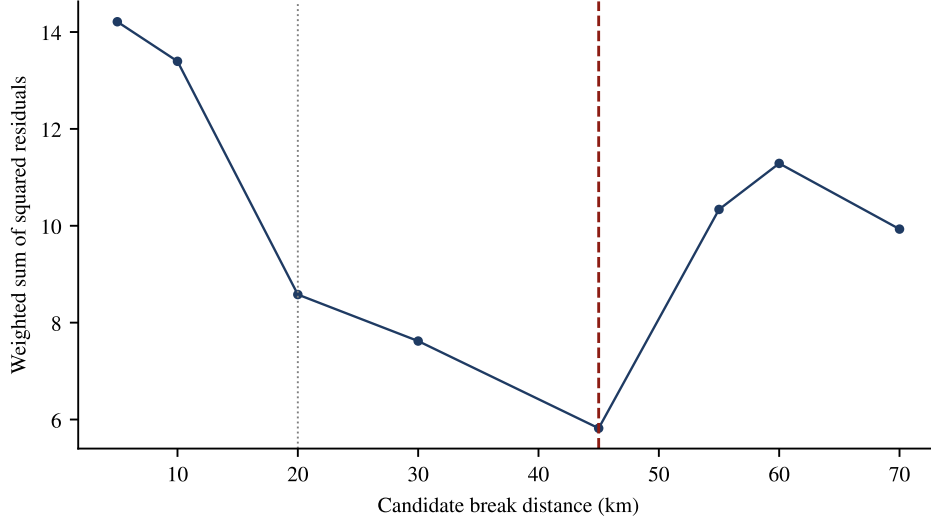


Figure 22: Where does the entry profile break? Single-break search

Notes: Weighted sum of squared residuals from fitting one constant on each side of a candidate break to the 5-km bin estimates for new domestic firms, weights equal to inverse variances. Dashed line: minimum. Dotted line: the main design's 20-km radius.

within 20, 20–40, or 40–80 km of the cell. Two pools are compared: the not-yet-treated pool without the exclusion rule, in which a cell is a control as long as the next zone within 40 km of it opens after g , and the pool the main design retains, in which that zone opens after $g + 3$. If a share s of controls receives the treatment effect τ , the estimate converges to $\tau(1 - s)$ (Butts 2023); Table 24 reports s and this factor by distance. Without the exclusion rule the shares within 20 and 20–40 km would be 2 and 4 percent, small to begin with; with it they are 0 and 0 percent by construction. The rule does not reach 40–80 km, where 16 percent of retained controls see a zone open during the window (15 percent without it). The factor 0.84 is a worst case, since controls there see a zone at a distance at which entry does not fall (Figure 6). If they received only the 40–80 km effect, which for entry is positive, their raised entry would overstate the decline by roughly the product of that share and that coefficient, small next to the main estimate.

D Entry: Additional Evidence

Alternative explanations. Four alternatives to admission rationing each predict a pattern that the data reject (Table 25, Panel B). (i) If the fence had occupied scarce land in a narrow physical sense, entry would fall most in land-intensive industries. Plot areas are not observed, so we classify two-digit industries by their median fixed assets per worker in 1998, before any zone in the sample opened; this measures capital intensity and serves as a proxy for land intensity. The difference between high- and low-intensity industries is 0.176 (s.e. 0.105),

Table 24: Contamination of the control pool

Band	Contaminated share s	Bias factor $1 - s$
<i>Not-yet-treated pool without the exclusion rule</i>		
0–20 km	2%	0.98
20–40 km	4%	0.96
40–80 km	15%	0.85
<i>Retained pool of the main design (next zone within 40 km opens after $g + 3$)</i>		
0–20 km	0%	1.00
20–40 km	0%	1.00
40–80 km	16%	0.84

Notes: Share s of candidate control cells for which a special customs zone of any type opens during $[g, g + 3]$ at the stated distance, averaged over opening cohorts, and the implied attenuation factor $1 - s$ if those controls received the full treatment effect. Top panel: controls whose next zone within 40 km opens after g . Bottom panel: controls retained by the main design, whose next zone within 40 km opens after $g + 3$; within 40 km their share is zero by construction.

with the larger loss in low-intensity industries. The test rules out a loss concentrated among capital-intensive entrants, not every form of competition for land. (ii) If fenced employers had drawn workers away from small firms through non-wage means, the loss would concentrate in labor-intensive industries and grow with fence activity; the labor-intensity difference is -0.098 (s.e. 0.105), and the loss is as large around empty fences. (iii) If the host government had shifted tax or subsidy favors toward the fence, nearby domestic firms would pay more or receive less; the effective tax rate of incumbents changes by -0.0020 (s.e. 0.0075), and entrants' subsidy rate by 1.655 percentage points (s.e. 1.474). Split by ownership, in a version of the design with a 40 km radius, neither foreign-invested nor domestic firms show a significant change in their value-added tax burden, interest rate, or subsidy rate, except a rise in domestic firms' subsidy rate that is matched by a pre-opening difference of the same size. (iv) If domestic firms had simply registered inside the fence, counting them would close the gap; excluding the few fence-registered domestic entrants leaves the estimate unchanged. Nor is the loss an anticipation effect: on the approval clock entry rises before the fence opens (Panel D).

Leader age. If prefecture leaders with stronger promotion incentives committed more of the zone's capacity to the fence, entry would fall more where they govern. Splitting treated cells by whether the prefecture's party secretary is younger than the sample median age, a common proxy for promotion incentives, entry falls *less* where the secretary is younger: the younger-minus-older difference is 0.376 (s.e. 0.178). Age is confounded with the administrative rank and the size of the city, which also shape how a zone is run, so we do not interpret

Table 25: Who is kept out, and why

	(1)	(2)	(3)
<i>Panel A. New domestic entrants, by birth-size quintile: (1) output (2) employment (3) assets</i>			
Bottom quintile	−0.572 [−44%] (0.187)	−0.242 [−21%] (0.206)	−0.237 [−21%] (0.168)
2nd quintile	−0.406 [−33%] (0.157)	−0.297 [−26%] (0.149)	−0.443 [−36%] (0.174)
3rd quintile	−0.043 [−4%] (0.172)	−0.272 [−24%] (0.144)	−0.252 [−22%] (0.154)
4th quintile	−0.181 [−17%] (0.158)	−0.315 [−27%] (0.131)	−0.332 [−28%] (0.143)
Top quintile	0.011 [1%] (0.101)	0.067 [7%] (0.134)	0.101 [11%] (0.138)
Top − bottom	0.583 [79%] (0.184)	0.309 [36%] (0.218)	0.337 [40%] (0.170)
Observations	129,025	131,907	131,650
<i>Panel B. Alternative channels</i>			
<i>C1. Land/capital intensity</i>			
High land intensity		−0.126 [−12%] (0.113)	
Low land intensity		−0.301 [−26%] (0.138)	
High − low		0.176 [19%] (0.105)	
<i>C2. Labor intensity</i>			
Low labor intensity		−0.169 [−16%] (0.115)	
High labor intensity		−0.266 [−23%] (0.142)	
High − low		−0.098 [−9%] (0.105)	
<i>C3. Effective tax and subsidy rates (percentage points)</i>			
Entrants' subsidy rate (subsidy / sales)		1.655 (1.474)	
Incumbents' effective tax rate (tax paid / profit)		−0.002 (0.007)	
<i>C4. Robustness to fence-registered entrants</i>			
Including fence-registered (main spec.)		−0.219 [−20%] (0.118)	
Excluding fence-registered		−0.220 [−20%] (0.118)	
<i>Panel C. Timing: approval-clock event time (years since the site was approved)</i>			
$e = 0$		0.348 [42%] (0.111)	
$e = 1$		0.222 [25%] (0.102)	
<i>Panel D. City-level factor prices (Callaway-Sant'Anna, not-yet-treated)</i>			
Wage (log points)		−0.037 [≤0.6%] (0.022)	
Industrial land price, 20 km (log points)		0.076 [≤19%] (0.060)	

Notes: Poisson coefficients on new domestic firms from equation (5), main design, with the proportional effect in brackets and prefecture-clustered standard errors in parentheses. Panel A: entrants by birth-size quintile (output, employment, fixed assets). Panel B: by fenced exports relative to nearby base-year output, for all entrants, the smallest quintile, and labor-intensive industries; siting. Panel C: alternative channels. Panel D: timing on the approval clock. Panel E: city wages and constant-quality industrial land prices; the land-price estimate comes from zones that opened in 2009–2013, not from the EPZs of the main design (see “Land prices” below).

this estimate either as support for or as evidence against a political channel. Section 9 and Appendix J examine political incentives with measures built for the purpose, rivalry between neighboring prefectures.

Approved area and siting. The split by approved area is less informative. Relative to the development-zone land within 20 km, the decline is 11 percent in the lowest tercile of approved area, 30 percent in the middle tercile, and 17 percent in the highest; the estimates are imprecise and not monotone, and when all initial exposure states are included the ordering changes. The data therefore reject crowding-out that scales with activity more firmly than they establish crowding-out that scales with the land assigned. The foregone-entry estimate is also similar around zones nested inside a parent development zone (19 percent) and around zones that adjoin one or stand alone (24 percent).

Timing on EPZs with an official acceptance date. Of the 59 EPZs, 36 have an official customs acceptance date; the other 23 open, by assumption, two years after approval (§M.11). For the latter the approval and operating clocks differ by construction, so Table 27 re-estimates both clocks keeping only the treated cells around EPZs with an official date (23 of them anchor treated cells in the main sample) and all control cells. Control cells are kept because a sub-experiment’s controls serve every EPZ that opens in its cohort, dated or not. On the operating clock the pooled coefficient is -0.238 (0.122) against -0.219 (0.118) in the full sample, and the event-study coefficients nearly coincide; on the approval clock entry still rises in the approval year and the year after. The 17 zones of the delay-lag test in Appendix E.4 are a narrower subset of these 36, approved late enough for three pre-approval years of customs records to exist; the two counts answer different questions.

Keeping 2004. The 2004 economic census added to the survey firms that were already above the size threshold but had been missed. Keeping them is correct—they met the same criterion as every other firm—but it matters for our design only if additions were more common near EPZs. They were somewhat more common: census additions account for 2.8 percentage points (s.e. 1.3) more of 2004 employment in treated cells than in reweighted controls (control mean 14 percent), and 6.0 points (s.e. 1.5) more in upstream industries. Because census additions appear as incumbents or entrants near the zone, this imbalance biases the estimated effects on entry and on upstream industries upward, against our findings. Excess 2004 entry relative to the average of 2003 and 2005 does not differ between treated and control cells (0.09, s.e. 0.18).

Table 26: Entry effect by approved area, fence activity, and siting

	(1)	(2)
	First zone	All initial states
<i>Panel A. By approved zone area relative to local development-zone land</i>		
Approved area share: lowest tercile	−0.116 (0.149)	−0.234 (0.132)
Approved area share: middle tercile	−0.361 (0.185)	−0.275 (0.169)
Approved area share: highest tercile	−0.192 (0.158)	−0.085 (0.155)
<i>Panel B. By fence activity (fenced exports relative to nearby base output)</i>		
Fence activity: empty (<0.1%)	−0.280 (0.233)	−0.184 (0.233)
Fence activity: small (0.1–2%)	−0.184 (0.150)	−0.094 (0.132)
Fence activity: large (>2%)	−0.400 (0.230)	−0.506 (0.263)
<i>Panel C. By siting</i>		
Nested in parent zone	−0.206 (0.134)	−0.246 (0.120)
Adjacent or stand-alone	−0.273 (0.155)	−0.048 (0.167)

Notes: Coefficients on $T_{sc}Post_{st}$ interacted with group indicators in equation (5), outcome new domestic firms; each panel is a separate regression. Approved area share: the EPZ’s approved area divided by the approved area of all development zones within 20 km that existed in the year before opening plus the EPZ itself, in terciles across treated cells. Fence activity: average fenced exports in operating years 0–3 relative to the base-year output of incumbents within 20 km; cells around zones without customs records of fenced firms are estimated as a separate group and not shown. Controls not yet treated, reweighted. Standard errors clustered by prefecture in parentheses.

The capacity share under alternative control pools. Treating each entrant’s choice of cell within a development-zone network as a conditional logit, the entry capacity assigned to the fence is $\rho = 1 - e^{\beta}$, where β is the Poisson coefficient on treated cells (Guimarães et al. 2003). In the main design β is the pooled coefficient of Table 5, so ρ is 20 percent, about 5.6 above-scale domestic entrants per zone-year in the ledger sample (Appendix Table 51); these are the values the paper uses. Table 69 re-estimates β with development-zone network $\times$ year effects and other control pools. With controls near future EPZ sites, ρ is 35 percent (95 percent interval 17–48), about 14 entrants per zone-year; with controls drawn from all locations that eventually receive any special zone, it is 11 percent and not distinguishable from zero, since these controls include inland sites whose entry dynamics differ from coastal

Table 27: Entry timing on EPZs with an official acceptance date

Specification	Estimate (SE)	% effect	95% CI	N
<i>Panel A. Opening-clock, pooled post effect (new domestic firms)</i>				
EPZs with official opening dates (23 in the entry sample)	−0.238 (0.122)	−21%	[−38, 0]%	35,774 (177 cells)
Main sample (main text)	−0.219 (0.118)	−20%	[−36, 1]%	44,263 (279 cells)
<i>Panel B. Opening-clock event study, $e = 0$ and $e = 1$</i>				
EPZs with official opening dates (23 in the entry sample), $e = 0$	−0.154 (0.139)	−	−	−
EPZs with official opening dates (23 in the entry sample), $e = 1$	−0.583 (0.146)	−	−	−
Main sample (main text), $e = 0$	−0.139 (0.111)	−	−	−
Main sample (main text), $e = 1$	−0.570 (0.121)	−	−	−
<i>Panel C. Approval-clock event study, $e = 0$ and $e = 1$</i>				
EPZs with official opening dates (23 in the entry sample), $e = 0$	0.500 (0.125)	−	−	−
All EPZs (main text), $e = 0$	0.348 (0.111)	−	−	−
EPZs with official opening dates (23 in the entry sample), $e = 1$	0.269 (0.116)	−	−	−
All EPZs (main text), $e = 1$	0.222 (0.102)	−	−	−

Notes: New domestic firms within 20 km, main design, Poisson coefficients in log points; standard errors clustered by prefecture in parentheses. Rows labelled “officially-dated” keep only treated cells around EPZs with an official customs acceptance date and all control cells; rows labelled “main text” are the estimates of Table 5 and of Table 25, Panel C. Panel A: pooled post-opening coefficient, $e = 0, \dots, 3$, with the percent effect and its 95 percent interval; N is cell-years, with treated cells in parentheses. Panels B and C: event-study coefficients against the year before opening (B) or before approval (C).

development zones. The future-site pool is chosen by the same state for the same kind of zone but is smaller and less balanced (Appendix B), which is why it serves as a check rather than the main estimate. In the same regressions untreated locations in the host city do not gain entrants (−14 percent, interval −36 to 16 percent).

Reallocation: how the rows of Table 7 are built. The five Reallocation rows answer five different questions and are on different scales. The first two are the ring estimates of Appendix C.5 for new domestic firms at 20–40 and 40–80 km, each ring with its own not-yet-treated controls and pre-opening test. The third repeats the pooled post-opening coefficient of Table 5 for the output of all surveyed firms within 20 km. The fourth is a composition share, not a treatment effect. For each zone with customs records of fenced firms, we take the exports of every firm registered inside the fence in operating years 0–3 and ask whether that firm was already active in the host prefecture in the two years before opening. A fenced exporter counts as relocated if its name, after legal-entity suffixes, country qualifiers, brackets, and spaces are removed, matches the name of a firm that exported through customs or appeared in the manufacturing survey in the host prefecture in those two years; or if the link between customs and survey records ties it to a survey firm present in the host prefecture before opening, or to another customs registration of the same firm that exported before opening. A firm that moves into the fence appears under a different customs code, because

the code records the special-supervision status of the registered location (Appendix M), so the name and survey links, not the customs code, are what identify it. The share is weighted by fenced exports across the 24 zones. It misses local firms that were neither exporters nor above the survey’s size threshold before they moved, and firms that changed their names when they moved; for such firms it is a lower bound. The fifth row comes from a conditional-logit model of destination choice for relocating firms, estimated as a Poisson regression with mover fixed effects (Guimarães et al. 2003). The relocation records cover county-to-county moves of above-scale manufacturing firms in 1999–2007 (Appendix M). Each move’s choice set is every county with above-scale firms within 150 km of the origin; the regressors are log distance, the log number of above-scale firms at the destination, and indicators for a destination in another prefecture of the same province and for one in another province. The reported number is the exponentiated coefficient on the first indicator, the relative probability of choosing another prefecture of the same province over an otherwise identical destination in the origin prefecture, in the sample that excludes the four centrally administered municipalities, whose districts are not comparable to prefectures. It is essentially unchanged with 100 or 200 km choice sets, with confirmed moves only, and with the municipalities included; cross-province destinations are chosen far less often still. Two limits apply. The model conditions on moving, so it says where movers go, not how many firms move; and a move across a provincial border that breaks the firm identifier is recorded as an exit and an entry, not as a move. We have not estimated the entry effect on other prefectures of the same province; a check in the spirit of Wang (2013) would relate provincial entry to the share of each province’s development-zone capacity taken by fences.

Land prices. We have no land-price estimate for the years in which the EPZs of this paper opened. Transaction records become comprehensive only with the disclosure rules of 2007 (Appendix Q.2), so a comparison of prices before and after opening needs zones that opened later. The only constant-quality estimate available uses industrial parcels sold in 2007–2013 within 20 km of special customs zones whose treatment cohorts open in 2009–2013 (5 cohorts; the zone directory lists 34 bonded zones and EPZs opening in those years), with not-yet-treated controls (Callaway and Sant’Anna 2021) and prices residualized on parcel characteristics (equation (15)). The main design’s EPZs open in 2001–2007, so the two samples share 0 zones. The post-opening average is 0.076 log points (s.e. 0.060), and a rise of up to 19 percent cannot be excluded; the pre-opening average difference is -0.136 ($p = 0.10$). The estimate describes the price response to later zones under a different land regime, not the price response to the EPZs studied here, and Section 6 does not use it.

Land parcels around later special zones. The parcel records also report each parcel’s area and whether it drew on new construction-land quota, so in principle they could show how much land a fence took from other applicants. Applying the main design and control definition to special customs zones that began operating in 2009–2017, with a 40 km radius, the area of industrial land granted from new quota rises after operation while the average industrial land price does not change. Timed instead from approval, the area does not move and the price rises, but with a pre-approval difference in the same direction. Both clocks show sizable pre-period differences and they disagree, so we read the parcels only as the absence of a robust land-price response to a zone. The later zones also operated after the move to tender, auction, and listing of industrial land, and the land granted around them after operation came from new quota, conditions under which a fence need not draw on existing capacity. The evidence therefore speaks to a different sample window and a different land regime; it cannot measure the capacity the EPZs took, and the main text does not use it.

Upstream responses by supply base. The split of upstream responses by the pre-opening local supply base, and the placebo checks showing that it reflects mean reversion, are in Appendix H.3; we do not use the split as evidence.

E Learning: Details

This appendix documents the measurement and designs behind Section 8 and collects the robustness checks to which the text points.

E.1 Measuring portable experience

Risk set. An observation is a firm f , an HS6 product p , a year $t \in \{2002, \dots, 2006\}$, and a candidate destination d . Candidates are the 40 largest destinations of Chinese exports in $t - 1$, so the choice set uses only information available before the choice. A tuple enters the risk set if f exported p in $t - 1$, exports in t , and did not export p to d under ordinary trade in $t - 1$; the outcome indicates that f exports p to d under ordinary trade in t . Groups (f, p, t) without any entry carry no information once firm $\times$ product $\times$ year effects are included and are dropped. The sample is a 5 percent random sample of exporting firms after excluding trade intermediaries, identified by name, because the new markets they enter belong to their clients.

Regressors and weights. In equation (7), $D_{fd,t-1}^r$ indicates that f exported to d itself under form $r \in \{O, I, A\}$ in $t - 1$ (any product), and $S_{fd,t-1}^r$ that f exported under form

r in $t - 1$ to a destination other than d that is similar to d : on the same continent, with the same official language, or within 2,000 km. The model is a linear probability model; the destination-specific weights $\kappa_r = a_r/a_O$ and the portability weights $\omega_r = b_r/b_O$ are reported with delta-method standard errors clustered by firm. Two tables below (Tables 29 and 30) print these weights as μ_r and λ_r ; the symbols differ from the text only because μ and λ have other meanings in the model of Section 3. Ownership is taken from the sixth digit of the customs registration code: domestic firms are state-owned and private firms; foreign-owned firms are joint ventures and wholly foreign-owned firms, including those invested from Hong Kong, Macao, and Taiwan.

The transition figure. Figure 9 uses the same sample, candidates, and risk set, restricted as in the regression to groups with at least one entry. For each form r it keeps observations of firms that exported under r in $t - 1$ and reports the entry rate separately for candidates similar and dissimilar to the firm’s destinations under r . It uses no fixed effects, so it describes the data before the comparisons within firm, product, and year that the regression makes.

What the weights measure. Organizational form is chosen by the firm and its foreign partner, so the weights measure how much each kind of experience predicts about where a firm expands next, not the causal effect of controlling a transaction. The fixed effects rule out the two most direct alternative readings: differences in ability across firms, because destinations are compared within the same firm, product, and year, and national demand shocks, because product $\times$ destination $\times$ year effects absorb them and the candidate set uses only information from $t - 1$.

E.2 Robustness of the portability weights

Table 28 completes Table 10: the tests of equal weights and the ordinary-trade coefficients by which the weights are normalized. Among all firms and among foreign-owned firms the two portability weights differ at conventional levels; among domestic firms they do not ($p = 0.230$), which is why the text rests its conclusion for local firms on the capability stock rather than on this difference.

Table 29 re-estimates equation (7) with one change at a time. Imported-materials experience is more portable than consigned experience with candidate sets of the 20 or 80 largest destinations, with similarity defined by continent alone, by distance alone, or by income group within 5,000 km, with a dose measure of the number of similar destinations served, and in the specification with experience dated by year. The one exception is similarity by official language alone, under which both processing weights are negative ($\omega_I = -0.26$,

Table 28: Portability weights: tests and normalization

	All firms (1)	Domestic (2)	Foreign-owned (3)
<i>Panel A. Tests (p-values)</i>			
$\omega_I = \omega_A$	<0.001	0.230	<0.001
$\omega_I = 1$	<0.001	<0.001	<0.001
$\omega_A = 0$	0.014	0.867	<0.001
$\kappa_I = \kappa_A$	0.067	0.067	0.006
<i>Panel B. Ordinary-trade coefficients (normalization)</i>			
Same destination, a_O	0.064 (0.003)	0.044 (0.004)	0.105 (0.004)
Similar destination, b_O	0.013 (0.001)	0.018 (0.002)	0.015 (0.002)
Candidate tuples	1,161,607	632,524	423,498
Firms	4,231	1,910	2,321
Entry rate	0.058	0.058	0.057

Notes: Same regressions as Table 10. Panel A: p -values of Wald tests on the relative weights (delta method, standard errors clustered by firm). Panel B: the ordinary-trade coefficients a_O and b_O of equation (7), by which κ_r and ω_r are normalized; standard errors clustered by firm in parentheses.

$\omega_A = -1.07$). Portability in these data is extended gravity in the sense of geographic and income proximity. A structural conditional logit of destination choice, estimated by maximum likelihood on a 10 percent random sample of firms, gives the same ranking ($\omega_I = 0.27$, $\omega_A = -0.07$) and rejects the restriction that experience in similar destinations carries the same relative weights as experience in the destination itself (Table 30).

E.3 Persistence and the link to learning in processing trade

Table 31 replaces the experience indicators of equation (7) by mutually exclusive indicators for the most recent year of experience under each form. Capability should outlast a relationship; the state of a relationship should not. Experience in the destination itself (Panel A) separates the two. Consigned experience stops mattering after one year: its coefficient is 0.051 when the experience dates from $t - 1$ and -0.004 (s.e. 0.007) when it dates only from $t - 2$. Imported-materials experience retains a share of 0.38 of its effect after a year, and ordinary-trade experience retains more. Consigned experience records a current contract; imported-materials experience leaves something the firm keeps. Experience in similar destinations (Panel B) is small at every lag for both processing forms.

Table 32 reproduces the predetermined risk-set design of Bai et al. (2021) on non-

Table 29: Portability weights under alternative specifications

Specification	λ_I (1)	λ_A (2)	$p(\lambda_I = \lambda_A)$ (3)	μ_I (4)	μ_A (5)
Baseline: top 40 destinations in $t - 1$	0.36 (0.15)	-0.44 (0.18)	<0.001	1.19 (0.11)	0.80 (0.22)
Candidate set: top 20	0.33 (0.20)	-0.43 (0.24)	0.017	1.05 (0.10)	1.01 (0.14)
Candidate set: top 80	0.28 (0.11)	-0.26 (0.15)	0.004	1.49 (0.13)	0.69 (0.29)
Similar = same continent	0.62 (0.14)	-0.08 (0.20)	0.003	1.19 (0.11)	0.80 (0.22)
Similar = within 2,000 km	0.41 (0.15)	-0.01 (0.13)	0.045	1.17 (0.11)	0.80 (0.22)
Similar = same income group, <5,000 km	0.31 (0.13)	-0.27 (0.24)	0.028	1.18 (0.11)	0.81 (0.22)
Similar = same official language	-0.26 (0.18)	-1.07 (0.41)	0.076	1.20 (0.11)	0.83 (0.21)
Dose: $\ln(1 + n)$ similar destinations	0.06 (0.09)	-0.27 (0.12)	0.040	1.20 (0.11)	0.82 (0.22)
Memory specification, 2003–2006	0.41 (0.18)	-0.56 (0.19)	<0.001	1.04 (0.09)	0.68 (0.21)

Notes: Each row re-estimates equation (7) with one change relative to the baseline of Table 10. In the dose row, S^r is replaced by $\ln(1 + n^r)$, where n^r is the number of similar destinations served under form r . The memory row reports the weights implied by the $t - 1$ indicators of Table 31. The table prints the destination-specific weights κ_r as μ_r and the portability weights ω_r as λ_r . Delta-method standard errors clustered by firm in parentheses.

intermediary exporters (839,612 observations). Pooled processing experience predicts later ordinary-trade entry with a coefficient of 0.035 (s.e. 0.007), of the same order as the original study. Split by form, the value-weighted effect comes from imported-materials processing (0.043, s.e. 0.008) rather than consigned processing (0.008, s.e. 0.009). Both forms predict a firm's return to its own destinations, but only imported-materials processing has positive predictive content for new markets, and only that content persists.

E.4 The portable-experience stock and trade-form switches: design and robustness

Treatment and comparison. Trade forms are observed only in the customs data, so Sections 8.2 and 8.3 use a customs-based counterpart of the main design. The unit is a 0.1-degree cell c and year t . Exposure is the pair of counts of operating special customs zones of

Table 30: Structural conditional logit of destination choice

Parameter	Estimate (1)	Standard error (2)
Destination-specific experience, ordinary (γ_k)	1.151	(0.023)
Relative weight, imported materials (μ_I)	0.757	(0.043)
Relative weight, consigned (μ_A)	0.730	(0.067)
Portable experience, ordinary (γ_g)	0.392	(0.020)
Relative weight, imported materials (λ_I)	0.265	(0.070)
Relative weight, consigned (λ_A)	-0.074	(0.089)
Market size, log (δ)	0.265	(0.005)
Market size zero (δ_0)	1.760	(0.054)
LR test: $\lambda_r = \mu_r$ (2 restrictions)	297.9	<0.001
LR test: $\lambda_I = \mu_I = 1$, $\lambda_A = \mu_A$ (3 restrictions)	699.5	<0.001
Choice situations	66,938	
Alternatives	2,457,050	
Firms	8,402	

Notes: Maximum-likelihood estimates on a 10 percent random sample of non-intermediary exporters, 2002–2006. Choice situations are firm $\times$ HS6 $\times$ year groups with at least one entry; alternatives are the 40 largest destinations in $t - 1$. The table prints the destination-specific weights κ_r as μ_r and the portability weights ω_r as λ_r . The likelihood-ratio statistics compare the unrestricted model with the restricted models named in each row. Standard errors clustered by firm.

any type within 40 km and within 40–150 km; EPZs enter at their official customs-acceptance dates. A cell is treated in year g when its 40 km count rises and at least one arriving zone is an EPZ. A sub-experiment collects treated cells with the same year, the same initial 40 km count, the same bin of the 40–150 km count, and the same increment; treated cells are followed until their 40 km count changes again. For event time e , comparison cells start in the same state, keep their 40 km count unchanged from $g - 1$ through $g + e$, and eventually have a special zone within 40 km. As in the main design, they are places that the state also selected but had not yet treated, and they are reweighted within the sub-experiment by the odds of treatment given pre-period size and growth.

Estimator. With Y the cell total of an outcome over non-fenced, non-intermediary exporters, the zero-inclusive proportional effect is

$$\hat{\theta}_e = \ln \sum_s Y_{s,g+e}^T - \ln \sum_s Y_{s,g-1}^T \frac{Y_{s,g+e}^C}{Y_{s,g-1}^C}, \quad (10)$$

Table 31: Persistence of export experience

	Ordinary	Imported materials	Consigned
Most recent experience	(1)	(2)	(3)
<i>Panel A. Experience in the destination itself</i>			
Year $t - 1$	0.075 (0.004)	0.078 (0.006)	0.051 (0.016)
Only year $t - 2$	0.047 (0.003)	0.029 (0.004)	-0.004 (0.007)
Only year $t - 3$	0.030 (0.004)	0.019 (0.004)	0.015 (0.011)
<i>Panel B. Experience in similar destinations</i>			
Year $t - 1$	0.015 (0.002)	0.006 (0.002)	-0.008 (0.003)
Only year $t - 2$	0.008 (0.003)	0.002 (0.003)	-0.001 (0.004)
Only year $t - 3$	0.007 (0.005)	0.001 (0.003)	-0.002 (0.004)
Candidate tuples		1,005,126	
Firms		4,052	

Notes: Linear probability model as in Table 10, 2003–2006, with the experience indicators replaced by mutually exclusive indicators for the most recent year of experience under each form. Coefficients are on the entry probability (mean entry rate 0.058). Standard errors clustered by firm in parentheses.

the log gap between the outcome of treated areas and the outcome implied by the growth of comparison areas. It keeps cells without exporters. Window averages are simple means of event-time estimates; inference is by bootstrap over prefectures.

Firm-level switches. Table 11 uses the same sub-experiments at the firm level, for zones opening in 2001–2006. The sample is firms that exported in the year before opening and are neither fenced nor intermediaries, located within 40 km of the arriving zone (treated) or in comparison cells. Outcomes compare the year before opening with a post-opening window of up to four years that ends by 2006. Comparison firms are reweighted within sub-experiments by the odds of treatment given pre-opening firm characteristics, and the regressions control for the change in exposure to zones 40–150 km away.

Robustness. Table 34 varies one element at a time. Adding cells that never receive a special zone to the comparison pool leaves the estimate unchanged (0.053, s.e. 0.041). Using as controls only cells near zones that were approved but not yet operating, so that only the

Table 32: Processing experience and ordinary-trade entry in a predetermined risk set

Experience in $t - 1$	Firm FE		Firm-year FE	
	Value-weighted (1)	Unweighted (2)	Value-weighted (3)	Unweighted (4)
<i>Panel A. Processing experience pooled</i>				
Ordinary trade	0.065 (0.006)	0.049 (0.002)	0.075 (0.006)	0.058 (0.002)
Processing (pooled)	0.035 (0.007)	0.045 (0.002)	0.034 (0.008)	0.042 (0.002)
<i>Panel B. Processing experience split by form</i>				
Ordinary trade	0.064 (0.006)	0.049 (0.002)	0.073 (0.006)	0.058 (0.002)
Imported materials	0.043 (0.008)	0.047 (0.002)	0.043 (0.009)	0.045 (0.002)
Consigned	0.008 (0.009)	0.024 (0.003)	0.011 (0.008)	0.022 (0.003)
Observations	839,612	839,612	832,882	832,882
Entry rate	0.067	0.067	0.067	0.067

Notes: Linear probability model of ordinary-trade entry into an HS6 $\times$ destination cell, 2002–2006, on a 15 percent random sample of non-intermediary exporters, with HS6 $\times$ destination $\times$ year fixed effects and firm (columns 1–2) or firm $\times$ year (columns 3–4) fixed effects. Value weights are the firm’s exports in t . Standard errors clustered by firm in parentheses.

timing of operation after approval is used, gives 0.063 (s.e. 0.072). The estimate is 0.034 (s.e. 0.040) for treated cells within 20 km and 0.069 (s.e. 0.054) at 20–40 km. In cells with an above-median electronics share of exports it is 0.066 (s.e. 0.043). In the other cells it is larger (0.189, s.e. 0.089), but their pre-opening estimate is also clearly negative (-0.069 , s.e. 0.032), so this larger estimate is not clean evidence of a gain. Panel B gives the long-run ordinary-trade outcomes discussed in the text, on the customs panel through 2011: from 2012 the customs records no longer carry the postcode that locates new firms, so the window is not extended further.

Timing of arrival. Among cells that will eventually receive an EPZ within 40 km, the year in which it begins operating is unrelated to the cell’s processing-export growth over the previous three years (0.005, s.e. 0.015) and to growth in electronics firm births (-0.006 , s.e. 0.012; Table 35); of the seven characteristics, only ordinary-export growth is statistically significant, with a small negative coefficient. The lag from approval to official acceptance can be related to pre-approval trends only for EPZs that have an official acceptance date and

Table 33: EPZs and the export experience of nearby firms

Outcome	Before opening $e = -4, \dots, -2$ (1)	After opening $e = 0, \dots, 3$ (2)
<i>Panel A. Capability: non-fenced exporters within 40 km</i>		
Portable-experience stock, K	-0.017 (0.045)	0.053 (0.041)
Ordinary-trade destinations, N^O	-0.018 (0.048)	0.063 (0.049)
Imported-materials destinations, N^I	-0.045 (0.053)	-0.002 (0.063)
Consigned destinations, N^A	-0.187 (0.089)	-0.007 (0.088)
Exporting firms	-0.079 (0.047)	0.047 (0.038)
<i>Panel B. Activity inside the fence</i>		
Fence exports in year 5 / local exports before opening		0.39
Sub-experiments	53	
Treated cells	1,132	
Firms (all cells)	151,839	
Zones in activity benchmark	37	

Notes: Panel A reports window averages of $\hat{\theta}_e$ (equation (10)) for cells whose count of special customs zones within 40 km rises with the arrival of an EPZ, 2001–2006. Outcomes are cell totals over non-fenced, non-intermediary exporters; the portable-experience stock is equation (8). Comparison cells are not yet treated and are reweighted within sub-experiments by the odds of treatment. Standard errors from a bootstrap over prefectures in parentheses. Panel B reports the ratio of fenced firms' exports in the fifth year after opening to the exports of non-fenced firms within 40 km in the year before opening, summed over zones observed for five years.

were approved in 2003 or later, so that three pre-approval years of customs records exist; for these 17 zones the lag is unrelated to pre-approval growth in processing exports, ordinary exports, or the number of exporters ($R^2 \leq 0.02$).

E.5 Signals from fenced firms and local peers

The signal regressions estimate the entry model as a conditional logit within firm $\times$ HS6 product $\times$ year groups, with destination $\times$ year effects, the log national market size for the product and destination, and the firm's own experience under each form. Fenced firms are those of an operating EPZ within 20 km, and a fenced signal means that they sold the same four-digit product to d in $t - 1$. City peers are other non-intermediary exporters in the same

Table 34: The portable-experience stock under alternative specifications

Specification	Portable stock K		Ordinary destinations N^O	
	Before (1)	After (2)	Before (3)	After (4)
<i>Panel A. Opening window, 2000–2006 forms</i>				
Baseline, 0–40 km	-0.017 (0.045)	0.053 (0.041)	-0.018 (0.048)	0.063 (0.049)
Never-treated in controls	-0.010 (0.043)	0.053 (0.041)	-0.013 (0.046)	0.065 (0.049)
Approved-not-operating controls	0.000 (0.174)	0.063 (0.072)	-0.007 (0.178)	0.072 (0.075)
Treated within 20 km	0.024 (0.049)	0.034 (0.040)	0.027 (0.051)	0.040 (0.041)
Treated 20–40 km	-0.063 (0.053)	0.069 (0.054)	-0.069 (0.058)	0.085 (0.069)
Electronics-intensive cells	-0.051 (0.062)	0.066 (0.043)	-0.050 (0.068)	0.075 (0.050)
Other cells	-0.069 (0.032)	0.189 (0.089)	-0.073 (0.031)	0.201 (0.093)
<i>Panel B. Long run, 2000–2011, ordinary trade only</i>				
Baseline	Ordinary exporters		New surviving markets	
	Before	After, $e = 4, \dots, 8$	Before	After, $e = 4, \dots, 8$
	-0.028 (0.028)	0.022 (0.127)	-0.008 (0.047)	0.077 (0.125)

Notes: Window averages of $\hat{\theta}_e$ (equation (10)); “before” is $e = -4, \dots, -2$ and “after” is $e = 0, \dots, 3$ unless stated. Never-treated controls add cells that never receive a special zone within 40 km. Approved-not-operating controls use cells near zones already approved whose operation begins after the window. The electronics split uses the median share of HS84–85 in the cell’s 2000 exports, computed among cells that eventually have a zone within 40 km. Panel B uses all located firms, 2000–2011, and ordinary-trade outcomes only; new surviving markets are firm $\times$ HS6 $\times$ destination combinations served for the first time under ordinary trade and still served the next year. Standard errors from a bootstrap over prefectures in parentheses.

city selling the same HS6 product to d under ordinary trade. Bins of the city’s total number of export links to d absorb city $\times$ destination shocks. Table 36 reports the main specification with the placebo of markets that fenced firms enter only in t , and the two counterfactuals of the text. Table 37 adds two placebos for the peer signal: peers whose first sale to d occurs only in t , which has the same coefficient as the peer signal itself, and exporters in other cities of the province. The design cannot separate information among local firms from shocks common to them, so the text credits only the fence signal, and only as an upper bound.

Table 35: Pre-determined characteristics and the timing of zone arrival

Pre-determined cell characteristic	Zone begins operating (1)	Zone is approved (2)
Processing exports, growth	0.005 (0.015)	0.003 (0.006)
Ordinary exports, growth	-0.024 (0.011)	-0.013 (0.008)
Ordinary exporters, growth	0.028 (0.017)	0.009 (0.006)
Electronics firm births, growth	-0.006 (0.012)	0.011 (0.006)
Processing exports, level (log)	0.003 (0.024)	0.012 (0.009)
Ordinary exports, level (log)	0.014 (0.019)	0.009 (0.015)
Ordinary exporters, level (log)	-0.005 (0.016)	-0.007 (0.006)
Mean of outcome	0.328	0.403
Cell-years	1,872	640
Cells	802	341

Notes: Linear probability models on cell-years, 2003–2011, of cells that eventually receive an EPZ within 40 km, observed until arrival. The outcome is one in the year the zone begins operating (column 1) or is approved (column 2). All characteristics enter jointly, standardized; growth rates are over the previous three years. The regressions include year fixed effects. Standard errors clustered by prefecture in parentheses.

E.6 Domestic firms that switched to consigned processing

A natural follow-up question is what became of the domestic firms that took on consigned work. We compare, among domestic exporters in treated cells within 40 km of an arriving zone whose post-opening window covers at least two years, the 330 firms that started consigned processing with the 5,900 that did not. In the window, 98.6 percent of switchers entered at least one new product–destination market under ordinary trade, against 81.3 percent of non-switchers; within sub-experiments the difference is 16.5 percentage points ($p < 0.001$). The number of new ordinary-trade markets relative to one plus the markets already served is 5.78 for switchers and 2.86 for non-switchers (difference within sub-experiments 2.95, $p < 0.001$).

This comparison describes selection, not an effect of consigned work. Firms that take on a new trade form are expanding firms, and expanding firms add business on every margin at once; switchers and non-switchers differ in ways that the comparison does not hold fixed. Nor does it conflict with the portability weights of Table 10, which compare destinations

Table 36: Signals from fenced firms and local exporters in firms' entry decisions

Signal	Coefficient (1)	Relative to own same-destination (2)	Share of decisions exposed (3)
Own experience, same destination	1.119 (0.024)	1.000	0.3238
Own experience, similar destination	0.394 (0.020)	0.352	0.8354
City peers sell the product to d	0.427 (0.015)	0.381	0.4270
Fenced firms sell the product to d	0.141 (0.070)	0.126	0.0046
Fenced firms sell only other products	-0.039 (0.037)	-0.034	0.0989
Fenced firms first sell it to d only in t	0.068 (0.068)	0.061	0.0018
<i>Counterfactual change in ordinary-trade entries</i>			
Fence signal off	-0.005		
Fence overlap at city-peer rate	0.062		
Alternatives		2,376,163	
Firms		7,827	

Notes: Conditional logit of ordinary-trade entry into candidate destination d , groups defined by firm $\times$ HS6 product $\times$ year, 2002–2006, on a 10 percent random sample of non-intermediary exporters; candidates are the 40 largest destinations in $t - 1$. The model also includes own experience under each processing form, destination $\times$ year effects, log national market size for the product and destination, and bins of the city's number of export links to d . Signals refer to year $t - 1$ unless stated. Fenced firms are those of an operating EPZ within 20 km, and “the product” is the same HS4 heading; city peers are other non-intermediary exporters in the same city selling the same HS6 product under ordinary trade. The last signal row is a placebo: fenced firms whose first sale of the product to d occurs only in t . Column 2 divides each coefficient by the coefficient on own same-destination experience; column 3 is the share of alternatives in which the signal is present. The counterfactual rows report the proportional change in predicted ordinary-trade entries of nearby firms. Standard errors clustered by firm in parentheses.

within the same firm, product, and year and so hold the firm's expansion fixed: switchers enter more markets because they are expanding, and their consigned experience does not tell us where they will go next.

F Estimator, Spatial Unit, and Radius

Table 38 re-estimates the main results with other estimators and other spatial units, holding the design fixed. New domestic firms fall by 12.4 percent with OLS on $\log(1 + y)$, 14.9 percent

Table 37: Signals from local exporters and their placebos

Signal	Coefficient (1)	Share of decisions exposed (2)
City peers sell the product to d	0.369 (0.014)	0.4270
City peers first sell it to d only in t	0.370 (0.014)	0.3135
Other cities in the province sell it to d	0.244 (0.015)	0.4724
Fenced firms sell the product to d	0.129 (0.070)	0.0046
Fenced firms sell only other products	-0.043 (0.037)	0.0989

Notes: Conditional logit as in Table 36, with two placebo signals added: city peers whose first sale of the product to d occurs only in t , and exporters in other cities of the same province selling the product to d in $t - 1$. Other signals are measured in $t - 1$ and defined as in Table 36. Standard errors clustered by firm in parentheses.

with the inverse hyperbolic sine, and 21.6 percent of the pre-period mean with OLS in levels, against 19.7 percent with Poisson; the probability that a cell has any new domestic firm falls by 7.9 percent. Incumbent output and total output within 20 km are indistinguishable from zero under every estimator. The loss is local: it is 17.9 percent within 10 km and 14.4 percent within 30 km, and zero in the 20–40 km ring. Coarser units do not dilute it: with 0.2-degree cells it is 22.4 percent, and with counties 36.1 percent, although both draw on far fewer candidate control units.

G Robustness

Table 39 varies every choice behind the entry estimate that a reader could question. Excluding the 2004 census year, dropping the reweighting, or restricting controls to future EPZ sites—whose full battery of 21 outcomes appears as column (2) of Appendix Table 18—makes the decline larger (25 to 39 percent, all intervals excluding zero); including locations with any initial exposure state (cells that already had a special zone) gives 19 percent. Clustering by zone rather than prefecture, because a 20 km circle can span prefectures, narrows the interval to -34 to -3 percent. Future-EPZ controls show no pre-trend (joint $p = 0.37$). The main specification is the most conservative of these.

Table 38: Entry and output under alternative estimators and spatial units

Specification	Coefficient	% effect	<i>N</i> (treated cells)
<i>Panel A. New domestic firms</i>			
Poisson (baseline)	−0.219 (0.118)	−19.7	44,263 (279)
OLS, log(1+y)	−0.132 (0.058)	−12.4	44,263 (279)
OLS, levels (% of pre-period mean)	−0.802 (0.329)	−21.6	44,263 (279)
OLS, inverse hyperbolic sine	−0.161 (0.073)	−14.9	44,263 (279)
Linear probability: any new domestic firm	−0.044 (0.028)	−7.9	55,826 (315)
<i>Panel B. Incumbent output</i>			
Poisson (baseline)	−0.005 (0.046)	−0.5	53,050 (307)
OLS, log(1+y)	−0.088 (0.052)	−8.4	53,050 (307)
OLS, levels (% of pre-period mean)	−250.563 (366.199)	−6.8	53,050 (307)
OLS, inverse hyperbolic sine	−0.094 (0.056)	−9.0	53,050 (307)
<i>Panel C. Total output within 20 km (all firms)</i>			
Poisson (baseline)	0.022 (0.035)	2.2	55,826 (315)
OLS, log(1+y)	−0.046 (0.052)	−4.5	55,826 (315)
OLS, levels (% of pre-period mean)	−5.236 (700.272)	−0.1	55,826 (315)
OLS, inverse hyperbolic sine	−0.047 (0.056)	−4.6	55,826 (315)
<i>Panel D. Spatial unit and radius (new domestic firms, Poisson)</i>			
Main: 20 km, 0.1° cells	−0.219 (0.118)	−19.7	44,263 (279)
10 km radius	−0.197 (0.131)	−17.9	34,487 (94)
30 km radius	−0.155 (0.116)	−14.4	46,739 (544)
20–40 km ring	0.007 (0.123)	0.7	51,881 (554)
0.2° cells, 20 km	−0.254 (0.113)	−22.4	17,320 (99)
Counties (districts), 20 km	−0.448 (0.131)	−36.1	10,959 (100)

Notes: Main design (not-yet-treated controls, reweighted; cell and cohort-year effects; prefecture clustering) with the estimator or unit changed as labeled. The percent effect is $e^\beta - 1$ for log-type models and the coefficient relative to the treated pre-period mean for levels. Standard errors in parentheses.

Table 39: Robustness of the entry estimate

Specification	Estimate	SE	% effect	95% CI	N (treated cells)
<i>Panel A. Initial-exposure / weighting variants</i>					
Main (not-yet-treated, reweighted)	−0.219	(0.118)	−20%	[−36, 1]%	44,263 (279 treated)
2004 additions excluded	−0.320	(0.118)	−27%	[−42, −9]%	34,818 (263 treated)
All initial exposure states	−0.206	(0.106)	−19%	[−34, 0]%	48,562 (434 treated)
Unweighted not-yet-treated	−0.287	(0.123)	−25%	[−41, −5]%	44,263 (279 treated)
Future-EPZ sites as controls	−0.492	(0.135)	−39%	[−53, −20]%	15,633 (279 treated)
<i>Panel B. Clustering</i>					
Prefecture (main)	−0.219	(0.118)	−20%	[−36, 1]%	44,263 (117 clusters)
EPZ zone	−0.219	(0.098)	−20%	[−34, −3]%	44,263 (71 clusters)
Zone × sub-experiment	−0.219	(0.099)	−20%	[−34, −2]%	44,263 (85 clusters)
Prefecture, score wild bootstrap	−0.219	—	−20%	[−35, 2] (p=0.081)	9,999 reps (117 clusters)
EPZ zone, score wild bootstrap	−0.219	—	−20%	[−33, −4] (p=0.038)	9,999 reps (71 clusters)
<i>Panel C. Pre-trend and alternative estimator</i>					
Future-EPZ pre-trend, joint Wald test	—	—	—	p = 0.3695	—

Notes: Outcome: new domestic firms within 20 km, equation (5). Rows vary the control group, sample, weighting, clustering, and estimator as labeled.

H What the Fence Left: Details

H.1 Local income and domestic value added

The fences year by year. Table 40 follows the firms registered inside an EPZ from 2001 to 2014. The domestic value-added ratio of fenced exports is one minus processing imports over processing exports (Kee and Tang 2016), defined in Appendix M.8. Within a zone, one firm often imports what another exports, and imports lead exports in the first months of operation, so the ratio is computed on totals rather than firm by firm: on zone totals over operating years 0–3 for the zone-level ratios of Section 7.1, and on national totals for the yearly ratios of Table 40. The ratio rose from 0.27 in 2003 to 0.48 in 2006. It includes wages, rents, and profits paid inside the fence as well as domestic inputs, and customs records do not identify carry-over transfers to outside firms, so it bounds local intermediate sourcing s_z from above. A ratio that divides all imports of fenced firms by all their exports, rather than processing trade only, is lower in 2003; the difference comes from which imports are counted—the second ratio adds duty-free equipment and other imports outside the processing regimes—and not from which firms are included.

The wage bill. The local factor share σ_z is the wage bill of fenced firms that can be matched to the firm survey, divided by their exports. Its median across the 17 zones with matched firms is 6 percent (interquartile range 3.5–9.2 percent). The ledger of Section 10 applies the median to every zone, because a zone-specific share would record zero wherever

Table 40: Activity inside the fences

Year	Fenced firms	Processing exports (US\$100m)	Processing imports (US\$100m)	DVAR $1 - M/X$	Imported- materials share (%)	Consigned- materials share (%)
2001	48	4.0	4.3	-0.084	100	0
2002	125	21.9	21.5	0.018	100	0
2003	230	100.6	73.1	0.273	100	0
2004	356	208.3	132.1	0.366	100	0
2005	485	343.8	204.2	0.406	100	0
2006	588	543.2	284.7	0.476	100	0
2007	767	757.9	447.2	0.410	—	—
2008	849	1014.7	508.5	0.499	—	—
2009	939	1014.2	498.3	0.509	—	—
2010	999	1301.6	687.7	0.472	—	—
2011	986	1017.3	557.3	0.452	—	—
2012	888	827.1	492.4	0.405	—	—
2013	902	762.1	450.3	0.409	—	—
2014	859	839.8	509.8	0.393	—	—

Notes: Customs data, firms registered inside an EPZ (fifth digit of the customs registration code), identified year by year. Processing exports and imports are trade under processing with imported materials and with supplied materials; imported equipment is excluded. DVAR is one minus processing imports over processing exports, an upper bound on local inputs and local factor payments combined, since it also contains the profits of foreign owners. The trade-mode split is not reported after 2006.

matching fails; Appendix I reports the ledger at the two quartiles.

H.2 Agglomeration and productivity: the designs

Table 41 reports each design of Section 7.2 in its own units, with the additional outcomes each design produces. The designs were built independently and share neither sample nor estimator; what they share is the question whether activity near a zone became more productive or more export-intensive.

In-place firms. Incumbent manufacturing firms near the zone, observed in both the customs and the firm records, in a grid event study with prefecture-clustered errors. Productivity is the residual of a Cobb–Douglas production function whose sector coefficients are fitted on the full firm census, so that firms are compared within their industry. The linked sample is small and tilted toward large firms, so the estimate (0.127, s.e. 0.102) is imprecise rather than a precise zero. The same design finds no larger return to export experience near the zone: the difference in export growth between experienced and new exporters is -0.096 (s.e. 0.261).

Lu–Wang–Zhu replication. We rerun the spillover specifications of [Lu et al. \(2019\)](#) on a zip-code $\times$ year panel covering the national special customs zones. The distance-free term, which pools all non-host locations in the zone’s city, is 0.067 (s.e. 0.0518) for employment and 0.023 (s.e. 0.0622) for output. In the joint concentric-ring version the nearest ring (0–20 km) is negative for output (−0.144, s.e. 0.0601), and the ring coefficients do not decay with distance, which is the opposite of the positive, decaying profile an agglomeration spillover would produce. Our assignment of zones to jurisdictions is coarser than in the original study, which could dilute a very local spillover; the grid designs of the main text are not subject to this concern and find no spillover either.

Peer export network. On the full customs records at the firm $\times$ product $\times$ destination $\times$ year level, a firm’s entry into a new market is regressed on the log of last year’s local stock of exporters already serving it, with city $\times$ product $\times$ destination effects absorbing long-run comparative advantage. The elasticity is −0.0019 (s.e. 0.0019). Without the city $\times$ product $\times$ destination effects the association is negative, so the raw correlation reflects which cities serve which markets rather than learning from neighbors.

Winner versus runner-up. Following [Greenstone et al. \(2010\)](#), a city panel for 2010–2018 compares cities approved for a comprehensive bonded zone with cities that applied in the same round and were not approved. City GDP changes by 0.022 (s.e. 0.029) and night lights within 3 km of the candidate site by −0.047 (s.e. 0.060). This design concerns the successor program and a later period; it is included because it identifies from a different source of variation, the approval decision among applicants.

Wage capitalization. The average wage in a city panel with not-yet-treated controls ([Callaway and Sant’Anna 2021](#)): −0.037 (s.e. 0.022), the same estimate that tests wage competition in Section 6.

Eight-year window. The longer horizon is reported once, with the export-experience evidence: Section 8.3 and Panel B of Appendix Table 34 give the number of ordinary-trade exporters within 40 km four to eight years after opening, on the customs panel through 2011.

H.3 Local suppliers and users

Upstream industries near the fence. Table 42 reports the binary version of equation (6), in which the panel is collapsed to three industry groups—the fence’s own industry (electronics), its top-quartile suppliers, and all others—and the exposure is the indicator of the supplier group, with electronics entered separately. Relative to other industries in the same cell and year, output of upstream supplier industries near the fence changes by 0.022 (s.e. 0.041) and employment by −0.081 (s.e. 0.043). The upper end of the confidence interval for output is 11

Table 41: Agglomeration and productivity spillovers: designs in their own units

Design	Data	Outcome	Estimate	Source
In-place firms: productivity	0.1° grid event study, customs-firm bridge, city-clustered s.e.	Industry-adjusted TFP (log points)	0.127 (0.102)	Cobb-Douglas OLS residual, sector coefficients fit on the full manufacturing census
In-place firms: learning	same design	Experienced – new exporters, export-value growth (log points)	−0.096 (0.261)	Return to pre-existing export experience, inside vs. outside the fence
Lu-Wang-Zhu (2019) replication, Eq. 3	Zip-code × year panel, 1998–2013, 174 national zones	Employment (log points)	0.067 (0.0518)	Distance-free, jurisdiction-blind spillover term
Lu-Wang-Zhu (2019) replication, Eq. 3	same design	Output (log points)	0.023 (0.0622)	Distance-free, jurisdiction-blind spillover term
Lu-Wang-Zhu (2019) replication, Eq. 4	same panel, joint concentric-ring regression	Output, nearest ring 0–20 km (log points)	−0.144 (0.0601)	Not monotonically decaying with distance
Peer export-network entry	Full customs microdata, firm × product × destination × year	Entry into a new market (elasticity to log local peer stock)	−0.0019 (0.0019)	City × product × destination fixed effects absorb long-run comparative advantage
Winner vs. runner-up contest	City panel, 2010–2018, 45 winners vs. same-round runner-up applicants	City GDP (log points)	0.022 (0.029)	Greenstone-Hornbeck-Moretti (2010) design
Winner vs. runner-up contest	same design	Night lights within 3 km of the candidate site (log points)	−0.047 (0.060)	Independent satellite-based outcome, same contest
Wage capitalization	City panel, Callaway-Sant’Anna not-yet-treated design	Average wage (log points)	−0.037 (0.022)	Same estimate as the wage test of Section 6

Notes: Each row is one estimate in the units of its own design, with the standard error, clustered as in that design, in parentheses. The first row of each design is the one reported in Section 7.2. Log points are Poisson or log-linear coefficients; the peer-network row is an elasticity of the probability of entering a market.

percent: upstream industries did not expand on average. The fence’s own industry grows somewhat faster near the zone than other industries (Panel B, 0.154, s.e. 0.107); Table 8 shows that this faster growth is already present before opening.

The exposure measures. Let w be the national industry mix of fenced output and a_{jk} the share of industry k ’s intermediate inputs bought from industry j in the 2002 input–output table. The horizontal exposure of industry j is $H_j = w_j$; the backward exposure is $B_j = \sum_k a_{jk}w_k$, industry j ’s weight in the fence’s input basket; and the forward exposure is

Table 42: Upstream industries near the fence: binary triple difference

	(1)	(2)	(3)
<i>Panel A. Upstream supplier industries relative to other industries</i>			
Output	0.022 (0.041)	0.022 (0.035)	0.030 (0.073)
Employment	−0.081 (0.043)	−0.054 (0.025)	−0.042 (0.060)
Incumbent output	−0.003 (0.049)	0.022 (0.035)	0.013 (0.072)
Incumbent employment	−0.068 (0.036)	−0.033 (0.020)	−0.016 (0.043)
<i>Panel B. Electronics (the fence’s own industry) relative to other industries</i>			
Output	0.154 (0.107)	0.229 (0.100)	0.132 (0.125)
Employment	0.036 (0.078)	0.078 (0.045)	0.094 (0.080)
<i>Panel C. Upstream industries by pre-opening supply base (thick minus thin)</i>			
Output, thin base	0.107 (0.056)	0.083 (0.046)	
Output, thick − thin	−0.121 (0.071)	−0.084 (0.053)	
Employment, thin base	0.123 (0.061)	0.081 (0.035)	
Employment, thick − thin	−0.309 (0.049)	−0.204 (0.039)	
Incumbent employment, thin base	0.026 (0.042)	0.015 (0.026)	
Incumbent employment, thick − thin	−0.141 (0.039)	−0.070 (0.031)	
Initial exposure	First zone	All	First zone
Controls	Not yet treated, reweighted		Future EPZ sites

Notes: Coefficients from the binary version of equation (6). Industry groups are fixed before opening from the 2002 input–output table: upstream suppliers are the top quartile of direct input requirements per unit of fence output (the national industry mix of fenced exports), excluding electronics; electronics is the fence’s own industry; all others form the comparison group. Outcomes are totals within cell $\times$ industry group $\times$ year. Panel C interacts the upstream term with an indicator for a thick supply base, an upstream employment share in the year before opening at or above the median across treated cells; the thin row reports the upstream coefficient for thin cells and the thick − thin row the interaction. Table 45 shows that the Panel C pattern reappears at a placebo opening date, so it is not used as evidence. Standard errors clustered by prefecture in parentheses.

$F_j = \sum_k w_k a_{kj}$, the weight of the fence’s industries in industry j ’s own inputs. The coefficients serve only as linkage weights, and no Leontief inverse is used. Each measure is standardized across the two-digit industries matched to the table, so β in equation (6) is the proportional effect of a one-standard-deviation difference in exposure.

The three exposures together. Table 43 repeats the single-exposure estimates of Table 8 in columns (1)–(3) and enters all three in column (4), as in Javorcik (2004). The panel is at the cell $\times$ industry $\times$ year level, but the fixed effects are defined for the three industry groups of Table 42, because a Poisson model with cell $\times$ industry effects at the full two-digit resolution does not converge; the continuous exposure is identified within and across groups. Collapsed to the three groups, the reconstructed panel reproduces the estimates of Table 42 almost exactly. Column (4) cannot separate the channels. Electronics ranks first on all three exposures, the horizontal measure is correlated 0.94 with the backward and 0.95 with the forward measure, and the horizontal and backward coefficients take large values of opposite sign (for output, 0.804 and -0.859) that offset each other. The forward coefficient in column (4) is 0.137 (s.e. 0.082) for employment and 0.051 (s.e. 0.085) for output.

New entrants by industry exposure. Table 44 gives the race for new domestic entrants. The point estimate for supplier industries is positive (0.120, s.e. 0.066), but its pre-opening coefficients are jointly different from zero ($p = 0.006$), so it does not separate the fence from where supplier entry was already heading. For this outcome the fixed-effect demeaning stops at its iteration limit before reaching the convergence tolerance, and the placebo with openings moved three years earlier cannot be estimated on the short pre-opening sample. We therefore do not use the entry estimates in the main text.

Thin and thick supply bases. Panel C of Table 42 splits treated cells at the median upstream employment share in the year before opening. Taken at face value, upstream employment grows in thin-base cells (0.123, s.e. 0.061) and falls in thick-base cells (thick minus thin -0.309 , s.e. 0.049). The split sorts cells on a share measured one year before opening, and a share that is unusually high in one year tends to fall back whether or not a zone arrives. Table 45 tests this. Measuring the base four years before opening removes the gain in thin cells and the output difference (thick minus thin 0.019, s.e. 0.058), while the employment difference remains (-0.244 , s.e. 0.078). Moving every opening three years earlier and keeping only years before the true opening, so that no year the placebo treats is treated, reproduces the employment difference at full size: -0.337 (s.e. 0.078). The event study (Figure 23) shows upstream employment in thick-base cells already falling in the years before

Table 43: Horizontal, backward, and forward exposure: one at a time and together

	(1)	(2)	(3)	(4)
	Horizontal	Backward	Forward	All three
<i>Panel: New domestic entrants</i>				
Horizontal H_j (own industry)	0.000 (0.055)			-0.821 (0.276)
Backward B_j (fence buys from j)		0.120 (0.066)		0.604 (0.195)
Forward F_j (j buys from the fence)			0.088 (0.059)	0.239 (0.104)
Observations	259,375	259,375	259,375	259,375
<i>Panel: Employment</i>				
Horizontal H_j (own industry)	0.043 (0.016)			0.068 (0.170)
Backward B_j (fence buys from j)		0.016 (0.020)		-0.169 (0.123)
Forward F_j (j buys from the fence)			0.076 (0.030)	0.137 (0.082)
Observations	458,623	458,623	458,623	458,623
<i>Panel: Output</i>				
Horizontal H_j (own industry)	0.049 (0.023)			0.804 (0.288)
Backward B_j (fence buys from j)		-0.069 (0.045)		-0.859 (0.254)
Forward F_j (j buys from the fence)			0.071 (0.035)	0.051 (0.085)
Observations	458,623	458,623	458,623	458,623

Notes: Poisson estimates of equation (6), main design, cell $\times$ industry $\times$ year panel; pooled post-opening coefficients, $e = 0, \dots, 3$, per standard deviation of exposure. Columns (1)–(3) enter one exposure at a time and are column (1) of Table 8; column (4) enters all three. Fixed effects: cell $\times$ industry group, cell $\times$ year, and industry group $\times$ year, all within sub-experiment. Standard errors clustered by prefecture in parentheses.

Table 44: Horizontal, backward, and forward exposure: new domestic entrants

	Estimate (1)	Pre-opening joint p (2)
<i>New domestic entrants</i>		
Horizontal H_j (own industry)	0.000 (0.055)	0.188
Backward B_j (fence buys from j)	0.120 (0.066)	0.006
Forward F_j (j buys from the fence)	0.088 (0.059)	0.054
Observations	259,375	259,375

Notes: Poisson estimates of equation (6) for new domestic entrants, one exposure at a time, specification as in Table 8. Column (1): pooled post-opening coefficient, $e = 0, \dots, 3$; the fixed-effect demeaning stopped at its iteration limit before reaching the convergence tolerance, so the estimates are indicative. Column (2): p -value of the joint test that the event-time coefficients $e = -4, \dots, -2$ are zero. Standard errors clustered by prefecture in parentheses.

opening. The divergence between thin and thick bases is therefore convergence between cells with high and low upstream shares, which occurs without a zone; it is mean reversion, not the fence’s demand or its draw on local workers, and the paper does not use it. For new entrants the placebo cannot be estimated because the Poisson estimator does not converge in the short pre-opening sample, and the thin–thick difference in entry is imprecise and changes sign when the base is measured earlier (columns 1 and 2).

New suppliers. Table 46 reports the formation of supporting firms near the zones. It counts births in three sources that do not share a firm identifier: the industrial-firm census, the customs records of first-time exporters, and business registrations. Each is expressed per above-scale incumbent firm in the base year, so the three are on the scale of the local industrial base, and compared between treated cells and reweighted comparison cells of the same sub-experiments before and after opening. Treated cells are those of the broader design described in Appendix H.4. Births of electronic-component and final-electronics firms were already higher near future zones and did not rise further; only registrations of logistics, customs-brokerage, and trading firms rise, imprecisely.

H.4 The local multiplier: data and construction

Registrations. The business registration records list firms registered from 1998 to 2016 that were still on the register in 2019 (4,481,974 records), with a free-text line of business. We assign each record to one of three groups by keywords in that text, checking manufacturing first

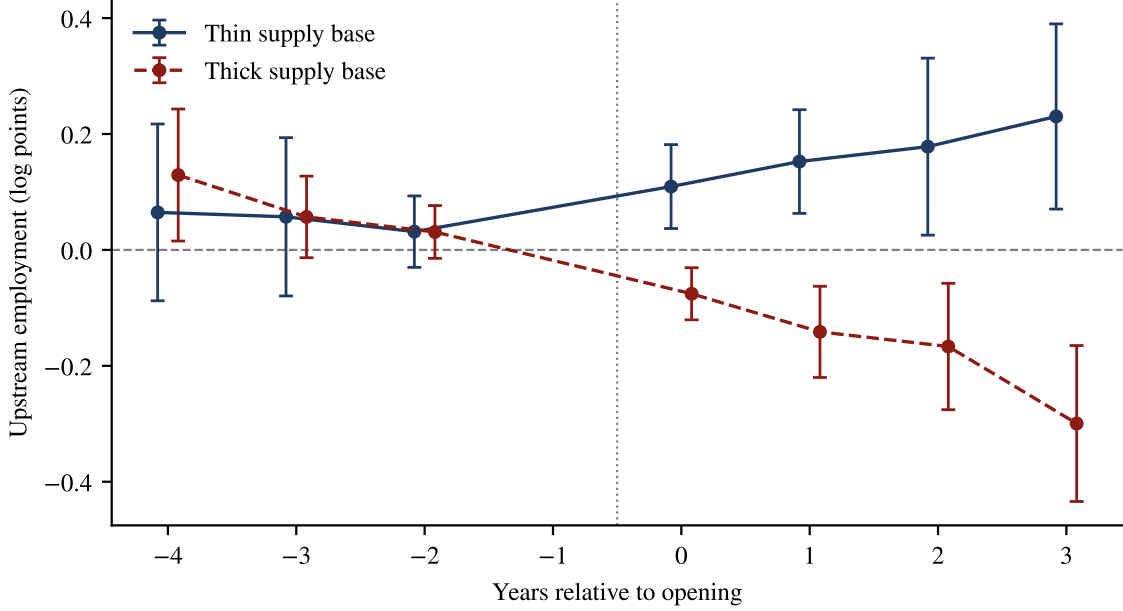


Figure 23: Upstream employment near an EPZ, by pre-opening supply base

Notes: Equation (6) event study for upstream industries in locations with a thin and a thick pre-opening local supply base. Bars are 95 percent confidence intervals clustered by prefecture. The divergence begins before opening; the placebo checks are in Table 45.

so that a factory that also lists wholesale or retail of its own output is not counted as a shop: non-tradable services (wholesale and retail, accommodation and catering, resident services, real estate, culture, sports, and entertainment), producer services (transport, warehousing, and logistics; leasing and business services), and manufacturing; the remainder (agriculture, construction, finance, and others) is not reported. This is a keyword approximation of the national industry classification, not an audited code. Records are located at the township for 45 percent of the sample and otherwise at the county centroid; records with neither are dropped. Because only survivors are observed, levels are understated in sectors with high exit, such as retail and catering; the comparison between treated and comparison cells is biased only if survival to 2019 differs between them.

Design. The multiplier and the supplier births of Table 46 use a broader version of the main design. Like the main design, it forms sub-experiments by opening year and initial exposure state, takes as controls not-yet-treated cells whose near-ring count does not change during the window, and ends a treated cell's window at its next near-ring change (Section 4). Unlike the main design, it treats a cell whenever an EPZ arrives within 20 km, including arrivals near a special customs zone that was already operating, not only the first zone within 40 km. This yields 624 treated cells within 20 km of 52 arriving EPZs, against 315 cells around 39 EPZs in the main design, so the estimates describe the average arrival rather

Table 45: Thin and thick supply bases: placebo checks

	(1)	(2)	(3)
	Original: real timing, $g - 1$ baseline	Real timing, $g - 4$ baseline	Fake $g - 3$, pre-opening years only
<i>Panel: Output</i>			
Thin (upstream)	0.107 (0.056)	0.016 (0.043)	0.206 (0.124)
Thick (upstream)	-0.014 (0.051)	0.035 (0.057)	0.021 (0.055)
Thick - thin (upstream)	-0.121 (0.071)	0.019 (0.058)	-0.185 (0.158)
<i>Panel: Employment</i>			
Thin (upstream)	0.123 (0.061)	0.016 (0.049)	0.118 (0.041)
Thick (upstream)	-0.186 (0.043)	-0.228 (0.073)	-0.219 (0.053)
Thick - thin (upstream)	-0.309 (0.049)	-0.244 (0.078)	-0.337 (0.078)
<i>Panel: New domestic entrants</i>			
Thin (upstream)	0.170 (0.205)	-0.151 (0.137)	
Thick (upstream)	-0.240 (0.134)	0.007 (0.177)	
Thick - thin (upstream)	-0.409 (0.247)	0.158 (0.217)	

Notes: Upstream-group coefficients from the binary version of equation (6) with the upstream term interacted with a thick-base indicator (upstream employment share at or above the median across treated cells). Column (1): real opening dates, base measured in the year before opening (Panel C of Table 42). Column (2): real opening dates, base measured four years before opening. Column (3): every opening moved three years earlier, sample restricted to the four years before the true opening, base measured in the first of them, so columns (2) and (3) use the same split; not estimable for new entrants. Main design; standard errors clustered by prefecture in parentheses.

than a first fence. Comparison cells are not-yet-treated cells of the same sub-experiments, reweighted by the odds of treatment. The outcome is new registrations per above-scale incumbent manufacturing firm in the cell in the base year. At each event time $e = -4, \dots, 6$ the effect is the difference between the treated rate and the reweighted comparison rate, averaged over sub-experiments with weights equal to the base-year incumbent count; no event time is normalized, so $e = -1$ is estimated like the others. Relative effects divide the pooled difference by the comparison mean. Standard errors come from a bootstrap over prefectures (199 draws), and the pre-opening test is a joint Wald test that the differences at $e = -4, \dots, -1$ are zero, using the bootstrap covariance.

Magnitude. The conversion to a per-job ratio is an order of magnitude, not an estimate. The median non-hub zone's fenced wage bill, 7.0 million yuan a year, divided by the average annual manufacturing wage in the firm survey (15,388 yuan), implies about 455 fenced jobs per zone-year, or 37.91 per treated cell-year given 12.0 treated cells per zone. The

Table 46: Formation of supporting firms near EPZs

	Before $e = -4, \dots, -1$	After $e = 0, \dots, 3$	After $e = 4, \dots, 6$	Change (2)–(1)
Births near the zone	(1)	(2)	(3)	(4)
<i>Panel A. Industrial-firm census, 4-digit industries</i>				
Electronic components	0.498 (0.374)	0.078 (0.317)	-0.107 (0.294)	-0.419 (0.330)
Final electronics	0.568 (0.365)	-0.051 (0.301)	-0.033 (0.311)	-0.619 (0.406)
General supporting industries	0.010 (0.147)	-0.165 (0.112)	-0.343 (0.108)	-0.175 (0.151)
Unrelated industries	-0.100 (0.070)	-0.288 (0.079)	-0.394 (0.090)	-0.188 (0.083)
<i>Panel B. Customs, all new exporters (4-digit HS baskets)</i>				
Inputs used by fenced firms	0.298 (0.373)	0.273 (0.342)	0.124 (0.372)	-0.025 (0.183)
Other products	-0.044 (0.202)	0.023 (0.202)	-0.152 (0.174)	0.067 (0.084)
<i>Panel C. Business registrations (number)</i>				
Electronics supporting	0.570 (0.601)	0.568 (0.521)	0.342 (0.493)	-0.002 (0.305)
General supporting	0.247 (0.304)	0.360 (0.267)	0.276 (0.300)	0.113 (0.184)
Logistics and customs brokerage	0.055 (0.271)	0.307 (0.264)	0.232 (0.298)	0.252 (0.150)
Import-export trading	0.021 (0.306)	0.312 (0.367)	0.371 (0.353)	0.291 (0.184)
Other	0.087 (0.242)	0.242 (0.228)	0.250 (0.210)	0.155 (0.114)
Registered capital, all	0.111 (0.242)	0.365 (0.283)	0.424 (0.324)	0.253 (0.164)

Notes: Each row is a separate estimate: the difference in birth intensity (births per above-scale incumbent firm in the year before opening) between treated cells within 20 km of an arriving EPZ and odds-weighted comparison cells of the same sub-experiments, averaged over the window and divided by the comparison mean. Panel A: births in the industrial-firm census, 1999–2009, GB/T 4754-2002 codes (electronic components 405–406; final electronics 401–404, 407, 409; general supporting industries are molds, plastic parts, wire and cable, batteries, metal structures and hardware, and packaging). Panel B: first-time exporters, 2001–2011, whose first-year exports are mostly in the HS4 headings that make up 95 percent of fenced firms’ imports. Panel C: business registrations, 2000–2016, surviving to 2019, classified by business scope; the last row is registered capital. Column (4) is column (2) minus column (1). Standard errors from a bootstrap over prefectures (199 draws, shared across rows) in parentheses.

Table 47: New business registrations near EPZs: relative effects and magnitude

Category	Pre-trend p	Pre-period (rel.)	Post-opening $e=0-3$ (rel.)	Post-opening $e=4-6$ (rel.)
<i>Panel A. Event-study relative effects (new registrations per baseline incumbent firm)</i>				
Non-tradable services	0.081	0.219 (0.326)	0.436 (0.316)	0.326 (0.269)
Producer services	0.308	-0.063 (0.423)	0.260 (0.443)	0.316 (0.369)
Manufacturing	0.823	0.129 (0.280)	0.141 (0.213)	0.118 (0.221)
<i>Panel B. Implied local-multiplier magnitude (illustrative order of magnitude, not a causal estimate)</i>				
Excess non-tradable registrations per treated grid cell-year			6.530	
Implied fenced jobs per treated grid cell-year (wage-bill-based)			37.909	
Implied ratio: non-tradable registrations per fenced job			0.172	

Notes: Panel A: pooled differences between treated and reweighted comparison cells in new registrations per base-year above-scale incumbent, divided by the comparison mean, for the four pre-opening years ($e = -4, \dots, -1$), years 0–3, and years 4–6; the first column is the p -value of the joint test that the four pre-opening differences are zero. Bootstrap standard errors over prefectures (199 draws) in parentheses. Panel B: the conversion described in the text; an order of magnitude, not a causal estimate.

post-opening difference in non-tradable registrations, converted from a rate to a level with the average base-year incumbent count, is 6.53 registrations per treated cell-year. The ratio is 0.172 registrations per fenced job, one for every 5.8 jobs. It counts registrations, not employment.

H.5 The 2005 carry-over reform

Were the zones enclaves because of the rules, or because of what local firms could do with the fence’s products? The rules changed once inside our window. Deep-processing carry-over—the transfer of fenced output to processing firms outside the fence for further processing and re-export, not general domestic-market access for the fence’s products—was permitted from 2002 in two pilot zones, Kunshan and Songjiang, and extended to all EPZs in May 2005. The reform opened the forward linkage from the fence to downstream users. It did not change the terms on which local suppliers sold into the fence: domestic inputs bought by zone firms regained export rebates only under a 2008 rule, and only for listed products. Downstream user industries—general and special equipment, transport equipment, electrical machinery, and instruments, which each use at least 0.02 units of the fence’s products per unit of output—should have gained relative to other industries near zones already in operation if local processors could use the fence’s components; upstream industries, on which the reform did not act, serve as a within-location comparison (Appendix O.11).

Table 48 shows that relaxing the carry-over restriction generated no detectable short-run expansion of eligible downstream industries over the 2005–2007 post-reform window available in the firm survey: for employment the estimates rule out increases above 10 percent (output: upper bound 23 percent). Relative to other industries, output of downstream users near

Table 48: The 2005 extension of carry-over transfers

	(1)	(2)	(3)	(4)
	Extension, 2005	Extension, excl. 2005	Placebo: bonded zones	Pilot zones, 2003
<i>Panel A. Downstream user industries</i>				
Output	−0.067 (0.140)	−0.083 (0.150)	0.095 (0.138)	0.029 (0.155)
Employment	−0.026 (0.061)	−0.048 (0.076)	0.127 (0.106)	0.227 (0.172)
Export delivery value	−0.220 (0.116)	−0.257 (0.122)	−0.220 (0.082)	−0.245 (0.195)
<i>Panel B. Upstream supplier industries</i>				
Output	−0.073 (0.108)	−0.096 (0.120)	−0.007 (0.145)	−0.046 (0.106)
Employment	−0.076 (0.085)	−0.065 (0.093)	0.056 (0.086)	−0.018 (0.135)
Export delivery value	0.088 (0.130)	0.053 (0.147)	−0.103 (0.136)	0.133 (0.157)
<i>Panel C. Electronics</i>				
Output	−0.105 (0.161)	−0.138 (0.172)	0.098 (0.136)	1.123 (0.155)
Employment	0.171 (0.151)	0.155 (0.166)	0.283 (0.148)	0.527 (0.250)
Export delivery value	−0.076 (0.147)	−0.133 (0.154)	0.062 (0.111)	1.048 (0.159)
Treated grid cells	267	267	21	24

Notes: Coefficients on $T_c \text{Post}_t \mathbb{I}[k]$ for each industry group k in a Poisson triple difference with cell $\times$ group, cell $\times$ year, and group $\times$ year fixed effects. Treated cells lie within 20 km of an EPZ operating by 2004, excluding the two pilot zones; controls are cells with no special customs zone within 40 km before 2008 but one by 2013. Column (2) drops the transition year 2005. Column (3) treats cells near 1990s bonded zones, to which the 2005 rule did not apply. Column (4) treats cells near the two pilot zones with 2003 as the first post year. Downstream users: industries with direct requirements of the fence's products of at least 0.02 per unit of output. Export delivery value is reported in the firm survey. Standard errors clustered by prefecture in parentheses.

operating zones changes by -0.067 (s.e. 0.140) and employment by -0.026 (s.e. 0.061). Their export delivery value falls (-0.220 , s.e. 0.116), but by the same amount near bonded zones, where the reform did not apply (-0.220 , s.e. 0.082); the decline reflects a common trend in how processing exporters report, not the reform. The event study in Figure 24 shows no break in 2005. Nearby firms' export capability did not change either: the stock of portable export experience (Section 8.3) in cells near operating zones moves by 0.004 log points after the rule relative to before (s.e. 0.085), and destinations served under processing with imported materials by 0.022 (s.e. 0.113).

The pilot zones show larger point estimates for downstream employment (0.227, s.e. 0.172)

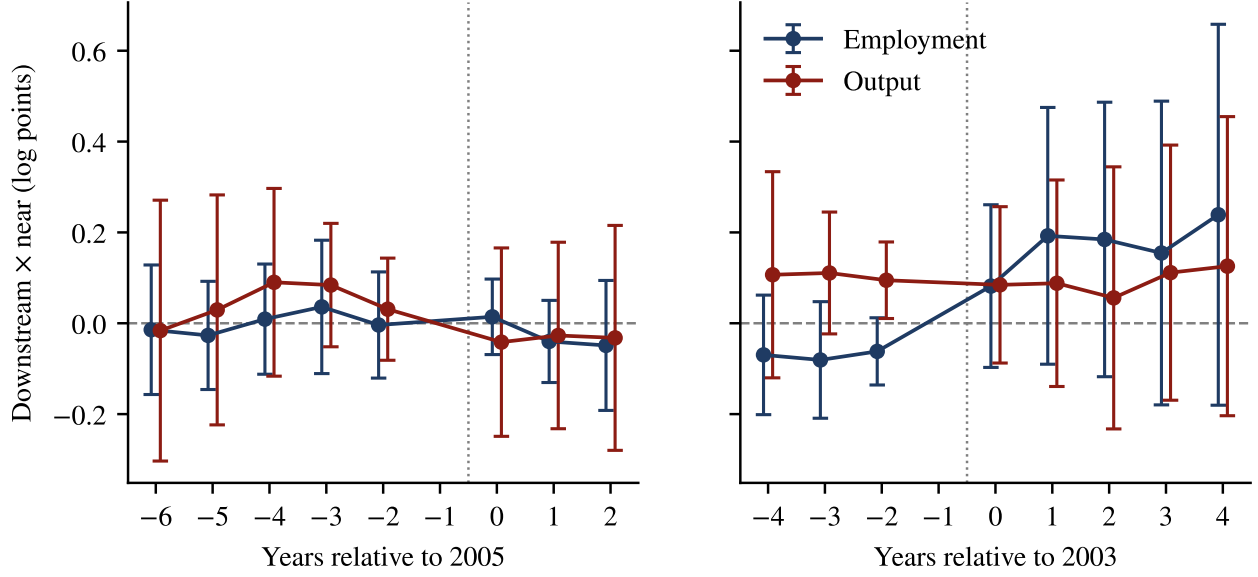


Figure 24: Downstream industries around the carry-over reform

Notes: Coefficients on treated cell $\times$ year $\times$ downstream industry from the triple difference in Table 48. Left: all operating zones except the pilots, reference year 2004. Right: Kunshan and Songjiang, reference year 2002. Bars are 95 percent confidence intervals clustered by prefecture.

and a large relative expansion of electronics after 2003. Two zones in one of the densest electronics clusters in China cannot separate the pilot rule from the scale of foreign investment that arrived there at the same time, and we do not read these estimates as the effect of carry-over.

The reform result sharpens the interpretation of Section 7.3. The forward-exposure estimate there is as large at a placebo date three years before opening as after it, and before 2005 fenced firms outside the pilot zones could not sell components to local processors at all; here, once they could, the users did not expand. Regulatory permission was therefore not sufficient to generate forward linkages in 2005–2007: local processing firms that could use the fence’s components did not expand once they were allowed to buy them. This does not mean that regulation never matters for these linkages—it shows that removing this particular restriction was not, on its own, enough. Later reforms that equalized tax treatment inside and outside the fence, and the conversion of EPZs into comprehensive bonded zones after 2015, fall outside the data (Appendix L.9).

I The Ledger: Estimates, Bounds, and Checks

The ledger of Section 10 values the projects the fence kept out with the estimated admission model. This appendix reports the reduced-form ledger, which values them with the entry

effect in each size quintile directly, and the bounds and checks built on it; the sensitivity of the model-based ledger; the national accounting; and the fiscal items that the ledger leaves out.

I.1 The reduced-form ledger

Inputs and units. Table 51 lists the inputs of the reduced-form ledger. Both sides are slot-year values. The entry estimate gives the proportional loss of domestic entry near the fence, 0.197 in the main specification. The entrants kept out in each birth-output quintile follow from the quintile-specific entry effects; each is valued with the survival and value-added profile, at ages zero to three, of domestic entrants of the same national quintile born in the treated locations in the first four operating years. The cost in the fourth year is therefore the value added of four lost cohorts, zero to three years old. Valuing the lost entrants as average entrants instead raises it from RMB 2,038 million to 7,675 million a year across the 24 zones. On the fence’s side, exports are averaged over the operating years observed, up to four. The local wage share is measured for the 17 zones whose fenced firms can be matched to the firm survey (interquartile range 3.5–9.2 percent); applying a zone-specific share would record zero wherever matching fails, so the ledger applies the median to every zone.

How it differs from the model-based ledger. The two ledgers share the fence’s side and differ only in how they value the projects kept out. The reduced form multiplies the entry effect in each birth-size quintile by the value of entrants of that size. Its point estimate for the largest quintile is slightly positive, so taken literally it says that the fence let in more of the largest projects, and because those projects are worth far more than the rest, this imprecise cell lowers the valuation. The model does not allow it: a higher admission threshold can only lower entry in every size class. The reduced-form median break-even scale is therefore lower, RMB 1,209 million of fenced exports a year (US\$147 million; 797–2,069 across the interquartile range of wage shares), against RMB 1,796 million in the model, and the median non-hub zone kept out RMB 59 million of value added a year, against 88 million. The shares of zones that fall short coincide at the median wage share, 84 percent of non-hub zones and 71 percent of all zones, although the two valuations are not the same calculation; if all of the fence’s domestic value added stayed local, the reduced form leaves 58 percent of non-hub zones short, against 84 percent in the model, because its valuation of the blocked projects is lower. Summed over zones, the reduced form’s value of blocked projects is RMB 2,038 million a year and its net change outside the hubs RMB –444 million (last row of Table 52). The hubs’ share of fenced exports is 90 percent when each zone’s exports are summed over its first four operating years, and 84 percent when they are averaged per year observed, the

convention of the main text.

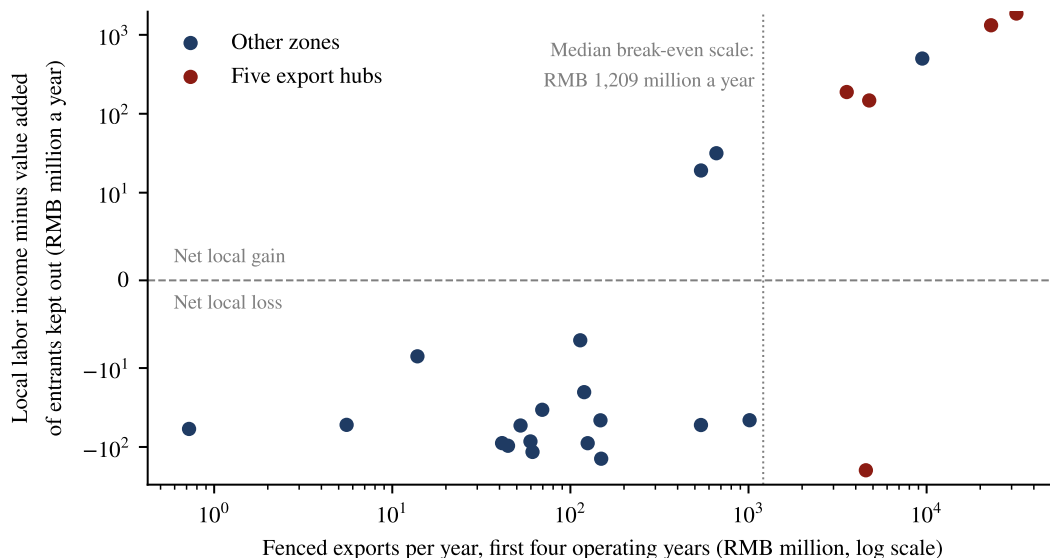


Figure 25: The local ledger of each export processing zone: reduced-form valuation

Notes: Each point is an EPZ with both entry and customs data (24 zones). Horizontal axis: mean annual fenced exports over the zone’s first operating years (up to four), log scale. Vertical axis, slot-year values in RMB million: local labor income of the fence (6 percent of fenced exports, the median wage share of fenced firms matched to the firm survey) minus the value added of the domestic entrants kept out, valued with the reduced-form entry effect in each birth-size quintile and the survival and value-added profile of entrants of that size. Linear between -10 and 10 and logarithmic beyond. Dashed line: zero net effect. Dotted line: the median of the zones’ break-even scales in this valuation, RMB 1,209 million a year; it is not a zone-by-zone boundary. Hubs as in Figure 13. The main-text ledger (Figure 13) uses the estimated admission model instead.

Bounds on what the fence leaves. The customs domestic value-added ratio contains local inputs, local wages, and the profits of the fence’s owners (equation (2)), so it bounds local retention from above but does not split it. Panel B of Table 51 reports the pooled net change at three points of that budget, each including the value added of local suppliers on the change in local purchases. If all domestic value added were paid to local factors, the net change would be RMB 19,600 million a year ($-3,994$ to $51,410$); if each fence paid its own wage bill locally and spent the rest of its domestic value added on local inputs, RMB 2,789 million; if nothing stayed local, RMB $-3,406$ million ($-14,394$ to $9,969$). The main text counts the wage bill alone and sets local purchases to zero, as the absence of any supplier response suggests (Section 7.3). Even if a fence spent all of its domestic value added on local inputs, its local purchases would fall short of those the lost entrants would have made in a share 0.67 of the zones (0.00 to 0.88). We do not use the difference between upstream industries in thin and thick pre-opening supply bases to discipline this term: the same difference appears at a placebo opening date, and it reflects mean reversion from sorting

locations on a pre-opening share (Section 7.3).

Host-city output. Adding the fence’s output, the output the lost entrants would have produced, and the upstream response predicts a change in host-city output of -0.20 percent of pre-opening output (-2.03 to 1.83) for the host cities of the zones with complete data. Only the part of fenced output that enters the firm survey (7.6 percent) is added, since city totals are built from the survey. The prediction has the same sign as the city estimates for industrial output (-0.020 , s.e. 0.032) and manufacturing output (-0.036 , s.e. 0.026) and lies inside both intervals, which are too wide to discriminate sharply between the ledger and a zero effect.

I.2 Sensitivity of the model-based ledger

Table 52 changes one element of the main specification at a time (Table 15).

- *Admission rate before opening.* The model does not identify it (Section 10.1). Over its range from 0.05 to 0.95 the value of blocked projects lies between RMB 2,734 and 3,350 million a year, the efficiency loss between 1,749 and 2,240 million, the median break-even scale between 1,623 and 1,989 million, and the share of zones that fall short stays at 71 percent. Rule (b), which takes the fence’s slots from the smallest admitted projects, gains between 783 and 1,281 million. A rule that also ranks every admission before opening by size is not identified: its gain ranges from RMB 32 to 3,241 million along the same range, so the main text does not report it.
- *Capacity.* The main specification measures a zone’s capacity in the years the fence occupies it, from post-opening entry and the entry estimate. Measuring it by mean annual entry before opening lowers the totals most of any assumption here but leaves most non-hub zones short (74 percent); Section 10.4 reports the numbers. Entry into the treated locations grew over the period, so pre-opening entry understates the capacity the fence took.
- *Local income.* At the quartiles of fenced firms’ wage shares the efficiency loss is RMB 2,203 and 1,757 million a year. If all of the fence’s domestic value added were local income, it would be 1,220 million, and 67 percent of zones (84 percent outside the hubs) would still fall short.
- *Timing convention.* Summing both sides over each zone’s operating years observed, instead of comparing slot-year values in the fourth year, gives an efficiency loss of RMB 1,648 million, with 67 percent of zones short.
- *Queue.* The specification in which admission follows birth fixed assets, rejected by the

out-of-sample test of Section 10.1, places more of the blocked projects in the largest quintile and values them at RMB 5,836 million a year, with an efficiency loss of 4,346 million; we do not use it.

- *Hubs.* Treating Nanjing, which is nearly tied with Wuxi for fifth place in cumulative fenced exports, as a hub in place of Wuxi leaves the share of non-hub zones that fall short at 84 percent, because both zones clear their break-even scale.

I.3 National accounting

Table 49 scales the value of blocked projects by $1 - \phi$, where ϕ is the share of them founded elsewhere in China (Section 10.7); the fence's side is unchanged. The program's net change rises from RMB 1,891 million a year at $\phi = 0$ to 2,647 million at $\phi = 0.25$ and 3,404 million at $\phi = 0.5$, and the efficiency loss falls from 1,981 to 1,378 and 775 million. It stays positive at every ϕ considered, because in the zones furthest below break-even the fence displaced more than it left even at half the cost. The data bound only the within-prefecture relocation share λ , not ϕ , so the table is an accounting exercise, not an estimate of the national effect.

Table 49: National accounting: blocked projects founded elsewhere in China

	Estimate	95 percent interval
<i>Share founded elsewhere in China $\phi = 0$ (the local ledger)</i>		
Value added of blocked projects not founded elsewhere	3,025	[820, 8,657]
Net change in value added	1,891	[−4,798, 6,745]
Efficiency loss	1,981	[350, 6,996]
<i>Share founded elsewhere in China $\phi = 0.25$</i>		
Value added of blocked projects not founded elsewhere	2,269	[615, 6,492]
Net change in value added	2,647	[−2,885, 7,139]
Efficiency loss	1,378	[246, 5,085]
<i>Share founded elsewhere in China $\phi = 0.5$</i>		
Value added of blocked projects not founded elsewhere	1,513	[410, 4,328]
Net change in value added	3,404	[−1,283, 7,650]
Efficiency loss	775	[147, 3,196]

Notes: Estimated admission model, main specification, 24 zones; slot-year values in RMB million. Each panel scales the value of blocked projects by $1 - \phi$; the fence's local labor income is unchanged. The efficiency loss sums, over zones, the scaled value of blocked projects in excess of local income. 95 percent intervals from 499 bootstrap replications over zones.

I.4 Fiscal items outside the ledger

Two fiscal flows are often part of the case for or against the zones: the tariffs and import-stage value-added tax the central government does not collect on bonded imports, and the land-lease revenue of the host government. We do not estimate either in this paper’s designs, and neither enters the ledger. Table 50 reports the values from a calibrated model of the zones’ fiscal flows, together with the reasons.

The central government’s forgone revenue is an accounting calculation: the statutory tariff and import value-added tax applied to each EPZ’s bonded imports, averaged over zone-years after opening, net of the exemption that processing trade outside the zones also receives. It is large relative to every item in the ledger, but it is a transfer from the central government to the fenced firms, mostly foreign-invested; its cost to the country depends on the cost of public funds, which place comparisons do not identify, and it is not a flow of local value added. The land-revenue figure comes from a ring design on land-transaction records, which are reliable only from 2007, so it covers special customs zones opened in 2009–2017, none of the 24 ledger zones. Restricted to the locations where land, firm, and customs data are all available, the same design gives a much smaller increase; the two estimates rest on different sets of zones and have not been reconciled. We report both items for completeness and draw no conclusion from them.

Table 50: Fiscal items outside the ledger

Party	Item (RMB million per zone-year)	Source, and why it is not in the ledger
Central government	Tariffs and import-stage value-added tax not collected on bonded imports, net of the exemption processing trade outside the zones also receives: –2,459 (gross of that exemption: –3,842)	Statutory rates applied to each EPZ’s bonded imports, mean over zone-years after opening. A transfer to the fenced firms, not local value added
Host government (land)	Increase in industrial land-lease revenue: 402 (88 to 859)	Ring design on land-transaction records for special customs zones opened 2009–2017, not the ledger zones; a much smaller increase on the locations with land, firm, and customs data together; not reconciled

Notes: Signs from the point of view of each party. Values from a calibrated model of the zones’ fiscal flows, in 2010 prices; they are not estimated with the designs of this paper and are not comparable in sample or unit with the rows of Table 16. The interval for the land item is the 95 percent interval of the ring design.

Table 51: Inputs and bounds of the reduced-form ledger

	Value
<i>Panel A. Inputs</i>	
Entry loss near the fence, ρ (main entry estimate)	0.197
Domestic entrants kept out per zone-year	5.6
Domestic entrants per zone-year before opening	18.4
Output of an entrant, mean over ages 0–3 (RMB million)	85.70
Value added of an entrant, mean over ages 0–3 (RMB million)	18.94
Domestic share of entrants' input purchases (survival-weighted)	0.794
Domestic value-added ratio of fenced exports (export-weighted)	0.291
Value added per unit of output of local suppliers, μ	0.214
<i>Panel B. Ledger of the zones, pooled (RMB million a year)</i>	
Value added of entrants kept out, size-specific profile	2,038
valued as average entrants	7,675
Net change, all domestic value added paid locally ($\sigma_z = \text{DVAR}$, $s_z = 0$)	19,600 [−3,994, 51,410]
Net change, own wage bill paid and rest of DVAR bought locally (21 zones)	2,789
Net change, nothing retained locally ($\sigma_z = s_z = 0$)	−3,406 [−14,394, 9,969]
Zones whose local purchases at $s_z = \text{DVAR}$ fall short of the entrants'	0.67 [0.00, 0.88]
<i>Panel C. Host-city output, not targeted</i>	
Predicted change, percent of pre-opening output	−0.20 [−2.03, 1.83]
Share of fenced output that enters the firm survey (percent)	7.6
<i>Panel D. Rules (RMB million a year)</i>	
Fence on land outside the admission capacity	2,038 [−5,951, 10,245]
Fence sources locally like processing firms outside it	1,719 to 7,500

Notes: 24 EPZs with both entry and customs data. Panel A: shares, counts per zone-year, and RMB million per entrant. Panels B and D: slot-year values in RMB million, pooled over zones. Each net change in Panel B includes the value added of local suppliers, at the ratio μ , on the change in local purchases (the fence's purchases minus those the lost entrants would have made); the middle case is available for the zones with a matched wage bill. Panel C: host cities of the zones with complete data; only fenced output matched to the firm survey is added. Intervals in brackets: zone bootstrap with a draw of the entry coefficient, 499 replications. The last row of Panel D is a range because the fence's local purchases are bounded only from above.

J The Cross-Border Design: Additional Evidence

This appendix supports Section 9.1: which zones the incumbent design covers, how the result changes with the control pool, pre-trend-robust inference, the effects in money, heterogeneity and the multiple comparisons it involves, relocation, and why the ratio of cross-border to host-side effects is not reported as the headline.

Table 52: Sensitivity of the model-based ledger

	Value of blocked projects	Efficiency loss	Zones short (percent) All	Non-hub	Break-even exports	Net change, non-hub zones
Main specification	3,025	1,981	71	84	1,796	−1,048
Admission rate before opening 0.05	3,350	2,240	71	84	1,989	−1,247
Admission rate before opening 0.95	2,754	1,764	71	84	1,634	−882
Capacity from pre-opening entry	1,208	545	58	74	553	−44
Local wage share 3.5 percent (25th percentile)	3,025	2,203	79	89	3,072	−1,382
Local wage share 9.2 percent (75th percentile)	3,025	1,757	71	84	1,183	−634
All domestic value added of the fence local	3,025	1,220	67	84	—	724
Both sides summed over the operating years observed	4,119	1,648	67	79	—	−308
Admission follows birth fixed assets	5,836	4,346	79	89	3,464	−2,768
Reduced-form valuation (quintile point estimates)	2,038	1,193	71	84	1,209	−444

Notes: Point estimates, 24 zones. Value of blocked projects, efficiency loss, break-even exports (median zone), and net change outside the five hubs in RMB million (slot-year values). Each row changes one element of the main specification (Table 15): the admission rate before opening, fixed at the value shown; capacity measured by mean annual entry before opening instead of post-opening entry scaled by the entry estimate; the local wage share at the quartiles across matched zones; local income equal to all of the fence’s domestic value added (break-even not defined by a single share); both sides summed over the operating years observed; admission following birth fixed assets; and the reduced-form valuation of Figure 25.

J.1 Which zones the design covers

The design uses EPZs that opened between 2001 and 2007 in prefectures that are not centrally administered municipalities, whose districts do not hold the powers of an ordinary prefecture (Appendix L.10), and that have enough incumbents across the border to compare. Table 53 shows that 43 zones qualify; 81 percent of them are in coastal provinces, and the median zone lies 14 km from its host prefecture’s border. The results speak to zones in dense, integrated regions where a prefecture border is close to the places from which the zone draws its resources, not to isolated inland zones.

Table 53: Which EPZs enter the incumbent design

Group	EPZs	Median opening year	Coastal share	Median firms, host	Median firms, other prefectures	Median km to border
Opened outside 2001–2007	7	2008	—	—	—	—
Host is a centrally administered municipality	8	2003	—	—	—	—
Too few same-province firms in other prefectures within 150 km	1	2002	0.00	101	0	77.1
In the main design	43	2005	0.81	1,515	3,938	14.3

Notes: Zones grouped by the reason they do or do not enter the stacked incumbent design. Firm counts are above-scale manufacturers in the year before opening: in the host prefecture, and in other prefectures of the same province within 150 km. Distance to border: from the zone to the nearest land border of its host prefecture with an adjacent prefecture.

J.2 Control pools

The main specification uses incumbents around sites of special zones that had not yet been *approved* three years after the treated zone opened. Defining the not-yet-treated group by operation instead would admit sites that were approved and under construction during the window and may already have affected their surroundings. Table 54 compares six pools. The two broad pools—not yet approved, and all zones opening after the window—give the same answers with and without reweighting: host incumbents near the zone are unaffected once controls are reweighted, and neighboring incumbents at 40–80 km lose survival in all four columns. The unweighted declines near the zone in columns (1) and (3) reflect differences in local firm density, coastal location, and distance to national development zones between treated and control sites, and disappear with reweighting. The broad pools share most of their control firms, so their agreement is not independent confirmation.

The two narrow pools differ. Using only EPZs that open after the window, host output near the zone rises (0.116, against -0.012 in the main column) and the neighbor’s survival loss disappears (0.004, s.e. 0.005). This pool is not comparable to the treated sites. Even after reweighting, treated incumbents had grown faster before opening than this pool’s controls (standardized difference in log output growth over the two years before opening of 0.27, against 0.02 for the main pool), its sites lie farther from national development zones (standardized difference 0.60, against -0.08), and its effective control sample is 4,785 firms against 33,149. A pre-opening growth difference of that sign, if it continued, would by itself push post-opening estimates up, the direction in which they move. The pool of zones approved but not yet operating is an anticipation check: its sites were being built during the window, and any response around them biases the comparison toward zero, which is where the neighbor’s survival coefficient lands (0.009, s.e. 0.008).

J.3 Pre-trend-robust inference

Table 55 reports relative-magnitude robust sets (Rambachan and Roth 2023) for the output and survival contrasts. Panel A shows that the output contrasts are not significant even under exact parallel trends, so no breakdown value is defined for them; host incumbents’ pre-opening decline is the reason not to read the host’s output coefficients as precise zeros. Panel B shows that the host’s retention of incumbents at 40–80 km survives allowances up to $\bar{M} = 1.98$, the host’s retention within the border band up to $\bar{M} = 0.59$, and the neighbor’s survival loss at 40–80 km, whose confidence interval barely excludes zero, only up to $\bar{M} = 0.42$. The robust sets are computed for each side’s coefficient, not for the neighbor-minus-host difference, and with a plug-in implementation that fixes the largest pre-opening change at

Table 54: Alternative control pools

Control pool	(1) Not yet approved	(2) Not yet approved, reweighted	(3) All later zones	(4) All later zones, reweighted	(5) Later EPZs only, reweighted	(6) Approved, not operating, reweighted
Aggregate output, host, 0–20 km	−0.092 (0.046)	−0.012 (0.062)	−0.081 (0.043)	−0.002 (0.057)	0.116 (0.067)	0.183 (0.085)
Aggregate output, host, 20–40 km	−0.165 (0.056)	−0.091 (0.079)	−0.156 (0.052)	−0.113 (0.069)	0.035 (0.107)	−0.221 (0.151)
Aggregate output, neighbor, 40–80 km	−0.096 (0.042)	−0.081 (0.051)	−0.099 (0.040)	−0.122 (0.050)	0.123 (0.053)	0.064 (0.085)
Survival, neighbor, 40–80 km	−0.036 (0.016)	−0.035 (0.017)	−0.037 (0.015)	−0.039 (0.016)	0.004 (0.005)	0.009 (0.008)
Control sites	56	56	67	67	24	32

Notes: Post-period averages on the common support of the pools. Pools: sites of special zones not yet approved three years after treatment (main); all special zones opening after the window; EPZs opening after the window; zones approved but not yet operating (anticipation check). Reweighted columns use odds weights estimated within stack on firm and site characteristics. Column (2) is the specification of Table 13. Standard errors clustered by prefecture in parentheses.

its estimate, which overstates robustness. Host survivors’ log output at 40–80 km, which Section 9.1 uses only as a consistency check, has a pre-period average of 0.070 (s.e. 0.032) and a breakdown value of 0.05.

Table 55: Pre-trend-robust inference for the incumbent contrasts

Contrast	Post-period average	(s.e.)	Pre-period average	Largest pre change	Robust 95% set ($\bar{M} = 1$)	Breakdown $\bar{M}^*$
<i>Panel A. Aggregate incumbent output</i>						
Host, 0–20 km	−0.012	(0.062)	0.043	0.060	[−0.284, 0.261]	0.00
Host, 20–40 km	−0.091	(0.079)	0.063	0.046	[−0.361, 0.180]	0.00
Host, 40–80 km	−0.022	(0.063)	0.091	0.072	[−0.325, 0.280]	0.00
Neighbor, 40–80 km	−0.081	(0.051)	−0.002	0.022	[−0.234, 0.073]	0.00
<i>Panel B. Survival</i>						
Host, 40–80 km	0.054	(0.014)	−0.006	0.005	[0.013, 0.094]	1.98
Neighbor, 40–80 km	−0.035	(0.017)	−0.001	0.002	[−0.074, 0.003]	0.42
Host, 25 km border band	0.051	(0.021)	−0.009	0.007	[−0.007, 0.109]	0.59
Neighbor, 25 km border band	0.005	(0.023)	−0.009	0.010	[−0.066, 0.075]	0.00

Notes: Post-period average ($e = 0, \dots, 3$) and standard error from Table 13 and the border-band sample; pre-period average of $e = -3, -2$; largest consecutive pre-period change m , including the normalization $\beta_{-1} = 0$. Robust sets allow the post-period violation of parallel trends to change by at most $\bar{M}m$ per period; $\bar{M}^*$ is the smallest value at which the set includes zero; 0.00 marks contrasts whose 95% interval already includes zero under exact parallel trends, for which no breakdown value is defined. For survival, pre-period coefficients compare presence in the above-scale universe before the base year, which is small by construction. Plug-in implementation of Rambachan and Roth (2023), coefficient by coefficient.

J.4 The border effects in money

Table 56 converts the proportional effects of Table 13 into money: each treated incumbent’s base-year output times the post-opening effect in its side $\times$ distance cell, summed over incumbents and stacks. Measured by the base-year output of incumbents that remain above scale (Panel B), the host side’s retained output changes by RMB +36 billion (s.e. 27) over 0–80 km and the neighboring side’s by –55 billion (s.e. 29) over 20–80 km, a difference of –91 billion (s.e. 30). Because the neighboring side has too few incumbents within 20 km to estimate, these two sums cover different distance ranges. Restricting the host side to 20–80 km as well, its retained output changes by 42 billion (s.e. 13) and the difference is –97 billion (s.e. 27), the number used in the main text. Measured by aggregate output (Panel A), the neighboring side changes by –142 billion (s.e. 92), about 7.6 percent of its base, and the host side by –83 billion (s.e. 99); neither these totals nor their difference (–59, s.e. 119; –74, s.e. 122, on the common 20–80 km range) is precisely estimated.

Table 56: Jurisdictional incidence within 80 km of the zone, in RMB

Side of the border	Base-year output (RMB billion)	Change (RMB billion) Estimate	(s.e.)
<i>Panel A. Aggregate incumbent output</i>			
Host prefecture	2,433	–82.9	(98.9)
Neighboring prefectures	1,875	–141.9	(92.1)
Neighbor minus host		–59.0	(119.1)
<i>Panel B. Base-year output of incumbents still above scale</i>			
Host prefecture	2,433	36.0	(27.3)
Neighboring prefectures	1,875	–54.6	(29.4)
Neighbor minus host		–90.6	(29.9)

Notes: Sums over treated incumbents (firm $\times$ stack) of base-year output times the post-period effect in their side $\times$ distance cell, in RMB billion at current prices. Panel A uses the effects on aggregate output; Panel B uses the effects on survival, so it measures the base-year output of incumbents retained in, or lost from, the above-scale universe. Host side: 0–80 km; neighboring side: 20–80 km. With the host side also restricted to 20–80 km, the neighbor-minus-host difference is –97 (27) in Panel B and –74 (122) in Panel A. Host and neighbor totals come from one regression; standard errors from prefecture-level influence functions, so each difference has its own standard error.

J.5 Heterogeneity and multiple comparisons

Table 57 splits the neighbor-minus-host difference by characteristics of the border pair. In the 20–80 km sample the survival gap is concentrated where the neighbor pays lower wages than the host, the direction in which workers move; the difference between poorer and richer neighbors is –0.137 (s.e. 0.042), but within the 25 km border band it is –0.005 (s.e. 0.046).

In the zones' own industries, host incumbents grow while their counterparts across the border shrink (-0.392 , s.e. 0.209), in samples of a few hundred firms per side.

We also asked whether jurisdictional competition shapes the gap, with the tests and the rule for reading them set down before estimation. If the host's shielding were strategic, neighbors with weaker fiscal capacity should lose more, and neighbors that are the host's rivals for promotion should defend their incumbents and lose less. Table 58 reports each difference in both samples, from one regression with side $\times$ group $\times$ period indicators, so that each difference has a joint standard error from prefecture-level influence functions. Fiscal capacity does not predict the gap in either sample. In the 20–80 km sample the gap is smaller between rivals under rank adjacency and promotion overlap and larger under the appointment cohort, and none of these differences is significant. In the band the ordering reverses for rank adjacency, and the only significant difference (same appointment cohort) has the opposite sign from its 20–80 km counterpart. Two further implications fail in the wrong direction: the host's own incumbents are retained less, not more, next to high-overlap rivals (-0.067 , s.e. 0.025); and rank-adjacent rivals expand their industrial land more slowly after the host's zone opens, not faster (-0.792 , s.e. 0.333 , in the change of log industrial land supplied from three years before to three years after). Whether rivals obtain their own special zones faster cannot be tested with any power, because such zones are rarely approved in any window. On the aggregate-output scale some rivalry splits are large—the gap for high-overlap pairs is -0.405 (s.e. 0.089)—but output weighted by base-year size is sensitive to a few large firms and its ordering differs from the survival scale, so we base judgments on survival.

Counting the 12 survival-scale group differences—the ten in Table 58 and a looser version of rank adjacency (at most two places apart) in both samples—a Bonferroni threshold at the 5 percent level requires $|z| > 2.87$. Only the wage split in the 20–80 km sample clears it ($z = -3.27$), and it does not replicate in the band. We report the remaining comparisons as evidence of the absence of systematic patterns, not as findings, and do not interpret the incidence as a strategic outcome. Two tests that would bear on the shielding more directly are not done (Section 9.2): an indirect test in the spirit of Bai et al. (2020), asking whether shielded incumbents sell more at home and exposed ones turn to exports, and a test of whether zone size predicts the promotion of the host's officials.

J.6 Relocation and the meaning of exit

Table 59 estimates where relocating firms go (the relocation records are described in Appendix M.16). A destination in another prefecture of the same province is chosen with a relative probability of about 0.024 compared with a destination at the same distance and of

Table 57: The cross-border gap by characteristics of the border pair

	Aggregate output, 20–80 km			Survival	Aggregate output	Treated firms
	Host	Neighbor	Neighbor – host	Neighbor – host 20–80 km	Neighbor – host, 25 km band	neighbor / host
<i>Panel A. Wage gap between the two sides</i>						
Neighbor poorer than host	0.072 (0.085)	−0.015 (0.071)	−0.086 (0.094)	−0.075 (0.021)	0.009 (0.097)	18,804 / 6,693
Similar wages	−0.115 (0.077)	0.014 (0.089)	0.129 (0.067)	−0.028 (0.027)	−0.151 (0.107)	3,788 / 5,940
Neighbor richer than host	0.082 (0.064)	0.279 (0.123)	0.197 (0.119)	0.062 (0.033)	0.046 (0.150)	1,469 / 3,619
<i>Panel B. Industry overlap with the zone</i>						
Zone industries (electronics, instruments)	0.102 (0.123)	−0.290 (0.180)	−0.392 (0.209)	0.004 (0.036)	−0.107 (0.203)	387 / 529
Other industries	−0.017 (0.053)	−0.057 (0.065)	−0.040 (0.071)	−0.072 (0.015)	0.022 (0.070)	24,087 / 16,682
<i>Panel C. Economic rivalry (same province)</i>						
Adjacent in provincial output rank	−0.091 (0.092)	−0.020 (0.101)	0.071 (0.123)	−0.043 (0.019)	−0.189 (0.129)	11,944 / 4,288
Not adjacent	−0.002 (0.058)	−0.030 (0.061)	−0.027 (0.074)	−0.074 (0.018)	0.058 (0.092)	11,819 / 10,845
<i>Panel D. Political rivalry (same province)</i>						
High promotion-opportunity overlap	0.101 (0.087)	−0.304 (0.104)	−0.405 (0.089)	−0.085 (0.021)	−0.087 (0.106)	9,381 / 3,773
Low overlap	0.023 (0.131)	0.097 (0.056)	0.074 (0.141)	−0.120 (0.030)	0.232 (0.100)	8,821 / 4,685

Notes: Post-period averages for treated incumbents by side of the border and group, from regressions of the form of Table 13 with side $\times$ group $\times$ period indicators; comparison cells include the group. Groups are defined on the border pair formed by the host prefecture and the incumbent’s neighboring prefecture (for host incumbents, the nearest adjacent prefecture), measured in the year before opening. Wage gap: log average wage of the neighbor minus the host, below -0.1 , within ± 0.1 , above 0.1 . Zone industries: harmonized two-digit codes 40 and 41. Rank adjacency and promotion-opportunity overlap are defined for pairs in the same province. Odds-reweighted not-yet-approved controls; standard errors clustered by prefecture in parentheses.

Table 58: Does jurisdictional competition predict the cross-border survival gap?

Split of border pairs (A versus B)	20–80 km from the zone			25 km border band		
	ΔG	(s.e.)	z	ΔG	(s.e.)	z
Neighbor poorer versus richer than host	−0.137	(0.042)	−3.27	−0.005	(0.046)	−0.11
Neighbor fiscally weaker versus stronger	−0.028	(0.036)	−0.76	0.006	(0.038)	0.16
Adjacent versus not in provincial GDP rank	0.075	(0.041)	1.85	−0.026	(0.039)	−0.67
High versus low promotion-opportunity overlap	0.034	(0.039)	0.88	0.015	(0.048)	0.30
Leaders appointed within one year versus not	−0.024	(0.040)	−0.60	0.102	(0.043)	2.37

Notes: G is the neighbor-minus-host difference in the post-period survival effect within a group of border pairs; $\Delta G = G_A - G_B$. Both come from one regression with side $\times$ group $\times$ period indicators, and the standard error of ΔG is computed from prefecture-level influence functions of that regression. Fiscal capacity: budget revenue per capita of the neighbor relative to the host. Rank adjacency: the two prefectures are at most one place apart in the province’s ranking by per-capita GDP. Promotion-opportunity overlap: the index of Yu et al. (2016), split at the median. Groups are measured in the year before opening where data allow. Under the rivalry hypothesis $\Delta G > 0$ for the three rivalry rows; under the fiscal hypothesis $\Delta G < 0$ for the fiscal row.

the same size in the firm’s own prefecture, and cross-province moves are rarer still, across choice-set radii and with or without the municipalities. Legal entities, and the tax base registered with them, stay in their prefecture, so incumbents that exit on the neighboring side are not reappearing as registered firms on the host side. Exit itself is departure from the above-scale universe, which combines closure, shrinking below the reporting threshold, and re-registration under a new identifier; the relocation evidence makes re-registration in the host an unlikely explanation for most neighboring exits, but the data cannot separate closure from shrinking within the window.

Table 59: Legal-entity relocation across prefecture borders

Choice set radius (km)	Sample	Other prefecture, same province γ_P	e^{γ_P}	Other province, γ_C	e^{γ_C}	Log distance	Moves
150	All candidate moves	−3.84 (0.29)	0.021	−7.68 (0.65)	0.0005	−1.35 (0.11)	6,233
150	Confirmed moves	−3.77 (0.32)	0.023	−7.88 (0.66)	0.0004	−1.07 (0.11)	3,250
150	Excluding municipalities	−3.75 (0.29)	0.024	not identified	—	−1.41 (0.12)	4,879
100	All candidate moves	−3.90 (0.33)	0.020	−7.11 (0.68)	0.0008	−1.35 (0.12)	6,195
100	Confirmed moves	−3.88 (0.36)	0.021	−7.28 (0.69)	0.0007	−1.05 (0.11)	3,221
100	Excluding municipalities	−3.82 (0.33)	0.022	not identified	—	−1.41 (0.12)	4,841
200	All candidate moves	−3.88 (0.28)	0.021	−7.91 (0.65)	0.0004	−1.36 (0.11)	6,246
200	Confirmed moves	−3.86 (0.30)	0.021	−8.18 (0.66)	0.0003	−1.07 (0.11)	3,259
200	Excluding municipalities	−3.78 (0.28)	0.023	not identified	—	−1.42 (0.11)	4,891

Notes: Conditional-logit destination choice of relocating firms, estimated as Poisson regressions with mover fixed effects. Choice sets are all counties with above-scale firms within the stated radius of the origin. e^γ is the probability of choosing a destination in another prefecture of the same province (another province), relative to a destination at the same distance and of the same size in the origin prefecture. Regressions also include log destination firm count. When no relocation in the sample crosses a province border, γ_C is not identified. Standard errors clustered by origin prefecture in parentheses.

J.7 Why the pass-through ratio is not the headline

A natural summary of the border result is the pass-through ratio $\tau = \beta_N(d)/\beta_H(d)$, the cross-border effect relative to the host-side effect at the same distance. It is not the right headline here. When the host shields its incumbents, $\beta_H(d)$ is close to zero and τ is not identified, even though the cost that lands outside the host, $\Delta Y_N = \sum_{i \in N} Y_{i0} \beta_N(d_i)$, is (Table 56). Table 60 reports τ for aggregate output of incumbents in the 25 km border band, with Anderson–Rubin sets, which remain valid when the denominator is weakly identified (Anderson and Rubin 1949): under every control pool the 90% set covers the whole unit interval, as the argument predicts. We therefore report the level of the cross-border cost rather than its ratio to a host effect that is close to zero.

Table 60: Pass-through ratio with weak-identification-robust sets

Control pool	Weights	Host side	Neighboring side	90% AR set in $[0, 1]$	$p(\tau = 0)$	$p(\tau = 1)$
Not yet approved	Unweighted	-0.111	-0.076	$[0.00, 1.00]$	0.174	0.622
Not yet approved	Reweighted	-0.014	-0.007	$[0.00, 1.00]$	0.920	0.911
All later zones	Unweighted	-0.106	-0.061	$[0.00, 1.00]$	0.267	0.455
All later zones	Reweighted	-0.006	-0.027	$[0.00, 1.00]$	0.614	0.732
Later EPZs only	Unweighted	-0.008	0.063	$[0.00, 1.00]$	0.222	0.244
Later EPZs only	Reweighted	0.068	-0.034	$[0.00, 1.00]$	0.663	0.225
Approved, not operating	Unweighted	0.011	0.073	$[0.00, 1.00]$	0.289	0.334
Approved, not operating	Reweighted	0.100	-0.015	$[0.00, 1.00]$	0.854	0.193

Notes: Host side and neighboring side: post-period effects on aggregate output of incumbents within 25 km of the host prefecture’s border, under each control pool and weighting. The 90% Anderson–Rubin set for $\tau = \beta_N/\beta_H$ is intersected with $[0, 1]$; $p(\tau = 0)$ and $p(\tau = 1)$ are Anderson–Rubin p -values.

K Channels beyond Geography: Measures, Designs, and Results

This appendix documents each row of Table 14 and each point of Figure 26: the measure, the data, the design, the validity check, and what the estimate can and cannot say. Channels are grouped as in the figure.

Figure 26 places every channel estimate on one scale, its t -statistic, and marks whether it clears the validity check its own design provides. No point clears its check and carries an effect beyond the zone’s ring. Two estimates are significant and not failed by a placebo, but act only inside the ring: upstream industries become more likely to be present near a zone without growing, and the loss within 20 km is larger where local industries would have sold to the zone’s sector. One estimate passes its placebo but not a change of estimator: employment in adjacent prefectures of the same province rises in the doubly robust staggered design, and the long-difference two-way fixed-effects version on the same panel is zero (Appendix K.1). Everything else is indistinguishable from zero or fails its own placebo or pre-trend test.

K.1 Geography and proximity on the prefecture panel

Design. The unit is the prefecture-year, 2003–2013, with employment, industrial output, average wage, and fiscal revenue from city yearbooks. Treatment here is the set on which this panel was built: the 174 national-level zones in the development-zone catalogue whose names identify them as bonded areas or export processing zones (136 bonded, 34 export processing, and 4 of other types by the catalogue’s type field), dated by approval year. The rest of the paper uses the directory of 142 special customs zones described in Section 4, of which 60 are EPZs, dated by customs acceptance. Matching on names picks up bonded port areas and

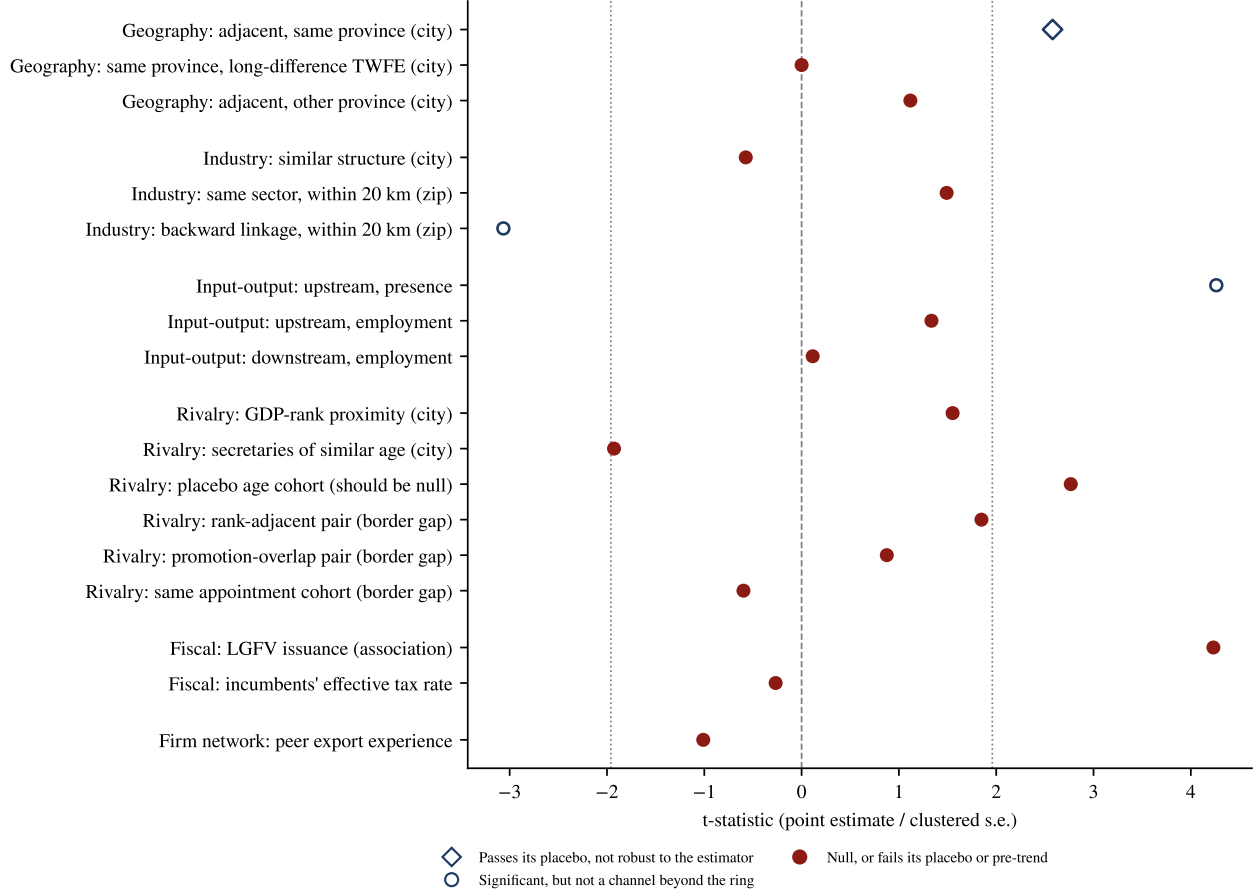


Figure 26: A race among channels: t -statistics of every channel estimate

Notes: Each point is one estimate from Table 14 or this appendix, divided by its clustered standard error; dotted lines at ± 1.96 . Hollow diamond: passes its placebo but is not robust to the estimator (the row below it is the long-difference two-way fixed-effects version on the same panel). Hollow circle: significant and not failed by a placebo or pre-trend, but not a channel that carries effects beyond the zone's own ring (a presence margin with no employment effect, or an interaction within 20 km). Filled: indistinguishable from zero, or significant but failing its own placebo or pre-trend test. “City”: prefecture panel; “zip”: zip-code panel; “border gap”: difference in the neighbor-minus-host survival gap between rival and other border pairs.

bonded logistics parks as well, so the panel's treatment set is broader, and its estimates are effects of bonded and export processing zones together, not of EPZs alone. For a proximity type k with neighbor set $N_k(c)$, a prefecture c that does not host a zone is treated from the first year in which one of its k -neighbors opens one. With g_n the year in which prefecture n opens its first zone,

$$G_c^k = \min_{n \in N_k(c)} g_n, \quad \widehat{ATT}_{\text{post}}^k = \frac{1}{4} \sum_{e=0}^3 \widehat{ATT}^k(e), \quad (11)$$

estimated with the Callaway and Sant'Anna (2021) estimator, not-yet-treated controls, and doubly robust adjustment for base-year (2003–2004) log GDP, log population, export density,

foreign-capital share, and log distance to the nearest port; standard errors are clustered by province. Host prefectures are excluded. The neighbor sets are: adjacent prefectures in the same province; adjacent prefectures in another province; prefectures in the same province within two places in the per-capita GDP ranking; prefectures in the same province in the lowest tercile of distance in industrial structure; prefectures in the same province whose party secretaries were born within two years of each other; and, as a placebo, secretaries born five to eight years apart. Because yearbook variables begin in 2003, cohorts run from 2005 to 2013 and prefectures exposed earlier are dropped as always treated.

Results. Table 61 reports the race. Only adjacency within the province shows an employment response with a flat pre-trend ($p = 0.42$) that also lies outside the distribution of 20 placebo draws permuting approval years across zones. Adjacency across a province line gives a zero with a minimum detectable effect (0.053) smaller than the within-province estimate. GDP-rank proximity is small and borderline on its pre-trend; industrial-structure proximity fails its pre-trend; the age-cohort row is negative, and its placebo cohort, which should show nothing, shows a larger output response than the real one with a clean pre-trend ($p = 0.77$). On a common sample standardized to one target population, the hypothesis that the five proximity types that fit on it (all but cross-province adjacency, which has too few treated prefectures there) have the same employment effect is rejected ($\chi^2(4) = 10.36$, $p = 0.035$).

Limits. Three limits govern how the race is read. First, the within-province estimate depends on the estimator: a long-difference two-way fixed-effects version gives 0.000 (s.e. 0.045; wild-bootstrap $p = 0.996$ with 26 province clusters), and a stacked triple difference of within- against cross-province adjacency on a common clean control pool gives -0.067 (s.e. 0.069), so the ordering “within above across” is not independently confirmed at the prefecture scale. Second, the non-geographic proximity sets are defined within a province, and many of the rank- and structure-treated prefectures are also within-province neighbors; versions that exclude overlap leave too few treated prefectures to estimate, so these rows cannot be separated from within-province geography and are exposed to common provincial shocks, which the failed placebo cohort makes visible. Third, the object differs from the border design of Section 9.1: it is total employment in an entire adjacent prefecture, including entrants and non-manufacturing, not the survival of incumbent manufacturers near the zone. The race is used for the ranking of channels it delivers. The same panel also shows that the host prefecture’s own employment effect, 0.067 (s.e. 0.026) without covariates, falls to 0.004 (s.e. 0.020) once base-year characteristics enter the adjustment: at the prefecture scale, the apparent effect of hosting a zone is observable siting.

Table 61: Proximity to a host prefecture: a race on the prefecture panel

Proximity to a host prefecture	Log employment			Log industrial output		Permutation	Long-difference TWFE	
	ATT	Pre-trend p	MDE	ATT	Pre-trend p	share	Estimate	Wild p
Adjacent, same province	0.066 (0.025)	0.423	0.071	0.071 (0.039)	0.050	0.00	0.000	0.996
Adjacent, other province	0.021 (0.019)	0.624	0.053	0.018 (0.069)	<0.001	0.50	0.027	0.524
GDP rank within two places	0.034 (0.022)	0.070	0.062	0.003 (0.030)	0.552	0.05	−0.008	0.862
Similar industrial structure	−0.035 (0.061)	0.008	0.171	0.038 (0.063)	0.003	0.70	−0.069	0.514
Secretaries born within two years	−0.059 (0.030)	0.658	0.085	0.040 (0.060)	<0.001	0.15	−0.083	0.387
Placebo: born five to eight years apart	0.026 (0.019)	0.616	0.053	0.082 (0.030)	0.771	—	0.033	0.361

Notes: Average post-period ($e = 0, \dots, 3$) effects on non-host prefectures exposed through each proximity type; every type except adjacency across a province line is defined within the province, equation (11); not-yet-treated controls, doubly robust with base-year covariates; standard errors clustered by province in parentheses. Pre-trend p : joint test of the pre-period coefficients. MDE: minimum detectable effect at 80% power and 5% size ($2.8 \times$ s.e.). Permutation share: fraction of placebo draws, permuting approval years across zones, whose employment effect is at least as large in absolute value as the estimate. Long-difference TWFE: static two-way fixed-effects coefficient on exposure with a wild cluster bootstrap p -value over provinces.

K.2 Industry and input–output linkages

The zip-code race. If the zones hurt nearby firms by competing in the same industry, the loss within 20 km would be larger where local industry overlaps the zone’s sector; if they helped through purchases, it would be smaller where local industries sell to the zone’s sector. On a zip-code $\times$ year panel of log industrial output, with zip and province $\times$ year effects and the zones’ own zip codes excluded, the same-province 0–20 km effect is -0.128 (s.e. 0.058). Interacting it with standardized horizontal, backward, and forward exposure to the zone’s sector, built from the 2002 provincial input–output table, moves the average to 0.008 (s.e. 0.069) and loads the loss onto the backward interaction, -0.136 (s.e. 0.051), while the horizontal interaction is 0.047 (s.e. 0.041); the employment version is in Table 14. The sign is the opposite of a supplier gain and matches the fence’s rules: fenced firms import their inputs duty-free, so local industries that would have supplied them get no demand but still face the zone’s draw on local factors. Two caveats apply: exposure is defined for a minority of zip-years, and the three exposure measures are strongly correlated, so the race separates them less sharply than its standard errors suggest. We read it as supporting evidence, and it concerns the effect inside the ring, not transmission beyond it.

Upstream and downstream industries. A second design groups industries by their input–output relation to the zones’ dominant sectors—upstream suppliers, downstream users,

and unrelated industries—and follows each group’s employment in zip-code transition cells, keeping cells with zero employment. The effect is the ratio

$$\theta = \frac{\sum_T y_{\text{post}} / \sum_T y_{\text{base}}}{\sum_C y_{\text{post}} / \sum_C y_{\text{base}}} - 1, \quad (12)$$

over treated (T) and control (C) cells, numerically equal to the treatment coefficient of a two-period Poisson regression with unit effects. Upstream employment changes by 0.032 (s.e. 0.024), with a minimum detectable effect of 6.7 percent; downstream by 0.005 (s.e. 0.040). The only precise response is a presence margin: upstream industries become more likely to be present in a zip code (1.8 percentage points, s.e. 0.42), with a clean placebo in the pre-period (0.1, s.e. 0.56), whereas the downstream presence margin already differs before opening (−1.4, s.e. 0.61). Averaging $\ln(1 + y)$ instead would show upstream employment gains; they come from zip codes where an upstream industry appears from zero, which is the presence margin above. Section 7.3 reports the paper’s main linkage design.

K.3 Political rivalry

Pair rivalry is measured for the host and each adjacent prefecture of the same province: adjacency in the provincial per-capita GDP ranking (at most one place apart), the promotion-opportunity overlap index of Yu et al. (2016) built from officials’ career moves (split at the median), and appointment of the two party secretaries within a year of each other. Each enters the incumbent border design as a split of border pairs, and the forest plot reports the difference in the neighbor-minus-host survival gap between rival and other pairs in the 20–80 km sample: 0.075 (s.e. 0.041) for rank adjacency, 0.034 (s.e. 0.039) for promotion overlap, and −0.024 (s.e. 0.040) for the same appointment cohort. Both samples, the fiscal-capacity split, and the multiple-comparison threshold are in Appendix Table 58. On the prefecture panel the cohort measure uses the secretaries’ birth years (Appendix K.1). None of these measures carries the zones’ effects: the pair differences are insignificant or reverse between samples, rivals’ land supply moves in the direction opposite to competition (−0.792, s.e. 0.333), and the prefecture-level cohort result is matched by its placebo. The measures do not isolate a channel through which a zone’s effects reach a rival’s firms.

K.4 Public finance

Issuance by local-government financing vehicles is merged to the prefecture panel for 2006–2018. In static regressions with prefecture and year effects, the first zone is associated with higher inverse-hyperbolic-sine annual issuance, 0.943 (s.e. 0.223). The near leads are flat,

but once earlier years are included the hypothesis of no pre-opening difference is rejected ($p = 0.0002$), and positive post-opening coefficients appear only two years after opening, so we report an association: the zones coincide with more borrowing by the host, not with a traceable fiscal transfer to other places. The effective tax rate of incumbents near the zones does not change (-0.0020 , s.e. 0.0075 ; Appendix D).

K.5 Firm networks

Peers' export experience. If local exporters learned from one another, a firm's entry into a product–destination market would respond to the cumulative experience of other firms in its city in that market. With the firm's own history and city $\times$ product $\times$ destination effects, the elasticity of entry with respect to the lagged local peer stock is -0.0019 (s.e. 0.0019 ; $p = 0.31$) on 863,218 observations; without the market effects the raw association is negative, so the pattern in levels reflects which markets cities serve, not learning. A revealed-preference comparison points the same way: relative to a size-matched alternative, the market a firm enters is more likely to have more local peers than fewer, by a ratio of 2.64 for firms far from any EPZ but only 1.92 for firms within 20 km of one. Firms near the zones follow local peers less, not more.

Suppliers. Three sources ask whether the zones seeded new local suppliers: births of above-scale firms in the component and electronics industries, first-time exporters in the product lines that make up most of fenced firms' imports, and new business registrations by scope. Electronics suppliers do not rise after opening in any source; registrations of logistics and customs-brokerage firms (0.252 , s.e. 0.150) and of trading firms (0.291 , s.e. 0.184) rise, imprecisely—the services a customs platform needs rather than production linkages.

K.6 Channels not tested

Two channels are not covered. Links through a common foreign parent—a fenced plant passing orders or know-how to affiliates elsewhere in China—would require group identifiers that the firm and customs data used here do not provide. Worker flows between prefectures, as distinct from the commuting implied by the border results, would require census microdata on migration, which we do not have. Both would test whether the zones' effects travel along links that neither geography nor the administrative map records.

L Institutional Background

L.1 Origins and approval batches

Export processing zones (EPZs) are a customs special supervision zone created by the State Council to concentrate processing-trade activity under unified customs supervision and to curb smuggling that scattered processing arrangements had made hard to police.² Section 2 gives the context; here we list the approval batches and the zone-level detail. The State Council approved the first batch of 15 zones on April 27, 2000: Beijing Tianzhu, Tianjin, Dalian, Hunchun, Yantai, Weihai, Kunshan, Suzhou Industrial Park, Songjiang, Hangzhou, Xiamen, Shenzhen, Guangzhou, Wuhan, and Chengdu. Further approvals followed in 2001–2003 (24 zones) and 2005–2006 (19 zones), and one zone each was approved in 2010 (Hefei) and 2011 (Jinggangshan), for 60 in total (Appendix P lists them). A zone’s planned area was capped at 3 km²; zones that grew beyond it received a separately approved second area, which is why several zone names carry “and Area B.” Our analysis panels exclude Jinqiao, whose directory entry dates to a 1990 development-zone designation rather than to a customs-registered EPZ, leaving 59 zones in the geographic and treatment-timing panel used for estimation.

L.2 From approval to customs operation

Approval by the State Council and the start of customs supervision are distinct dates: for example, Shenzhen was approved in the first batch but its zone was accepted into customs operation only on March 31, 2001, and Tianjin’s on June 29, 2001. How the operating date is set for every zone, including how many have an official acceptance date rather than an imputed one, is in §M.11. The median zone began operating 2 years after approval (Figure 15), and approval itself set off construction and investment promotion, so approval and operation are not interchangeable timing conventions for design purposes: the location design dates treatment by operation, and the cross-border design builds its controls on the approval clock (Appendix O).

L.3 Investment promotion and firm admission

Once customs accepted the zone, the host prefecture’s zone administration recruited firms into the fence; investment promotion, like land and infrastructure, is the host prefecture’s own responsibility (§L.10). The founding customs order allowed three kinds of firms inside: export processors, warehousing firms serving them, and transport firms licensed to move

²General Office of the State Council (2002); see [Kee and Tang \(2016\)](#) and [Koopman et al. \(2012\)](#) for the broader processing-trade regime that EPZs sit inside.

goods in and out of the zone. The resulting activity is concentrated and foreign: among firms with customs records inside an EPZ, foreign-invested firms account for 99 percent of fenced exports in 2000–2006, and processing with imported materials—in which the zone firm itself owns the imported inputs and the exported output, as opposed to processing on a foreign principal’s account—accounts for 100 percent of fenced processing exports. The fence’s industry mix, dominated nationally by electronics but concentrated in a handful of hub zones, is examined in §L.8.

Admission rules broadened over time. The founding customs order restricted zone firms to export processing only, unchanged through a 2011 revision. Successive entry guidelines in 2012, 2018, and 2021 progressively widened the menu of admissible activity to include bonded logistics, research, testing and repair, entrepôt trade, domestic sales, and processing on commission for firms outside the fence, classifying admissible projects by the combination of input sources and output destinations that fits the tax regime; the 2018 guideline, for instance, lists projects using only domestic inputs as admissible. None of the guidelines conditions entry on local sourcing or on linkages with local firms.

L.4 Siting inside existing development zones

In practice EPZs were placed inside national economic and technological development zones or high-technology zones, one per development zone, with Qingdao and Beihai among the few sited outside one (Section 2 gives the count by siting type). Land, investment promotion, and infrastructure for the EPZ are the host prefecture’s responsibility, so an EPZ occupies development-zone land and entry quotas that the host government could otherwise have allocated to other firms. Because the customs fence is layered onto a pre-existing development zone rather than a bare site, distinguishing a newly established EPZ from the mere upgrading, renaming, or later revocation of an existing special-zone designation matters for treatment timing; designs that build control pools from “newly established” special zones exclude sites whose change in status is an upgrade, retention, or renaming of an existing zone rather than a first customs-supervision event (§M.9).

L.5 Rules on transactions inside and outside the fence

The regulatory detail behind Section 2’s description of what the fence restricted: deep-processing carry-over, piloted only in the Kunshan and Songjiang zones in 2002, was extended to all zones by General Administration of Customs Order No. 126, effective May 1, 2005—the only reform inside the firm-survey window that identifiably opened a channel between zone and non-zone firms. A firm subcontracting a stage of production to a supplier outside the

fence had to post a bond equal to the tariffs and import value-added tax that would otherwise have been due on the subcontracted materials. The 2007 deposit requirement on processing trade in restricted products, which applied to processing firms elsewhere, did not apply inside customs special supervision zones. The design built around the 2005 reform, and its estimated effect on downstream industries, are in [Appendix H.5](#).

L.6 Five zone types compared

Table 1 in the main text compares the EPZ with four other zone types. Three points qualify it. First, national economic and technological development zones and high-technology zones are labels for bundles of preferences whose content varied across places and years; the rules we can document from primary legal sources are income-tax rates—15 percent for productive foreign-invested firms inside a national economic and technological development zone (1991–2007) and for certified high-technology enterprises inside a national high-technology zone (1991–2006)—and each applies to a firm inside the zone’s legal boundary that meets the stated conditions. These zones are not customs special supervision zones, so no customs rule separates their firms from the rest of the economy. Second, the customs special supervision zones form a sequence: bonded zones from 1987, EPZs from 2000, bonded logistics parks and bonded ports in the mid-2000s, and comprehensive bonded zones from the Suzhou Industrial Park pilot of December 2006. A 2012 State Council guideline made the comprehensive bonded zone the only type for new special customs zones, and the 2015 integration plan converted the existing EPZs zone by zone. Third, the comprehensive bonded zone reversed the EPZ’s defining restrictions in stages ([Table 63](#)) and added a performance evaluation in 2020 under which a zone graded lowest for two consecutive years and ranked among the bottom five nationally starts an exit procedure; the only quantitative linkage indicator in that evaluation measures goods moving between the zone and the rest of China and carries a tenth of the score.

[Table 62](#) completes the main-text comparison with each type’s approval authority and the preferences it offered to firms inside.

L.7 Where China’s exports were registered

The fifth digit of a customs registration code records the kind of zone in which a firm is registered. Pooling all exports over 2000–2011, firms registered in no designated zone shipped 69.8 percent of China’s exports. Firms in national economic and technological development zones shipped 9.4 percent, special economic zones 6.1 percent, EPZs 5.8 percent, national high-technology zones 4.1 percent, bonded zones 3.4 percent, comprehensive bonded zones

Table 62: Approval and main preferences of the five zone types

Zone type (first year)	Approval	Main preferences for firms inside
Economic and technological development zone (1984)	State Council	Tax, land, and approval preferences varying by place and year; e.g., 15% income tax for productive foreign-invested firms (1991–2007)
High-technology development zone (1988)	State Council	Same kind of bundle; e.g., 15% income tax for certified high-technology firms (1991–2006)
Bonded zone (1987)	State Council	Bonded warehousing, processing, and trade behind a customs fence
Export processing zone (2000)	State Council, on application through the province	Duty-free imports of materials and equipment; tax-free exports; export processing only
Comprehensive bonded zone (2006)	State Council, zone by zone	Bonded processing, logistics, research, testing, and repair; general-taxpayer VAT status piloted from 2016

Notes: National-level zones; development and high-technology zones also exist at the provincial level, approved by provinces. First year: earliest approval year of a zone of that type in the official Catalogue of China’s Development Zones; for comprehensive bonded zones, the Suzhou Industrial Park pilot (December 2006). Bonded zones and EPZs were exempt from the 2007 deposit on processing trade in restricted products. Sources and detail: §L.6 and Table 63.

and bonded ports 0.5 percent, and bonded logistics parks 0.2 percent; the remaining 0.7 percent carry no zone code. All customs special supervision zones together account for 10.0 percent. The EPZs were thus one instrument for concentrating processing trade, not the place where most of it happened: the fences shipped a small minority of national processing exports throughout (Figure 1).

L.8 The fence’s industry mix: two measures

The fence’s industry composition is used twice in the paper: to describe what the fence holds, and to define the fence’s suppliers and users in the input–output analysis (§O.9). Products are mapped from six-digit HS codes to Chinese industries and then to the 42 sectors of the 2002 input–output table; a manual rule at the four-digit HS level covers the few codes the concordance misses. Electronics is the sector “communication equipment, computers, and other electronic equipment,” which contains computers and their parts (HS 8471 and 8473) and the telecommunications, audio-visual, electronic-component, and semiconductor headings of chapter 85 (HS 8517–8529, 8534, 8540–8542); general and special-purpose machinery, office machinery, and electrical machinery are separate sectors.

Two measures answer different questions. Weighted by exports and pooled over all fenced firms in 2000–2006, electronics is 87 percent of fenced exports; this is the composition the input–output analysis uses. Zone by zone, over each zone’s first three operating years, the

picture differs: among the 37 zones whose fenced exports can be assigned to a named zone, the median electronics share is 21 percent, 14 zones have more than half their fenced exports in electronics, and 9 export none. The national share is high because most of the largest fences—the hubs—are electronics fences. A third measure, the share of fenced firms whose main export is in HS chapters 84, 85, or 90, is 55 percent (Table 2); it counts firms rather than exports and includes general machinery and instruments. None of these product-based measures is affected by the 2003 revision of the Chinese industry classification (Appendix N).

L.9 Later reforms and the conversion to comprehensive bonded zones

The customs regime above was not static. Between 2002 and 2019 the central government issued a sequence of rules that progressively lowered the cost of transactions between firms inside a zone and firms outside it (Table 63). Two features of the sequence matter for empirical work. First, the reforms that most directly open the fence—integration into comprehensive bonded zones, value-added-tax general-taxpayer status for zone firms, and selective tariffs on domestic sales—date from 2015 or later. Second, only two reforms change the rules within the period covered by firm-level customs records with registration codes (2000–2016): the 2005 national extension of carry-over, and the upgrading of individual zones into comprehensive bonded zones, which occurred zone by zone rather than all at once.

No central document diagnoses weak local linkages of export processing zones as such: the recurring official diagnosis of zones through 2011–2012 concerns land use, operating returns, and the proliferation of zone types, and local linkage enters policy documents as an objective only from 2011–2012. Official statements in the same period also assert that processing trade had already “driven the development of supporting industries,” citing a rise in domestic purchases and value added from 32 to 44 percent of processing exports between 2002 and 2012; that ratio is measured by place of production and includes foreign-owned suppliers operating in China, so it is the domestic-content concept of [Kee and Tang \(2016\)](#) and [Koopman et al. \(2012\)](#), not a measure of local firms’ capabilities, and is not the domestic value-added ratio used in this paper’s own accounting (§M.8).

The official *Catalogue of China’s Development Zones* lists every zone with its approval date and tracks the conversion into comprehensive bonded zones: the 2006 edition lists 58 export processing zones; the 2018 edition lists 135 special customs zones, in which zones that had been upgraded appear under their comprehensive-bonded-zone name and approval date, and 27 export processing zones appear unchanged. Matching the two editions by city and name gives the upgrade date of each zone; zones whose successor cannot be identified

Table 63: Rules governing transactions across the fence of export processing zones, 2000–2020

Year	Document	Content
2000	Customs General Administration Order 81 (revised 2003, 2011)	Zone firms may only conduct export processing; general trade, entrepôt trade, and retail are prohibited. The prohibition is unchanged in the 2011 revision.
2002	State Council General Office letter [2002] 64	Pilot of deep-processing carry-over (bonded sales of intermediate goods from zone firms to processing firms outside the fence) in the Kunshan and Songjiang zones.
2005	Customs General Administration Order 126	Carry-over extended to all export processing zones from May 1, 2005.
2008	Ministry of Finance, Customs, and Tax Administration notice [2008] 10	Listed domestic inputs purchased by zone firms regain export rebates when they enter the zone.
2011	Ministry of Commerce and five ministries, [2011] 448	Processing trade described as of “low growth quality and returns”; zones instructed to attract “large firms with strong capacity to drive local supporting industries.”
2012	State Council [2012] 58	First stand-alone task to “cultivate supporting capacity outside the zones” and to “bring qualified firms into processing-trade production and supply chains”; zones with low land use and poor performance to be rectified.
2015	State Council General Office [2015] 66	Integration of special customs zones into comprehensive bonded zones; zone firms allowed to take processing orders from firms outside the fence.
2016	Tax Administration, Finance, Customs announcement 2016 no. 65 (extended 2018, 2019)	Value-added-tax general-taxpayer status for zone firms, starting with seven zones: purchases from domestic suppliers outside the fence become creditable.
2016	Finance and Customs [2016] 40; announcement 2020 no. 20	Selective tariffs on domestic sales of zone output, extended to all comprehensive bonded zones in 2020.
2019	State Council [2019] 3	Twenty-one measures, including processing on commission for firms outside the fence.
2020	Customs and five ministries, performance evaluation rules	The only quantitative linkage indicator (“radiation and service”: second-line trade with the rest of China) receives 10 percent of the score.

Notes: Document titles are translated; the 2009 functional expansion of export processing zones (bonded logistics, research and development, testing, and repair) is documented indirectly in the 2011 and 2012 texts and its originating document number is omitted.

unambiguously are treated as unmatched. By 2024 no zone of the export-processing type remained in official counts.

L.10 The prefecture as the decision unit

In China’s administrative hierarchy, the prefecture-level city is the effective decision-making unit for the policy instruments relevant to a zone: land allocation, investment promotion, most industrial policy, and the fiscal revenue that a zone generates, and the unit along which local officials are evaluated. An export processing zone is proposed and operated by the local government—ordinarily the prefecture—and approved by the State Council; once approved, the zone’s land, infrastructure, and investment promotion are the host prefecture’s responsibility, and the zone’s tax revenue accrues to the host prefecture. Municipalities directly under the central government (Beijing, Tianjin, Shanghai, Chongqing) are a special case: their districts and counties do not hold the same administrative powers as an ordinary prefecture-level city, so designs that use the prefecture as the unit of jurisdiction either drop the municipalities or treat an entire municipality as a single prefecture-and-province, ignoring its internal district boundaries (§M.13).

L.11 Firm registration and the cost of moving

A firm’s legal identity is a registration in a specific county-level jurisdiction, and it pays tax where it is registered. Moving a firm’s registration across a prefecture boundary requires changing or re-filing its tax registration; moving it across a provincial boundary ordinarily requires deregistering the firm and registering it anew, which can break the cross-year identifier that links a firm’s records over time (§M.16). By contrast, the channels through which a place-based policy transmits an economic shock across a jurisdictional boundary—commuting, input purchases, and output sales—do not require any change in registration. Migrant labor inflows are a further source of asymmetry: they make local labor supply close to perfectly elastic at the scale of an EPZ’s labor demand, so a zone’s effect on nearby wages need not resemble its effect on nearby firm entry or the allocation of development-zone land.

M Data

M.1 The firm survey: coverage and the above-scale threshold

The firm-level data are China’s Annual Survey of Industrial Firms (“above-scale” manufacturing firms), covering 1998–2007, with output for 2008–2009 drawn from a separate factor-input

survey that extends the panel window. In every year the survey is restricted to firms above the “above-scale” threshold used throughout this literature: annual sales (or output) of at least RMB 5 million and at least eight employees (Brandt et al. 2012). We apply that threshold uniformly across years and analyses, because the raw survey’s own threshold shifted over time and unifying it is necessary for entry and exit to be measured on a constant scale (§M.3). Firms are followed across years using the data provider’s cross-year firm lineage, and are geocoded to the village level for 74 percent of observations.

M.2 The 2004 census year

The 2004 economic census wave is treated specially. In our main specification 2004 observations are *retained*: only census-year firms that appear solely in 2004 (economic-census stragglers with no other-year presence) are dropped, while firms whose 2004 record reflects legitimate supplementary registration are kept, because the resulting excess of new registrations in 2004 balances across treated and control locations under the reweighted not-yet-treated design (§O.4). A version of the sample that excludes 2004 entirely is reported alongside as a robustness check (Appendix G).

M.3 Entry and exit

A firm is coded as a new entrant in year t if it appears in the above-scale sample for the first time in t and its self-reported registered opening year is no earlier than $t - 1$; the second condition excludes firms that are new to the above-scale sample only because they grew past the size threshold in t , rather than because they were newly established. A firm is coded as exiting when it leaves the above-scale sample—because its output falls below RMB 5 million, its employment falls below eight, or it ceases operating—which is not equivalent to the firm shutting down.

M.4 Industry-code harmonization

Each firm is assigned a single two-digit industry code that does not change over time, in the 2003 revision of the national industry classification (GB/T 4754-2002). The classification changed in 2003: several manufacturing codes in the tail of the schedule shifted by one digit, so that a firm coded 40 under the pre-2003 classification (“electrical machinery”) corresponds to code 39 under the 2002 classification, while code 40 under the 2002 classification is instead “electronics and telecommunications equipment.” The full correspondence used is $\{40 \rightarrow 39, 41 \rightarrow 40, 42 \rightarrow 41, 43 \rightarrow 42\}$. Firms with any post-2003 record are assigned the mode of

their post-2003 codes; firms observed only before 2003 are assigned the mode of their pre-2003 codes after applying this correspondence. Grouping firms by their *current-year* reported code instead—without this harmonization—misclassifies pre-2003 electrical-machinery firms as electronics, because 94.5 percent of firms reporting the old code 40 in a pre-2003 year report code 39 from 2003 onward. Any analysis that groups firms by industry, or that defines treatment or an outcome using an industry code, uses the harmonized code rather than the current-year code.

M.5 Customs transactions: coverage

The customs data are the universe of Chinese customs transactions, recorded at the firm $\times$ product $\times$ destination $\times$ year level for 2000–2011. Trade regime (§M.7) is reported separately in the raw data only through 2006; analyses that use trade regime as a treatment or outcome are accordingly restricted to 2000–2006, while analyses of total trade value extend to 2011 using a coarse trade-value variable that carries total export value once the regime-level breakdown is no longer separately reported.

M.6 Identifying firms inside a fence

Each customs registration code carries a fifth digit that identifies the customs special-supervision status of the firm’s registered location: 1, special economic zone; 2, economic and technological development zone; 3, high-technology zone; 4, bonded zone; 5, export processing zone; 6, comprehensive bonded zone; 7, bonded logistics park. This fifth digit is a legal fact about where a firm is registered, not an inference from its address or postal code, and is the basis on which a firm is classified as legally located inside an EPZ (§M.9).

M.7 Processing with imported vs. supplied materials

Customs records distinguish two organizational forms of processing trade. Under *processing with imported materials*, the exporting firm itself purchases the imported inputs and owns both the inputs and the resulting output. Under *processing with supplied materials*, a foreign principal owns the imported inputs and the output throughout, and the domestic firm performs only the processing step under contract. This distinction is a legal and contractual fact recorded in the trade regime field, not a measure of production technology: two firms performing identical physical transformations can be recorded under either regime depending on who holds title to the goods.

M.8 Domestic value-added ratio (DVAR)

For a set of fenced (zone) firms, the domestic value-added ratio of their exports is defined as one minus the ratio of processing imports to processing exports,

$$\text{DVAR} = 1 - \frac{\text{processing imports}}{\text{processing exports}}.$$

Because processing exports also embed domestic wages and profits that are not themselves evidence of local intermediate-input sourcing, DVAR is an upper bound on the share of a fence’s exports actually sourced from local suppliers, and customs records do not identify carry-over transfers to outside firms, so it bounds local intermediate sourcing from above rather than measuring it directly. DVAR is computed at the level of an individual fence or aggregated across fences, and rose in the aggregate from 0.27 in 2003 to 0.48 in 2006, consistent with the economy-wide rise in the domestic content of Chinese processing exports documented by [Kee and Tang \(2016\)](#).

M.9 The zone directory

The project’s zone directory records, for each customs special supervision zone, an approval year, an operating (customs-acceptance) date, and a geographic location, together with the zone’s type: export processing zone, comprehensive bonded zone, bonded port, bonded logistics park, or (plain) bonded zone. These non-EPZ customs special supervision zones are not the object of study but enter the design in two ways: as the population from which not-yet-treated control locations are drawn (§O.4), and as the source of the time-varying exposure controls that count how many such zones are already operating within a given ring of a location (§O.3).

M.10 Geolocating a zone

A zone’s location is constructed, in order of preference, as: (i) the median coordinates of its legally registered member firms under the customs fifth-digit code, provided at least 5 member firms are matched to coordinates and their median distance to that median point does not exceed 15 km; (ii) failing that, an empirical footprint centroid computed from firms independently attributed to the zone, provided it is based on at least 10 firms with a median spread of at most 5 km; (iii) failing that, a zone-directory “recommended” coordinate; and (iv) failing all of the above, the zone directory’s raw coordinate. This fallback hierarchy privileges coordinates revealed by firms that are legally, rather than merely nominally, inside the fence.

M.11 Operation dates

A zone’s operating date is its official customs acceptance date where one is recorded, and the approval year plus two years otherwise (§L.2); approval years come from the approval documents underlying the three EPZ batches (§L.1). Official acceptance dates are available for 36 of the 59 zones in the estimation panel; the remaining 23 are dated by the approval-year-plus-two imputation. Non-EPZ special zones’ approval years are taken from the zone directory’s own recorded start year.

M.12 Upgraded vs. newly established zones

A change in a location’s customs special-supervision status is not always a newly established zone. A site can instead be *upgraded* from one special-zone type to another, *retained* under a new administrative category, or *renamed*, without a first customs-supervision event actually taking place there; a site’s special-zone status can also later be *revoked*. Because these status changes do not represent a first opening, control pools built from “newly established” special zones exclude sites whose status change is an upgrade, retention, renaming, or revocation of an existing designation.

M.13 Prefecture polygons and border distance

Prefecture-level administrative boundaries are taken from GADM version 4.1, level 2 (prefecture level), simplified with a tolerance of approximately 200 meters. Each polygon is assigned to a prefecture by taking the modal administrative code among the firms whose coordinates fall inside it, rather than trusting the polygon dataset’s own administrative labels directly; the firm-level administrative code agrees with the polygon it falls inside for 99.6 percent of firms, and firms whose code disagrees with their enclosing polygon are excluded from any border-based design. Distance from a firm to a jurisdictional border is computed only to the land border with an adjacent prefecture; distance to the coastline is not treated as a border distance.

M.14 Border segments and relative border position

A *border segment* pairs a host prefecture with one specific adjacent prefecture; designs that compare the same border segment over time (rather than pooling all of a host prefecture’s neighbors) hold fixed which two jurisdictions are being compared. Because a fixed-radius border band treats a small prefecture and a large prefecture asymmetrically, a *relative* border position is also constructed: for a firm at distance d_b from its own prefecture’s boundary

and distance d_c from its own prefecture’s output-weighted industrial centroid, the relative position is $d_b/(d_b + d_c)$, making border proximity comparable across jurisdictions of very different size and shape.

M.15 Municipalities in the boundary design

Because a municipality directly under the central government does not allocate the relevant policy instruments at the district level (§L.10), the main boundary design excludes zones located inside a municipality. An extension that reintroduces municipalities treats each entire municipality as a single prefecture and, simultaneously, as a single province, ignoring internal district boundaries.

M.16 Firm relocation records

The project’s firm relocation records cover county-to-county moves of above-scale manufacturing firms for 1999–2007, with 9,102 candidate moves and 3,982 confirmed moves.

M.17 How a relocation is identified

A *candidate* relocation is a change in a firm’s recorded county-level administrative code between consecutive appearances in the firm-level data. A *confirmed* relocation is a candidate that has been independently verified as an actual physical move of the firm’s operations, as opposed to a change in recorded location arising from administrative boundary reclassification or a data-entry correction. As in §L.10, municipalities are treated as a single prefecture and a single province for the purpose of classifying a relocation’s origin and destination.

M.18 Known limitation

Because a cross-province move ordinarily requires a firm to deregister in its origin province and register anew in its destination province (§L.11), a firm’s cross-year identifier can fail to survive a cross-province relocation. The relocation records therefore capture within-province moves more completely than cross-province moves: a firm that relocates across a provincial boundary and re-registers under a new identifier will generally not appear in these records as a relocation at all, but instead as an exit at the origin and an unrelated entry at the destination.

N Data Traps and How We Handle Them

Chinese firm and customs microdata contain structural breaks and coding conventions that can manufacture results. This appendix lists the eight that matter for this paper. For each we state the trap, how it could mislead, how we handle it, and what our handling does to the estimates.

1. The 2003 revision of industry codes. *Trap.* The national industry classification was revised in 2003, and several two-digit manufacturing codes shifted by one: code 40 means electrical machinery before 2003 and electronics and telecommunications equipment from 2003 on. *Risk.* Grouping firms by the code they report in each year relabels pre-2003 electrical-machinery firms as electronics, producing a jump in “electronics” near the zones exactly when the fence—itsself mostly electronics—arrives; left uncorrected, this misclassification roughly doubles apparent electronics output near the zones. *Handling.* Every firm gets one harmonized two-digit code that does not change over time (§M.4), used in every analysis that groups firms by industry, including the electronics output share in the reweighting and the input-output groups. *Effect.* With harmonized codes the fence’s own industry grows only 17 percent faster near the zones than other industries, an imprecise estimate (s.e. 0.107 in logs; Appendix H.3).

2. The 2004 economic census. *Trap.* The 2004 survey wave was assembled from the economic census and contains two kinds of new records: census stragglers that appear in 2004 only, and established firms registered for the first time that stay in the survey afterwards. *Risk.* Counting both as entrants inflates entry in 2004; dropping the whole year removes legitimate firms and shortens every event window that crosses it. *Handling.* We drop only firms observed in 2004 and in neither 2003 nor 2005, and keep the supplementary registrants, whose excess entry in 2004 balances across treated and reweighted control locations (§M.2). Value added is not collected in 2004 and is coded missing rather than zero. *Effect.* The main entry estimate is -0.219 (a decline of 20 percent); dropping 2004 altogether gives -0.320 (27 percent). We report the smaller decline as the headline (Section G).

3. Output recorded as zero in 2010. *Trap.* In the 2010 wave of the firm survey, gross output is stored as zero for every firm rather than as missing, while employment is recorded normally. *Risk.* A Poisson regression treats these zeros as real, so any design that reaches 2010 would record a collapse of output in that year. *Handling.* Our firm panel ends in 2007, with output for 2008–2009 taken from a separate factor-input survey; no analysis uses 2010 output. Negative reported value added, an accounting artifact in a small share of firm-years,

is set to zero before summing. *Effect.* None on the reported estimates; it limits how far the firm panel can be extended, and the long-run results rely on customs data instead.

4. Zone types and dates in the zone directory. *Trap.* The official directory lists 26 of the 60 EPZs, including 8 of the 15 in the first batch, under bonded zones, and for one zone, Caohejing, it records 1992, the year of its parent high-technology park; the EPZ was approved in March 2003 and began operating in March 2004. *Risk.* Splitting EPZs by the directory’s type field compares two arbitrary groups of genuine EPZs and invites a spurious “true versus mislabeled” interpretation; a parent-zone year would date treatment more than a decade too early. *Handling.* EPZs are identified by name and checked against the approval documents; treatment is dated as in §M.11, and Caohejing’s approval year is corrected in every table. *Effect.* The estimates already used the corrected dates; only displayed approval years changed, and Caohejing is outside the location sample in any case because other special zones were already operating within 40 km of it (Appendix P).

5. Two measures of the fence’s electronics share. *Trap.* “Share of the fence in electronics” can be computed by export value or by firm count, nationally or zone by zone, and with a narrow or broad product definition. *Risk.* The export-weighted national share, 87 percent, describes the hubs, not a typical zone; read as a typical zone, it would suggest that the fence is an electronics enclave everywhere, and that nearby electronics firms are the natural suppliers or rivals of every fence. *Handling.* We report the distribution across zones: the median zone’s electronics share is 21 percent and 14 of 37 measurable zones are mainly electronics (§L.8); the firm-count share on a broad definition is 55 percent (Table 2). The product classification is by HS code and is therefore unaffected by trap 1. *Effect.* The input–output groups use the national mix, fixed before any zone opens; aggregate facts are reported zone by zone and without the hubs.

6. Two scopes for the domestic value-added ratio. *Trap.* The ratio $1 - M/X$ for fenced firms depends on which flows and which firms enter it. Our ratio uses processing imports and processing exports of firms assigned to a named EPZ: 0.27 in 2003, rising to 0.48 in 2006. Counting all imports and exports of every firm registered inside an EPZ—including duty-free equipment and other imports outside the processing regimes—gives 0.20 in 2003; the two firm sets export almost the same amount, so the gap comes from the imports counted. Adding firms of the comprehensive bonded zones and bonded ports that succeeded the EPZs changes nothing before 2007, when they barely traded, but lowers the ratio afterwards: in 2011 it is 0.41 with them and 0.46 without. *Risk.* Comparing ratios across studies without

noting the scope attributes a difference in definitions to a difference in the fence. *Handling.* Every ratio is reported with its scope; the ledger uses the processing-only ratio, the higher of the two, and treats it as an upper bound on local sourcing, since it also includes wages and profits paid inside the fence and customs records do not show carry-over sales (§M.8). *Effect.* The choice favors the fence: any other scope would leave even less locally.

7. Identifying fenced exports year by year. *Trap.* After 2006 customs no longer separate the two processing regimes, and as zones were converted into comprehensive bonded zones, fenced firms re-registered under new codes. *Risk.* A series built from the regime columns stops in 2006, and a series built from a fixed list of fenced firms shows a spurious decline after 2010; plotted on the same linear axis as national exports, either suggests that the zones faded. *Handling.* Fenced firms are identified each year from that year’s registration code, and processing exports are exports other than general trade in every year. The resulting series reproduces the official EPZ export total for 2012 closely; we stop it in 2014, before the conversion wave recodes most fenced firms, and plot it on a log scale (Figure 1). *Effect.* Descriptive only: the estimates use fenced firms’ records from 2000–2006, before re-registration became common.

8. Where firms and zones are. *Trap.* Firm locations come from geocoded addresses, and the official coordinates of several national zones—including the largest EPZ by exports—lie far from where their member firms actually are. In customs records a minority of firms, mostly small and short-lived, report an address and a name that point to different provinces. *Risk.* A misplaced zone puts treated firms in the wrong distance ring and misassigns cells to treatment; a misplaced firm does the same at the firm level. *Handling.* A zone is located from the median coordinates of its legally registered member firms where enough of them are geocoded, and from the directory only as a last resort (§M.10). Whether a firm is inside the fence is read from its registration code, not from its location. Customs firms whose address and name conflict are flagged, not dropped; they carry a small share of exports. *Effect.* The treated radius (20 km) and cell size (0.1 degree, about 10 km) are wide relative to the remaining coordinate error, and updating customs coordinates to their final version moved very few firms across ring boundaries; results at 10 and 30 km radii and with 0.2-degree cells or counties are in Appendix F.

O Design and Inference

O.1 Stacking by opening year

Because customs special supervision zones were approved and opened in different years, and the same location can be exposed to more than one such zone over time, both designs in the paper treat each opening as a separate event and stack the events into a single estimating sample, following the stacked-cohort literature (Cengiz et al. 2019; Wing et al. 2024; de Chaisemartin and D’Haultfoeuille 2024; Callaway and Sant’Anna 2021). The *location design*, behind equation (5) and every grid-cell result, stacks sub-experiments of treated and control cells (§O.2). The *anchor design*, behind the analysis of incumbents across prefecture borders in Section 9, stacks zones—a treated zone and control zones with the firms around them (§O.4). In both, event time is $e = t - g_s$ and the reference period is $e = -1$.

O.2 The location design

Exposure counts. For every 0.1-degree cell c and year t we count the customs special supervision zones of any type (EPZs, bonded zones, bonded logistics parks, bonded ports, comprehensive bonded zones) operating within 40 km of the cell’s centroid, N_{ct} (the near ring), and between 40 and 150 km, F_{ct} (the far ring). An EPZ operates from its official customs acceptance year, counted from the following year if acceptance fell in the second half of a year (operating dates and their imputation where no official date is recorded follow §M.11); other special zones operate from the directory’s operating year. Zone locations are the geolocations of §M.10.

Treated cells. Cell c is treated in year g if $N_{cg} > N_{c,g-1}$, at least one of the arriving zones is an EPZ, the nearest arriving EPZ lies within 20 km of the centroid, and the cell held at least one above-scale firm in $g - 1$. Its post window runs from g to the year before the next change in N_{ct} , capped at four years and at the end of the firm data; the effect estimated is therefore that of one change in exposure, and a later zone arriving nearby ends the window rather than accumulating into the estimate. The main sample keeps treated cells with $N_{c,g-1} = 0$: the EPZ is the first special zone within 40 km.

Sub-experiments and controls. Treated cells are grouped into sub-experiments $s = (g, n_0, f_0, \Delta N, L)$ by opening year, near-ring count in $g - 1$, far-ring tier in $g - 1$ ($F = 0, 1-2, 3-5, 6-9, \geq 10$), number of zones arriving in g , and window length. The controls of sub-experiment s are cells with at least one above-scale firm in $g - 1$ whose near-ring count

equals n_0 in every year from $g - 1$ to $g + L - 1$ and whose far-ring tier in $g - 1$ equals f_0 ; they must lie within 40 km of some special zone that eventually opens, and in the main (not-yet-treated) pool the next zone within 40 km of them opens after $g + 3$. A sub-experiment needs at least five controls. The second pool used for comparison in the robustness appendix replaces the not-yet-treated screen with the requirement that an EPZ later opens within 40 km.

Weights. Controls are reweighted by the odds of treatment, $\hat{p}/(1 - \hat{p})$, from one ridge-penalized logit (§O.7) of treatment on sub-experiment indicators and nine pre-period cell characteristics: log number of above-scale firms and log output in $g - 1$, foreign-invested and electronics output shares, the domestic entry rate over $g - 3$ to $g - 1$, far-ring count in $g - 1$, log distance to the nearest national development zone existing in $g - 1$ and to the nearest major port, and an eastern-province indicator. Control weights are trimmed at the 99th percentile and rescaled within each sub-experiment to sum to its number of treated cells. Equation (5) is then estimated with cell-by-sub-experiment and sub-experiment-by-year effects (§O.6).

O.3 Overlapping exposure: how the location design handles it

Four features of the location design address overlap, each corresponding to a failure of a binary one-zone design. (i) *Initial-state stratification*: treated and control cells share the exposure state before opening, so the comparison is not between locations that already differed in how many zones they saw. (ii) *Contamination exclusion*: a control cell whose near ring changes during the window is not a control; in a binary design such cells would stay in the control pool and bias the estimate toward zero. (iii) *Truncation of dose accumulation*: the post window of a treated cell ends when a further zone arrives within 40 km. (iv) *Far-ring exposure*: the far ring enters through the initial tier and the odds weights; year-to-year changes in the far ring after opening are not a regressor in the main specification, and Section 5 shows that adding them as a time-varying control leaves the estimates unchanged. Section 5 also reports how the estimate moves from a pure binary design to the main design as these features are switched on one at a time, and the contribution of each.

Is exposure endogenous? Table 64 regresses the exposure counts of treated cells on predetermined geography: distance to the nearest of 23 major ports, distance to the provincial capital, the log number of above-scale firms in the prefecture in 1998, an indicator for a national development zone established before 2000 within 5 km, and the approval year of the EPZ. Column (1) has no fixed effects; column (2) demeans every variable within the design's

strata (opening year $\times$ near-ring count in $g - 1 \times$ far-ring tier in $g - 1$). Geography explains a sizable share of exposure across strata and almost none within them. The near-ring count in $g - 1$ defines a stratum, so its within-stratum R^2 is not defined.

Table 64: Does predetermined geography explain exposure? R^2 across and within design strata

	(1) No fixed effect	(2) Within $g \times n_0 \times$ far tier
Near-ring count, year $g - 1$ (mechanically pinned by the FE; within-FE n.a.)	0.174	n.a.
Far-ring count, year $g - 1$	0.365	0.020
Near-ring count, year g	0.201	0.020
Far-ring count, year g	0.397	0.057

Notes: Each row is a regression of a treated cell’s count of operating special customs zones within 40 km (near ring) or 40–150 km (far ring), in the year before opening ($g - 1$) or the opening year (g), on five predetermined characteristics: distance to the nearest major port, distance to the provincial capital, log 1998 above-scale firm count of the prefecture, a pre-2000 national development zone within 5 km, and the EPZ’s approval year. Column (2) demeans dependent and independent variables within opening year $\times$ near-ring count in $g - 1 \times$ far-ring tier in $g - 1$, the strata of the location design. Treated cells of all initial exposure states. n.a.: the near-ring count in $g - 1$ is constant within a stratum.

O.4 Anchor-level controls for the cross-border design

The analysis of incumbents across prefecture borders (Section 9) builds its stacks from zones rather than cells. A *treated anchor* is a customs special supervision zone that becomes operational in year g (§M.11). The main control population is defined on the *approval* clock rather than the operating clock: a *not-yet-treated control anchor* is a newly established customs special supervision zone (§M.12) that has not yet been *approved* by $g + H$, where H is the length of the post-period window ($H = 3$ throughout). The approval clock is used because approval itself triggers construction and investment promotion at a site: a control defined as “not yet *operating*” by $g + H$ can already have been approved and be under active development within the event window, so its neighborhood may already be responding in anticipation. A candidate control anchor is further excluded if it is an upgraded, retained, renamed, or revoked special-zone designation rather than a first customs-supervision event, or if another customs special supervision zone is already operating within 20 km of it by $g + H$, since in either case the site’s neighborhood is not a clean not-yet-treated observation.

O.5 Alternative control pools

Because the choice of control population is itself a design decision, the cross-border analysis reports the main not-yet-treated pool alongside three alternative pools, each of which relaxes or tightens one element of the definition: a pool of anchors whose *operation* (rather than

approval) is later than $g + H$, drawn from any customs special supervision zone type; the intersection of that pool with EPZs only; and a pool restricted to anchors that have been *approved* but are not yet operating by $g + H$, which serves as a direct test for anticipation effects in the main pool. Most of the control units in the broadest pool are anchored to the same control zones as the main pool, so the two are not independent checks of one another, and the EPZ-only pool typically has too few anchors to be balanced by reweighting; we treat agreement between the main pool and the broadest not-yet-treated pool as informative, and the EPZ-only and approved-but-not-operating pools mainly as diagnostics rather than as equally credible alternative point estimates. Table 69 and Table 39 report how the headline estimates move across pools. In the anchor design, an incumbent firm is dropped from a stack if a customs special supervision zone other than its assigned anchor becomes operational within 40 km of it during the event window $[g - 3, g + H]$, and exposure to special zones farther away enters as a time-varying control: the number of other special zones already operating within 0–20, 20–40, 40–80, and 80–150 km of the firm, updated year by year.

O.6 Fixed effects

In the location design, treated and control cells are compared only within the same sub-experiment and calendar year. Equation (5) nests cell-by-sub-experiment effects α_{sc} and sub-experiment-by-year effects δ_{st} , so every common shock to locations that started from the same exposure state in the same year is absorbed, and is estimated as a stacked (quasi-)Poisson regression, weighted following Wing et al. (2024) so that coefficients are proportional effects and zero outcomes are retained rather than dropped (Wooldridge 2023; Chen and Roth 2024).

O.7 Odds reweighting

Because treated and not-yet-treated locations are not randomly assigned—denser, more coastal locations closer to national development zones are systematically more likely to receive a zone—controls are reweighted by the estimated odds of treatment. For a control unit with pre-period covariates X_i ,

$$w_i = \begin{cases} 1, & T_i = 1 \\ \frac{\hat{p}_i}{1 - \hat{p}_i}, & T_i = 0, \end{cases} \quad \hat{p}_i = \Lambda(\gamma_s + X_i' \delta), \quad (13)$$

where $\hat{p}_i$ is fit by a ridge-penalized logistic regression (ridge parameter 10^{-3} , iteratively reweighted least squares) with stack indicators γ_s , and control weights are trimmed at the

99th percentile of the control weight distribution to limit the influence of any single poorly matched control. The covariates of the location design are listed in §O.2; the anchor design uses anchor-level characteristics (the number of above-scale firms within 40 km, a coastal indicator, and the distance to the nearest national development zone). After reweighting, balance is assessed by the standardized difference between treated and (weighted) control means for each covariate (Table 4), and a control pool whose effective (Kish) sample size collapses after reweighting is flagged as an extrapolation rather than a matched comparison.

O.8 Common support

Because the control pools of §O.5 do not always offer a control in every stack, a treated unit is included in a comparison across pools only if its stack contains at least one control unit under *every* pool being compared, so that cross-pool comparisons differ because of the pools rather than because of sample composition.

O.9 Input–output linkage measures

The fence’s national industry mix—the share of each 2002 input–output sector in the exports of firms legally registered inside an EPZ, pooled over all fenced exports of 2000–2006 and identified by the fifth digit of the customs registration code (§M.6)—and the upstream-supplier, downstream-user, and electronics industry groups built from it by backward and forward linkage intensity are defined together with the resulting estimates in §H.3. The national mix, rather than each zone’s own mix, fixes the groups before any zone opens; the same groups, collapsed within each cell and year because a fully disaggregated Poisson does not converge, are used unchanged in the 2005 reform design below (§O.11).

O.10 Supply base and fence activity

The pre-opening supply base used to split treated and control cells as thick or thin—the share of the cell’s above-scale employment in upstream-supplier industries in the year before the zone opens ($e = -1$)—is defined in §H.3, together with the placebo checks on that split. For each zone, fence activity is average annual exports of its fenced firms in operating years 0–3 (through 2006) divided by the base-year output of treated incumbents within 20 km, converted to US dollars at 8.28 RMB per dollar; zones are grouped as empty (below 0.1 percent), small (0.1–2 percent), and large (above 2 percent), with zones whose fenced firms cannot be assigned from customs records forming a separate group. The approved-area share is a zone’s approved area divided by the sum of the approved areas of all development zones

within 20 km that existed in the year before opening plus the zone itself, measuring the capacity committed to the fence relative to the local stock at the time of approval. The domestic value-added ratio of fenced exports (§M.8) bounds the fence’s local intermediate sourcing from above, because domestic value added also includes wages, rents, and profits paid inside the fence.

O.11 The 2005 carry-over reform: design

The rule change is described in Appendix L (§L.5). Treated cells are 0.1° grid cells within 20 km of an EPZ that was operating by 2004, excluding the two pilot zones (Kunshan and Songjiang); control cells have no special customs zone of any type operating within 40 km before 2008 but have one by 2013, the same window logic as the not-yet-treated controls of §O.2. The sample is the firm survey for 1999–2007 under the uniform size threshold and the 2004 rule of §M.2. Industry groups are downstream users, upstream suppliers (excluding downstream users), electronics, and other, as in §O.9. For outcome y in cell c , industry group k , and year t ,

$$E[y_{ckt}] = \exp \left(\alpha_{ck} + \delta_{ct} + \mu_{kt} + \sum_{k \in \{D, U, H\}} \beta_k T_c \text{Post}_t \mathbb{I}[k] \right), \quad \text{Post}_t = \mathbb{I}[t \geq 2005], \quad (14)$$

estimated by Poisson pseudo-maximum likelihood, with cell $\times$ year effects absorbing all location-level shocks (including the zone’s own ramp-up) and group $\times$ year effects absorbing national industry trends; β_D is the response of downstream users relative to other industries near operating zones, relative to the same difference in control cells. Because 2005 is a transition year (the rule took effect in May), a version drops it; the event-study version replaces Post_t with year indicators, with 2004 as the reference year. The placebo treats cells within 20 km of a 1990s bonded zone and not within 40 km of an EPZ before 2007, since Order No. 126 did not apply to bonded zones; the pilot design treats cells near Kunshan and Songjiang with 2003 as the first post-reform year and 2002 as the reference year, though with two zones it cannot separate the rule from other changes specific to those locations.

O.12 Inference

Standard errors are clustered by prefecture throughout, on the grounds that the shocks relevant to this paper’s design—a zone’s approval, its operation, and the local conditions that make a location more or less likely to receive one—are correlated within a prefecture and largely independent across prefectures; Table 39 reports clustering by zone as an alternative, since a 20 km circle can span prefectures.

P Sample Flow: Which EPZs Enter Each Analysis

The paper uses four overlapping samples of EPZs. The *location sample* (Sections 5 to 8) contains the zones that anchor at least one treated grid cell in the main location design, that is, zones that were the first special customs zone within 40 km of at least one cell holding an above-scale firm, and that opened by 2007. The *ledger sample* (Section 10) further requires that the zone’s fenced exports can be matched in customs records. The *cross-border sample* (Section 9) comes from the anchor design (§O.4), which excludes zones in the four province-level municipalities and needs incumbent firms on both sides of a prefecture border. The last column records whether the zone’s fenced exports are matched at all.

Table 65 counts the zones by approval batch and hub status. Of the 60 EPZs, 39 are in the location sample; 13 are not because every cell near them was already exposed to an operating special customs zone, 7 because they opened after the firm survey ends, and 1 because its directory entry duplicates another zone. Of the location-sample zones, 15 have no matched fenced exports, which leaves 24 in the ledger sample. The five hubs—Kunshan, Suzhou Industrial Park, Wuxi, Hangzhou, and Songjiang, which together account for most fenced exports—are in both the location and the ledger samples (5 and 5 of 5), and 4 of them are in the cross-border sample; Songjiang drops out of it because Shanghai is a municipality. The entry decline and the fence’s exports are therefore measured on overlapping sets of zones, not on disjoint ones. Table 66 lists every zone.

Table 66: All 60 EPZs and their sample membership

Zone	City	Approval year	Opening year	Hub	Entry sample	Ledger sample	Border sample	Fence export
Beijing Tianzhu EPZ	Beijing	2000	2001		Y	Y	N	Y
Sichuan Chengdu EPZ (incl. West Area)	Chengdu	2000	2002		Y	Y	Y	Y
Liaoning Dalian EPZ	Dalian	2000	2001		N	N	Y	Y
<i>every candidate cell already had an operating special customs zone within 40 km at opening</i>								
Guangzhou EPZ	Guangzhou	2000	2002		N	N	Y	Y
<i>every candidate cell already had an operating special customs zone within 40 km at opening</i>								
Zhejiang Hangzhou EPZ	Hangzhou	2000	2001	Y	Y	Y	Y	Y
Shanghai Songjiang EPZ (incl. Area B)	Shanghai	2000	2001	Y	Y	Y	N	Y
Guangdong Shenzhen EPZ	Shenzhen	2000	2001		N	N	Y	Y
<i>every candidate cell already had an operating special customs zone within 40 km at opening</i>								
Jiangsu Kunshan EPZ	Suzhou	2000	2001	Y	Y	Y	Y	Y

Zone	City	Approval year	Opening year	Hub	Entry sample	Ledger sample	Border sample	Fence export
Jiangsu Suzhou Industrial Park EPZ (incl. Area B)	Suzhou	2000	2001	Y	Y	Y	Y	Y
Tianjin EPZ	Tianjin	2000	2001		N	N	N	N
<i>every candidate cell already had an operating special customs zone within 40 km at opening</i>								
Shandong Weihai EPZ	Weihai	2000	2001		Y	Y	Y	Y
Hubei Wuhan EPZ	Wuhan	2000	2001		Y	Y	Y	Y
Fujian Xiamen EPZ	Xiamen	2000	2002		N	N	Y	Y
<i>every candidate cell already had an operating special customs zone within 40 km at opening</i>								
Jilin Hunchun EPZ	Yanbian	2000	2002		Y	Y	N	Y
Shandong Yantai EPZ (incl. Area B)	Yantai	2000	2002		Y	Y	Y	Y
Chongqing EPZ	Chongqing	2001	2003		Y	N	N	N
<i>no customs record of fenced exports could be matched to this zone</i>								
Jinqiao EPZ (duplicate entry)	Shanghai	2001	2002		N	N	N	N
<i>duplicate directory entry (the name dates to a 1990 development-zone designation); excluded</i>								
Shanghai Jinqiao EPZ (South Area)	Shanghai	2001	2003		N	N	N	N
<i>every candidate cell already had an operating special customs zone within 40 km at opening</i>								
Inner Mongolia Hohhot EPZ	Hohhot	2002	2008		N	N	N	N
<i>opened after the 1998–2007 firm-survey window</i>								
Jiangsu Nantong EPZ	Nantong	2002	2004		Y	Y	Y	Y
Zhejiang Ningbo EPZ	Ningbo	2002	2004		N	N	Y	Y
<i>every candidate cell already had an operating special customs zone within 40 km at opening</i>								
Hebei Qinhuangdao EPZ	Qinhuangdao	2002	2004		Y	Y	Y	Y
Anhui Wuhu EPZ	Wuhu	2002	2003		Y	Y	Y	Y
Jiangsu Wuxi EPZ	Wuxi	2002	2003	Y	Y	Y	Y	Y
Shaanxi Xi'an EPZ	Xi'an	2002	2004		Y	Y	Y	Y
Henan Zhengzhou EPZ	Zhengzhou	2002	2004		Y	Y	Y	Y
Guangxi Beihai EPZ	Beihai	2003	2005		Y	Y	Y	Y
Zhejiang Jiaxing EPZ	Jiaxing	2003	2005		Y	N	Y	N
<i>no customs record of fenced exports could be matched to this zone</i>								
Jinan EPZ	Jinan	2003	2006		Y	Y	Y	Y
Jiangsu Lianyungang EPZ	Lianyungang	2003	2004		Y	Y	Y	Y
Jiangsu Nanjing EPZ (incl. South Area)	Nanjing	2003	2005		Y	Y	Y	Y
Shandong Qingdao EPZ	Qingdao	2003	2004		Y	Y	Y	Y
Shanghai Caohejing EPZ	Shanghai	2003	2004		N	N	N	Y

Zone	City	Approval year	Opening year	Hub	Entry sample	Ledger sample	Border sample	Fence export
<i>every candidate cell already had an operating special customs zone within 40 km at opening</i>								
Shanghai Qingpu EPZ	Shanghai	2003	2004		N	N	N	Y
<i>every candidate cell already had an operating special customs zone within 40 km at opening</i>								
Shenyang EPZ	Shenyang	2003	2007		Y	N	Y	N
<i>no customs record of fenced exports could be matched to this zone</i>								
Jiangsu Suzhou New & Hi-Tech Zone EPZ	Suzhou	2003	2005		N	N	Y	Y
<i>every candidate cell already had an operating special customs zone within 40 km at opening</i>								
Xinjiang Urumqi EPZ	Urumqi	2003	2005		Y	Y	Y	Y
Weifang EPZ	Weifang	2003	2006		Y	N	Y	N
<i>no customs record of fenced exports could be matched to this zone</i>								
Jiangsu Zhenjiang EPZ	Zhenjiang	2003	2004		Y	Y	Y	Y
Jiangsu Changzhou EPZ	Changzhou	2005	2007		Y	N	Y	N
<i>no customs record of fenced exports could be matched to this zone</i>								
Hunan Chenzhou EPZ	Chenzhou	2005	2008		N	N	N	N
<i>opened after the 1998–2007 firm-survey window</i>								
Fujian Fuqing EPZ	Fuzhou	2005	2007		Y	N	Y	N
<i>no customs record of fenced exports could be matched to this zone</i>								
Fujian Fuzhou EPZ	Fuzhou	2005	2007		N	N	Y	N
<i>every candidate cell already had an operating special customs zone within 40 km at opening</i>								
Guangdong Nansha EPZ	Guangzhou	2005	2007		Y	N	Y	N
<i>no customs record of fenced exports could be matched to this zone</i>								
Guangdong Huizhou EPZ	Huizhou	2005	2007		Y	N	Y	N
<i>no customs record of fenced exports could be matched to this zone</i>								
Jiangxi Jiujiang EPZ	Jiujiang	2005	2006		Y	Y	Y	Y
Yunnan Kunming EPZ	Kunming	2005	2008		N	N	N	N
<i>opened after the 1998–2007 firm-survey window</i>								
Hebei Langfang EPZ	Langfang	2005	2007		Y	N	Y	N
<i>no customs record of fenced exports could be matched to this zone</i>								
Sichuan Mianyang EPZ	Mianyang	2005	2007		Y	N	Y	N
<i>no customs record of fenced exports could be matched to this zone</i>								
Zhejiang Cixi EPZ	Ningbo	2005	2007		Y	N	Y	N
<i>no customs record of fenced exports could be matched to this zone</i>								
Fujian Quanzhou EPZ	Quanzhou	2005	2007		Y	N	Y	N
<i>no customs record of fenced exports could be matched to this zone</i>								
Shanghai Jiading EPZ	Shanghai	2005	2007		N	N	N	N
<i>every candidate cell already had an operating special customs zone within 40 km at opening</i>								

Zone	City	Approval year	Opening year	Hub	Entry sample	Ledger sample	Border sample	Fence export
Liaoning Shenyang (Zhangshi) EPZ	Shenyang	2005	2007		Y	N	Y	N
<i>no customs record of fenced exports could be matched to this zone</i>								
Jiangsu Changshu EPZ	Suzhou	2005	2007		N	N	Y	N
<i>every candidate cell already had an operating special customs zone within 40 km at opening</i>								
Jiangsu Wujiang EPZ	Suzhou	2005	2007		Y	N	Y	N
<i>no customs record of fenced exports could be matched to this zone</i>								
Jiangsu Wuzhong EPZ	Suzhou	2005	2008		N	N	N	N
<i>opened after the 1998–2007 firm-survey window</i>								
Jiangsu Yangzhou EPZ	Yangzhou	2005	2007		Y	N	Y	N
<i>no customs record of fenced exports could be matched to this zone</i>								
Shandong Qingdao West Coast EPZ	Qingdao	2006	2008		N	N	N	N
<i>opened after the 1998–2007 firm-survey window</i>								
Anhui Hefei EPZ	Hefei	2010	2012		N	N	N	N
<i>opened after the 1998–2007 firm-survey window</i>								
Jiangxi Jinggangshan EPZ	Ji'an	2011	2013		N	N	N	N
<i>opened after the 1998–2007 firm-survey window</i>								

Notes: Opening year: official customs acceptance (acceptance in the second half of a year counts from the next year), otherwise approval year plus two. Y: in the sample; N: not in the sample. An italic line under a zone gives the reason it is outside the entry or ledger sample. The duplicate directory entry for Jinqiao is kept in the list for completeness; its South Area is a separate zone.

Q Factor Prices

Q.1 Wages

The city-level average wage is the prefecture's total manufacturing payroll divided by its total employment in the firm survey (payable wages over the number of employees), computed separately for all firms and for firms above the uniform size threshold; it is an employment-weighted average rather than an average of firm-level wages. Municipalities are treated as single cities. The series is nominal and covers 1998–2007. A second series, the average wage of staff and workers from the city statistical yearbooks (whole-prefecture scope, 2003–2019), is used as a check.

Two gaps limit what can be measured. The payroll variable is missing for every firm in

Table 65: EPZs in each analysis sample, by approval batch and hub status

	Total	Entry	Ledger	Border	Fence export
		sample	sample	sample	matched
<i>By approval batch (State Council approval year)</i>					
2000	15	10	10	11	14
2001	3	1	0	0	0
2002	8	6	6	7	7
2003	13	10	7	11	10
2005	18	12	1	14	1
2006	1	0	0	0	0
2010	1	0	0	0	0
2011	1	0	0	0	0
<i>By hub status</i>					
Hub (5 named zones)	5	5	5	4	5
Non-hub (55 zones)	55	34	19	39	27
All 60 EPZs	60	39	24	43	32

Notes: Number of EPZs. Entry sample: zones anchoring at least one treated grid cell in the main location design. Ledger sample: entry-sample zones with matched fenced exports. Border sample: zones in the anchor design of the cross-border analysis. Fence export matched: zones whose fenced exports are matched in customs records. Approval years follow the State Council documents; the directory’s year for Caohejing (1992) belongs to its parent zone and is corrected to 2003. Hubs: the five zones with the largest fenced exports.

2008 and 2009, so the firm survey cannot measure wages after 2007; and the postcode field used for sub-city geography is missing in 2000 and 2008–2010. Wage effects for zones that opened after 2007 are therefore measurable only in yearbook data at the city level.

Q.2 Industrial land prices

Source. Land prices come from parcel-level transaction records of the China Land Market Network, 2000–2021, with price, area, use, transfer channel, coordinates, floor-area ratio, land grade, and lease term. A parcel is industrial if its stated use is industrial. The price outcome is the log price per hectare of industrial parcels transferred through market channels; commercial and residential parcels are measured separately. Parcels priced more than fifty times the median of their city, year, and use are dropped.

Quality adjustment. Because parcels enter the data only when they are sold, the average transaction price mixes changes in the price of comparable land with changes in which land

the government chooses to sell. We residualize log prices on parcel characteristics,

$$\ln p_j = \gamma_1 \ln \text{area}_j + \gamma_2 \text{FAR}_j + \gamma_3 \text{grade}_j + \gamma_4 \text{lease}_j + \gamma_5 \text{newquota}_j + \delta_{\text{year}(j)} + u_j, \quad (15)$$

with year fixed effects only, so that spatial price differences, which carry the treatment effect, are not absorbed. The residual $\hat{u}_j$ is the constant-quality log price, aggregated to cell $\times$ year. Standard errors in the second stage are not corrected for estimation of γ .

Coverage. Disclosure of land transactions became mandatory with the tender, auction, and listing rules of 2007: the number of cities with priced industrial transactions rises from 227 in 2006 to 340 in 2007. Near zones approved before 2007, coverage jumps from under a fifth before approval to more than two-thirds after, which would bias any comparison that spans the break. Land-price comparisons therefore use transactions from 2007 onward. All five parcel characteristics are observed for 29 percent of paid industrial parcels (167,863 of 574,400), so the quality-adjusted sample is narrower than the raw price sample. Coordinates for most parcels are address-level, which supports comparisons at kilometer scale but not finer.

R Model Derivations

R.1 Admission under capacity rationing

Applicants to a development zone arrive at rate N with priority index $x \sim F$, which rises with project size. The zone admits projects in decreasing order of x until its $\bar{K}$ slots are filled, so the admission threshold solves $N[1 - F(\bar{x})] = \bar{K}$. When the fence occupies m_z slots, a share $m_z/\bar{K}$ of capacity, the threshold $\bar{x}'$ solves $N[1 - F(\bar{x}')] = \bar{K} - m_z$ and admissions fall from $\bar{K}$ to $\bar{K} - m_z$. A share λ of the refused projects locates elsewhere in the prefecture, so domestic entry into the prefecture falls by $(1 - \lambda)m_z$, a proportional loss $\rho = (1 - \lambda)m_z/\bar{K}$, which equals the slot share $m_z/\bar{K}$ when $\lambda = 0$. The projects lost are those with $x \in [\bar{x}, \bar{x}']$; because the index rises with project size, they are mostly the smallest admitted projects, and the mean size of admitted projects rises. Neither $\bar{x}'$ nor ρ depends on the fence's output Q_z . Because industrial land is granted at administered prices, the adjustment is in the threshold, not in the price of land; with elastically supplied migrant labor, wages need not move either. Incumbents hold their plots and are not affected by the threshold.

Under factor-price competition instead, the fence raises local wages and land rents in proportion to its factor demand, and entry falls by $d \ln E = -\zeta (\alpha_L d \ln w + \alpha_T d \ln r)$, which scales with Q_z and requires prices to rise, and which affects incumbents as much as entrants.

The data distinguish the two through the dependence on Q_z (empty versus busy fences), the response of prices, the size composition of the lost entrants, and the response of incumbents.

From the Poisson coefficient to ρ . Let an entrant choose cell c in development-zone network m with conditional-logit probability proportional to the slots open in c . [Guimarães et al. \(2003\)](#) show that the conditional-logit likelihood equals a Poisson likelihood for cell counts with network-by-year fixed effects, so the coefficient on treated cells is $\beta = \ln(1 - \rho)$. We report $1 - e^\beta$ as the proportional loss of entry ρ . With $\lambda = 0$ it is also the share of slots that would generate the observed loss under the model, $m_z/\bar{K}$; it is not a direct measurement of land. If crowded-out entrants instead chose other cells in the same network ($\lambda > 0$), those cells would gain; untreated cells in treated cities show no such gain (Appendix D), which is the evidence that λ is close to zero.

R.2 What the fence leaves locally

Fenced output Q_z uses imported inputs M_z (in bond), local inputs $s_z Q_z$, and local factors paid $\sigma_z Q_z$, with the remainder accruing to foreign owners. Customs records give $1 - M_z/Q_z = \text{DVAR}_z$, which contains local inputs, local factor payments, and foreign profits, so $s_z + \sigma_z \leq \text{DVAR}_z$. Local inputs are a CES aggregate of n_c symmetric varieties with elasticity $\epsilon > 1$, each produced with fixed cost F_s ; with free entry $n_c = D_c/(\epsilon F_s)$, where D_c is local spending on intermediates ([Rodríguez-Clare 1996](#)). The fence raises D_c by $s_z Q_z$, so upstream output rises in proportion to $s_z Q_z/D_c$, and the value added it generates is $\mu s_z Q_z$, with μ the value-added share of local suppliers. The carry-over reform of 2005 allows the fence to sell components to local processors; it raises local value added only if those processors can use the components, that is, only if it raises the effective forward counterpart of s_z .

R.3 The net effect and break-even scale

Summing the two sides gives equation (4). The cost $C(m_z)$ is the value added the lost entrants would have produced; with an entry flow lost at rate ρE_c per year and an age profile of value added v_a and survival S_a , the cost H years after opening is $\rho \sum_{k=0}^H E_{c,k} S_{H-k} v_{H-k}$. It is fixed once the fence occupies its slots. The return is linear in Q_z with slope $\sigma_z + \mu s_z$, so $\Delta VA_z > 0$ if and only if $Q_z > Q_z^*$. At the upper bound $\sigma_z = \text{DVAR}_z$, $s_z = 0$, the ledger is most favorable to the zone; setting σ_z to the fence's wage bill per unit of exports gives a central case. The model is the counterpart of Proposition 1 of [Rodríguez-Clare \(1996\)](#) with capacity in place of labor: the zone deepens the local economy only if what it leaves exceeds what the displaced local activity would have left.

R.4 Estimating the admission model

The estimable model, the moments, and the identification argument are in Section 10.1. The priority index is $x = a + u$, with size a the normal score of birth output (main specification) or birth fixed assets (alternative), ranked within two-digit industry and birth year among all domestic entrants nationally, so the national quintiles are the fixed intervals $B_q = [\Phi^{-1}(\frac{q-1}{5}), \Phi^{-1}(\frac{q}{5})]$. The probabilities in equation (9) are bivariate normal rectangle probabilities, computed exactly. The second size measure enters the out-of-sample tests through its joint distribution with the first among pre-opening entrants, estimated directly; selection acts only on the queued measure, so the conditional distribution of the other is the same among applicants and entrants. Estimation is by minimum distance with diagonal weights; the overidentification statistic uses the full covariance of the moments, with the covariance of the quintile effects approximated by the correlation between the bottom and top quintiles. Inference is by 499 bootstrap replications over zones, each re-estimating the model on resampled pre-opening entrants and a draw of the quintile effects from their sampling distribution. The share of applicants admitted before opening is not identified. In the main specification, along its range from 0.05 to 0.95 the objective varies by at most 0.9, the entry loss stays between 19 and 20 percent, the queue noise σ_u ranges from 0.94 to 1.39, the threshold shift Δ from 0.25 to 1.60, and the correlation of the priority index with size from 0.62 to 0.92. Quantities that depend on the admission rate are reported over this range.

R.5 Portability

A firm serves destination d if its capability for d exceeds an entry cost, where capability accumulates as $K_{fd} = \sum_r a_r D_{fd}^r + \sum_r b_r S_{fd}^r$. Normalizing by ordinary trade, the weight $\omega_r = b_r/b_O$ measures how much experience in form r with similar destinations substitutes for ordinary-trade experience. Under consigned processing, the foreign principal holds the buyer relationship and assigns the market, so the processor's experience in destination d' conveys information about the principal's buyers, not about demand in markets similar to d' ; the model predicts $\omega_A \approx 0$ for firms that do not control the transaction and $\omega_I > 0$ for firms that do.

S Additional Tables

Table 68: The admission model: estimates, fit, and out-of-sample tests

	(1)	(2)	(3)	(4)	(5)
	Model			p-value of the gap	
	Fixed-asset index	Output index	Data	(1) vs (3)	(2) vs (3)
<i>Panel A. Parameters (admission rate before opening normalized to one half)</i>					
Dispersion of applicant size, s	1.16	1.45			
	[0.96, 1.44]	[1.09, 1.93]			
Ranking noise, σ_u	1.44	1.06			
	[0.58, 6.78]	[0.64, 1.98]			
Threshold before opening, k_0	-0.63	-0.74			
	[-1.11, -0.16]	[-1.20, -0.24]			
Threshold shift, Δ	0.47	0.44			
	[0.01, 2.10]	[0.19, 0.87]			
Threshold shift, log points of size	0.72	0.42			
	[0.01, 3.19]	[0.18, 0.83]			
Entry loss, ρ	0.200	0.193	0.197		
	[0.005, 0.366]	[0.079, 0.322]	[0.001, 0.366]		
range over admission rates 0.05–0.95	[0.197, 0.207]	[0.186, 0.198]			
range of σ_u over admission rates 0.05–0.95	[1.01, 3.17]	[0.94, 1.39]			
<i>Panel B. Fit to the targeted moments</i>					
Objective, 9 moments, diagonal weights	6.90	2.83			
Overidentification statistic, 5 df	12.75	8.39			
p-value	0.026	0.136			
Largest studentized gap, targeted moments	1.32	1.00			
<i>Panel C. Out-of-sample predictions against the data</i>					
Output quintile 1 (smallest)	-0.305	targeted	-0.572	0.246	
	[-0.565, -0.009]		(0.187)		
Output quintile 5 (largest)	-0.158	targeted	0.011	0.241	
	[-0.377, -0.003]		(0.101)		
Output, top minus bottom	0.147	targeted	0.583	0.024	
	[0.000, 0.247]		(0.184)		
Fixed assets, top minus bottom	targeted	0.217	0.337		0.511
		[0.073, 0.347]	(0.170)		
Surviving entrants, mean log output	0.059	0.174	0.174	0.133	0.999
	[0.000, 0.099]	[0.059, 0.280]	(0.073)		
Land-intensive minus other industries	0.004	0.005	0.176	0.102	0.106
	[-0.010, 0.018]	[-0.009, 0.025]	(0.105)		
ranking on a common scale across industries	0.038	0.017	0.176	0.197	0.136
	[0.004, 0.076]	[-0.008, 0.051]	(0.105)		
Empty fences	-0.205	-0.229	-0.280	0.770	0.835
	[-0.439, -0.005]	[-0.416, -0.079]	(0.233)		
Large fences	-0.220	-0.192	-0.400	0.486	0.392
	[-0.449, -0.006]	[-0.377, -0.071]	(0.230)		
Joint test, five output quintiles (p-value)				0.115	
Joint test, five fixed-asset quintiles (p-value)					0.121
Joint test, three fence-activity groups (p-value)				0.871	0.843

Notes: Column (2) is the main specification (the priority index follows birth output); column (1) is the alternative (it follows birth fixed assets). Panel A: estimates with 95 percent bootstrap intervals in brackets, the admission rate before opening normalized to one half; the last two rows give the range of ρ and σ_u as that rate varies from 0.05 to 0.95. The capacity-share row is the entry loss ρ ; with $\lambda \approx 0$ (Section 6) it equals the slot share $m_z/\bar{K}$. Panel B: fit to the nine targeted moments. Panel C: predictions for moments the model was not fitted to, against the reduced-form estimates (standard errors in parentheses), and the p -value of the gap; “targeted” marks moments used in estimation.

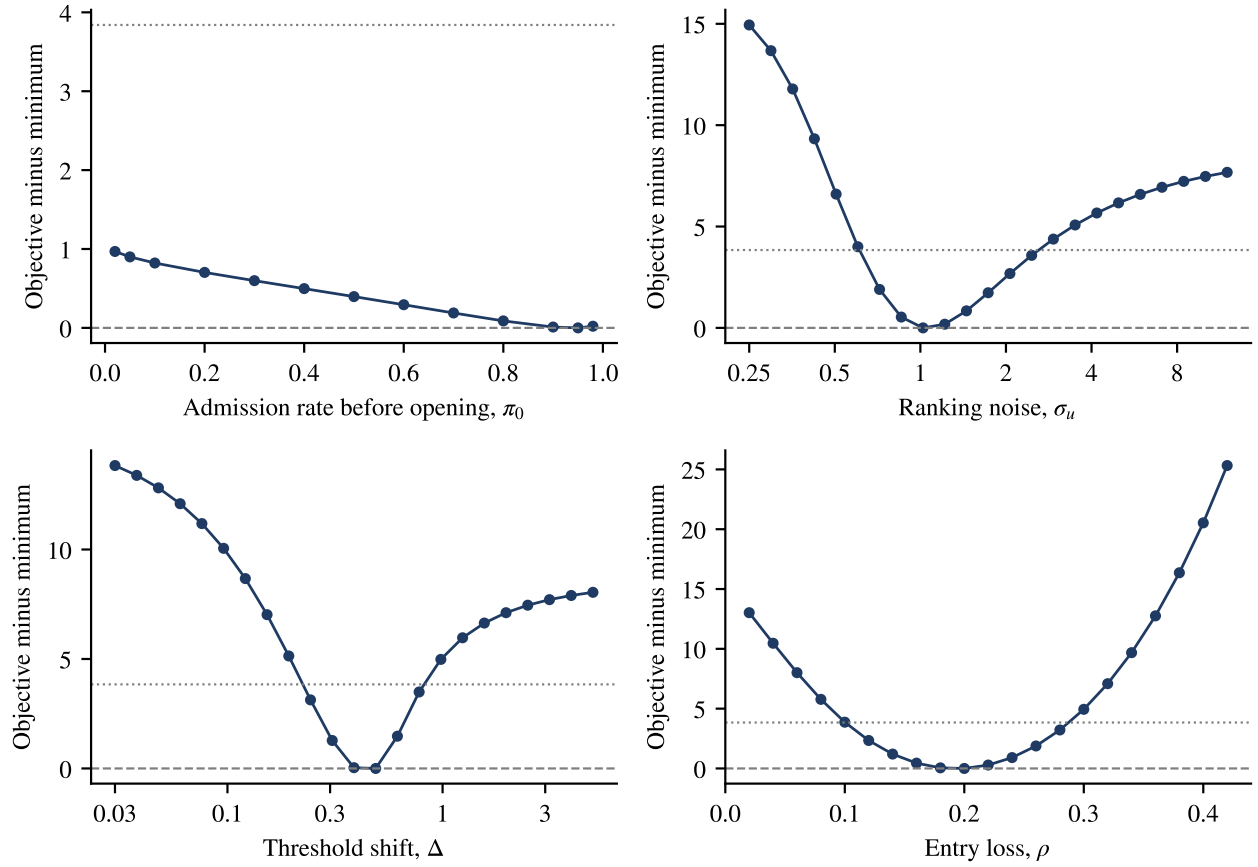


Figure 27: Profiles of the minimum-distance objective

Notes: Objective function of the main specification, in which the priority index follows birth output (Table 68, column 2), minimized over the other parameters, as each parameter is fixed on a grid. The horizontal line marks the approximate 95 percent threshold. The entry loss, the queue noise, and the threshold shift are bounded on both sides; the admission rate before opening is flat over its whole range and is not identified.

Table 69: Location-choice estimates of the entry capacity assigned to the fence

	(1)	(2)	(3)
	Parent-zone matched	Future EPZ sites	All eventual sites
<i>Panel A. Domestic entrants</i>			
ρ : share of entry capacity assigned to the fence	0.352 [0.150, 0.506]	0.348 [0.174, 0.485]	0.111 [−0.107, 0.285]
$e^\theta - 1$: same-city untreated locations	−0.064 [−0.348, 0.343]	−0.136 [−0.356, 0.160]	−0.119 [−0.363, 0.217]
ρ , empty fences	0.430 [0.064, 0.653]	0.444 [0.129, 0.645]	0.254 [−0.143, 0.513]
ρ , small fences	0.414 [0.152, 0.595]	0.392 [0.197, 0.539]	0.155 [−0.035, 0.311]
ρ , large fences	0.353 [0.129, 0.519]	0.324 [0.110, 0.486]	0.068 [−0.564, 0.444]
<i>Panel B. Foreign and HMT entrants</i>			
ρ : share of entry capacity assigned to the fence	0.277 [−0.030, 0.492]	0.192 [−0.071, 0.390]	0.020 [−0.340, 0.284]

Notes: Poisson regressions of entrant counts on treated cells with development-zone network $\times$ year fixed effects, which are equivalent to a conditional logit of entrants' location choice (Guimarães et al. 2003); $\rho = 1 - e^\beta$ (Appendix R.1). Column (1): controls are cells in parent development zones of future special zones; column (2): cells near future EPZ sites; column (3): cells near all sites that eventually receive any special customs zone. $e^\theta - 1$ is the change in entry at untreated cells in treated cities. Fence activity groups as in Table 26. 95 percent confidence intervals in brackets, clustered by prefecture.